

```

1 classdef FUNMAP_Input < matlab.apps.AppBase
2
3     % Properties that correspond to app components
4     properties (Access = public)
5         UIFigure                                matlab.ui.Figure
6         Friction_____Label_9    matlab.ui.control.Label
7
8         Panel_32                                matlab.ui.container.Panel
9         Button_41                                matlab.ui.control.Button
10        Mpif90EditField                        matlab.ui.control.EditField
11        Mpif90EditFieldLabel                    matlab.ui.control.Label
12        DoublePrecisionCheckBox                matlab.ui.control.CheckBox
13        IntelCompilerCheckBox                  matlab.ui.control.CheckBox
14        ParallelModeCheckBox                    matlab.ui.control.CheckBox
15        MakefileEditField                      matlab.ui.control.EditField
16        MakefileEditFieldLabel                  matlab.ui.control.Label
17        Button_42                                matlab.ui.control.Button
18        StartsimulationButton                    matlab.ui.control.Button
19        CreateMakefileButton                    matlab.ui.control.Button
20        TabGroup                                matlab.ui.container.TabGroup
21        InputTab                                matlab.ui.container.Tab
22        Friction_____Label_10    matlab.ui.control.Label
23
24        Panel_33                                matlab.ui.container.Panel
25        SteadyTimeEditField                    matlab.ui.control.NumericEditField
26        SteadyTimeEditFieldLabel                matlab.ui.control.Label
27        TimeintervalsecEditField                matlab.ui.control.NumericEditField
28        TimeintervalsecEditFieldLabel            matlab.ui.control.Label
29        UseSmagorinskyCheckBox                  matlab.ui.control.CheckBox
30        GenerateinputtxtButton                    matlab.ui.control.Button
31        Friction_____Label_7    matlab.ui.control.Label
32
33        Panel_30                                matlab.ui.container.Panel
34        C2EditField                            matlab.ui.control.NumericEditField
35        C2EditFieldLabel                        matlab.ui.control.Label
36        C1EditFieldLabel                        matlab.ui.control.Label
37        C1EditField                            matlab.ui.control.NumericEditField
38        ButtonGroup_23                          matlab.ui.container.ButtonGroup
39        ViscositybreakingButton                    matlab.ui.control.RadioButton
40        ShockwavecapturingButton                    matlab.ui.control.RadioButton
41        DimensionLabel                            matlab.ui.control.Label
42        DepthLabel                                matlab.ui.control.Label
43        Friction_____Label_6    matlab.ui.control.Label
44
45        Panel_29                                matlab.ui.container.Panel
46        SavefilestoEditField                    matlab.ui.control.EditField
47        SavefilestoLabel                        matlab.ui.control.Label
48        Button_14                                matlab.ui.control.Button
49        Friction_____Label_5    matlab.ui.control.Label
50
51        Panel_28                                matlab.ui.container.Panel
52        SpacingEditField                        matlab.ui.control.NumericEditField
53        SpacingEditFieldLabel                    matlab.ui.control.Label
54        BoundarylimitsButton                    matlab.ui.control.Button
55        WestEditField_2                          matlab.ui.control.NumericEditField
56        SouthEditField_2                        matlab.ui.control.NumericEditField
57        EastEditField_2                          matlab.ui.control.NumericEditField
58        NorthEditField_2                        matlab.ui.control.NumericEditField
59    end
60 end

```

54	ImportgaugelistsCheckBox	matlab.ui.control.CheckBox
55	LoadFileEditField	matlab.ui.control.EditField
56	Button_7	matlab.ui.control.Button
57	CreatenestedgridboundaryCheckBox	matlab.ui.control.CheckBox
58	BottomFrictionLabel	matlab.ui.control.Label
59	Panel_26	matlab.ui.container.Panel
60	ConstantFrictionValue	matlab.ui.control.NumericEditField
61	ValueEditFieldLabel	matlab.ui.control.Label
62	Button_10	matlab.ui.control.Button
63	FrictionFileEditField	matlab.ui.control.EditField
64	FrictionFileEditFieldLabel	matlab.ui.control.Label
65	ButtonGroup_8	matlab.ui.container.ButtonGroup
66	NonconstantButton	matlab.ui.control.RadioButton
67	ConstantCoefficientButton	matlab.ui.control.RadioButton
68	PhysicsLabel	matlab.ui.control.Label
69	ObstaclesLabel	matlab.ui.control.Label
70	Panel_24	matlab.ui.container.Panel
71	AddobstaclesCheckBox	matlab.ui.control.CheckBox
72	ObstacleFilename	matlab.ui.control.EditField
73	ObstacleLoadFile	matlab.ui.control.Button
74	SpongeBoundaryLabel	matlab.ui.control.Label
75	Panel_23	matlab.ui.container.Panel
76	MaximumCspEditField	matlab.ui.control.NumericEditField
77	MaximumCspEditFieldLabel	matlab.ui.control.Label
78	MaximumCdEditField	matlab.ui.control.NumericEditField
79	MaximumCdEditFieldLabel	matlab.ui.control.Label
80	DecayTypeDropDown	matlab.ui.control.DropDown
81	DecayTypeDropDownLabel	matlab.ui.control.Label
82	SouthLabel	matlab.ui.control.Label
83	WestLabel	matlab.ui.control.Label
84	EastLabel	matlab.ui.control.Label
85	NorthLabel	matlab.ui.control.Label
86	WestEditField	matlab.ui.control.NumericEditField
87	SouthEditField	matlab.ui.control.NumericEditField
88	EastEditField	matlab.ui.control.NumericEditField
89	NorthEditField	matlab.ui.control.NumericEditField
90	DiffusionCheckBox	matlab.ui.control.CheckBox
91	FrictionCheckBox	matlab.ui.control.CheckBox
92	DirectCheckBox	matlab.ui.control.CheckBox
93	LayerwidthLabel	matlab.ui.control.Label
94	GridLabel	matlab.ui.control.Label
95	Panel_22	matlab.ui.container.Panel
96	GridSizeLabel	matlab.ui.control.Label
97	SouthwestcornerLabel	matlab.ui.control.Label
98	ButtonGroup_4	matlab.ui.container.ButtonGroup
99	Button_2	matlab.ui.control.RadioButton
100	CartesianButton	matlab.ui.control.RadioButton
101	SphericalButton	matlab.ui.control.RadioButton
102	LatitudeEditField	matlab.ui.control.NumericEditField
103	LatitudeEditFieldLabel	matlab.ui.control.Label
104	LongitudeEditField	matlab.ui.control.NumericEditField
105	LongitudeEditFieldLabel	matlab.ui.control.Label
106	YEditField	matlab.ui.control.NumericEditField
107	YgridsizeLabel	matlab.ui.control.Label
108	XEditField	matlab.ui.control.NumericEditField
109	XEditFieldLabel	matlab.ui.control.Label
110	TimeLabel_2	matlab.ui.control.Label
111	Panel_21	matlab.ui.container.Panel

112	TotalEditField	matlab.ui.control.NumericEditField
113	TotalEditFieldLabel	matlab.ui.control.Label
114	SimulationIntervalEditField	matlab.ui.control.NumericEditField
115	SimulationIntervalEditFieldLabel	matlab.ui.control.Label
116	GaugeIntervalEditField	matlab.ui.control.NumericEditField
117	GaugeIntervalEditFieldLabel	matlab.ui.control.Label
118	StartEditField	matlab.ui.control.NumericEditField
119	ScreenLoadingIntervalLabel	matlab.ui.control.Label
120	PrintLabel	matlab.ui.control.Label
121	Panel_20	matlab.ui.container.Panel
122	ResultsfolderEditField	matlab.ui.control.EditField
123	ResultsfolderEditFieldLabel	matlab.ui.control.Label
124	Panel_19	matlab.ui.container.Panel
125	XEditField_2	matlab.ui.control.NumericEditField
126	XEditField_2Label	matlab.ui.control.Label
127	YEditField_2	matlab.ui.control.NumericEditField
128	YEditField_2Label	matlab.ui.control.Label
129	Panel_18	matlab.ui.container.Panel
130	AngleEditField	matlab.ui.control.NumericEditField
131	AngleEditFieldLabel	matlab.ui.control.Label
132	StartatXEditField	matlab.ui.control.NumericEditField
133	StartatXEditFieldLabel	matlab.ui.control.Label
134	DepthmEditFieldLabel	matlab.ui.control.Label
135	DepthmEditField	matlab.ui.control.NumericEditField
136	InvertvaluesCheckBox	matlab.ui.control.CheckBox
137	WaterlevelmEditFieldLabel	matlab.ui.control.Label
138	WaterlevelmEditField	matlab.ui.control.NumericEditField
139	Button_8	matlab.ui.control.Button
140	DepthFileEditField	matlab.ui.control.EditField
141	ButtonGroup_7	matlab.ui.container.ButtonGroup
142	Button4	matlab.ui.control.RadioButton
143	SlopeButton	matlab.ui.control.RadioButton
144	FlatButton	matlab.ui.control.RadioButton
145	ImportDataButton	matlab.ui.control.RadioButton
146	Friction_____Label_2	matlab.ui.control.Label
Label		
147	ParallelInfoLabel	matlab.ui.control.Label
148	Panel_17	matlab.ui.container.Panel
149	Panel_5	matlab.ui.container.Panel
150	ProcessorY	matlab.ui.control.NumericEditField
151	YLabel	matlab.ui.control.Label
152	ProcessorX	matlab.ui.control.NumericEditField
153	ProcessornumbersXLabel	matlab.ui.control.Label
154	Panel_25	matlab.ui.container.Panel
155	ButtonGroup_3	matlab.ui.container.ButtonGroup
156	FroudeCapEditField	matlab.ui.control.NumericEditField
157	FroudeCapEditFieldLabel	matlab.ui.control.Label
158	NonlinearShallowWaterEquationButton	matlab.ui.control.RadioButton
159	Button	matlab.ui.control.RadioButton
160	FullyNonlinearBoussinesqEquationButton	matlab.ui.control.RadioButton
161	LinearShallowWaterEquationButton	matlab.ui.control.RadioButton
162	Panel_27	matlab.ui.container.Panel
163	BottomFrictionEditField	matlab.ui.control.NumericEditField
164	BottomFrictionEditFieldLabel	matlab.ui.control.Label
165	WetDrySchemeEditField	matlab.ui.control.NumericEditField
166	WetDrySchemeEditFieldLabel	matlab.ui.control.Label
167	InitialConditionTab	matlab.ui.container.Tab
168	SedimentTransportCheckBox	matlab.ui.control.CheckBox

169	GenerateinputtxtButton_4	matlab.ui.control.Button
170	ButtonGroup_18	matlab.ui.container.ButtonGroup
171	VesselButton	matlab.ui.control.RadioButton
172	MeteotsunamiButton	matlab.ui.control.RadioButton
173	Button_31	matlab.ui.control.RadioButton
174	WavemakerButton	matlab.ui.control.RadioButton
175	ImportUVZButton	matlab.ui.control.RadioButton
176	SelectInitialConditonLabel	matlab.ui.control.Label
177	TabGroup2	matlab.ui.container.TabGroup
178	Tab_5	matlab.ui.container.Tab
179	Tab	matlab.ui.container.Tab
180	Button_33	matlab.ui.control.Button
181	VVelocityEditField	matlab.ui.control.EditField
182	VVelocityEditFieldLabel	matlab.ui.control.Label
183	UVelocityEditField	matlab.ui.control.EditField
184	UVelocityEditFieldLabel	matlab.ui.control.Label
185	SurfaceHeightEditField	matlab.ui.control.EditField
186	SurfaceHeightEditFieldLabel	matlab.ui.control.Label
187	Button_34	matlab.ui.control.Button
188	Button_32	matlab.ui.control.Button
189	ImportUVZfilesLabel	matlab.ui.control.Label
190	Label_9	matlab.ui.control.Label
191	Tab_2	matlab.ui.container.Tab
192	WaveHeightmEditField	matlab.ui.control.NumericEditField
193	WaveHeightmEditFieldLabel	matlab.ui.control.Label
194	MaximumEditField	matlab.ui.control.NumericEditField
195	MaximumEditFieldLabel	matlab.ui.control.Label
196	MinimumEditField	matlab.ui.control.NumericEditField
197	MinimumEditFieldLabel	matlab.ui.control.Label
198	PeakEditField	matlab.ui.control.NumericEditField
199	FrequencyPeakLabel	matlab.ui.control.Label
200	FrequencyLabel	matlab.ui.control.Label
201	WaveCompCountEditField	matlab.ui.control.NumericEditField
202	WavecomponentcountEditFieldLabel	matlab.ui.control.Label
203	WavecomponentfileEditField	matlab.ui.control.EditField
204	WavecomponentfileEditFieldLabel	matlab.ui.control.Label
205	ThetadegreesEditField	matlab.ui.control.NumericEditField
206	ThetadegreesEditFieldLabel	matlab.ui.control.Label
207	PeriodsecEditField	matlab.ui.control.NumericEditField
208	PeriodsecEditFieldLabel	matlab.ui.control.Label
209	WidthDeltaEditField	matlab.ui.control.NumericEditField
210	WidthDeltaEditFieldLabel	matlab.ui.control.Label
211	TimerampsecEditField	matlab.ui.control.NumericEditField
212	TimerampsecEditFieldLabel	matlab.ui.control.Label
213	WidthmEditField	matlab.ui.control.NumericEditField
214	WidthmEditFieldLabel	matlab.ui.control.Label
215	YcoordinatemEditField	matlab.ui.control.NumericEditField
216	YcoordinatemEditFieldLabel	matlab.ui.control.Label
217	XcoordinatemEditField	matlab.ui.control.NumericEditField
218	XcoordinatemEditFieldLabel	matlab.ui.control.Label
219	LagtimesecEditField	matlab.ui.control.NumericEditField
220	LagtimesecEditFieldLabel	matlab.ui.control.Label
221	WaterdepthmEditField	matlab.ui.control.NumericEditField
222	WaterdepthmEditFieldLabel	matlab.ui.control.Label
223	AmplitudemEditField	matlab.ui.control.NumericEditField
224	AmplitudemEditFieldLabel	matlab.ui.control.Label
225	WavemakertypeDropDown	matlab.ui.control.DropDown
226	WavemakertypeDropDownLabel	matlab.ui.control.Label

227	Button_20	matlab.ui.control.Button
228	WavemakerParametersLabel_2	matlab.ui.control.Label
229	Label_10	matlab.ui.control.Label
230	Tab_4	matlab.ui.container.Tab
231	WindLabel	matlab.ui.control.Label
232	PressureLabel	matlab.ui.control.Label
233	Button_43	matlab.ui.control.Button
234	PressurefileEditField	matlab.ui.control.EditField
235	PressurefileEditFieldLabel	matlab.ui.control.Label
236	StormfileEditFieldLabel	matlab.ui.control.Label
237	StormfileEditField	matlab.ui.control.EditField
238	WindwaveinteractionSwitch	matlab.ui.control.Switch
239	WindwaveinteractionSwitchLabel	matlab.ui.control.Label
240	HollandstormmodelSwitch	matlab.ui.control.Switch
241	HollandstormmodelSwitchLabel	matlab.ui.control.Label
242	ConstantwindfileEditField	matlab.ui.control.EditField
243	ConstantwindfileEditFieldLabel	matlab.ui.control.Label
244	ConstantwindfieldSwitch	matlab.ui.control.Switch
245	ConstantwindfieldSwitchLabel	matlab.ui.control.Label
246	CrestRatioEditField	matlab.ui.control.NumericEditField
247	CrestRatioEditFieldLabel	matlab.ui.control.Label
248	WindstresscoefficientEditFieldLabel	matlab.ui.control.Label
249	WindstresscoefficientEditField	matlab.ui.control.NumericEditField
250	AirpressureSwitch	matlab.ui.control.Switch
251	AirpressureSwitchLabel	matlab.ui.control.Label
252	WindforceSwitch	matlab.ui.control.Switch
253	WindforceSwitchLabel	matlab.ui.control.Label
254	Button_30	matlab.ui.control.Button
255	Button_29	matlab.ui.control.Button
256	MeteotsunamiLabel	matlab.ui.control.Label
257	Label_11	matlab.ui.control.Label
258	Tab_3	matlab.ui.container.Tab
259	AddrollereffectsCheckBox	matlab.ui.control.CheckBox
260	TimeStepSedToMorphlEditField	matlab.ui.control.NumericEditField
261	TimeStepSedToMorpEditFieldLabel	matlab.ui.control.Label
262	ButtonGroup_21	matlab.ui.container.ButtonGroup
263	Button_37	matlab.ui.control.RadioButton
264	SedimentbedButton	matlab.ui.control.RadioButton
265	HardbednoerosionButton	matlab.ui.control.RadioButton
266	BottomTypeLabel	matlab.ui.control.Label
267	Button_35	matlab.ui.control.Button
268	FileEditField	matlab.ui.control.EditField
269	FileEditFieldLabel	matlab.ui.control.Label
270	RungeKuttaparam2EditField_2	matlab.ui.control.NumericEditField
271	RungeKuttaparam2EditFieldLabel_2	matlab.ui.control.Label
272	ButtonGroup_22	matlab.ui.container.ButtonGroup
273	Button_38	matlab.ui.control.RadioButton
274	NoButton_7	matlab.ui.control.RadioButton
275	YesButton_8	matlab.ui.control.RadioButton
276	ConsideravalancheLabel	matlab.ui.control.Label
277	MorphologicalChangeLabel_2	matlab.ui.control.Label
278	AvalancheLabel	matlab.ui.control.Label
279	TimeStepBouss	matlab.ui.control.NumericEditField
280	EditFieldLabel	matlab.ui.control.Label
281	BedChangeLabel	matlab.ui.control.Label
282	ButtonGroup_20	matlab.ui.container.ButtonGroup
283	Button_36	matlab.ui.control.RadioButton
284	NobedchangeButton	matlab.ui.control.RadioButton

285	UpdatedepthButton	matlab.ui.control.RadioButton
286	MinDepthSedPickUpEditField	matlab.ui.control.NumericEditField
287	MinDepthSedPickUpEditFieldLabel	matlab.ui.control.Label
288	RungeKuttaparameter2EditField	matlab.ui.control.NumericEditField
289	RungeKuttaparameter2EditFieldLabel	matlab.ui.control.Label
290	RungeKuttaparameter1EditField	matlab.ui.control.NumericEditField
291	RungeKuttaparameter1EditFieldLabel	matlab.ui.control.Label
292	ShieldsparmBedloadEditField	matlab.ui.control.NumericEditField
293	ShieldsparmeterbedloadEditFieldLabel	matlab.ui.control.Label
294	CriticalShieldsEditField	matlab.ui.control.NumericEditField
295	CriticalShieldsEditFieldLabel	matlab.ui.control.Label
296	SettlingvelocitymsEditField	matlab.ui.control.NumericEditField
297	SettlingvelocitymsEditFieldLabel	matlab.ui.control.Label
298	SedimentporosityEditField	matlab.ui.control.NumericEditField
299	SedimentporosityEditFieldLabel	matlab.ui.control.Label
300	SedimentdensityEditField	matlab.ui.control.NumericEditField
301	SedimentdensityEditFieldLabel	matlab.ui.control.Label
302	MediagrainDMEditField	matlab.ui.control.NumericEditField
303	MediagrainDMEditFieldLabel	matlab.ui.control.Label
304	NumericalschemeLabel	matlab.ui.control.Label
305	ButtonGroup_19	matlab.ui.container.ButtonGroup
306	TVDButton	matlab.ui.control.RadioButton
307	UpwindingButton	matlab.ui.control.RadioButton
308	SedimentLabel	matlab.ui.control.Label
309	Label_12	matlab.ui.control.Label
310	Label_5	matlab.ui.control.Label
311	Label_6	matlab.ui.control.Label
312	Tab_6	matlab.ui.container.Tab
313	SelectMethodLabel	matlab.ui.control.Label
314	ValueEditField	matlab.ui.control.NumericEditField
315	ValueEditFieldLabel_2	matlab.ui.control.Label
316	FrictionCoefficientEditField	matlab.ui.control.NumericEditField
317	FrictionCoefficientEditFieldLabel	matlab.ui.control.Label
318	ViscosityCheckBox	matlab.ui.control.CheckBox
319	FrictionCheckBox_2	matlab.ui.control.CheckBox
320	ShockcapturingCheckBox	matlab.ui.control.CheckBox
321	ActivateCheckBox	matlab.ui.control.CheckBox
322	IncludeSedimentEffectintheModelCheckBox	matlab.ui.control.CheckBox
323	MinimumClearanceEditField	matlab.ui.control.NumericEditField
324	MinimumClearanceLabel	matlab.ui.control.Label
325	DeepDraftVesselsLabel	matlab.ui.control.Label
326	Label_8	matlab.ui.control.Label
327	VesselLabel	matlab.ui.control.Label
328	Button_40	matlab.ui.control.Button
329	ImportvesselfilesTextArea	matlab.ui.control.TextArea
330	ImportVesselFilesLabel	matlab.ui.control.Label
331	Label_13	matlab.ui.control.Label
332	ExpectedOutputTab	matlab.ui.container.Tab
333	GridLayout	matlab.ui.container.GridLayout
334	ThresholdEditField	matlab.ui.control.NumericEditField
335	ThresholdEditFieldLabel	matlab.ui.control.Label
336	MaximumWaveHeightCheckBox	matlab.ui.control.CheckBox
337	MinimumWaveHeightCheckBox	matlab.ui.control.CheckBox
338	VVelocitymeanCheckBox	matlab.ui.control.CheckBox
339	UVelocitymeanCheckBox	matlab.ui.control.CheckBox
340	GenerateinputtxtButton_2	matlab.ui.control.Button
341	OthersLabel	matlab.ui.control.Label
342	TimeLabel	matlab.ui.control.Label

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343         FluxLabel                                matlab.ui.control.Label
344         MaskLabel                                matlab.ui.control.Label
345         SourceLabel                             matlab.ui.control.Label
346         WaveHeightLabel                        matlab.ui.control.Label
347         VelocityLabel                          matlab.ui.control.Label
348         UndertowCheckBox                      matlab.ui.control.CheckBox
349         MaximumVelocityCheckBox               matlab.ui.control.CheckBox
350         BreakingAgeCheckBox                   matlab.ui.control.CheckBox
351         GyFluxCheckBox                       matlab.ui.control.CheckBox
352         GxFluxCheckBox                       matlab.ui.control.CheckBox
353         FyFluxCheckBox                       matlab.ui.control.CheckBox
354         FxFluxCheckBox                       matlab.ui.control.CheckBox
355         QmomentumFluxCheckBox                 matlab.ui.control.CheckBox
356         PMomentumFluxCheckBox                 matlab.ui.control.CheckBox
357         BreakingLocationCheckBox              matlab.ui.control.CheckBox
358         YSourceCheckBox                      matlab.ui.control.CheckBox
359         PressureFieldCheckBox                 matlab.ui.control.CheckBox
360         XSourceCheckBox                      matlab.ui.control.CheckBox
361         RollerinducedFluxCheckBox             matlab.ui.control.CheckBox
362         WetdrymaskforBoussinesqNSWECheckBox  matlab.ui.control.CheckBox
363         WaveHeightCheckBox                   matlab.ui.control.CheckBox
364         WetdrymaskCheckBox                   matlab.ui.control.CheckBox
365         SurfaceElevationattimetCheckBox      matlab.ui.control.CheckBox
366         ArrivalCheckBox                      matlab.ui.control.CheckBox
367         VVelocityCheckBox                    matlab.ui.control.CheckBox
368         MaxMomentumFluxCheckBox               matlab.ui.control.CheckBox
369         UVelocityCheckBox                    matlab.ui.control.CheckBox
370         MaxVorticityCheckBox                  matlab.ui.control.CheckBox
371         DepthCheckBox                        matlab.ui.control.CheckBox
372         TabGroup3                            matlab.ui.container.TabGroup
373         LogReportTab                         matlab.ui.container.Tab
374         Preview_LogReport                    matlab.ui.control.TextArea
375         inputtxtTab                          matlab.ui.container.Tab
376         Preview_Input                       matlab.ui.control.TextArea
377         Makefile                             matlab.ui.container.Tab
378         Preview_Makefile                     matlab.ui.control.TextArea
379     end
380
381
382     properties (Access = private)
383         filename
384         path
385         inputpath
386         inputdefault
387         outputfile
388         ouputdirectoryfinal
389         SWE
390         DspongeSwitch
391         DspongeDecay
392         initialHeight
393         initialXvelocity
394         initialYvelocity
395         Mglob
396         Nglob
397         NumberStations
398         STATION_FILE
399         bathymetryinputdata
400         NestStation

```

401	WORKFOLDER
402	GAUGEFILE
403	DEPTHFILE
404	GaugeListGrids
405	GaugeFname
406	Cd_file
407	filenamefriction
408	SurfaceHeightNAME
409	XVelocityNAME
410	YVelocityNAME
411	SurfaceHeightLOC
412	XVelocityLOC
413	YVelocityLOC
414	obstacleLOC
415	obstacleNAME
416	WaveFileName
417	WaveFileLOC
418	wavemaker
419	ConsWindNAME
420	ConsWindLOC
421	StormFileName
422	StormFileLOC
423	PressureFileName
424	PressureFileLOC
425	HardBottomFileName
426	HardBottomFileLOC
427	VesselNumber
428	VesselLOC
429	VesselNAME
430	HardbottomLOC
431	HardbottomNAME
432	CouplingLOC
433	CouplingNAME
434	mpif90val
435	directory
436	logreport
437	MakefileContent
438	processPID
439	makefile
440	depthfilelocation
441	depthdata
442	depthtype
443	rasterdegrees
444	depthfinal
445	GAUGEFILElocation
446	colsChildLayer
447	rowsChildLayer
448	directorylog
449	CouplingLOC2
450	fortranFilePath0
451	gfortranReport
452	resultCompile
453	resultRun
454	inputFolder
455	CouplingX
456	CouplingY
457	CoupX
458	CoupY


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459         GAUGEFILElocation
460         latGauge
461         gaugetext0
462         CoupMatrixON
463         gridX
464         gridY
465         row1
466         col1
467     end
468
469
470     % Callbacks that handle component events
471     methods (Access = private)
472
473         % Code that executes after component creation
474         function startupFcn(app)
475             %Set the screenshot mode to 'manual' to disable automatic
screenshots,
476             %to improve app efficiency
477             currentAppModel.MetadataModel.ScreenshotMode = 'manual';
478         end
479
480         % Button pushed function: Button_14
481         function Button_14Pushed(app, event)
482             workingfolder = uigetdir;
483             workingfolder = fullfile(workingfolder);
484
485             %Create the 'INPUT_FILES' folder in the working directory
486             app.WORKFOLDER = fullfile(string(workingfolder), 'INPUT_FILES');
487             app.SavefilestoEditField.Value = fullfile(string(workingfolder),
'INPUT_FILES');
488             app.SavefilestoEditField.FontColor = 'k';
489         end
490
491         % Button pushed function: Button_8
492         function Button_8Pushed(app, event)
493             %Load the bathymetry file and display its filename in the textbox
494             [bathymetryfile,path2] = uigetfile('*.txt;*.tif');
495             app.depthfilelocation = string(path2);
496             app.bathymetryinputdata = string(path2) + string(bathymetryfile);
497             app.DepthFileEditField.BackgroundColor = 'w';
498             app.DepthFileEditField.FontColor = 'k';
499
500             %
501             try
502                 if contains(string(app.bathymetryinputdata),'.txt')
503                     app.depthtype = 1;
504                     A1 = readmatrix(app.bathymetryinputdata);
505                     app.depthdata = A1;
506                     app.DEPTHFILE = string(bathymetryfile);
507
508                     %Extract M and N global dimensions
509                     [row,col] = size(A1);
510                     app.XEditField_2.Value = col;
511                     app.YEditField_2.Value = row;
512                     app.XEditField_2.FontColor = 'k';
513                     app.YEditField_2.FontColor = 'k';
514                     app.ButtonGroup_4.SelectedObject = app.Button_2;

```

```

515         elseif contains(string(app.bathymetryinputdata), '.tif')
516             %This works with a .tif file generated using QGIS
517             %Extract information from the bathymetry
518             app.depthtype = 2;
519             [A1,R2] = readgeoraster(app.bathymetryinputdata);
520             A = size(A1);
521             app.depthdata = flipud(A1); %FUNWAVE reads the depth file
starting from the southwest corner
522             %The readgeoraster function generates a matrix with the
first point representing the northwest corner
523
524             bathname= erase(bathymetryfile, '.tif'); % Remove .tif
extension
525             app.DEPTHFILE = [bathname, '.txt']; % Append .txt
extension
526
527
528
529             if strcmpi(R2.CoordinateSystemType, 'geographic')
530                 southwestLongitude = min(R2.LongitudeLimits);
531                 southwestLatitude = min(R2.LatitudeLimits);
532                 gridSize = R2.CellExtentInLatitude;
533                 gridSize = R2.CellExtentInLongitude;
534
535                 app.ButtonGroup_4.SelectedObject = app.
SphericalButton;
536                 app.SphericalButton.FontColor = 'k';
537                 app.rasterdegrees = R2;
538
539             elseif strcmpi(R2.CoordinateSystemType, 'planar')
540                 southwestLongitude = min(R2.XWorldLimits);
541                 southwestLatitude = min(R2.YWorldLimits);
542                 gridSize = R2.CellExtentInWorldX;
543                 gridSize = R2.CellExtentInWorldY;
544
545                 app.ButtonGroup_4.SelectedObject = app.
CartesianButton;
546                 app.CartesianButton.FontColor = 'k';
547             end
548
549
550             %Extract lat and long of the southwest corner
551             app.LongitudeEditField.Value = southwestLongitude;
552             app.LatitudeEditField.Value = southwestLatitude;
553             app.LongitudeEditField.FontColor = 'k';
554             app.LatitudeEditField.FontColor = 'k';
555             app.LatitudeEditField.Enable = "on";
556             app.LongitudeEditField.Enable = "on";
557
558             %Extract grid size
559             app.XEditField.Value = gridSize;
560             app.YEditField.Value = gridSize;
561             app.gridX = gridSize;
562             app.gridY = gridSize;
563             app.XEditField.FontColor = 'k';
564             app.YEditField.FontColor = 'k';
565
566             %Extract M and N global dimensions

```

```

567         app.XEditField_2.Value = A(1,2);
568         app.YEditField_2.Value = A(1,1);
569         app.col1 = A(1,2);
570         app.row1 = A(1,1);
571         app.XEditField_2.FontColor = 'k';
572         app.YEditField_2.FontColor = 'k';
573         app.XEditField_2.Enable = "on";
574         app.YEditField_2.Enable = "on";
575
576         % if R2.AngleUnits == "degrees"
577         %     app.ButtonGroup_4.SelectedObject = app.
SphericalButton;
578         %     app.SphericalButton.FontColor = 'k';
579         %
580         % end
581         % app.rasterdegrees = R2;
582     else
583         app.ButtonGroup_4.SelectedObject = app.Button_2;
584         app.DepthFileEditField.Value = '';
585     end
586     app.DepthFileEditField.Value = app.DEPTHFILE; %Display the
filename in the textbox
587 %     end
588
589     end
590
591     % Callback function
592     function TotalTimeEditFieldValueChanged(app, event)
593
594     end
595
596     % Callback function
597     function PlotintervalEditFieldValueChanged(app, event)
598
599
600     end
601
602     % Button pushed function: GenerateinputtxtButton
603     function GenerateinputtxtButtonPushed(app, event)
604 %Please cite this paper when you use the applications:
605 %Felix, R., Tan, E. H. Z., Watanabe, M., Verolino, A., Puah, J. Y., & Switzer,
A. D. (2025). Funwave-based open-access mapping applications (FUNMAP) applied to
Tsunami modelling from the Manila Trench to Manila Bay, Philippines. Geoscience
Letters. https://doi.org/10.1186/s40562-025-00422-5
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680
681
682
683 %This is the start of the app's code:
684
685
686          %-----
687          %-----Remove old Makefile, funwave.exe-----
688          %-----
689          %In reclicking the button, delete existing Makefile and funwave. ✓
exe
690          if exist(fullfile(app.directory, "funwave"), 'file')
691              delete(fullfile(app.directory, "funwave"));
692          end
693
694          if exist(fullfile(app.directory, "Makefile"), 'file')
695              delete(fullfile(app.directory, "Makefile"));
696          end
697
698          %Disable the Makefile  buttons
699          app.CreateMakefileButton.Enable = 'off';
700          app.StartsimulationButton.Enable = 'off';
701
702          %Remove preview in the Makefile Tab
703          app.Preview_Makefile.Value = '';
704
705          %-----
706          %-----Output Directory-----
707          %-----
708          %Check if the output directory has been manually set by the user
709          if app.SavefilestoEditField.Value == string(app.WORKFOLDER)
```



```

710         inputtxt_directory = fullfile(app.WORKFOLDER);
711         app.SavefilestoEditField.FontColor = 'k';
712
713         %Check if the 'INPUT_FILES' directory exists, and create it if it
it doesn't
714         if ~exist(inputtxt_directory, 'dir')
715             mkdir(inputtxt_directory); % Creates the directory
716         end
717     else
718         %Set the default directory to the Desktop
719         if ismac
720             %macOS
721             defaultDir = fullfile(getenv('HOME'), 'Desktop');
722         elseif ispc
723             %Windows
724             defaultDir = fullfile(getenv('USERPROFILE'), 'Desktop');
725         else
726             %Handle other platforms
727             defaultDir = pwd;
728         end
729
730         Dir1 = fullfile(defaultDir, 'INPUT_FILES'); %Create
'INPUT_FILES' folder
731
732         % Check if the 'INPUT_FILES' directory exists, and create
it if it doesn't
733         if ~exist(Dir1, 'dir')
734             mkdir(Dir1); % Creates the directory
735         end
736
737         app.SavefilestoEditField.Value = deblank(string(Dir1)); %"
Current Folder";
738         inputtxt_directory = deblank(string(Dir1));
739         app.SavefilestoEditField.FontColor = 'k';
740     end
741
742     app.directory = inputtxt_directory;
743
744
745     %-----
746     %-----CREATE INPUT.TXT HEADER-----
747     %-----
748     %Create the input.txt
749     fileInputPath = fullfile(inputtxt_directory, 'input.txt');
750     fileInputId = fopen(fileInputPath, 'w');
751
752     %Template of the first few lines of the input.txt
753     templateStr = [
754         '! INPUT FILE FOR BOUSS_TVD\n' ...
755         '! Note: all input parameter are capital sensitive\n\n' ...
756         '! Generated using FUNMAP by Felix, et al. (2025) \n\n'...
757         '! -----
TITLE-----\n' ...
758         'TITLE = FUNWAVE\n\n' ...
759         ' ! -----HOT
START-----\n' ...
760         'HOT_START = F\n' ...
761         'FileNumber HOTSTART = HotStartMTLB\n'];

```

```

762
763         %Add the template into the text file
764         fprintf(fileInputId, templateStr);
765
766         %Create Input_Data subfolder
767         inputFolder = fullfile(inputtxt_directory, 'Input_Data');
768         app.inputFolder = inputFolder;
769
770         %Check if the 'INPUT_FILES/Input_Data' directory exists, create it
if it doesn't
771         if ~exist(app.inputFolder, 'dir')
772             mkdir(app.inputFolder); % Creates the 'INPUTFOLDER' directory
773         end
774
775
776         %-----
777         %-----LOG REPORT-----
778         %-----
779         %Create 'Log_Files' subfolder
780         LogFolder = fullfile(inputtxt_directory, 'Log_Files');
781         LogFile = fullfile(LogFolder, 'Log_Report_InputPrep.txt');
782
783         % Check if the 'INPUT_FILES/Log_Files' directory exists,
create it if it doesn't
784         if ~exist(LogFolder, 'dir')
785             mkdir(LogFolder);
786         end
787
788         %Create the log text file
789         fileId = fopen(fullfile(LogFolder, 'Log_Report_InputPrep.
txt'), 'w');
790
791         %Template for the default section of the log report
792         fprintf(fileId, '%s\n', '***** LOG REPORT
***** ');
793         fprintf(fileId, '%-30s%-s\n', 'Type:', 'input.txt, Makefile,
funwave executable');
794         timestamp = datestr(now, 'yyyy-mm-dd HH:MM:SS'); % Get the
current timestamp
795         fprintf(fileId, '%-30s%-s\n', 'Timestamp:', timestamp); %Add
timestamp in the header
796         fprintf(fileId, '%s\n', '***** ');
797         fprintf(fileId, '%-s\n', '');
798         fprintf(fileId, '%-s\n\n', '***** INPUT.TXT SECTION
***** ');
799
800         error = 0; % General error counter
801         error_mkfile = 0; %Makefile error counter
802
803         %-----
804         %-----PARALLEL INFO-----
805         %-----
806         tempParallel = sprintf(['\n! -----PARALLEL
INFO-----\n' ...
807             '! PX,PY - processor numbers in X and Y\n' ...
808             '! NOTE: make sure consistency with mpirun -np n
(px*py)\n' ...

```

```

809         'PX = %.0f\n' ...
810         'PY = %.0f\n'], ...
811         app.ProcessorX.Value, ...
812         app.ProcessorY.Value);
813
814     %Add these lines in the input.txt
815     fprintf(fileInputId, '%s', tempParallel);
816
817     %-----
818     %-----DEPTH-----
819     %-----
820     %If no option is selected
821     if app.Button4.Value
822         app.ImportDataButton.FontColor = 'r';
823         app.FlatButton.FontColor = 'r';
824         app.SlopeButton.FontColor = 'r';
825         fprintf(fileId, '%-30s%-s\n', 'DEPTH:', 'No data is
selected');
826         error = error + 1; %to note an error occurred
827     end
828
829     %-----
830     %-----DEPTH: Import Data-----
831     %-----
832     if app.ImportDataButton.Value
833         % Check if no file is uploaded
834         Depthconditions = [...
835             isempty(app.DepthFileEditField.Value), ...
836             ~strcmp(app.DepthFileEditField.Value, app.DEPTHFILE)
...
837             str2double(app.DepthFileEditField.Value) == 0 ...
838             ];
839
840         %Display 'NO FILE' in the textbox and note the issue in
the log report
841         if any(Depthconditions)
842             app.DepthFileEditField.BackgroundColor = 'r';
843             app.DepthFileEditField.Value = 'NO FILE';
844             app.DepthFileEditField.FontColor = 'w';
845             fprintf(fileId, '%-30s%-s\n', 'DEPTH:', 'No file is
uploaded');
846             error = error + 1;
847
848         else %Read the bathymetry file
849             if app.InvertvaluesCheckBox.Value
850                 %Convert water depth values to positive values
851                 depth = -app.depthdata;
852             else
853                 depth = app.depthdata;
854             end
855
856             % Round off depth to the nearest five decimal places
857             if isnumeric(depth) && isscalar(depth)
858                 %Scalar numeric input
859                 app.depthfinal = round(depth, 5);
860
861             elseif ischar(depth) || isstring(depth)
862                 %String input

```

```

863         depthNum = str2double(depth);
864         if ~isnan(depthNum)
865             app.depthfinal = round(depthNum, 5);
866         end
867     elseif isnumeric(depth) && ~isscalar(depth)
868         %Numeric array input
869         app.depthfinal = round(double(depth), 5);
870     else
871         app.depthfinal = round(depth, 5);
872     end
873
874     %Save the bathymetry file in 'INPUT_FILES/Input_Data/'
875     outputFilePath = fullfile(app.inputFolder,app.
DEPTHFILE);
876
877
878     % Open the file for writing
879     fileID0 = fopen(outputFilePath, 'w');
880
881     % Define the format for each row (e.g., '%.5f' for two
decimal places)
882     formatSpec = [repmat('%%.5f\t', 1, size(app.depthfinal,
2) - 1), '%.5f\n'];
883
884     % Write each row of depthfinal to the file
885     for row = 1:size(app.depthfinal, 1)
886         fprintf(fileID0, formatSpec, app.depthfinal(row,
:));
887     end
888
889     % Close the file
890     fclose(fileID0);
891
892
893
894     tempDepth = sprintf([
895         '\n ! -----
DEPTH-----\n' ...
896         'DEPTH_TYPE = DATA\n' ...
897         'DEPTH_FILE = Input_Data/%s\n' ...
898         'DepthFormat = ELE\n' ...
899         'WaterLevel = %.1f\n'], ...
900         app.DEPTHFILE,app.WaterlevelmEditField.Value);
901     end
902 end
903
904     %-----
905     %-----DEPTH: Flat-----
906     %-----
907     if app.FlatButton.Value
908         tempDepth = sprintf([
909             '\n ! -----
DEPTH-----\n' ...
910             'DEPTH_TYPE = FLAT\n' ...
911             'DEPTH_FLAT = %.2f\n'], ...
912             app.DepthmEditField.Value);
913
914     % Add warning note in the log report if the 0 value is

```

```

unchanged
915         if app.DepthmEditField.Value == 0
916             app.DepthmEditField.FontColor = 'r';
917             fprintf(fileId, '%-30s%-s\n', 'DEPTH:', 'Flat depth is
zero');
918             error = error + 1;
919         end
920     end
921
922     %-----
923     %-----DEPTH: Slope-----
924     %-----
925     if app.SlopeButton.Value
926         tempDepth = sprintf([
927             '\n ! -----
DEPTH-----\n' ...
928             'DEPTH_TYPE = SLOPE\n' ...
929             'DEPTH_FLAT = %.2f\n' ...
930             'SLP = %.2f\n' ...
931             'Xslp = %.2f\n'], ...
932             app.DepthmEditField.Value, ...
933             app.AngleEditField.Value, ...
934             app.StartatXEditField.Value);
935
936
937     %If one of the depth options is selcted
938     if exist('tempDepth', 'var') == 1
939         fprintf(fileInputId, '%s', tempDepth); %Add the Depth
info in the input.txt
940     end
941
942     % Add warning note in the log report if 0 value is
unchanged
943     if app.DepthmEditField.Value == 0
944         app.DepthmEditField.FontColor = 'r';
945         fprintf(fileId, '%-30s%-s\n', 'DEPTH:', 'Flat depth is
zero');
946         error = error + 1;
947     end
948
949     if app.DepthmEditField.Value == 0
950         app.DepthmEditField.FontColor = 'r';
951         fprintf(fileId, '%-30s%-s\n', 'DEPTH:', 'Slope depth is
zero');
952         error = error + 1;
953     end
954
955     if app.AngleEditField.Value == 0
956         app.AngleEditField.FontColor = 'r';
957         fprintf(fileId, '%-30s%-s\n', 'DEPTH:', 'Slope angle is
zero');
958         error = error + 1;
959     end
960
961 end
962
963 if exist("tempDepth", 'var')
964     fprintf(fileInputId, '%s', tempDepth);

```

```

965         end
966
967         %-----
968         %-----DIMENSION-----
969         %-----
970         tempDim = sprintf([
971             '\n ! -----↵
DIMENSION-----\n' ...
972             ' ! global grid dimension\n' ...
973             'Mglob = %d\n' ...
974             'Nglob = %d\n'], ...
975             app.XEditField_2.Value, ...
976             app.YEditField_2.Value);
977
978         %Add these lines in the input.txt
979         fprintf(fileInputId, '%s', tempDim);
980
981         %Add warning note in the log report if the 0 value is↵
unchanged
982         if app.XEditField_2.Value == 0
983             app.XEditField_2.FontColor = 'r';
984             fprintf(fileId, '%-30s%-s\n', 'DIMENSION:', 'X is zero');
985             error = error + 1;
986         end
987
988         if app.YEditField_2.Value == 0
989             app.YEditField_2.FontColor = 'r';
990             fprintf(fileId, '%-30s%-s\n', 'DIMENSION:', 'Y is zero');
991             error = error + 1;
992         end
993
994         %-----
995         %-----PRINT-----
996         %-----
997         % Identify the correct directory separator based on the↵
operating system
998         if ismac
999             dirSeparator = "/";
1000         elseif ispc
1001             dirSeparator = "\";
1002         else
1003             dirSeparator = "/";
1004         end
1005
1006         if isempty(app.ResultsfolderEditField.Value)
1007             %Use 'Simulation_Results' as the default folder name
1008             RESULT_FOLDER = "Simulation_Results" + dirSeparator;
1009             app.ResultsfolderEditField.Value = RESULT_FOLDER;
1010             app.ResultsfolderEditField.FontColor = 'k';
1011         else %Use the user's input
1012             % if endsWith(app.ResultsfolderEditField.Value,↵
dirSeparator)
1013                 RESULT_FOLDER = app.ResultsfolderEditField.Value;
1014             % else
1015             %     RESULT_FOLDER = app.ResultsfolderEditField.Value +↵
dirSeparator;
1016             %     app.ResultsfolderEditField.Value = RESULT_FOLDER;
1017             % end

```

```

1018         end
1019
1020         tempPrint = sprintf([
1021             '\n ! -----↵
PRINT-----\n' ...
1022             'RESULT_FOLDER = %s\n'], ...
1023             RESULT_FOLDER); %
1024
1025         %Add these lines in the input.txt
1026         fprintf(fileInputId, '%s', tempPrint);
1027
1028         %-----
1029         %-----TIME-----
1030         %-----
1031         tempTime = sprintf([
1032             '\n ! -----↵
TIME-----\n' ...
1033             ' ! time: total computational time / plot time / screen↵
interval \n' ...
1034             ' ! all in seconds\n' ...
1035             'PLOT_START_TIME = %.1f\n' ...
1036             'TOTAL_TIME = %.1f\n' ...
1037             'PLOT_INTV = %.1f\n' ...
1038             'PLOT_INTV_STATION = %.1f\n' ...
1039             'SCREEN_INTV = %.1f\n'], ...
1040             app.StartEditField.Value, ...
1041             app.TotalEditField.Value, ...
1042             app.SimulationIntervalEditField.Value, ...
1043             app.GaugeIntervalEditField.Value, ...
1044             app.SimulationIntervalEditField.Value);
1045
1046         %Add these lines in the input.txt
1047         fprintf(fileInputId, '%s', tempTime);
1048
1049         %-----
1050         %-----GRID-----
1051         %-----
1052         tempGrid = sprintf([
1053             '\n ! -----↵
GRID-----\n' ...
1054             'StretchGrid = F\n']);
1055
1056         fprintf(fileInputId, '%s', tempGrid);
1057
1058         %-----
1059         %-----GRID: Spherical-----
1060         %-----
1061         %Changing the parameter variable for Grid Size based on the↵
coordinate system
1062         if app.SphericalButton.Value
1063             tempGrid2 = sprintf([
1064                 '!spherical grid, in decimal degrees\n' ...
1065                 'Lon_West = %.4f\n' ...
1066                 'Lat_South = %.4f\n' ...
1067                 'Dphi = %.4f\n' ...
1068                 'Dtheta = %.4f\n'], ...
1069                 app.LongitudeEditField.Value, app.LatitudeEditField.↵
Value, app.XEditField.Value, app.YEditField.Value);

```


[illegible]

```

1123         app.XEditField.FontColor = 'r';
1124         fprintf(fileId, '%-30s%-s\n', 'GRID:', 'X is zero');
1125         error = error + 1;
1126     end
1127     if app.YEditField.Value == 0
1128         app.YEditField.FontColor = 'r';
1129         fprintf(fileId, '%-30s%-s\n', 'GRID:', 'Y is zero');
1130         error = error + 1;
1131     end
1132
1133     if app.XEditField.Value == 0 && app.YEditField.Value == 0
1134         error_mkfile = error_mkfile +1;
1135         error_mkfiletext3 = sprintf('Grid');
1136     end
1137
1138     %-----
1139     %-----GRID: None-----
1140     %-----
1141     elseif app.Button_2.Value
1142         %If no Coordinate system is selected, add an error in the
log report
1143         app.SphericalButton.FontColor = 'r';
1144         app.CartesianButton.FontColor = 'r';
1145         fprintf(fileId, '%-30s%-s\n', 'GRID:', 'No coordinate system
is selected ');
1146         tempGrid2 = '';
1147         error = error + 1;
1148         error_mkfile = error_mkfile +1;
1149         error_mkfiletext3 = sprintf('Grid');
1150     end
1151
1152     %Add these lines in the input.txt
1153     fprintf(fileInputId, '%s', tempGrid2);
1154
1155     %-----
1156     %-----INITIAL CONDITION-----
1157     %-----
1158     if app.Button_31.Value
1159         fprintf(fileId, '%-30s%-s\n', 'INITIAL CONDITION:', 'None
selected. Set it up in Tab 2');
1160         error = error + 1;
1161     end
1162
1163     %-----
1164     %-----INITIAL CONDITION: Import UVZ-----
1165     %-----
1166     if app.ImportUVZButton.Value
1167         tempUVZ1 = sprintf([
1168             '\n ! ----- INITIAL UVZ
-----\n' ...
1169             ' ! INI_UVZ - initial UVZ e.g., initial
deformation\n' ...
1170             ' ! must provide three (3) files \n' ...
1171             'INI_UVZ = T\n']);
1172
1173         %Set up the initial values for the three parameters to
avoid errors when creating the text file
1174         tempUVZ2 = sprintf('ETA FILE = \n');

```

```

1175         tempUVZ3 = sprintf('U_FILE = \n');
1176         tempUVZ4 = sprintf('V_FILE = \n');
1177
1178         %Check if the eta file / Z file is loaded
1179         ETAconditions = [...
1180             isempty(app.SurfaceHeightEditField.Value), ...
1181             ~any(strcmp(app.SurfaceHeightEditField.Value, app.
SurfaceHeightNAME)), ...
1182             str2double(app.SurfaceHeightEditField.Value) == 0 ...
1183             ];
1184
1185         if any(ETAconditions)
1186             %Display 'NO FILE' in the textbox and note the issue
in the log report
1187             fprintf(fileId, '%-30s-s\n', 'INITIAL CONDITION:',
'Import UVZ:');
1188             fprintf(fileId, '%-30s-s\n', '', 'No file uploaded in
'Surface Height'');
1189             app.SurfaceHeightEditField.Value = "NO FILE";
1190             app.SurfaceHeightEditField.FontColor = 'w';
1191             app.SurfaceHeightEditField.BackgroundColor = 'r';
1192             error = error + 1;
1193         elseif strcmp(app.SurfaceHeightEditField.Value, app.
SurfaceHeightNAME)
1194             %If the filename matches the name in the textbox, add
it to input.txt
1195             tempUVZ2 = sprintf('ETA_FILE = Input_Data/%s\n', app.
SurfaceHeightNAME);
1196
1197             %If the file is not in 'INPUT_FILES/Input_Data', copy
it there
1198             if ~exist(fullfile(app.inputFolder, app.
SurfaceHeightNAME), 'file')
1199                 copyfile(app.SurfaceHeightLOC, app.inputFolder);
1200             end
1201         end
1202
1203         % Checking if U vector file is loaded
1204         uConditions = [...
1205             isempty(app.UVelocityEditField.Value), ... % Check if
the field is empty
1206             ~any(strcmp(app.UVelocityEditField.Value, app.
XVelocityNAME)), ... % Check if the value does not match expected names
1207             str2double(app.UVelocityEditField.Value) == 0 ... %
Check if the value is zero (assuming numeric comparison is relevant)
1208             ];
1209
1210         if any(uConditions)
1211             %Display 'NO FILE' in the textbox and note the issue
in the log report
1212             fprintf(fileId, '%-30s-s\n', 'INITIAL CONDITION:',
'Import UVZ:');
1213             fprintf(fileId, '%-30s-s\n', '', 'No file uploaded in
'U Velocity'');
1214             app.UVelocityEditField.Value = "NO FILE";
1215             app.UVelocityEditField.FontColor = 'w';
1216             app.UVelocityEditField.BackgroundColor = 'r';
1217             error = error + 1;

```

```

1218         else
1219             %If the filename matches the name in the textbox, add
it to input.txt
1220             tempUVZ3 = sprintf('U_FILE = Input_Data/%s\n', app.XVelocityNAME);
1221
1222             %If the file is not in 'INPUT_FILES/Input_Data', copy
it there
1223             if ~exist(fullfile(app.inputFolder, app.XVelocityNAME), 'file')
1224                 copyfile(app.XVelocityLOC, app.inputFolder);
1225             end
1226         end
1227
1228         % Checking if V vector file is loaded
1229         vConditions = [...
1230             isempty(app.VVelocityEditField.Value), ... % Check if
the field is empty
1231             ~any(strcmp(app.VVelocityEditField.Value, app.YVelocityNAME)), ... % Check if the value does not match expected names
1232             str2double(app.VVelocityEditField.Value) == 0 ... %
Check if the value is zero (assuming numeric comparison is relevant)
1233         ];
1234
1235         if any(vConditions)
1236             %Display 'NO FILE' in the textbox and note the issue
in the log report
1237             fprintf(fileId, '%-30s-s\n', 'INITIAL CONDITION:',
'Import UVZ:');
1238             fprintf(fileId, '%-30s-s\n', '', 'No file uploaded in
'V Velocity');
1239             app.VVelocityEditField.Value = "NO FILE";
1240             app.VVelocityEditField.FontColor = 'w';
1241             app.VVelocityEditField.BackgroundColor = 'r';
1242             error = error + 1;
1243         else
1244             %If the filename matches the name in the textbox, add
it to input.txt
1245             tempUVZ4 = sprintf('V_FILE = Input_Data/%s\n', app.YVelocityNAME);
1246
1247             %If the file is not in 'INPUT_FILES/Input_Data', copy
it there
1248             if exist(fullfile(inputFolder, app.YVelocityNAME),
'file') == 2
1249                 %do nothing
1250             else
1251                 %Copy the file to the OUTPUT_FOLDER
1252                 copyfile(app.YVelocityLOC, inputFolder);
1253             end
1254         end
1255
1256         % Combine all the the parts
1257         tempUVZ = [tempUVZ1,tempUVZ2,tempUVZ3,tempUVZ4];
1258
1259         %Add these lines in the input.txt
1260         fprintf(fileInputId, '%s', tempUVZ);
1261     else

```

```

1262             fprintf(fileInputId, [
1263                 '\n ! ----- INITIAL UVZ ↵
-----\n' ...
1264                 'INI_UVZ = F\n']]);
1265         end
1266
1267         %-----
1268         %-----INITIAL CONDITION: Wavemaker-----
1269         %-----
1270         %To list all the errors in the log report
1271         function checkAndReport(field, fieldName, fileId)
1272             if field.Value == 0
1273                 field.FontColor = 'r';
1274                 fprintf(fileId, '%-30s%-s\n', 'INITIAL CONDITION:', ↵
'Wavemaker:');
1275                 fprintf(fileId, '%-35s%-s\n', '', [fieldName ' is zero']);
1276                 error = error + 1;
1277             else
1278                 field.FontColor = 'k';
1279             end
1280         end
1281
1282         if app.WavemakerButton.Value
1283             %Define variables
1284             Xc = app.XcoordinatemEditField.Value;
1285             Yc = app.YcoordinatemEditField.Value;
1286             WID = app.WidthmEditField.Value;
1287             AMP = app.AmplitudemEditField.Value;
1288             DEP = app.WaterdepthmEditField.Value;
1289             LAGTIME = app.LagtimesecEditField.Value;
1290             Tperiod = app.PeriodsecEditField.Value;
1291             Theta_WK = app.ThetadegreesEditField.Value;
1292             Time_ramp = app.TimerampsecEditField.Value;
1293             NumWaveComp = app.WaveCompCountEditField.Value;
1294             FreqPeak = app.PeakEditField.Value;
1295             FreqMin = app.MinimumEditField.Value;
1296             FreqMax = app.MaximumEditField.Value;
1297             Hmo = app.WaveHeightmEditField.Value;
1298
1299
1300             tempWavemaker1 = sprintf(['\n! ----- ↵
WAVEMAKER-----\n' ]]);
1301
1302             %Add only the necessary parameters based on the dropdown ↵
selection
1303             switch app.WavemakertypeDropDown.Value
1304
1305                 case "INI_REC"
1306                     %-----
1307                     %.....INI_REC.....
1308                     %-----
1309                     tempWavemaker2 = sprintf('WAVEMAKER = INI_REC \nXc = ↵
%.4f\nYc = %.4f\nWID = %.2f\n', Xc, Yc, WID);
1310
1311                     %Log report
1312                     checkAndReport(app.XcoordinatemEditField, 'X ↵
Coordinate', fileId);
1313                     checkAndReport(app.YcoordinatemEditField, 'Y ↵

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```

Coordinate', fileId);
1314                                     checkAndReport(app.WidthmEditField, 'Width (m)', ↵
fileId);
1315
1316
1317                                     case "JON_1D"
1318                                     %-----
1319                                     %.....JON_1D.....
1320                                     %-----
1321                                     tempWavemaker2 = sprintf('WAVEMAKER = JON_1D\nXc_WK = ↵
%.4f\nYc_WK = %.4f\nYwidth_WK = %.2f\nDEP_WK = %.2f\nTime_ramp = %.2f\nDelta_WK = ↵
%.2f\nFreqPeak = %.2f\nFreqMin = %.2f\nFreqMax = %.2f\nHmo = %.2f\nGammaTMA = 3.3 ↵
\nNfreq = 45.0\n', Xc, Yc, WID, DEP, Time_ramp, app.WidthDeltaEditField.Value, ↵
FreqPeak, FreqMin, FreqMax, Hmo);
1322
1323                                     %Log report
1324                                     checkAndReport(app.XcoordinatemEditField, 'X'↵
Coordinate', fileId);
1325                                     checkAndReport(app.YcoordinatemEditField, 'Y'↵
Coordinate', fileId);
1326                                     checkAndReport(app.WidthmEditField, 'Width (m)', ↵
fileId);
1327                                     checkAndReport(app.WaterdepthmEditField, 'Water depth'↵
(m)', fileId);
1328                                     checkAndReport(app.TimerampsecEditField, 'Time ramp'↵
(sec)', fileId);
1329                                     checkAndReport(app.WidthDeltaEditField, 'Width Delta', ↵
fileId);
1330                                     checkAndReport(app.MinimumEditField, 'Frequency'↵
Minimum', fileId);
1331                                     checkAndReport(app.MaximumEditField, 'Frequency'↵
Maximum', fileId);
1332                                     checkAndReport(app.PeakEditField, 'Frequency Peak', ↵
fileId);
1333                                     checkAndReport(app.WaveHeightmEditField, 'Wave height'↵
(hmo)', fileId);
1334
1335                                     case "JON_2D"
1336                                     %-----
1337                                     %.....JON_2D.....
1338                                     %-----
1339                                     tempWavemaker2 = sprintf('WAVEMAKER = JON_2D\nXc_WK = ↵
%.4f\nYc_WK = %.4f\nYwidth_WK = %.2f\nDEP_WK = %.2f\nTime_ramp = %.2f\nDelta_WK = ↵
%.2f\nFreqPeak = %.2f\nFreqMin = %.2f\nFreqMax = %.2f\nHmo = %.2f\nGammaTMA = 3.3 ↵
\nThetaPeak = 0.0\nNfreq = 45.0\nNtheta = 24.0\n', Xc, Yc, WID, DEP, Time_ramp, app.↵
WidthDeltaEditField.Value, FreqPeak, FreqMin, FreqMax, Hmo);
1340
1341                                     %Log report
1342                                     checkAndReport(app.XcoordinatemEditField, 'X'↵
Coordinate', fileId);
1343                                     checkAndReport(app.YcoordinatemEditField, 'Y'↵
Coordinate', fileId);
1344                                     checkAndReport(app.WidthmEditField, 'Width (m)', ↵
fileId);
1345                                     checkAndReport(app.WaterdepthmEditField, 'Water depth'↵
(m)', fileId);
1346                                     checkAndReport(app.TimerampsecEditField, 'Time ramp'↵
(sec)', fileId);

```

```

1347         checkAndReport(app.WidthDeltaEditField, 'Width Delta',
fileId);
1348         checkAndReport(app.MinimumEditField, 'Frequency
Minimum', fileId);
1349         checkAndReport(app.MaximumEditField, 'Frequency
Maximum', fileId);
1350         checkAndReport(app.PeakEditField, 'Frequency Peak',
fileId);
1351         checkAndReport(app.WaveHeightmEditField, 'Wave height
(hmo)', fileId);
1352
1353         case "TMA_1D"
1354             %-----
1355             %.....TMA_1D.....
1356             %-----
1357             tempWavemaker2 = sprintf('WAVEMAKER = TMA_1D\nXc_WK =
%.4f\nYc_WK = %.4f\nYwidth_WK = %.2f\nDEP_WK = %.2f\nTime_ramp = %.2f\nDelta_WK =
%.2f\nFreqPeak = %.2f\nFreqMin = %.2f\nFreqMax = %.2f\nHmo = %.2f\nGammaTMA = 3.3
\nNfreq = 45.0\n', Xc, Yc, WID, DEP, Time_ramp, app.WidthDeltaEditField.Value,
FreqPeak, FreqMin, FreqMax, Hmo);
1358
1359             %Log report
1360             checkAndReport(app.XcoordinatemEditField, 'X
Coordinate', fileId);
1361             checkAndReport(app.YcoordinatemEditField, 'Y
Coordinate', fileId);
1362             checkAndReport(app.WidthmEditField, 'Width (m)',
fileId);
1363             checkAndReport(app.WaterdepthmEditField, 'Water depth
(m)', fileId);
1364             checkAndReport(app.TimerampsecEditField, 'Time ramp
(sec)', fileId);
1365             checkAndReport(app.WidthDeltaEditField, 'Width Delta',
fileId);
1366             checkAndReport(app.MinimumEditField, 'Frequency
Minimum', fileId);
1367             checkAndReport(app.MaximumEditField, 'Frequency
Maximum', fileId);
1368             checkAndReport(app.PeakEditField, 'Frequency Peak',
fileId);
1369             checkAndReport(app.WaveHeightmEditField, 'Wave height
(hmo)', fileId);
1370
1371         case "LEF_SOL"
1372             %-----
1373             %.....LEF_SOL.....
1374             %-----
1375             tempWavemaker2 = sprintf('WAVEMAKER = LEF_SOL\nAMP =
%.2f\nDEP = %.2f\nLAGTIME = %.2f\n', AMP, DEP, LAGTIME);
1376
1377             %Log report
1378             checkAndReport(app.AmplitudemEditField, 'Amplitude
(m)', fileId);
1379             checkAndReport(app.WaterdepthmEditField, 'Water depth
(m)', fileId);
1380             checkAndReport(app.LagtimesecEditField, 'Lag time
(sec)', fileId);
1381

```

```

1382         case "INI_SOL"
1383             %-----
1384             %.....INI_SOL.....
1385             %-----
1386             tempWavemaker2 = sprintf('WAVEMAKER = INI_SOL\nAMP = %
%.2f\nDEP = %.2f\nXWAVEMAKER = %.4f\n', AMP, DEP, Xc);
1387
1388             %Log report
1389             checkAndReport(app.AmplitudemEditField, 'Amplitude
(m)', fileId);
1390             checkAndReport(app.WaterdepthmEditField, 'Water depth
(m)', fileId);
1391             checkAndReport(app.XcoordinatemEditField, 'X
Coordinate', fileId);
1392
1393         case "WK_IRR"
1394             %-----
1395             %.....WK_IRR.....
1396             %-----
1397             tempWavemaker2 = sprintf('WAVEMAKER = WK_IRR\nXc_WK = %
%.4f\nYc_WK = %.4f\nYwidth_WK = %.2f\nDEP_WK = %.2f\nTime_ramp = %.2f\nDelta_WK = %
%.2f\nFreqPeak = %.2f\nFreqMin = %.2f\nFreqMax = %.2f\nHmo = %.2f\nGammaTMA = 3.3
\nThetaPeak = 0.0\nNfreq = 45.0\nNtheta = 24.0\nEqualEnergy = TRUE', Xc, Yc, WID,
DEP, Time_ramp, app.WidthDeltaEditField.Value, FreqPeak, FreqMin, FreqMax, Hmo);
1398
1399             %Log report
1400             checkAndReport(app.XcoordinatemEditField, 'X
Coordinate', fileId);
1401             checkAndReport(app.YcoordinatemEditField, 'Y
Coordinate', fileId);
1402             checkAndReport(app.WidthmEditField, 'Width (m)',
fileId);
1403             checkAndReport(app.WaterdepthmEditField, 'Water depth
(m)', fileId);
1404             checkAndReport(app.TimerampsecEditField, 'Time ramp
(sec)', fileId);
1405             checkAndReport(app.WidthDeltaEditField, 'Width Delta',
fileId);
1406             checkAndReport(app.MinimumEditField, 'Frequency
Minimum', fileId);
1407             checkAndReport(app.MaximumEditField, 'Frequency
Maximum', fileId);
1408             checkAndReport(app.PeakEditField, 'Frequency Peak',
fileId);
1409             checkAndReport(app.WaveHeightmEditField, 'Wave height
(hmo)', fileId);
1410
1411         case "WK_REG"
1412             %-----
1413             %.....WK_REG.....
1414             %-----
1415             tempWavemaker2 = sprintf('WAVEMAKER = WK_REG\nXc_WK = %
%.4f\nYc_WK = %.4f\nYwidth_WK = %.2f\nTperiod = %.2f\nAMP_WK = %.2f\nDEP_WK = %
%.2f\nTheta_WK = %.2f\nTime_ramp = %.2f\nDelta_WK = %.2f\n!The Delta_WK option is
activated as it is included in the simple_cases/sediment_rip file\n', Xc, Yc, WID,
Tperiod, AMP, DEP, Theta_WK, Time_ramp, app.WidthDeltaEditField.Value);
1416
1417             %Log report

```



```

1418             checkAndReport(app.XcoordinatemEditField, 'X'
Coordinate', fileId);
1419             checkAndReport(app.PeriodsecEditField, 'Period (sec)',
fileId);
1420             checkAndReport(app.AmplitudemEditField, 'Amplitude'
(m)', fileId);
1421             checkAndReport(app.WaterdepthmEditField, 'Water depth'
(m)', fileId);
1422             checkAndReport(app.ThetadegreesEditField, 'Theta'
(degrees)', fileId);
1423             checkAndReport(app.TimerampsecEditField, 'Time ramp'
(sec)', fileId);
1424
1425             case "WK_TIME_SERIES"
1426                 %-----
1427                 %.....WK_TIME_SERIES.....
1428                 %-----
1429                 tempWavemaker2 = sprintf('WAVEMAKER = WK_TIME_SERIES\n'
WaveCompFile = Input_Data/%s\n NumWaveComp = %.2f\nPeakPeriod = %.2f\nDEP_WK = %.
2f\nXc_WK = %.4f\nYwidth_WK = %.4f\n', app.WavecomponentfileEditField.Value,
NumWaveComp, Tperiod, DEP, Xc, WID);
1430
1431             checkAndReport(app.WaveCompCountEditField, 'Wave'
component count', fileId);
1432             checkAndReport(app.PeriodsecEditField, 'Period (sec)',
fileId);
1433             checkAndReport(app.WaterdepthmEditField, 'Water depth'
(m)', fileId);
1434             checkAndReport(app.XcoordinatemEditField, 'X'
Coordinate', fileId);
1435             checkAndReport(app.WidthmEditField, 'Width (m)',
fileId);
1436
1437             %Check if the file is loaded
1438             wavecomconditions = [...
1439                 isempty(app.WavecomponentfileEditField.Value ),
... % Check if the field is empty
1440                 ~any(strcmp(app.WavecomponentfileEditField.Value ,
app.WaveFileName)), ... % Check if the value does not match expected names
1441                 str2double(app.WavecomponentfileEditField.Value )
== 0 ... % Check if the value is zero (assuming numeric comparison is relevant)
1442                 ];
1443
1444             if any(wavecomconditions)
1445                 %Display 'NO FILE' in the textbox and note the
issue in the log report
1446                 fprintf(fileId, '%-30s%-s\n', 'INITIAL'
CONDITION:', 'Wavemaker:');
1447                 fprintf(fileId, '%-35s%-s\n', '', 'No file'
uploaded in 'Wave component file');
1448                 app.WavecomponentfileEditField.Value = "NO FILE";
1449                 app.WavecomponentfileEditField.FontColor = 'w';
1450                 app.WavecomponentfileEditField.BackgroundColor =
'r';
1451                 error = error + 1;
1452             elseif strcmp(app.WavecomponentfileEditField.Value ,
app.WaveFileName)
1453                 %If the file is not in 'INPUT FILES/Input Data',

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```

copy it there
1454             if ~exist(fullfile(app.inputFolder, app.WaveFile
WaveFileName), 'file')
1455                 copyfile(app.WaveFileLOC, app.inputFolder);
1456             end
1457
1458         end
1459
1460         case "WK_DATA2D"
1461             %-----
1462             %.....WK_DATA2D.....
1463             %-----
1464             tempWavemaker2 = sprintf('WAVEMAKER = WK_DATA2D\n
WaveCompFile = Input_Data/%s\n Xc_WK = %.2f\n Yc_WK = %.2f\nDEP_WK = %.2f\nDelta_WK
= %.2f\n ', app.WavecomponentfileEditField.Value,Xc,Yc,DEP,app.WidthDeltaEditField.
Value);
1465
1466             %Log report
1467             checkAndReport(app.XcoordinatemEditField, 'X
Coordinate', fileId);
1468             checkAndReport(app.YcoordinatemEditField, 'Y
Coordinate', fileId);
1469             checkAndReport(app.WidthDeltaEditField, 'Width Delta',
fileId);
1470             checkAndReport(app.WaterdepthmEditField, 'Water depth
(m)', fileId);
1471
1472             %Check if the file is loaded
1473             wavecompconditions = [...
1474                 isempty(app.WavecomponentfileEditField.Value ),
...
1475                 ~any(strcmp(app.WavecomponentfileEditField.Value ,
app.WaveFileName)), ...
1476                 str2double(app.WavecomponentfileEditField.Value )
== 0 ...
1477             ];
1478
1479             if any(wavecompconditions)
1480                 %Display 'NO FILE' in the textbox and note the
issue in the log report
1481                 fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Wavemaker:');
1482                 fprintf(fileId, '%-35s%-s\n', '', 'No file
uploaded in ''Wave component file''');
1483                 app.WavecomponentfileEditField.Value = "NO FILE";
1484                 app.WavecomponentfileEditField.FontColor = 'w';
1485                 app.WavecomponentfileEditField.BackgroundColor =
'r';
1486                 error = error + 1;
1487             elseif strcmp(app.WavecomponentfileEditField.Value ,
app.WaveFileName)
1488                 % Check if the uploaded file is not in the input
folder
1489                 if ~exist(fullfile(app.inputFolder, app.WaveFile
WaveFileName), 'file')
1490                     % If the file is not in
'INPUT_FILES/Input_Data', copy it there
1491                     copyfile(app.WaveFileLOC, app.inputFolder);

```

```

1492         end
1493     end
1494
1495     case "INI_GAUSSIAN"
1496         %-----
1497         %.....INI_GAUSSIAN.....
1498         %-----
1499         tempWavemaker2 = sprintf('WAVEMAKER =\n
INI_GAUSSIAN\nAMP = %.2f\nXc = %.4f\nYc = %.4f\nWID = %.1f\n', AMP,Xc, Yc, WID);
1500
1501         %Log report
1502         checkAndReport(app.AmplitudemEditField, 'Amplitude\
(m)', fileId);
1503         checkAndReport(app.XcoordinatemEditField, 'X\
Coordinate', fileId);
1504         checkAndReport(app.YcoordinatemEditField, 'Y\
Coordinate', fileId);
1505         checkAndReport(app.WidthmEditField, 'Width (m)',\
fileId);
1506
1507     case "See options"
1508         fprintf(fileId, '%-30s%-s\n', 'INITIAL CONDITION:', 'No\
Wavemaker type selected');
1509         tempWavemaker2 = sprintf('!**No Wavemaker type\
selected\n');
1510         error = error + 1;
1511     end
1512
1513     %Combine all lines
1514     tempWavemaker = [tempWavemaker1, tempWavemaker2];
1515
1516     %Add these lines in the input.txt
1517     fprintf(fileInputId, '%s', tempWavemaker);
1518 end
1519
1520 %-----
1521 %-----INITIAL CONDITION: Meteotsunami-----
1522 %-----
1523 if app.MeteotsunamiButton.Value
1524     tempMeteo = sprintf('\n! -----\
METEOTSUNAMI-----\n');
1525
1526 %-----
1527 %.....Air Pressure.....
1528 %-----
1529 if strcmp(app.AirpressureSwitch.Value, "On")
1530     tempMeteo = [tempMeteo, sprintf('MeteoGaussian = T\n')];
1531
1532 %Check if the pressure file exists
1533 Pressureconditions = [...
1534     isempty(app.PressurefileEditField.Value), ... % Check\
if the field is empty
1535     ~any(strcmp(app.PressurefileEditField.Value, app.\
PressureFileName)), ... % Check if the value does not match expected names
1536     str2double(app.PressurefileEditField.Value) == 0 ... %\
Check if the value is zero (assuming numeric comparison is relevant)
1537 ];
1538

```

```

1539         if any(Pressureconditions)
1540             %Display 'NO FILE' in the textbox and note the issue
in the log report
1541             app.PressurefileEditField.BackgroundColor = 'r';
1542             app.PressurefileEditField.Value = "NO FILE";
1543             app.PressurefileEditField.FontColor = 'w';
1544             tempMeteo = [tempMeteo, sprintf('METEO_GAUSSIAN_FILE =
Input_Data/\n')]];
1545             fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Meteotsunami');
1546             fprintf(fileId, '%-35s%-s\n', '', 'No file uploaded in
''Pressure file''');
1547             error = error + 1;
1548             error_mkfile = error_mkfile + 1;
1549             error_mkfiletext9 = sprintf('INITIAL CONDITION:
Meteotsunami');
1550         else
1551             tempMeteo = [tempMeteo, sprintf('METEO_GAUSSIAN_FILE =
Input_Data/%s\n', string(app.PressureFileName))]];
1552             %If the file is not in 'INPUT_FILES/Input_Data', copy
it there
1553             if ~exist(fullfile(app.inputFolder, app.
PressureFileName), 'file')
1554                 copyfile(app.PressureFileLOC, app.inputFolder);
1555             end
1556         end
1557     else
1558         %Set the value to False
1559         tempMeteo = [tempMeteo, sprintf('MeteoGaussian = F\n')]];
1560     end
1561
1562     %-----
1563     %.....Constant Wind Field.....
1564     %-----
1565     if strcmp(app.ConstantwindfieldSwitch.Value, "On")
1566         tempMeteo = [tempMeteo, sprintf('WindConstantField =
T\n')]];
1567
1568         %Check if the uploaded Constant wind file exists
1569         WindWaveconditions = [...
1570             isempty( app.ConstantwindfileEditField.Value), ... %
Check if the field is empty
1571             ~any(strcmp( app.ConstantwindfileEditField.Value,
app.ConsWindNAME)), ... % Check if the value does not match expected names
1572             str2double( app.ConstantwindfileEditField.Value) == 0 %
... % Check if the value is zero (assuming numeric comparison is relevant)
1573             ];
1574
1575         if any(WindWaveconditions)
1576             tempMeteo = [tempMeteo, sprintf('CONSTANT_WIND_FILE =
Input_Data/\n')]];
1577
1578             %Display 'NO FILE' in the textbox and note the issue
in the log report
1579             app.ConstantwindfileEditField.BackgroundColor = 'r';
1580             app.ConstantwindfileEditField.Value = "NO FILE";
1581             app.ConstantwindfileEditField.FontColor = 'w';
1582

```

```

1583             fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Meteotsunami');
1584             fprintf(fileId, '%-35s%-s\n', '', 'No file uploaded in
''Pressure file'');
1585             error = error + 1;
1586             error_mkfile = error_mkfile + 1;
1587             error_mkfiletext9 = sprintf('INITIAL CONDITION:
Meteotsunami');
1588             else
1589                 tempMeteo = [tempMeteo, sprintf('CONSTANT_WIND_FILE =
Input_Data/%s\n', string(app.ConsWindNAME))];
1590                 %If the file is not in 'INPUT_FILES/Input_Data', copy
it there
1591                 if ~exist(fullfile(app.inputFolder, app.ConsWindNAME),
'file')
1592                     copyfile(app.ConsWindLOC, app.inputFolder);
1593                 end
1594             end
1595         else
1596             %Set the value to False
1597             tempMeteo = [tempMeteo, sprintf('WindConstantField =
F\n')];
1598         end
1599
1600         %-----
1601         %.....Wind Wave Interaction.....
1602         %-----
1603         if strcmp(app.WindforceSwitch.Value, "On")
1604             tempMeteo = [tempMeteo, sprintf('WindForce = T\nCdw = %.
3g\n', app.WindstresscoefficientEditField.Value)];
1605             app.WindstresscoefficientEditField.FontColor = 'k';
1606
1607             % Wind Wave Interaction settings
1608             if strcmp(app.WindwaveinteractionSwitch.Value, "On")
1609                 tempMeteo = [tempMeteo, sprintf('WindWaveInteraction =
T\nWindCrestPercent = %.2f\n', app.CrestRatioEditField.Value)];
1610                 app.CrestRatioEditField.FontColor = 'k';
1611             else
1612                 %Set the value to False
1613                 tempMeteo = [tempMeteo, sprintf('WindWaveInteraction =
F\n')];
1614             end
1615         else
1616             %Set the value to False
1617             tempMeteo = [tempMeteo, sprintf('WindForce = F\n')];
1618         end
1619
1620         %-----
1621         %.....Holland Storm Model.....
1622         %-----
1623         if strcmp(app.HollandstormmodelSwitch.Value, "On")
1624             tempMeteo = [tempMeteo, sprintf('WindHollandModel =
T\n')];
1625
1626             %Check if the uploaded Storm file exists
1627             Stormconditions = [...
1628                 isempty(app.StormfileEditField.Value), ... % Check if
the field is empty

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```

1629             ~any(strcmp(app.StormfileEditField.Value, app.
StormFileName)), ... % Check if the value does not match expected names
1630             str2double(app.StormfileEditField.Value) == 0 ... %
Check if the value is zero (assuming numeric comparison is relevant)
1631         ];
1632
1633         if any(Stormconditions)
1634             tempMeteo = [tempMeteo, sprintf('STORM_FILE =
Input_Data/\n')]];
1635
1636             %Display 'NO FILE' in the textbox and note the issue
in the log report
1637             app.StormfileEditField.BackgroundColor = 'r';
1638             app.StormfileEditField.Value = "NO FILE";
1639             app.StormfileEditField.FontColor = 'w';
1640
1641             fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Meteotsunami');
1642             fprintf(fileId, '%-35s%-s\n', '', 'No file uploaded in
'Storm file'');
1643             error = error + 1;
1644             error_mkfile = error_mkfile + 1;
1645             error_mkfiletext9 = sprintf('INITIAL CONDITION:
Meteotsunami');
1646         else
1647             tempMeteo = [tempMeteo, sprintf('STORM_FILE =
Input_Data/%s\n', string(app.StormFileName))]];
1648             % Check if the uploaded file is not from the
INPUT_FOLDER
1649             if ~exist(fullfile(inputFolder, app.StormFileName),
'file')
1650                 copyfile(app.StormFileLOC, inputFolder);
1651             end
1652         end
1653     else
1654         %Set parameter to False
1655         tempMeteo = [tempMeteo, sprintf('WindHollandModel =
F\n')]];
1656     end
1657
1658     %Add these lines in the input.txt
1659     fprintf(fileInputId, '%s', tempMeteo);
1660 end
1661
1662
1663
1664     %-----
1665     %-----INITIAL CONDITION: Sediment Transport-----
1666     %-----
1667     tempSediment = sprintf('\n! ----- SEDIMENT TRANSPORT
----- \n');
1668
1669     if app.SedimentTransportCheckBox.Value
1670         %-----
1671         %.....Sediment.....
1672         %-----
1673         if app.TVDButton.Value
1674             tempSediment = [tempSediment, sprintf('Sed Scheme = TVD

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\n')]];
1675         elseif app.UpwindingButton.Value
1676             tempSediment = [tempSediment, sprintf('Sed_Scheme =↵
Upwinding \n')]];
1677         end
1678
1679         editFields = [app.TimeStepSedToMorphlEditField, app.↵
MinDepthSedPickUpEditField, ...
1680             app.RungeKuttaparameter2EditField, app.↵
RungeKuttaparameter1EditField, ...
1681             app.ShieldsparamBedloadEditField, app.↵
CriticalShieldsEditField, ...
1682             app.SettlingvelocitymsEditField, app.↵
SedimentporosityEditField, ...
1683             app.SedimentdensityEditField, app.MediagrainDMEditField];
1684
1685         for field1 = editFields
1686             field1.FontColor = 'k';
1687         end
1688
1689         %Median Grain diameter
1690         tempSediment = [tempSediment, sprintf('D50 = %.3f \n', app.↵
MediagrainDMEditField.Value)];
1691
1692         %Sediment density
1693         tempSediment = [tempSediment, sprintf('Sdensity = %.3f \n',↵
app.SedimentdensityEditField.Value)];
1694
1695         %Sediment porosity
1696         tempSediment = [tempSediment, sprintf('n_porosity = %.3f \n',↵
app.SedimentporosityEditField.Value)];
1697
1698         %Settling velocity
1699         tempSediment = [tempSediment, sprintf('WS = %.3f \n', app.↵
SettlingvelocitymsEditField.Value)];
1700
1701         %Shields parameters (suspended load)
1702         tempSediment = [tempSediment, sprintf('Shields_cr = %.3f \n',↵
app.CriticalShieldsEditField.Value)];
1703
1704         %Shields parameters (bedload)
1705         tempSediment = [tempSediment, sprintf('Shields_cr_bedload = %.↵
3f \n', app.ShieldsparamBedloadEditField.Value)];
1706
1707         %Runge-Kutta pameters 1 and2
1708         tempSediment = [tempSediment, sprintf('Kappal = %.4f \n', app.↵
RungeKuttaparameter1EditField.Value)];
1709
1710         tempSediment = [tempSediment, sprintf('Kappa2 = %.3f \n', app.↵
RungeKuttaparameter2EditField.Value)];
1711
1712         %Minimum depth for sediment pickup action
1713         tempSediment = [tempSediment, sprintf('MinDepthPickup = %.3f↵
\n', app.MinDepthSedPickUpEditField.Value)];
1714
1715         %-----
1716         %.....Morphological Changes.....
1717         %-----

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```

1718         if app.UpdateddepthButton.Value
1719             tempSediment = [tempSediment, sprintf('Bed_Change = T\
\nBedLoad = T\n')]];
1720
1721         elseif app.NobedchangeButton.Value
1722             tempSediment = [tempSediment, sprintf('Bed_Change = F\
\n')]];
1723
1724         elseif app.Button_36.Value
1725             tempSediment = [tempSediment, sprintf('Bed_Change = \n')]];
1726
1727             %Log report
1728             fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Sediment:');
1729             fprintf(fileId, '%-35s%-s\n', '', 'No type of Bed Change is
selected');
1730
1731             app.UpdateddepthButton.FontColor = 'r';
1732             app.NobedchangeButton.FontColor = 'r';
1733             error = error + 1;
1734         end
1735
1736         if app.HardbednoerosionButton.Value
1737             if app.FileEditField.Value == string(app.HardbottomNAME)
1738                 tempSediment = [tempSediment, sprintf('Hard_bottom =
T\n')]];
1739
1740                 tempSediment = [tempSediment, sprintf(
('Hard_bottom_file = Input_Data/%s\n', string(app.HardbottomNAME))]];
1741
1742                 if ~exist(fullfile(app.inputFolder, app.
HardbottomNAME), 'file')
1743                     %If the file is not in 'INPUT_FILES/Input_Data',
copy it there
1744                     copyfile(app.HardbottomLOC, app.inputFolder);
1745                 end
1746             else
1747                 tempSediment = [tempSediment, sprintf(
('Hard_bottom_file = Input_Data/\n')]];
1748
1749                 %Display 'NO FILE' in the textbox and note the issue
in the log report
1750                 app.FileEditField.Value = "NO FILE";
1751                 app.FileEditField.FontColor = 'w';
1752                 app.FileEditField.BackgroundColor = 'r';
1753                 fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Sediment:');
1754                 fprintf(fileId, '%-35s%-s\n', '', 'No file uploaded in
'Hard Bottom File');
1755
1756                 error = error + 1;
1757                 error_mkfile = error_mkfile + 1;
1758                 error_mkfiletext11 = sprintf('Initial Condition:
Sediment');
1759             end
1760         elseif app.SedimentbedButton.Value
1761             %Set value to False
1762             tempSediment = [tempSediment, sprintf('Hard_bottom =
F\n')]];

```



```

1762         elseif app.Button_37.Value
1763             tempSediment = [tempSediment, sprintf('Hard_bottom = \n')];
1764
1765             %Note the issue in the log report
1766             fprintf(fileId, '%-30s%-s\n', 'INITIAL
CONDITION:', 'Sediment:');
1767             fprintf(fileId, '%-35s%-s\n', '', 'No Bed Type is selected');
1768             app.HardbednoerosionButton.FontColor = 'r';
1769             app.SedimentbedButton.FontColor = 'r';
1770             error = error + 1;
1771         end
1772
1773         %-----
1774         %.....Avalanche.....
1775         %-----
1776         if app.YesButton_8.Value
1777             app.RungeKuttaparam2EditField_2.FontColor = 'k';
1778             tempSediment = [tempSediment, sprintf('Avalanche = T
\n')];
1779             tempSediment = [tempSediment, sprintf('Tan_phi = %.3g \n',
app.RungeKuttaparam2EditField_2.Value)];
1780         end
1781
1782         %Add these lines in the input.txt
1783         fprintf(fileInputId, '%s', tempSediment);
1784
1785     end
1786
1787
1788     %-----
1789     %-----INITIAL CONDITION: Vessel-----
1790     %-----
1791     if app.VesselButton.Value
1792         %-----
1793         %.....Vessel.....
1794         %-----
1795         tempVessel1 = sprintf('\n! ----- SHIP WAKES \n');
1796
1797         %Check if the file is loaded
1798         Vesselconditions = [...
1799             isempty(app.ImportvesselfilesTextArea.Value), ...
1800             ~any(strcmp( app.ImportvesselfilesTextArea.Value, app.
VesselNAME)), ...
1801             str2double( app.ImportvesselfilesTextArea.Value) == 0 ...
1802             ];
1803
1804         if any(Vesselconditions)
1805             tempVessel2 = sprintf(['VESSEL_FOLDER = ./\n' ...
1806                 'NumVessel = \n']);
1807             %Display 'NO FILE' in the textbox and note the issue in
the log report
1808             fprintf(fileId, '%-30s%-s\n', 'INITIAL CONDITION:',
'Vessel:');
1809             fprintf(fileId, '%-35s%-s\n', '', 'No file uploaded in
'Import vessel file/s:');
1810             app.ImportvesselfilesTextArea.Value = "NO FILE";

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```

1811         app.ImportvesselfilesTextArea.FontColor = 'w';
1812         app.ImportvesselfilesTextArea.BackgroundColor = 'r';
1813
1814         error = error + 1;
1815         error_mkfile = error_mkfile + 1;
1816         error_mkfiletext8 = sprintf('Initial Condition: Vessel');
1817     else
1818         tempVessel2 = sprintf(['VESSEL_FOLDER = %s\n' ...
1819             'NumVessel = %s\n'], fullfile('.', 'Input_Data/'),
num2str(app.VesselNumber));
1820
1821         for g = 1:length(app.ImportvesselfilesTextArea.Value)
1822             %Copy files to'INPUT_FILES/Input_Data'
1823             copyfile(app.VesselLOC(g), app.inputFolder);
1824         end
1825     end
1826
1827     %Combine all the lines
1828     tempVessel = [tempVessel1,tempVessel2];
1829
1830     if app.ActivateCheckBox.Value
1831         %-----
1832         %.....Deep Draft Vessels.....
1833         %-----
1834         %Options for Methods:
1835
1836         %Friction
1837         if app.FrictionCheckBox_2.Value
1838             frictionCoefficient = app.
FrictionCoefficientEditField.Value;
1839             tempVessel = [tempVessel, sprintf('FrictionMethod =
T\nCdDeepDraft = %.1f\n', frictionCoefficient)];
1840             app.FrictionCoefficientEditField.FontColor = 'k';
1841         else
1842             tempVessel = [tempVessel, sprintf('FrictionMethod =
F\n')]];
1843         end
1844
1845         %Viscosity
1846         if app.ViscosityCheckBox.Value
1847             viscosityValue = app.ValueEditField.Value;
1848             tempVessel = [tempVessel, sprintf('ViscosityMethod =
T\nVisDeepDraft = %.1f\n', viscosityValue)];
1849             app.ValueEditField.FontColor = 'k';
1850         else
1851             tempVessel = [tempVessel, sprintf('ViscosityMethod =
F\n')]];
1852         end
1853
1854         %Shock capturing
1855         if app.ShockcapturingCheckBox.Value
1856             tempVessel = [tempVessel, sprintf('MaskMethod =
T\n')]];
1857         else
1858             tempVessel = [tempVessel, sprintf('MaskMethod =
F\n')]];
1859         end
1860

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```

1861         minimumClearance = app.MinimumClearanceEditField.Value;
1862
1863         %Add another line at the end of tempVessel
1864         tempVessel = [tempVessel, sprintf('CLEARANCE = %.1f\n',
minimumClearance)];
1865         app.MinimumClearanceEditField.FontColor = 'k';
1866     end
1867
1868     %Add these lines in the input.txt
1869     fprintf(fileInputId, '%s', tempVessel);
1870 end
1871
1872 %-----
1873 %-----PERIODIC BOUNDARY CONDITION-----
1874 %-----
1875 tempPBC = sprintf([
1876     '\n ! ----- PERIODIC BOUNDARY CONDITION -----
\n' ...
1877     ' ! South-North periodic boundary condition\n' ...
1878     'PERIODIC = F\n']);
1879
1880 %Add these lines in the input.txt
1881 fprintf(fileInputId, '%s', tempPBC);
1882
1883 %-----
1884 %-----SPONGE BOUNDARY-----
1885 %-----
1886 tempSponge = sprintf([
1887     '\n ! ----- SPONGE LAYER -----
\n' ...
1888     ' ! DHI type sponge layer\n' ...
1889     ' ! need to specify widths of four boundaries and
parameters\n' ...
1890     ' ! set width=0.0 if no sponge\n' ...
1891     ' ! R_sponge: decay rate\n' ...
1892     ' ! A_sponge: maximum decay rate\n' ...
1893     ' ! e.g., sharp: R=0.85 \n' ...
1894     ' !           mild: R=0.90, A=5.0 \n' ...
1895     ' !           very mild, R=0.95, A=5.0 \n' ]);
1896
1897
1898 %Add these lines in the input.txt
1899 fprintf(fileInputId, '%s', tempSponge);
1900
1901 % Set sponge dimensions if any sponge setting is active
1902 if app.DirectCheckBox.Value || app.FrictionCheckBox.Value || app.
DiffusionCheckBox.Value
1903     spongeOn = sprintf('SPONGE_ON = T \n');
1904 else
1905     spongeOn = sprintf('SPONGE_ON = F \n');
1906 end
1907 fprintf(fileInputId, '%s', spongeOn);
1908
1909 %Direct Sponge
1910 if app.DirectCheckBox.Value
1911     fprintf(fileInputId, 'DIRECT_SPONGE = T \n');
1912
1913     % Determine the decay rate and associated sponge settings

```

```

1914         switch app.DecayTypeDropDown.Value
1915             case 'Very mild'
1916                 decayRate = sprintf([ ...
1917                     'R_sponge = 0.95 \n' ...
1918                     'A_sponge = 5.0 \n']);
1919             case 'Mild'
1920                 decayRate = sprintf([ ...
1921                     'R_sponge = 0.9 \n' ...
1922                     'A_sponge = 5.0 \n']);
1923             case 'Sharp'
1924                 decayRate = sprintf('R_sponge = 0.85 \n');
1925         end
1926
1927         %Add these lines in the input.txt
1928         fprintf(fileInputId, '%s', decayRate);
1929     end
1930
1931     %Friction
1932     spongeSetting = '';
1933     if app.FrictionCheckBox.Value
1934         spongeSetting = sprintf('FRICTION_SPONGE = T\nCDsponge = %.3f \n',
1935 \n', app.MaximumCdEditField.Value);
1936         %Add these lines in the input.txt
1937         fprintf(fileInputId, '%s', spongeSetting);
1938     end
1939
1940     %Diffusion
1941     if app.DiffusionCheckBox.Value
1942         spongeSetting = sprintf('DIFFUSION_SPONGE = T\nCsp = %.3f \n',
1943 app.MaximumCspEditField.Value);
1944         %Add these lines in the input.txt
1945         fprintf(fileInputId, '%s', spongeSetting);
1946     end
1947
1948     %Add these lines in the input.txt
1949     % fprintf(fileInputId, '%s', spongeSetting);
1950
1951     % Set sponge dimensions if any sponge setting is active
1952     if app.DirectCheckBox.Value || app.FrictionCheckBox.Value || app.
DiffusionCheckBox.Value
1953         spongeBound = sprintf([
1954             'Sponge_west_width = %.1f\n' ...
1955             'Sponge_east_width = %.1f\n' ...
1956             'Sponge_south_width = %.1f\n' ...
1957             'Sponge_north_width = %.1f\n'], ...
1958 app.WestEditField.Value, app.EastEditField.Value, app.
SouthEditField.Value, app.NorthEditField.Value);
1959
1960         %Log warnings when the sponge layer is turned ON but the
boundaries are set to zero
1961         if app.NorthEditField.Value == 0
1962             app.NorthEditField.FontColor = 'r';
1963             fprintf(fileId, '%-30s%-s\n', 'SPONGE BOUNDARY:', 'North
buffer is zero');
1964         end
1965         if app.EastEditField.Value == 0
1966             app.EastEditField.FontColor = 'r';

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```

1966             fprintf(fileId, '%-30s%-s\n', 'SPONGE BOUNDARY:', 'East'
buffer is zero');
1967         end
1968         if app.WestEditField.Value == 0
1969             app.WestEditField.FontColor = 'r';
1970             fprintf(fileId, '%-30s%-s\n', 'SPONGE BOUNDARY:', 'West'
buffer is zero');
1971         end
1972         if app.SouthEditField.Value == 0
1973             app.SouthEditField.FontColor = 'r';
1974             fprintf(fileId, '%-30s%-s\n', 'SPONGE BOUNDARY:', 'South'
buffer is zero');
1975         end
1976
1977     else
1978         spongeBound = sprintf([
1979             'Sponge_west_width = 0.0\n' ...
1980             'Sponge_east_width = 0.0\n' ...
1981             'Sponge_south_width = 0.0\n' ...
1982             'Sponge_north_width = 0.0\n']);
1983
1984         app.NorthEditField.FontColor = '0.65,0.65,0.65';
1985         app.EastEditField.FontColor = '0.65,0.65,0.65';
1986         app.WestEditField.FontColor = '0.65,0.65,0.65';
1987         app.SouthEditField.FontColor = '0.65,0.65,0.65';
1988     end
1989
1990     %Add these lines in the input.txt
1991     fprintf(fileInputId, '%s', spongeBound);
1992
1993     %-----
1994     %-----OBSTACLES-----
1995     %-----
1996     if app.AddobstaclesCheckBox.Value
1997         if isempty(app.ObstacleFilename.Value) || ~strcmp(app.
ObstacleFilename.Value, app.obstacleNAME) || ~exist(app.obstacleLOC, "file")
1998             % Display 'NO FILE' in the textbox
1999             app.ObstacleFilename.Value = "NO FILE";
2000             app.ObstacleFilename.FontColor = 'w';
2001             app.ObstacleFilename.BackgroundColor = 'r';
2002             fprintf(fileId, '%-30s%-s\n', 'OBSTACLES:', 'No file is
uploaded');
2003             error = error + 1;
2004         else
2005             if strcmp(app.ObstacleFilename.Value, string(app.
obstacleNAME))
2006                 tempObstacle = sprintf([
2007                     ' \n! -----'
OBSTACLES-----\n' ...
2008                     ' ! obstacle structures using mask_struc file\n'
...
2009                     ' ! mask_struc =0 means structure element\n' ...
2010                     ' ! give a file contains a mask array with Mloc X
Nloc\n' ...
2011                     ' OBSTACLE_FILE= %s\n'], ...
2012                     app.ObstacleFilename.Value);
2013
2014                 % Add these lines in the input.txt

```

```

2015         fprintf(fileInputId, '%s', tempObstacle);
2016
2017         if exist(app.obstacleLOC, "file")
2018             obstacle = readmatrix(app.obstacleLOC);
2019             if isnumeric(obstacle)
2020                 if ~all(obstacle(:) == 0 | obstacle(:) == 1)
2021                     fprintf(fileId, '%-30s%-s\n', '
'OBSTACLES:', 'File should only contain values of 1 and 0');
2022                     end
2023                 end
2024             else
2025                 fprintf(fileId, '%-30s%-s\n', 'OBSTACLES:', '
'Uploaded file is empty');
2026             end
2027
2028             % If the file is not in 'INPUT_FILES/Input_Data', copy
it there
2029             if ~exist(fullfile(inputFolder, app.obstacleNAME), '
file')
2030                 copyfile(app.obstacleLOC, inputFolder);
2031             end
2032         end
2033     end
2034 end
2035
2036 %-----
2037 %-----PHYSICS-----
2038 %-----
2039 tempPhysics1 = sprintf([
2040     '\n ! -----PHYSICS-----
'
2041     ' ! parameters to control type of equations\n' ...
2042     ' ! dispersion: all dispersive terms\n' ...
2043     ' ! gamma1=1.0,gamma2=0.0: NGs equations\n' ...
2044     ' ! gamma1=1.0,gamma2=1.0: Fully nonlinear equations\n']);
2045
2046 %Initial set up for the second part of the lines for this section
2047 tempPhysics2 = '';
2048
2049 %-----
2050 %.....Linear Shallow Water Equation.....
2051 %-----
2052 if app.LinearShallowWaterEquationButton.Value
2053     app.LinearShallowWaterEquationButton.FontColor = 'k';
2054     tempPhysics2 = sprintf([
2055         'Gamma1 = 0.0\n' ...
2056         'Gamma2 = 0.0\n' ...
2057         'Gamma3 = 0.0\n']);
2058
2059 %-----
2060 %.....Nonlinear Shallow Water Equation.....
2061 %-----
2062 elseif app.NonlinearShallowWaterEquationButton.Value
2063     app.NonlinearShallowWaterEquationButton.FontColor = 'k';
2064     tempPhysics2 = sprintf([
2065         'Gamma1 = 0.0\n' ...
2066         'Gamma2 = 0.0\n' ...
2067         'Gamma3 = 1.0\n' ...

```

```

2068         'Beta_ref=-0.531\n']]);
2069
2070         %-----
2071         %.....Nonlinear Boussinesq Equation.....
2072         %-----
2073     elseif app.FullyNonlinearBoussinesqEquationButton.Value
2074         app.FullyNonlinearBoussinesqEquationButton.FontColor = 'k';
2075         tempPhysics2 = sprintf([
2076             'Gamma1 = 1.0\n' ...
2077             'Gamma2 = 1.0\n' ...
2078             'Gamma3 = 1.0\n' ...
2079             'Beta_ref=-0.531\n' ...
2080             'SWE_ETA_DEP = 0.8\n'...
2081             'DISPERSION = T\n']]);
2082     end
2083
2084
2085
2086     if app.LinearShallowWaterEquationButton.Value || app.
FullyNonlinearBoussinesqEquationButton.Value || app.
NonlinearShallowWaterEquationButton.Value
2087         %Combine all the lines
2088         tempPhysics = [tempPhysics1 tempPhysics2];
2089
2090         %Add these lines in the input.txt
2091         fprintf(fileInputId, '%s', tempPhysics);
2092     elseif app.Button.Value
2093         %Display 'NO FILE' in the textbox and note the issue in the
log report
2094         app.FullyNonlinearBoussinesqEquationButton.FontColor = 'r';
2095         app.LinearShallowWaterEquationButton.FontColor = 'r';
2096         app.NonlinearShallowWaterEquationButton.FontColor = 'r';
2097         fprintf(fileId, '%-30s%-s\n', 'PHYSICS:', 'No equation is
selected');
2098         error = error + 1;
2099     end
2100
2101     %-----
2102     %-----BOTTOM FRICTION-----
2103     %-----
2104     % Initialize base friction information
2105     tempFriction1 = sprintf(' \n !-----
Friction-----\n');
2106
2107     tempFriction2 = ''; %Initial set up for the second part of the
lines for this section
2108
2109     if app.ConstantCoefficientButton.Value
2110         %-----
2111         %.....Constant Friction.....
2112         %-----
2113         tempFriction2 = sprintf([
2114             'Friction_Matrix= F\n' ...
2115             'Cd = %.3f\n' ...
2116             'Cd_fixed = %.3f\n'], ...
app.ConstantFrictionValue.Value, ...
app.ConstantFrictionValue.Value);
2117
2118
2119

```

```

2120         if app.ConstantFrictionValue.Value == 0
2121             tempFriction2 = sprintf('Friction_Matrix= F \n Cd =\n
0.0');
2122         end
2123
2124     elseif app.NonconstantButton.Value
2125         %-----
2126         %.....Nonconstan Friction.....
2127         %-----
2128         if isempty(app.FrictionFileEditField.Value) || ~strcmp(app.\n
FrictionFileEditField.Value, app.filenamefriction) || ~exist(app.Cd_file, "file")
2129             %Display 'NO FILE' in the textbox and note the issue in\n
the log report
2130             app.FrictionFileEditField.BackgroundColor = 'r';
2131             app.FrictionFileEditField.Value = "NO FILE";
2132             app.FrictionFileEditField.FontColor = 'w';
2133             fprintf(fileId, '%-30s%-s\n', 'BOTTOM FRICTION:', 'No file\n
is uploaded');
2134
2135             error = error + 1;
2136             error_mkfile = error_mkfile + 1;
2137             error_mkfiletext7 = sprintf('Friction');
2138         else
2139             tempFriction2 = sprintf('Friction_Matrix= T\nCd_file=\n
Input_Data/%s\n', app.FrictionFileEditField.Value);
2140
2141             if ~exist(fullfile(app.inputFolder, app.Cd_file), 'file')
2142                 % If the file is not in 'INPUT_FILES/Input_Data', copy\n
it there
2143                 copyfile(app.Cd_file, app.inputFolder);
2144             end
2145         end
2146     end
2147
2148     % Combine the parts
2149     tempFriction = [tempFriction1 tempFriction2];
2150
2151     %Add these lines in the input.txt
2152     fprintf(fileInputId, '%s', tempFriction);
2153
2154     %-----
2155     %-----NUMERICS-----
2156     %-----
2157     if app.LinearShallowWaterEquationButton.Value || app.\n
NonlinearShallowWaterEquationButton.Value
2158         tempNumerics = sprintf([\n ! -----
2159         NUMERICS-----\n' ...
2160         'HIGH_ORDER = SECOND \n' ...
2161         'CFL = 0.5 \n' ...
2162         ' ! Froude Number Cap (to avoid jumping drop, set 10) \n'\n
...
2163         'FroudeCap = %.2f \n'], app.FroudeCapEditField.Value);
2164     else
2165         tempNumerics = sprintf([\n ! -----
2166         NUMERICS-----\n' ...
2167         'HIGH ORDER = THIRD \n' ...

```



```

2168         'CFL = 0.5 \n' ...
2169         ' ! Froude Number Cap (to avoid jumping drop, set 10) \n'↵
...
2170         'FroudeCap = %.2f \n'],app.FroudeCapEditField.Value);
2171     end
2172
2173     %Add these lines in the input.txt
2174     fprintf(fileInputId, '%s', tempNumerics);
2175
2176     %-----
2177     %-----MINIMUM DEPTH-----
2178     %-----
2179     tempWetDry1 = sprintf('\n ! -----WET-↵
DRY-----\n');
2180
2181     tempWetDry2 = sprintf([
2182         ' ! MinDepth for wetting-drying\n' ...
2183         'MinDepth= %.2f\n'], ...
2184         app.WetDrySchemeEditField.Value);
2185
2186     if app.ConstantCoefficientButton.Value || app.NonconstantButton.↵
Value
2187         tempWetDry3 = sprintf([
2188             ' ! MinDepthfrc to limit bottom friction\n' ...
2189             'MinDepthFrc = %.2f \n'], ...
2190             app.BottomFrictionEditField.Value);
2191
2192         tempWetDry = [tempWetDry1, tempWetDry2, tempWetDry3];
2193     else
2194         tempWetDry = [tempWetDry1, tempWetDry2];
2195     end
2196
2197     %Add these lines in the input.txt
2198     fprintf(fileInputId, '%s', tempWetDry);
2199
2200     %-----
2201     %-----WAVE BREAKING-----
2202     %-----
2203     if app.FullyNonlinearBoussinesqEquationButton.Value
2204         tempShowBreaking1 = sprintf([
2205             '\n ! ----- SHOW BREAKING↵
-----\n' ...
2206             ' ! breaking is calculated using shock wave capturing↵
scheme\n' ...
2207             ' ! the criteria is only for demonstration or bubble↵
calculation\n' ...
2208             ' ! Cbrk1=0.45,Cbrk2=0.35, for irregular waves, there are↵
much small!\n']]);
2209
2210         if app.ShockwavecapturingButton.Value
2211             tempShowBreaking2 = sprintf([
2212                 'SHOW_BREAKING = T\n' ...
2213                 'VISCOSITY_BREAKING = F\n']);
2214         elseif app.ViscositybreakingButton.Value
2215             tempShowBreaking2 = sprintf([
2216                 'SHOW_BREAKING = F\n' ...
2217                 'VISCOSITY_BREAKING = T\n'...
2218                 'Cbrk1 = %.2f\n' ...

```

```

2219         'Cbrk2 = %.2f\n'], ...
2220         app.C1EditField.Value, ...
2221         app.C2EditField.Value);
2222     else
2223         tempShowBreaking2 = '';
2224     end
2225
2226     tempShowBreaking = [tempShowBreaking1, tempShowBreaking2];
2227     fprintf(fileInputId, '%s', tempShowBreaking);
2228 end
2229
2230 if app.AddrollereffectsCheckBox.Value && ~app.
FullyNonlinearBoussinesqEquationButton.Value
2231     tempShowBreaking1 = sprintf([
2232         '\n ! ----- SHOW BREAKING
-----\n']);
2233     tempShowBreaking2 = sprintf('ROLLER_EFFECT = T');
2234     app.UndertowCheckBox.Value = 1; %Set the expected output
UNDERTOW = T
2235     app.RollerinducedFluxCheckBox.Value = 1; %Set the expected
output Roller = T
2236
2237     %Combine all lines
2238     tempShowBreaking = [tempShowBreaking1, tempShowBreaking2];
2239
2240     %Add these lines in the input.txt
2241     fprintf(fileInputId, '%s\n', tempShowBreaking);
2242
2243 elseif app.AddrollereffectsCheckBox.Value && app.
FullyNonlinearBoussinesqEquationButton.Value
2244     fprintf(fileInputId, '%s\n', 'ROLLER_EFFECT = T');
2245 end
2246
2247 %-----
2248 %-----MIXING-----
2249 %-----
2250 if app.UseSmagorinskyCheckBox.Value
2251     tempMixing = sprintf([...
2252         '\n ! ----- MIXING
-----\n' ...
2253         ' ! if use smagorinsky mixing, have to set -DMIXING in
Makefile\n' ...
2254         ' ! and set averaging time interval, T_INTV_mean,
default: 20s\n' ...
2255         'T_INTV_mean = %.2f\n' ...
2256         'STEADY_TIME = %.2f\n' ...
2257         'C_smg = 0.25\n'], ...
2258         app.TimeintervalsecEditField.Value, ...
2259         app.SteadyTimeEditField.Value);
2260     fprintf(fileInputId, '%s', tempMixing);
2261 end
2262
2263 %-----
2264 %-----EXPECTED OUTPUT-----
2265 %-----
2266     tempOutput1 = sprintf('\n! -----
OUTPUT-----\n');
2267

```

```

2268 %-----
2269 %-----GAUGES-----
2270 %-----
2271 if app.ImportgaugelistCheckBox.Value
2272     %Use the loaded file as regular virtual gauges, not to be used
for nested grid
2273
2274     %Check if gauge file is loaded
2275     loadFileNum = str2double(app.LoadFileEditField.Value);
2276
2277     conditions = [...
2278         isempty(app.LoadFileEditField.Value), ...
2279         loadFileNum == 0, ...
2280         ~any(strcmp(app.LoadFileEditField.Value, app.GAUGEFILE))
...
2281     ];
2282
2283     if any(conditions)
2284         tempOutput2 = sprintf('NumberStations = 0\n');
2285
2286         %Display 'NO FILE' in the textbox and note the issue in
the log report
2287         app.LoadFileEditField.BackgroundColor = 'r';
2288         app.LoadFileEditField.Value = "NO FILE";
2289         app.LoadFileEditField.FontColor = 'w';
2290         fprintf(fileId, '%-30s%-s\n', 'GAUGES:', 'No file is
uploaded');
2291         tempOutput = sprintf('NumberStations = 0\n');
2292         error = error + 1;
2293
2294         app.SpacingEditField.FontColor = '0.65,0.65,0.65';
2295         app.WestEditField_2.FontColor = '0.65,0.65,0.65';
2296         app.SouthEditField_2.FontColor = '0.65,0.65,0.65';
2297         app.EastEditField_2.FontColor = '0.65,0.65,0.65';
2298         app.NorthEditField_2.FontColor = '0.65,0.65,0.65';
2299
2300     elseif app.LoadFileEditField.Value == string(app.GAUGEFILE)
2301         tempOutput2 = sprintf([
2302             'NumberStations = %s\n' ...
2303             'STATIONS_FILE = Input_Data/%s\n'], ...
2304             app.NumberStations, ...
2305             app.STATION_FILE);
2306
2307     if contains(app.LoadFileEditField.Value, '.shp')
2308         % If the file type is a shapefile, convert it to a
tab-delimited text file
2309         outputTEXTFILE = fullfile(app.inputFolder, string(app.
GaugeFname) + '.txt');
2310         writematrix(app.GaugeListGrids, outputTEXTFILE,
'Delimiter', '\t');
2311
2312     elseif contains(app.LoadFileEditField.Value, '.txt')
2313         % Check if all items are numeric
2314         if isnumeric(app.gaugetext0)
2315             % Log Report
2316             if size(app.gaugetext0, 2) ~= 2
2317                 fprintf(fileId, '%-30s%-s\n', 'GAUGES:', 'Make
sure that the .txt file contains 2 columns (Lat Long)');

```

```

2318             error = error + 1;
2319         end
2320
2321         if any(isnan(app.gaugetext0(:)))
2322             fprintf(fileId, '%-30s%-s\n', 'GAUGES:', 'Text
file contains NaN values');
2323             error = error + 1;
2324         end
2325     else
2326         if ischar(app.gaugetext0) || isstring(app.
gaugetext0)
2327             if any(isletter(app.gaugetext0))
2328                 fprintf(fileId, '%-30s%-s\n', 'GAUGES:',
'Uploaded .txt file contains non-numeric values');
2329                 error = error + 1;
2330             end
2331         end
2332     end
2333
2334     % If the file is not in 'INPUT_FILES/Input_Data', copy
it there
2335     if ~exist(fullfile(inputFolder, app.GAUGEFILE),
'file')
2336         copyfile(app.GAUGEFILElocation, inputFolder);
2337     end
2338 end
2339 end
2340 else
2341     %Set value to zero
2342     tempOutput2 = sprintf('NumberStations = 0\n');
2343 end
2344
2345
2346
2347 if app.CreatenestedgridboundaryCheckBox.Value
2348     %Use the gauge stations as the boundary of the nested grid
2349
2350     %Create nested grid boundary code from Shi et al. (2012)
2351     % FUNWAVE version 3.6: simple cases > nesting tools >
make_stations_single_block.m
2352
2353     dx= app.SpacingEditField.Value;
2354     id_count=0;
2355     icount=0;
2356
2357     x1 = app.WestEditField_2.Value + (dx/2);
2358     x2 = app.EastEditField_2.Value;
2359     y1 = app.SouthEditField_2.Value + (dx/2);
2360     y2 = app.NorthEditField_2.Value;
2361
2362
2363     % if app.CoupMatrixON == 0 %Use the coordinates of the
uploaded .tif file
2364     %           %Assuming the coordinate of the SW corner of the tif
file is measured at the pixel's corner.
2365     %           %This reads the coordinate at the centre of the pixel
2366     %           x1 = app.WestEditField_2.Value; %+ (dx/2);
2367     %           x2 = app.EastEditField_2.Value; %- (dx/2);

```

```

2368         %     y1 = app.SouthEditField_2.Value; %+ (dx/2);
2369         %     y2 = app.NorthEditField_2.Value; %- (dx/2);
2370         % end
2371
2372         %Create the list of coordinates
2373         x=[x1:dx:x2];
2374         y=[y1:dx:y2];
2375
2376
2377         XCouplingLength = length(x);
2378         YCouplingLength = length(y);
2379         id_count=id_count+1;
2380
2381         %Arrangement of E,W,S,N is based on the requirement of
convert.f
2382         for j=1:length(y)
2383             icount=icount+1;
2384             app.NestStation(icount,1)=y(j);
2385             app.NestStation(icount,2)=x(end);
2386             app.NestStation(icount,3)=id_count;
2387         end
2388
2389         id_count=id_count+1;
2390         % west
2391         for j=1:length(y)
2392             icount=icount+1;
2393             app.NestStation(icount,1)=y(j);
2394             app.NestStation(icount,2)=x(1);
2395             app.NestStation(icount,3)=id_count;
2396         end
2397
2398
2399         id_count=id_count+1;
2400         % south
2401         for i=1:length(x)
2402             icount=icount+1;
2403             app.NestStation(icount,1)=y(1);
2404             app.NestStation(icount,2)=x(i);
2405             app.NestStation(icount,3)=id_count;
2406         end
2407
2408         id_count=id_count+1;
2409         % north
2410         for i=1:length(x)
2411             icount=icount+1;
2412             app.NestStation(icount,1)=y(end);
2413             app.NestStation(icount,2)=x(i);
2414             app.NestStation(icount,3)=id_count;
2415         end
2416
2417         fullFilePathNest = fullfile(app.inputFolder, 'nestingboundary.
txt');
2418         icount_last=icount;
2419
2420
2421         fid2 = fopen(fullfile(app.inputFolder, 'nestingboundary.
txt'), 'wt');
2422         fprintf(fid2, ['%f %f %d', '\n'], app.NestStation');
2423         fclose(fid2);

```

```

2424
2425
2426         tempOutput2 = sprintf([
2427             'NumberStations = %d\n' ...
2428             'STATIONS_FILE = Input_Data/nestingboundary.txt \n'], ...
2429             icount_last);
2430
2431         roundup = ceil(icount_last / 500) * 500; %round up to the
nearest 500;
2432
2433
2434         %Setting up the info to be displayed in the log report
2435         formattedCouplingDetails = sprintf([
2436             '%-30s%-s\n', ...
2437             '%-30s%-s\n', ...
2438             '%-30s%-s\n', ...
2439             '%-30s%-s\n', ...
2440             '%-30s%-s\n', ...
2441             '%-30s%-s\n', ...
2442             '%-30s%-s\n\n'], ...
2443             'GAUGES:', ''nestingboundary.txt' is created', ...
2444             'COUPLING:', 'Details to input in convert.f for your next
nested grid run:', ...
2445             '', ['nsta_tot and Nb: ' num2str(roundup)], ...
2446             '', ['nsta: ' num2str(icount_last)], ...
2447             '', ['neast and nwest: ' num2str(YCouplingLength)], ... %
num2str(app.CouplingX)], ... %refers to the N matrix dimension
2448             '', ['nnorth and nsouth: ' num2str(XCouplingLength)], ...
%, ... %num2str(app.CouplingY)]) %refers to the M matrix dimension
2449             '', 'Ntotal refers to the total number of rows contained in
each output `sta_` file generated after running your model',...
2450             '', 'Update `fdir` to the directory path where your
`sta_` files and convert.f file are located',...
2451             '', 'The `convert.f` file is located in
benchmarks/sph_nesting/make_nest_file');
2452
2453         %Formatting
2454         fprintf(fileId, formattedCouplingDetails);
2455
2456         %Log Report
2457         fields = {
2458             app.SpacingEditField, 'Gauge spacing is zero';
2459             app.WestEditField_2, 'West grid coordinate is zero';
2460             app.EastEditField_2, 'East grid coordinate is zero';
2461             app.SouthEditField_2, 'South grid coordinate is zero';
2462             app.NorthEditField_2, 'North grid coordinate is zero'
2463         };
2464
2465         %Loop through each field and check its value
2466         for i = 1:size(fields, 1)
2467             field = fields{i, 1};
2468             message = fields{i, 2};
2469
2470             if field.Value == 0
2471                 field.FontColor = 'r';
2472                 fprintf(fileId, '%-30s%-s\n', 'GAUGES:', message);
2473                 error = error + 1;
2474             end

```

```

2475         end
2476
2477
2478     end
2479
2480     %Combine all the lines and add them in the input.txt
2481     tempOutput = [tempOutput1,tempOutput2];
2482     fprintf(fileInputId, '%s', tempOutput);
2483
2484
2485     %-----
2486     %-----Expected OUTPUT-----
2487     %-----
2488     commands = struct( ...
2489         'DepthCheckBox', 'DEPTH_OUT = T \n', ...
2490         'UVelocityCheckBox', 'U = T \n', ...
2491         'VVelocityCheckBox', 'V = T \n', ...
2492         'UVelocitymeanCheckBox', 'Umean = T \n', ...
2493         'VVelocitymeanCheckBox', 'Vmean = T \n', ...
2494         'SurfaceElevationattimetCheckBox', 'ETA = T \n', ...
2495         'WetdrymaskCheckBox', 'MASK = T \n', ...
2496         'WetdrymaskforBoussinesqNSWECheckBox', 'MASK9 = T \n', ...
2497         'XSourceCheckBox', 'SourceX = T \n', ...
2498         'YSourceCheckBox', 'SourceY = T \n', ...
2499         'PMomentumFluxCheckBox', 'P = T \n', ...
2500         'QmomentumFluxCheckBox', 'Q = T \n', ...
2501         'FxFluxCheckBox', 'Fx = T \n', ...
2502         'GxFluxCheckBox', 'Gx = T \n', ...
2503         'GyFluxCheckBox', 'Gy = T \n', ...
2504         'BreakingAgeCheckBox', 'AGE = T \n', ...
2505         'MaximumWaveHeightCheckBox', 'Hmax = T \n', ...
2506         'FyFluxCheckBox', 'Fy = T \n', ...
2507         'MinimumWaveHeightCheckBox', 'Hmin = T \n', ...
2508         'MaximumVelocityCheckBox', 'Umax = T \n', ...
2509         'UndertowCheckBox', 'UNDERTOW = T \n', ...
2510         'MaxVorticityCheckBox', 'VORmax = T \n', ...
2511         'MaxMomentumFluxCheckBox', 'MFmax = T \n', ...
2512         'ArrivalCheckBox', 'OUT_Time = T \n', ...
2513         'WaveHeightCheckBox', 'WaveHeight = T \n', ...
2514         'RollerinducedFluxCheckBox', 'ROLLER = T \n', ...
2515         'PressureFieldCheckBox', 'OUT_METEO = T \n', ...
2516         'BreakingLocationCheckBox', 'OUT_NU = T \n' ...
2517     );
2518
2519
2520     %Loop through each item. If the values are checked in the app,
2521     list each variable and set them to 'T' in the input.txt
2522     fields = fieldnames(commands);
2523     for i = 1:length(fields)
2524         fieldName = fields{i};
2525         if app.(fieldName).Value
2526             fprintf(fileInputId, commands.(fieldName));
2527         end
2528     end
2529
2530     %-----
2531     %-----GENERAL ERRORS-----
2532     %-----

```

```

2532         if error ~=0 %Errors are present
2533             %Updating the log report
2534             fprintf(fileId, '%s\n', '');
2535             fprintf(fileId, '%s\n', '> Status: Creating input.txt...␣
Completed WITH ISSUES');
2536             fprintf(fileId, '%s\n', '> Resolve the warnings above to␣
prevent modeling errors later');
2537             app.Preview_LogReport.Value = fileread(LogFile); %show the log␣
report
2538             drawnow
2539
2540         else
2541             %Updating the log report
2542             fprintf(fileId, '%s\n', '');
2543             fprintf(fileId, '%s\n\n', '> Status: Creating input.txt...␣
Successfully completed!');
2544             app.Preview_LogReport.Value = fileread(LogFile); %show the log␣
report
2545             drawnow
2546         end
2547
2548         %-----
2549         %-----MAKEFILE ERRORS-----
2550         %-----
2551         %Checks if there are issues associated with Makefile creation
2552         if error_mkfile ~= 0
2553             %Updating the log report
2554             fprintf(fileId, '%s\n', '> Status: The Makefile section will be␣
enabled once the raised issues are resolved');
2555             fprintf(fileId, '%s\n', '    Specifically, check the following␣
sections:');
2556
2557             % Define a list of variables to check
2558             variablesToCheck = {'error_mkfiletext3', 'error_mkfiletext5',␣
'error_mkfiletext7', 'error_mkfiletext8', 'error_mkfiletext9',␣
'error_mkfiletext11'};
2559             flags = {' ', ' ', ' ', ' ', 'flag10', ' '}; % Associated flags␣
for each variable
2560
2561             % Loop through each variable and check if it exists
2562             for i = 1:length(variablesToCheck)
2563                 variableName = variablesToCheck{i};
2564                 flag = flags{i};
2565
2566                 % Use eval to check if variable exists and is not empty
2567                 if eval(sprintf('exist('%s', 'var') && ~isempty(%s)',␣
variableName, variableName))
2568                     % Print the variable's value
2569                     fprintf(fileId, '\t%2s\n', eval(variableName));
2570
2571                     % Check and print associated flag if it exists and is␣
not empty
2572                     if ~isempty(flag) && eval(sprintf('exist('%s',␣
'var') && ~isempty(%s)', flag, flag))
2573                         fprintf(fileId, '\t%2s\n', eval(flag));
2574                     end
2575                 end
2576             end

```



```

2577
2578         %Deactivate the Makefile section
2579         app.Mpif90EditField.Enable = 'off';
2580         app.Mpif90EditFieldLabel.Enable = 'off';
2581         app.Button_41.Enable = 'off';
2582         app.CreateMakefileButton.Enable = 'off';
2583         app.DoublePrecisionCheckBox.Enable = 'off';
2584         app.IntelCompilerCheckBox.Enable = 'off';
2585         app.ParallelModeCheckBox.Enable = 'off';
2586         app.MakefileEditField.Enable = 'off';
2587         app.MakefileEditFieldLabel.Enable = 'off';
2588         app.Button_42.Enable = 'off';
2589     else
2590         %Updating the log report
2591         fprintf(fileId, '%s\n', '');
2592         fprintf(fileId, '%s\n\n', '> Status: The Makefile section is
now enabled');
2593
2594         app.CreateMakefileButton.Enable = 'on';
2595         app.DoublePrecisionCheckBox.Enable = 'on';
2596         app.IntelCompilerCheckBox.Enable = 'on';
2597         app.ParallelModeCheckBox.Enable = 'on';
2598         app.MakefileEditField.Enable = 'on';
2599         app.MakefileEditFieldLabel.Enable = 'on';
2600         app.Button_42.Enable = 'on';
2601     end
2602
2603     %Close the log report and show it in the preview panel
2604     fclose(fileId);
2605     app.Preview_LogReport.Value = fileread(LogFile);
2606
2607     %Close the input.txt and show it in the preview panel
2608     fclose(fileInputId);
2609     fileContent = fileread(fileInputPath);
2610     app.Preview_Input.Value = fileContent;
2611
2612     drawnow
2613 end
2614
2615 % Callback function
2616 function ImportButton_4Pushed(app, event)
2617
2618     %
2619 end
2620
2621 % Value changed function: SimulationIntervalEditField
2622 function SimulationIntervalEditFieldValueChanged(app, event)
2623     app.SimulationIntervalEditField.FontColor = 'k';
2624 end
2625
2626 % Value changed function: GaugeIntervalEditField
2627 function GaugeIntervalEditFieldValueChanged(app, event)
2628     app.GaugeIntervalEditField.FontColor = 'k';
2629 end
2630
2631 % Value changed function: StartEditField
2632 function StartEditFieldValueChanged(app, event)
2633     app.StartEditField.FontColor = 'k';

```

```

2634
2635     end
2636
2637     % Value changed function: XEditField
2638     function XEditFieldValueChanged(app, event)
2639         app.XEditField.FontColor = 'k';
2640
2641     end
2642
2643     % Value changed function: YEditField
2644     function YEditFieldValueChanged(app, event)
2645         app.YEditField.FontColor = 'k';
2646     end
2647
2648     % Value changed function: LongitudeEditField
2649     function LongitudeEditFieldValueChanged(app, event)
2650         app.LongitudeEditField.FontColor = 'k';
2651
2652
2653     end
2654
2655     % Value changed function: LatitudeEditField
2656     function LatitudeEditFieldValueChanged(app, event)
2657         app.LatitudeEditField.FontColor = 'k';
2658     end
2659
2660     % Callback function
2661     function ButtonGroupSelectionChanged(app, event)
2662         selectedButton = app.ButtonGroup.SelectedObject;
2663         if selectedButton == app.SphericaldegreesButton
2664             app.SphericaldegreesButton.FontColor = 'k';
2665             app.CartesianmetersButton.FontColor = '0.50,0.50,0.50';
2666             value1 = app.XEditField;
2667             value2 = app.YEditField;
2668             value3 = '1.0';
2669             value4 = '1.0';
2670         elseif selectedButton == app.CartesianmetersButton
2671             app.CartesianmetersButton.FontColor = 'k';
2672             app.SphericaldegreesButton.FontColor = '0.50,0.50,0.50';
2673             value3 = app.XEditField;
2674             value4 = app.YEditField;
2675             value1 = '1.0';
2676             value2 = '1.0';
2677         end
2678
2679         app.Dphi = value1;
2680         app.Dtheta = value2;
2681         app.DX = value3;
2682         app.DY = value4;
2683
2684     end
2685
2686     % Callback function
2687     function ButtonGroup_2SelectionChanged(app, event)
2688
2689         selectedButton = app.ButtonGroup_2.SelectedObject;
2690         app.SWE = selectedButton;
2691

```

```

2692     end
2693
2694     % Selection changed function: ButtonGroup_3
2695     function ButtonGroup_3SelectionChanged(app, event)
2696         app.FullyNonlinearBoussinesqEquationButton.FontColor = 'k';
2697         app.LinearShallowWaterEquationButton.FontColor = 'k';
2698         app.NonlinearShallowWaterEquationButton.FontColor = 'k';
2699
2700
2701         if app.FullyNonlinearBoussinesqEquationButton.Value
2702             app.ShockwavecapturingButton.Enable = "on";
2703             app.ViscositybreakingButton.Enable = "on";
2704             drawnow
2705             if app.ViscositybreakingButton.Value
2706                 app.C2EditField.Enable = "on";
2707                 app.C2EditFieldLabel.Enable = "on";
2708                 app.C1EditField.Enable = "on";
2709                 app.C1EditFieldLabel.Enable = "on";
2710             end
2711         else
2712             app.ShockwavecapturingButton.Enable = "off";
2713             app.ViscositybreakingButton.Enable = "off";
2714             app.C2EditField.Enable = "off";
2715             app.C2EditFieldLabel.Enable = "off";
2716             app.C1EditField.Enable = "off";
2717             app.C1EditFieldLabel.Enable = "off";
2718         end
2719     end
2720
2721     % Selection changed function: ButtonGroup_4
2722     function ButtonGroup_4SelectionChanged(app, event)
2723         app.SphericalButton.FontColor = 'k';
2724         app.CartesianButton.FontColor = 'k';
2725
2726         if app.SphericalButton.Value || app.CartesianButton.Value
2727             app.LatitudeEditField.Enable = 'on';
2728             app.LongitudeEditField.Enable = 'on';
2729             app.XEditField_2.Enable = "on";
2730             app.YEditField_2.Enable = "on";
2731         end
2732
2733     end
2734
2735     % Value changed function: DepthCheckBox
2736     function DepthCheckBoxValueChanged(app, event)
2737
2738     end
2739
2740     % Value changed function: UVelocityCheckBox
2741     function UVelocityCheckBoxValueChanged(app, event)
2742
2743
2744     end
2745
2746     % Value changed function: VVelocityCheckBox
2747     function VVelocityCheckBoxValueChanged(app, event)
2748
2749     end

```

```
2750
2751 % Value changed function: SurfaceElevationattimetCheckBox
2752 function SurfaceElevationattimetCheckBoxValueChanged(app, event)
2753
2754 end
2755
2756 % Value changed function: WetdrymaskCheckBox
2757 function WetdrymaskCheckBoxValueChanged(app, event)
2758
2759 end
2760
2761 % Value changed function: WetdrymaskforBoussinesqNSWECheckBox
2762 function WetdrymaskforBoussinesqNSWECheckBoxValueChanged(app, event)
2763
2764 end
2765
2766 % Value changed function: XSourceCheckBox
2767 function XSourceCheckBoxValueChanged(app, event)
2768
2769 end
2770
2771 % Value changed function: YSourceCheckBox
2772 function YSourceCheckBoxValueChanged(app, event)
2773
2774 end
2775
2776 % Value changed function: PMomentumFluxCheckBox
2777 function PMomentumFluxCheckBoxValueChanged(app, event)
2778
2779 end
2780
2781 % Value changed function: QmomentumFluxCheckBox
2782 function QmomentumFluxCheckBoxValueChanged(app, event)
2783
2784 end
2785
2786 % Value changed function: FxFluxCheckBox
2787 function FxFluxCheckBoxValueChanged(app, event)
2788
2789 end
2790
2791 % Value changed function: FyFluxCheckBox
2792 function FyFluxCheckBoxValueChanged(app, event)
2793
2794 end
2795
2796 % Value changed function: GxFluxCheckBox
2797 function GxFluxCheckBoxValueChanged(app, event)
2798
2799 end
2800
2801 % Value changed function: GyFluxCheckBox
2802 function GyFluxCheckBoxValueChanged(app, event)
2803
2804 end
2805
2806 % Value changed function: BreakingAgeCheckBox
2807 function BreakingAgeCheckBoxValueChanged(app, event)
```

```

2808
2809     end
2810
2811     % Value changed function: MaximumWaveHeightCheckBox
2812     function MaximumWaveHeightCheckBoxValueChanged(app, event)
2813
2814     end
2815
2816     % Value changed function: MinimumWaveHeightCheckBox
2817     function MinimumWaveHeightCheckBoxValueChanged(app, event)
2818
2819     end
2820
2821     % Value changed function: MaximumVelocityCheckBox
2822     function MaximumVelocityCheckBoxValueChanged(app, event)
2823
2824     end
2825
2826     % Value changed function: UndertowCheckBox
2827     function UndertowCheckBoxValueChanged(app, event)
2828
2829     end
2830
2831     % Value changed function: MaxVorticityCheckBox
2832     function MaxVorticityCheckBoxValueChanged(app, event)
2833
2834     end
2835
2836     % Value changed function: MaxMomentumFluxCheckBox
2837     function MaxMomentumFluxCheckBoxValueChanged(app, event)
2838
2839     end
2840
2841     % Value changed function: ArrivalCheckBox
2842     function ArrivalCheckBoxValueChanged(app, event)
2843     if app.ArrivalCheckBox.Value
2844         app.ThresholdEditFieldLabel.Visible = "on";
2845         app.ThresholdEditField.Visible = "on";
2846     else
2847         app.ThresholdEditFieldLabel.Visible = "off";
2848         app.ThresholdEditField.Visible = "off";
2849         app.ThresholdEditField.FontColor = '0.65,0.65,0.65';
2850     end
2851
2852
2853     end
2854
2855     % Value changed function: WaveHeightCheckBox
2856     function WaveHeightCheckBoxValueChanged(app, event)
2857
2858     end
2859
2860     % Value changed function: RollerinducedFluxCheckBox
2861     function RollerinducedFluxCheckBoxValueChanged(app, event)
2862
2863     end
2864
2865     % Value changed function: PressureFieldCheckBox

```

```

2866     function PressureFieldCheckBoxValueChanged(app, event)
2867
2868     end
2869
2870     % Value changed function: BreakingLocationCheckBox
2871     function BreakingLocationCheckBoxValueChanged(app, event)
2872
2873     end
2874
2875     % Callback function
2876     function EastlimitmEditFieldValueChanged(app, event)
2877         app.EastEditField.FontColor = 'k';
2878     end
2879
2880     % Callback function
2881     function WestbountmEditFieldValueChanged(app, event)
2882         app.WestEditField.FontColor = 'k';
2883     end
2884
2885     % Value changed function: SouthEditField
2886     function SouthEditFieldValueChanged(app, event)
2887         app.SouthEditField.FontColor = 'k';
2888     end
2889
2890     % Value changed function: NorthEditField
2891     function NorthEditFieldValueChanged(app, event)
2892         app.NorthEditField.FontColor = 'k';
2893     end
2894
2895     % Callback function
2896     function ButtonGroup_5SelectionChanged(app, event)
2897
2898     end
2899
2900     % Callback function
2901     function DirectSpongeSwitchValueChanged(app, event)
2902
2903         if strcmp(app.DirectSpongeSwitch.Value, 'On')
2904             app.DecayTypeDropDown.Visible = "on";
2905             app.DecayTypeDropDownLabel.Visible = "on";
2906             app.Panel_16.Enable = "on";
2907         elseif strcmp(app.DirectSpongeSwitch.Value, 'Off')
2908             app.DecayTypeDropDown.Visible = "off";
2909             app.DecayTypeDropDownLabel.Visible = "off";
2910         end
2911
2912
2913
2914     end
2915
2916     % Callback function
2917     function Button_9Pushed(app, event)
2918         maindir = uigetdir;
2919         maindir = fullfile(maindir, ' ');
2920         app.ResultsfolderEditField.Value = deblank(string(maindir));
2921         app.ResultsfolderEditField.FontColor = 'k';
2922     end
2923

```

```

2924 % Button pushed function: Button_7
2925 function Button_7Pushed2(app, event)
2926     [filename3,path3] = uigetfile('*.txt;*.shp');
2927     fullname = fullfile(path3,filename3);
2928     try
2929         [~, name, ~] = fileparts(filename3);
2930         app.GaugeFname = name;
2931         app.LoadFileEditField.Value = filename3;
2932         app.GAUGEFILE = filename3;
2933         app.GAUGEFILElocation = fullname;
2934         app.LoadFileEditField.FontColor = 'k';
2935         app.LoadFileEditField.BackgroundColor = 'w';
2936
2937         if contains(filename3, '.shp')
2938             gauge = shaperead(fullname);
2939             latGT = [];
2940             lonGT = [];
2941             app.latGauge = latGT;
2942
2943             for k = 1:length(gauge)
2944                 latGT = [latGT, gauge(k).X]; % Concatenate X coordinates
2945                 lonGT = [lonGT, gauge(k).Y]; % Concatenate Y coordinates
2946             end
2947
2948             list = [round(lonGT,4)',round(latGT,4)'];
2949             list = unique(list, 'rows'); %remove duplicate lat-long↙
combination
2950             app.NumberStations = num2str(height(list));
2951             app.STATION_FILE = string(name) + ".txt";
2952
2953             %Save as text file
2954             app.GaugeListGrids = list;
2955
2956         elseif contains(filename3, '.txt')
2957             gaugetext = readmatrix(fullname);
2958             app.gaugetext0 = gaugetext;
2959             app.NumberStations = num2str(height(gaugetext));
2960             app.STATION_FILE = string(filename3);
2961
2962         end
2963
2964         catch
2965             %do nothing
2966         end
2967
2968     end
2969
2970 % Value changed function: EastEditField
2971 function EastEditFieldValueChanged(app, event)
2972     app.EastEditField.FontColor = 'k';
2973 end
2974
2975 % Value changed function: WestEditField
2976 function WestEditFieldValueChanged(app, event)
2977     app.WestEditField.FontColor = 'k';
2978 end
2979
2980 % Value changed function: ResultsfolderEditField

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```

2981     function ResultsfolderEditFieldValueChanged(app, event)
2982
2983
2984     end
2985
2986     % Value changed function: ProcessorX
2987     function ProcessorXValueChanged(app, event)
2988         app.ProcessorX.FontColor = 'k';
2989
2990     end
2991
2992     % Value changed function: ProcessorY
2993     function ProcessorYValueChanged(app, event)
2994         app.ProcessorY.FontColor = 'k';
2995
2996     end
2997
2998     % Value changed function: TotalEditField
2999     function TotalEditFieldValueChanged2(app, event)
3000         app.TotalEditField.FontColor = 'k';
3001
3002     end
3003
3004     % Button pushed function: Button_32
3005     function Button_4Pushed(app, event)
3006         [bathymetryfile,path2] = uigetfile('*.');
3007         app.SurfaceHeightEditField.Value = string(bathymetryfile);
3008         app.SurfaceHeightEditField.FontColor = 'k';
3009         app.SurfaceHeightEditField.BackgroundColor = 'w';
3010         app.SurfaceHeightLOC = string(path2) + string(bathymetryfile);
3011         app.SurfaceHeightNAME = string(bathymetryfile);
3012
3013
3014     end
3015
3016     % Button pushed function: Button_34
3017     function Button_5Pushed(app, event)
3018         [file,path2] = uigetfile('*.');
3019         app.UVelocityEditField.Value = string(file);
3020         app.UVelocityEditField.FontColor = 'k';
3021         app.UVelocityEditField.BackgroundColor = 'w';
3022         app.XVelocityLOC = string(path2) + string(file);
3023         app.XVelocityNAME = string(file);
3024
3025     end
3026
3027     % Button pushed function: Button_33
3028     function Button_6Pushed(app, event)
3029         [file,path2] = uigetfile('*.');
3030         app.VVelocityEditField.Value = string(file);
3031         app.VVelocityEditField.FontColor = 'k';
3032         app.VVelocityEditField.BackgroundColor = 'w';
3033         app.YVelocityLOC = string(path2) + string(file);
3034         app.YVelocityNAME = string(file);
3035     end
3036
3037     % Selection changed function: ButtonGroup_18
3038     function ButtonGroup_6SelectionChanged(app, event)

```



```

3039
3040         if app.ImportUVZButton.Value
3041             app.TabGroup2.SelectedTab = app.Tab;
3042
3043         end
3044
3045         if app.WavemakerButton.Value
3046             app.TabGroup2.SelectedTab = app.Tab_2;
3047
3048         end
3049
3050         if app.MeteotsunamiButton.Value
3051             app.TabGroup2.SelectedTab = app.Tab_4;
3052             app.MinimumWaveHeightCheckBox.Value = 1;
3053             app.PressureFieldCheckBox.Value = 1;
3054         else
3055             %uncheck the expected output values
3056             app.MinimumWaveHeightCheckBox.Value = 0;
3057             app.PressureFieldCheckBox.Value = 0;
3058         end
3059
3060
3061         if app.VesselButton.Value
3062             app.TabGroup2.SelectedTab = app.Tab_6;
3063
3064         end
3065
3066     end
3067
3068     % Value changed function: DepthFileEditField
3069     function DepthFileEditFieldValueChanged(app, event)
3070         app.DepthFileEditField.BackgroundColor = 'w';
3071         app.DepthFileEditField.FontColor = 'k';
3072
3073     end
3074
3075     % Selection changed function: ButtonGroup_7
3076     function ButtonGroup_7SelectionChanged(app, event)
3077         %List all the items to be displayed or hidden
3078         function setVisibility(app, depthVisible, angleVisible, ↵
startXVisible, button8Visible, depthFileVisible, waterLevelVisible, ↵
invertValuesVisible)
3079             app.DepthmEditFieldLabel.Visible = depthVisible;
3080             app.DepthmEditField.Visible = depthVisible;
3081             app.AngleEditFieldLabel.Visible = angleVisible;
3082             app.AngleEditField.Visible = angleVisible;
3083             app.StartatXEditFieldLabel.Visible = startXVisible;
3084             app.StartatXEditField.Visible = startXVisible;
3085             app.Button_8.Visible = button8Visible;
3086             app.DepthFileEditField.Visible = depthFileVisible;
3087             app.WaterlevelmEditFieldLabel.Visible = waterLevelVisible;
3088             app.WaterlevelmEditField.Visible = waterLevelVisible;
3089             app.InvertvaluesCheckBox.Visible = invertValuesVisible;
3090         end
3091
3092
3093         if app.ImportDataButton.Value
3094             setVisibility(app, "off", "off", "off", "on", "on", "on", ↵

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```

3095         elseif app.FlatButton.Value
3096             setVisibility(app, "on", "off", "off", "off", "off", "off", "off",
"off");
3097         elseif app.SlopeButton.Value
3098             setVisibility(app, "on", "on", "on", "off", "off", "off",
"off");
3099         end
3100
3101         % Set font color to black
3102         if ~app.Button4.Value
3103             app.ImportDataButton.FontColor = 'k';
3104             app.SlopeButton.FontColor = 'k';
3105             app.FlatButton.FontColor = 'k';
3106         end
3107     end
3108
3109     % Size changed function: InputTab
3110     function InputTabSizeChanged(app, event)
3111
3112     end
3113
3114     % Value changed function: DepthmEditField
3115     function DepthmEditFieldValueChanged(app, event)
3116
3117         app.DepthmEditField.FontColor = 'k';
3118         app.DepthmEditField.BackgroundColor = 'w';
3119
3120     end
3121
3122     % Value changed function: AngleEditField
3123     function AngleEditFieldValueChanged(app, event)
3124
3125         app.AngleEditField.FontColor = 'k';
3126         app.AngleEditField.BackgroundColor = 'w';
3127     end
3128
3129     % Value changed function: StartatXEditField
3130     function StartatXEditFieldValueChanged(app, event)
3131
3132         app.StartatXEditField.FontColor = 'k';
3133         app.StartatXEditField.BackgroundColor = 'w';
3134     end
3135
3136     % Value changing function: ResultsfolderEditField
3137     function ResultsfolderEditFieldValueChanging(app, event)
3138
3139         app.ResultsfolderEditField.BackgroundColor = 'w';
3140         app.ResultsfolderEditField.FontColor = 'k';
3141     end
3142
3143     % Value changed function: YEditField_2
3144     function YEditField_2ValueChanged(app, event)
3145
3146         app.YEditField_2.FontColor = 'k';
3147     end
3148
3149     % Button pushed function: Button 10

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```

3150     function Button_10Pushed(app, event)
3151         [frictionfile,frictionpath] = uigetfile('*.txt;*.tif');
3152         app.FrictionFileEditField.Value = string(frictionfile);
3153         app.Cd_file = fullfile(string(frictionpath) + string(
(frictionfile)));
3154         app.filenamefriction = string(frictionfile);
3155         app.FrictionFileEditField.FontColor = 'k';
3156         app.FrictionFileEditField.BackgroundColor = 'w';
3157     end
3158
3159     % Value changing function: FrictionFileEditField
3160     function FrictionFileEditFieldValueChanging(app, event)
3161         app.FrictionFileEditField.Value = '';
3162         app.FrictionFileEditField.FontColor = 'k';
3163         app.FrictionFileEditField.BackgroundColor = 'w';
3164     end
3165
3166     % Selection changed function: ButtonGroup_8
3167     function ButtonGroup_8SelectionChanged(app, event)
3168         if app.ConstantCoefficientButton.Value
3169             app.FrictionFileEditField.Visible = "off";
3170             app.FrictionFileEditFieldLabel.Visible = "off";
3171             app.Button_10.Visible = "off";
3172             app.ConstantFrictionValue.Visible = "on";
3173             app.ValueEditFieldLabel.Visible = "on";
3174             % app.BottomFrictionEditField.Enable = "on";
3175             % app.BottomFrictionEditFieldLabel.Enable = "on";
3176         elseif app.NonconstantButton.Value
3177             app.FrictionFileEditField.Visible = "on";
3178             app.FrictionFileEditFieldLabel.Visible = "on";
3179             app.Button_10.Visible = "on";
3180             app.ConstantFrictionValue.Visible = "off";
3181             app.ValueEditFieldLabel.Visible = "off";
3182             % app.BottomFrictionEditField.Enable = "on";
3183             % app.BottomFrictionEditFieldLabel.Enable = "on";
3184         end
3185
3186     end
3187
3188     % Value changed function: ConstantFrictionValue
3189     function ConstantFrictionValueValueChanged(app, event)
3190         app.ConstantFrictionValue.FontColor = 'k';
3191         app.ConstantFrictionValue.BackgroundColor = 'w';
3192     end
3193
3194     % Value changed function: C2EditField
3195     function C2EditFieldValueChanged(app, event)
3196         app.C2EditField.FontColor = 'k';
3197
3198
3199     end
3200
3201     % Value changed function: C1EditField
3202     function C1EditFieldValueChanged(app, event)
3203
3204         app.C1EditField.FontColor = 'k';
3205     end
3206

```

```

3207 % Value changed function: WetDrySchemeEditField
3208 function WetDrySchemeEditFieldValueChanged(app, event)
3209     app.WetDrySchemeEditField.FontColor = 'k';
3210
3211
3212 end
3213
3214 % Value changed function: BottomFrictionEditField
3215 function BottomFrictionEditFieldValueChanged(app, event)
3216     app.BottomFrictionEditField.FontColor = 'k';
3217
3218 end
3219
3220 % Value changing function: SavefilestoEditField
3221 function SavefilestoEditFieldValueChanging(app, event)
3222     app.SavefilestoEditField.Value = '';
3223     app.SavefilestoEditField.FontColor = 'k';
3224     app.SavefilestoEditField.BackgroundColor = 'w';
3225
3226 end
3227
3228 % Value changed function: DecayTypeDropDown
3229 function DecayTypeDropDownValueChanged(app, event)
3230
3231 end
3232
3233 % Button pushed function: ObstacleLoadFile
3234 function Button_19Pushed(app, event)
3235     [filename,path] = uigetfile('*.txt;*.tif');
3236     app.ObstacleFilename.Value = string(filename);
3237     app.ObstacleFilename.FontColor = 'k';
3238     app.ObstacleFilename.BackgroundColor = 'w';
3239     app.obstacleLOC = string(path) + string(filename);
3240     app.obstacleNAME = string(filename);
3241 end
3242
3243 % Value changed function: MaximumCdEditField
3244 function MaximumCdEditFieldValueChanged(app, event)
3245     app.MaximumCdEditField.FontColor = 'k';
3246
3247 end
3248
3249 % Value changed function: MaximumCspEditField
3250 function MaximumCspEditFieldValueChanged(app, event)
3251     app.MaximumCspEditField.FontColor = 'k';
3252
3253 end
3254
3255 % Value changed function: WavemakertypeDropDown
3256 function WavemakertypeDropDownValueChanged(app, event)
3257 %Disable and reset all fields and buttons first
3258 allFields = [app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthmEditField, app.WidthDeltaEditField, app.AmplitudemEditField, app.↵
WaterdepthmEditField, app.LagtimesecEditField, app.ThetadegreesEditField, app.↵
TimerampsecEditField, app.PeriodsecEditField, app.WavecomponentfileEditField, app.↵
WaveCompCountEditField, app.MinimumEditField, app.MaximumEditField, app.PeakEditField, ↵
app.WaveHeightmEditField];
3259 for field = allFields

```

```

3260     field.Enable = 'off';
3261     field.FontColor = [0.8, 0.8, 0.8];
3262     if isprop(field, 'Value') && isnumeric(field.Value)
3263         field.Value = 0;
3264     elseif isprop(field, 'Value')
3265         field.Value = "";
3266     end
3267 end
3268 app.Button_20.Enable = 'off'; %Disable import button of 'Save Component File'
3269
3270
3271 %List of the paremeters to be enabled based on the wavemaker type selected
3272 wavemakerSettings = struct(...
3273     'INI_REC', {{app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthmEditField}}, ...
3274     'LEF_SOL', {{app.AmplitudemEditField, app.WaterdepthmEditField, app.↵
LagtimesecEditField}}, ...
3275     'INI_SOL', {{app.AmplitudemEditField, app.WaterdepthmEditField, app.↵
XcoordinatemEditField}}, ...
3276     'WK_IRR', {{app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthmEditField, app.TimerampsecEditField, app.WidthDeltaEditField, app.↵
MinimumEditField, app.MaximumEditField, app.PeakEditField, app.WaveHeightmEditField, ↵
app.WaterdepthmEditField}}, ...
3277     'JON_1D', {{app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthmEditField, app.TimerampsecEditField, app.WidthDeltaEditField, app.↵
MinimumEditField, app.MaximumEditField, app.PeakEditField, app.WaveHeightmEditField, ↵
app.WaterdepthmEditField}}, ...
3278     'JON_2D', {{app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthmEditField, app.TimerampsecEditField, app.WidthDeltaEditField, app.↵
MinimumEditField, app.MaximumEditField, app.PeakEditField, app.WaveHeightmEditField, ↵
app.WaterdepthmEditField}}, ...
3279     'TMA_1D', {{app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthmEditField, app.TimerampsecEditField, app.WidthDeltaEditField, app.↵
MinimumEditField, app.MaximumEditField, app.PeakEditField, app.WaveHeightmEditField, ↵
app.WaterdepthmEditField}}, ...
3280     'WK_REG', {{app.XcoordinatemEditField, app.AmplitudemEditField, app.↵
WaterdepthmEditField, app.ThetadegreesEditField, app.TimerampsecEditField, app.↵
WidthDeltaEditField}}, ...
3281     'WK_TIME_SERIES', {{app.WavecomponentfileEditField, app.Button_20, app.↵
PeriodsecEditField, app.WaterdepthmEditField, app.XcoordinatemEditField, app.↵
WidthmEditField, app.WaveCompCountEditField}}, ...
3282     'WK_DATA2D', {{app.XcoordinatemEditField, app.YcoordinatemEditField, app.↵
WidthDeltaEditField, app.WaterdepthmEditField, app.WavecomponentfileEditField, app.↵
Button_20}}, ...
3283     'INI_GAUSSIAN', {{app.AmplitudemEditField, app.XcoordinatemEditField, app.↵
YcoordinatemEditField, app.WidthmEditField}} ...
3284 );
3285
3286 %Enable paremeters based on the wavemaker type selected
3287 selectedType = app.WavemakertypeDropDown.Value;
3288 if isfield(wavemakerSettings, selectedType)
3289     fieldsToEnable = wavemakerSettings.(selectedType);
3290     for fieldCell = fieldsToEnable
3291         field = fieldCell{:};
3292         field.Enable = 'on';
3293     end
3294 end
3295

```

```

3296 if ~strcmp(app.WavemakertypeDropDown.Value, 'WK_TIME_SERIES')
3297     app.WavecomponentfileEditField.BackgroundColor = 'w';
3298     app.WavecomponentfileEditField.Value = '';
3299     app.WavecomponentfileEditField.FontColor = 'k';
3300 end
3301
3302     end
3303
3304     % Drop down opening function: WavemakertypeDropDown
3305     function WavemakertypeDropDownOpening(app, event)
3306         app.WavemakertypeDropDown.FontColor = '0.00,0.00,0.00';
3307     end
3308
3309     % Value changed function: AmplitudemEditField
3310     function AmplitudemEditFieldValueChanged(app, event)
3311         app.AmplitudemEditField.FontColor = 'k';
3312     end
3313
3314     % Value changed function: WaterdepthmEditField
3315     function WaterdepthmEditFieldValueChanged(app, event)
3316         app.WaterdepthmEditField.FontColor = 'k';
3317
3318
3319     end
3320
3321     % Value changed function: LagtimesecEditField
3322     function LagtimesecEditFieldValueChanged(app, event)
3323         app.LagtimesecEditField.FontColor = 'k';
3324
3325     end
3326
3327     % Value changed function: XcoordinatemEditField
3328     function XcoordinatemEditFieldValueChanged(app, event)
3329         app.XcoordinatemEditField.FontColor = 'k';
3330
3331
3332     end
3333
3334     % Callback function
3335     function XcentercoordinateEditFieldValueChanged(app, event)
3336
3337
3338     end
3339
3340     % Value changed function: YcoordinatemEditField
3341     function YcoordinatemEditFieldValueChanged(app, event)
3342         app.YcoordinatemEditField.FontColor = 'k';
3343
3344
3345     end
3346
3347     % Value changed function: WidthmEditField
3348     function WidthmEditFieldValueChanged(app, event)
3349         app.WidthmEditField.FontColor = 'k';
3350
3351
3352     end
3353

```

```

3354 % Value changed function: TimerampsecEditField
3355 function TimerampsecEditFieldValueChanged(app, event)
3356     app.TimerampsecEditField.FontColor = 'k';
3357
3358
3359 end
3360
3361 % Value changed function: WidthDeltaEditField
3362 function WidthDeltaEditFieldValueChanged(app, event)
3363     app.WidthDeltaEditField.FontColor = 'k';
3364
3365
3366 end
3367
3368 % Value changed function: PeriodsecEditField
3369 function PeriodsecEditFieldValueChanged(app, event)
3370     app.PeriodsecEditField.FontColor = 'k';
3371
3372
3373 end
3374
3375 % Button pushed function: Button_20
3376 function Button_20Pushed(app, event)
3377     [file,path] = uigetfile('*.txt'); %*.txt;
3378     app.WavecomponentfileEditField.Value = string(file);
3379     app.WaveFileName = string(file);
3380     app.WaveFileLOC = string(path) + string(file);
3381     app.WavecomponentfileEditField.BackgroundColor = 'w';
3382     app.WavecomponentfileEditField.FontColor = 'k';
3383
3384 end
3385
3386 % Value changed function: WavecomponentfileEditField
3387 function WavecomponentfileEditFieldValueChanged(app, event)
3388
3389
3390 end
3391
3392 % Value changed function: ThetadegreesEditField
3393 function ThetadegreesEditFieldValueChanged(app, event)
3394     app.ThetadegreesEditField.FontColor = 'k';
3395
3396 end
3397
3398 % Value changed function: XEditField_2
3399 function XEditField_2ValueChanged(app, event)
3400     app.XEditField_2.FontColor = 'k';
3401
3402
3403 end
3404
3405 % Callback function
3406 function TimeintervalsecEditFieldValueChanged(app, event)
3407     app.TimeintervalsecEditField.FontColor = 'k';
3408
3409 end
3410
3411 % Callback function

```

```

3412     function SteadyTimeEditFieldValueChanged(app, event)
3413         app.SteadyTimeEditField.FontColor = 'k';
3414
3415     end
3416
3417     % Callback function
3418     function ButtonGroup_12SelectionChanged(app, event)
3419
3420
3421     end
3422
3423     % Callback function
3424     function SchemeDropDownValueChanged(app, event)
3425
3426     end
3427
3428     % Value changing function: UVelocityEditField
3429     function UVelocityEditFieldValueChanging(app, event)
3430
3431     end
3432
3433     % Value changing function: SurfaceHeightEditField
3434     function SurfaceHeightEditFieldValueChanging(app, event)
3435
3436
3437     end
3438
3439     % Callback function
3440     function Button5Pushed(app, event)
3441         app.TabGroup2.SelectedTab = app.Tab_3;           %app.InputsTab;
3442     end
3443
3444     % Value changing function: VVelocityEditField
3445     function VVelocityEditFieldValueChanging(app, event)
3446
3447
3448     end
3449
3450     % Value changed function: WindforceSwitch
3451     function WindforceSwitchValueChanged(app, event)
3452         if strcmp(app.WindforceSwitch.Value, "On")
3453             app.WindwaveinteractionSwitch.Enable = "on";
3454             app.WindwaveinteractionSwitchLabel.Enable = "on";
3455             app.WindstresscoefficientEditField.Enable = "on";
3456             app.WindstresscoefficientEditFieldLabel.Enable = "on";
3457             app.CrestRatioEditField.Enable = "on";
3458             app.CrestRatioEditFieldLabel.Enable = "on";
3459
3460         elseif strcmp(app.WindforceSwitch.Value, "Off")
3461             app.WindwaveinteractionSwitch.Enable = "off";
3462             app.WindwaveinteractionSwitchLabel.Enable = "off";
3463             app.WindstresscoefficientEditField.Enable = "off";
3464             app.WindstresscoefficientEditFieldLabel.Enable = "off";
3465             app.CrestRatioEditField.Enable = "off";
3466             app.CrestRatioEditFieldLabel.Enable = "off";
3467             app.WindstresscoefficientEditField.FontColor = ⚡
3468             '0.65,0.65,0.65';
3469             app.WindstresscoefficientEditField.FontColor = ⚡

```



```

'0.65,0.65,0.65';
3469         app.CrestRatioEditField.FontColor = '0.65,0.65,0.65';
3470     end
3471
3472 end
3473
3474 % Value changed function: ConstantwindfieldSwitch
3475 function ConstantwindfieldSwitchValueChanged(app, event)
3476
3477     if strcmp(app.ConstantwindfieldSwitch.Value, "On")
3478         app.ConstantwindfileEditFieldLabel.Enable = "on";
3479         app.ConstantwindfileEditField.Enable = "on";
3480         app.Button_29.Enable = "on";
3481         app.WindforceSwitch.Value = "On";
3482
3483         app.ConstantwindfieldSwitch.Enable = "on";
3484         app.ConstantwindfieldSwitchLabel.Enable = "on";
3485         app.WindwaveinteractionSwitch.Enable = "on";
3486         app.WindwaveinteractionSwitchLabel.Enable = "on";
3487         app.WindstresscoefficientEditField.Enable = "on";
3488         app.WindstresscoefficientEditFieldLabel.Enable = "on";
3489         app.CrestRatioEditField.Enable = "on";
3490         app.CrestRatioEditFieldLabel.Enable = "on";
3491
3492     elseif strcmp(app.ConstantwindfieldSwitch.Value, "Off")
3493         app.ConstantwindfileEditFieldLabel.Enable = "off";
3494         app.ConstantwindfileEditField.Enable = "off";
3495         app.Button_29.Enable = "off";
3496         app.WindforceSwitch.Value = "Off";
3497         app.WindwaveinteractionSwitch.Enable = "off";
3498         app.WindwaveinteractionSwitchLabel.Enable = "off";
3499         app.WindstresscoefficientEditField.Enable = "off";
3500         app.WindstresscoefficientEditFieldLabel.Enable = "off";
3501         app.CrestRatioEditField.Enable = "off";
3502         app.CrestRatioEditFieldLabel.Enable = "off";
3503     end
3504
3505 end
3506
3507 % Button pushed function: Button_29
3508 function Button_29Pushed(app, event)
3509     [file,path2] = uigetfile('*.');
3510     app.ConstantwindfileEditField.Value = string(file);
3511     app.ConstantwindfileEditField.FontColor = 'k';
3512     app.ConstantwindfileEditField.BackgroundColor = 'w';
3513     app.ConsWindLOC = string(path2) + string(file);
3514     app.ConsWindNAME = string(file);
3515 end
3516
3517 % Value changed function: CrestRatioEditField
3518 function CrestRatioEditFieldValueChanged(app, event)
3519
3520 end
3521
3522 % Value changed function: WindstresscoefficientEditField
3523 function WindstresscoefficientEditFieldValueChanged(app, event)
3524     app.WindstresscoefficientEditField.FontColor = 'k';
3525

```

```

3526     end
3527
3528     % Button pushed function: Button_30
3529     function Button_30Pushed(app, event)
3530         [file,path2] = uigetfile('*.');
3531         app.StormfileEditField.Value = string(file);
3532         app.StormfileEditField.FontColor = 'k';
3533         app.StormfileEditField.BackgroundColor = 'w';
3534         app.StormFileLOC = string(path2) + string(file);
3535         app.StormFileName = string(file);
3536     end
3537
3538     % Callback function
3539     function Button_39Pushed(app, event)
3540         [file,path2] = uigetfile('*.');
3541         app.PressurefileEditField.Value = string(file);
3542         app.PressurefileEditField.FontColor = 'k';
3543         app.PressurefileEditField.BackgroundColor = 'w';
3544         app.PressureFileLOC = string(path2) + string(file);
3545         app.PressureFileName = string(file);
3546     end
3547
3548     % Value changed function: WindwaveinteractionSwitch
3549     function WindwaveinteractionSwitchValueChanged(app, event)
3550
3551
3552     end
3553
3554     % Value changed function: AirpressureSwitch
3555     function AirpressureSwitchValueChanged(app, event)
3556         if strcmp(app.AirpressureSwitch.Value,"On")
3557             app.PressurefileEditField.Enable = "on";
3558             app.PressurefileEditFieldLabel.Enable = "on";
3559             app.Button_43.Enable = "on";
3560         else
3561             app.PressurefileEditField.Enable = "off";
3562             app.PressurefileEditFieldLabel.Enable = "off";
3563             app.Button_43.Enable = "off";
3564         end
3565     end
3566
3567     % Value changed function: HollandstormmodelSwitch
3568     function HollandstormmodelSwitchValueChanged(app, event)
3569         if strcmp(app.HollandstormmodelSwitch.Value,"On")
3570             app.StormfileEditFieldLabel.Enable = "on";
3571             app.StormfileEditField.Enable = "on";
3572             app.Button_30.Enable = "on";
3573         else
3574             app.StormfileEditFieldLabel.Enable = "off";
3575             app.StormfileEditField.Enable = "off";
3576             app.Button_30.Enable = "off";
3577         end
3578     end
3579
3580     % Value changed function: MediangrainDMEditField
3581     function MediangrainDMEditFieldValueChanged(app, event)
3582         app.MediangrainDMEditField.FontColor = 'k';
3583

```

```

3584     end
3585
3586     % Value changed function: SedimentdensityEditField
3587     function SedimentdensityEditFieldValueChanged(app, event)
3588         app.SedimentdensityEditField.FontColor = 'k';
3589
3590     end
3591
3592     % Value changed function: SedimentporosityEditField
3593     function SedimentporosityEditFieldValueChanged(app, event)
3594         app.SedimentporosityEditField.FontColor = 'k';
3595
3596     end
3597
3598     % Value changed function: SettlingvelocitymsEditField
3599     function SettlingvelocitymsEditFieldValueChanged(app, event)
3600         app.SettlingvelocitymsEditField.FontColor = 'k';
3601
3602     end
3603
3604     % Value changed function: CriticalShieldsEditField
3605     function CriticalShieldsEditFieldValueChanged(app, event)
3606         app.CriticalShieldsEditField.FontColor = 'k';
3607
3608     end
3609
3610     % Value changed function: ShieldsparamBedloadEditField
3611     function ShieldsparamBedloadEditFieldValueChanged(app, event)
3612         app.ShieldsparamBedloadEditField.FontColor = 'k';
3613
3614     end
3615
3616     % Value changed function: RungeKuttaparameter1EditField
3617     function RungeKuttaparameter1EditFieldValueChanged(app, event)
3618         app.RungeKuttaparameter1EditField.FontColor = 'k';
3619
3620     end
3621
3622     % Value changed function: RungeKuttaparameter2EditField
3623     function RungeKuttaparameter2EditFieldValueChanged(app, event)
3624         app.RungeKuttaparameter2EditField.FontColor = 'k';
3625
3626     end
3627
3628     % Value changed function: MinDepthSedPickUpEditField
3629     function MinDepthSedPickUpEditFieldValueChanged(app, event)
3630         app.MinDepthSedPickUpEditField.FontColor = 'k';
3631
3632     end
3633
3634     % Value changed function: TimeStepSedToMorphlEditField
3635     function TimeStepSedToMorphlEditFieldValueChanged(app, event)
3636         app.TimeStepSedToMorphlEditField.FontColor = 'k';
3637
3638     end
3639
3640     % Callback function
3641     function PressurefileEditFieldValueChanged(app, event)

```

```

3642         app.PressurefileEditField.FontColor = 'k';
3643
3644     end
3645
3646     % Value changed function: TimeStepBouss
3647     function TimeStepBoussValueChanged(app, event)
3648         app.TimeStepBouss.FontColor = 'k';
3649
3650     end
3651
3652     % Value changed function: RungeKuttaparam2EditField_2
3653     function RungeKuttaparam2EditField_2ValueChanged(app, event)
3654         app.RungeKuttaparam2EditField_2.FontColor = 'k';
3655
3656     end
3657
3658     % Button pushed function: Button_35
3659     function Button_35Pushed(app, event)
3660         [file,path2] = uigetfile('*.*.');
3661         app.FileEditField.Value = string(file);
3662         app.FileEditField.FontColor = 'k';
3663         app.FileEditField.BackgroundColor = 'w';
3664         app.HardBottomFileLOC = string(path2) + string(file);
3665         app.HardBottomFileNAME = string(file);
3666     end
3667
3668     % Selection changed function: ButtonGroup_21
3669     function ButtonGroup_21SelectionChanged(app, event)
3670         if app.HardbednoerosionButton.Value
3671             app.Button_35.Visible= "on";
3672             app.FileEditField.Visible = "on";
3673             app.FileEditFieldLabel.Visible = "on";
3674         elseif app.SedimentbedButton.Value
3675
3676             app.Button_35.Visible = "off";
3677             app.FileEditField.Visible = "off";
3678             app.FileEditFieldLabel.Visible = "off";
3679         end
3680
3681         if ~app.Button_37.Value
3682             app.SedimentbedButton.FontColor = 'k';
3683             app.HardbednoerosionButton.FontColor = 'k';
3684         end
3685
3686     end
3687
3688     % Selection changed function: ButtonGroup_20
3689     function ButtonGroup_20SelectionChanged(app, event)
3690         if ~app.Button_36.Value
3691             app.NobedchangeButton.FontColor = 'k';
3692             app.UpdatedepthButton.FontColor = 'k';
3693         end
3694     end
3695
3696     % Selection changed function: ButtonGroup_19
3697     function ButtonGroup_19SelectionChanged(app, event)
3698
3699     end

```

```

3700
3701 % Selection changed function: ButtonGroup_22
3702 function ButtonGroup_22SelectionChanged(app, event)
3703     if app.YesButton_8.Value
3704         app.RungeKuttaparam2EditField_2.Enable = "on";
3705         app.RungeKuttaparam2EditFieldLabel_2.Enable = "on";
3706     elseif app.NoButton_7.Value
3707         app.RungeKuttaparam2EditField_2.Enable = "off";
3708         app.RungeKuttaparam2EditFieldLabel_2.Enable = "off";
3709     end
3710
3711
3712 end
3713
3714 % Button pushed function: Button_40
3715 function Button_40Pushed(app, event)
3716
3717     [files, path] = uigetfile('*..*','Select the files','MultiSelect',↵
'on');
3718     app.ImportvesselfilesTextArea.Value = string(files);
3719     app.VesselLOC = string(path) + string(files);
3720     app.VesselNAME = string(files); %string(path);
3721     app.ImportvesselfilesTextArea.FontColor = 'k';
3722     app.ImportvesselfilesTextArea.BackgroundColor = 'w';
3723
3724     if ischar(files)
3725         % Only one file was selected
3726         app.VesselNumber = 1;
3727     elseif iscell(files)
3728         % Multiple files were selected
3729         app.VesselNumber = numel(files);
3730     end
3731
3732
3733 end
3734
3735 % Value changed function: FileEditField
3736 function FileEditFieldValueChanged(app, event)
3737     [file,path] = uigetfile('*..*');
3738     app.FileEditField.Value = string(file);
3739     app.FileEditField.FontColor = 'k';
3740     app.FileEditField.BackgroundColor = 'w';
3741     app.HardbottomLOC = string(path) + string(file);
3742     app.HardbottomtNAME = string(file);
3743 end
3744
3745 % Value changed function: DirectCheckBox
3746 function DirectCheckBoxValueChanged(app, event)
3747 if app.DirectCheckBox.Value
3748     app.DecayTypeDropDown.Visible = "on";
3749     app.DecayTypeDropDownLabel.Visible = "on";
3750 else
3751     app.DecayTypeDropDown.Visible = "off";
3752     app.DecayTypeDropDownLabel.Visible = "off";
3753 end
3754
3755
3756 if app.DirectCheckBox.Value || app.FrictionCheckBox.Value || app.↵

```

```

DiffusionCheckBox.Value
3757     %if there are other buttons checked in Sponge Boundary, the N,E,W,S
textboxes are not disabled
3758     app.WestEditField.Enable = "on";
3759     app.SouthEditField.Enable = "on";
3760     app.EastEditField.Enable = "on";
3761     app.NorthEditField.Enable = "on";
3762 else
3763     %Include disable the N,E,W,S textboxes
3764     app.DecayTypeDropDown.Visible = "off";
3765     app.DecayTypeDropDownLabel.Visible = "off";
3766     app.WestEditField.Enable = "off";
3767     app.SouthEditField.Enable = "off";
3768     app.EastEditField.Enable = "off";
3769     app.NorthEditField.Enable = "off";
3770 end
3771
3772
3773     end
3774
3775     % Value changed function: FrictionCheckBox
3776     function FrictionCheckBoxValueChanged(app, event)
3777         if app.FrictionCheckBox.Value
3778             app.MaximumCdEditFieldLabel.Visible = "on";
3779             app.MaximumCdEditField.Visible = "on";
3780         else
3781             app.MaximumCdEditField.Visible = "off";
3782             app.MaximumCdEditFieldLabel.Visible = "off";
3783         end
3784
3785         if app.DirectCheckBox.Value || app.FrictionCheckBox.Value || app.
DiffusionCheckBox.Value
3786             %if there are other buttons checked in Sponge Boundary, the N,
E,W,S textboxes are not disabled
3787             app.WestEditField.Enable = "on";
3788             app.SouthEditField.Enable = "on";
3789             app.EastEditField.Enable = "on";
3790             app.NorthEditField.Enable = "on";
3791         else
3792             %Include disable the N,E,W,S textboxes
3793             app.DecayTypeDropDown.Visible = "off";
3794             app.DecayTypeDropDownLabel.Visible = "off";
3795             app.WestEditField.Enable = "off";
3796             app.SouthEditField.Enable = "off";
3797             app.EastEditField.Enable = "off";
3798             app.NorthEditField.Enable = "off";
3799         end
3800
3801     end
3802
3803     % Value changed function: DiffusionCheckBox
3804     function DiffusionCheckBoxValueChanged(app, event)
3805         if app.DiffusionCheckBox.Value
3806             app.MaximumCspEditField.Visible = "on";
3807             app.MaximumCspEditFieldLabel.Visible = "on";
3808         else
3809             app.MaximumCspEditFieldLabel.Visible = "off";
3810             app.MaximumCspEditField.Visible = "off";

```

```

3811         end
3812
3813         if app.DirectCheckBox.Value || app.FrictionCheckBox.Value || app.↵
DiffusionCheckBox.Value
3814             %if there are other buttons checked in Sponge Boundary, the N,↵
E,W,S textboxes are not disabled
3815             app.WestEditField.Enable = "on";
3816             app.SouthEditField.Enable = "on";
3817             app.EastEditField.Enable = "on";
3818             app.NorthEditField.Enable = "on";
3819         else
3820             %Include disable the N,E,W,S textboxes
3821             app.DecayTypeDropDown.Visible = "off";
3822             app.DecayTypeDropDownLabel.Visible = "off";
3823             app.WestEditField.Enable = "off";
3824             app.SouthEditField.Enable = "off";
3825             app.EastEditField.Enable = "off";
3826             app.NorthEditField.Enable = "off";
3827         end
3828
3829     end
3830
3831     % Callback function
3832     function UseBreakingIndexCheckBoxValueChanged(app, event)
3833
3834         if app.UseBreakingIndexCheckBox.Value
3835             app.C1EditField.Visible = "on";
3836             app.C1EditFieldLabel.Visible = "on";
3837             app.C2EditField.Visible = "on";
3838             app.C2EditFieldLabel.Visible = "on";
3839             app.SchemeDropDown.Visible = "on";
3840             app.SchemeDropDownLabel.Visible = "on";
3841         else
3842             app.C1EditField.Visible = "off";
3843             app.C1EditFieldLabel.Visible = "off";
3844             app.C2EditField.Visible = "off";
3845             app.C2EditFieldLabel.Visible = "off";
3846             app.SchemeDropDown.Visible = "off";
3847             app.SchemeDropDownLabel.Visible = "off";
3848         end
3849     end
3850
3851     % Callback function
3852     function UseSmagorinskymixingCheckBoxValueChanged(app, event)
3853         if app.UseSmagorinskyCheckBox.Value
3854             app.TimeintervalsecEditField.Visible = "on";
3855             app.TimeintervalsecEditFieldLabel.Visible = "on";
3856             app.SteadyTimeEditField.Visible = "on";
3857             app.SteadyTimeEditFieldLabel.Visible = "on";
3858         else
3859             app.TimeintervalsecEditField.Visible = "off";
3860             app.TimeintervalsecEditFieldLabel.Visible = "off";
3861             app.SteadyTimeEditField.Visible = "off";
3862             app.SteadyTimeEditFieldLabel.Visible = "off";
3863         end
3864     end
3865
3866     % Value changed function: AddobstaclesCheckBox

```

```

3867     function AddobstaclesCheckBoxValueChanged(app, event)
3868         if app.AddobstaclesCheckBox.Value
3869             app.ObstacleFilename.Visible = "on";
3870             app.ObstacleLoadFile.Visible = "on";
3871         else
3872             app.ObstacleFilename.Visible = "off";
3873             app.ObstacleLoadFile.Visible = "off";
3874         end
3875
3876
3877     end
3878
3879     % Value changed function: CreatenestedgridboundaryCheckBox
3880     function CreatenestedgridboundaryCheckBoxValueChanged(app, event)
3881         %Define the fields to be enabled/disabled under the nested grid
3882         option
3883             fieldsToToggle = {
3884                 app.SpacingEditFieldLabel, app.SpacingEditField, app.
3885                 WestEditField_2, ...
3886                 app.SouthEditField_2, app.EastEditField_2, app.
3887                 NorthEditField_2, app.BoundarylimitsButton
3888             };
3889
3890         if app.CreatenestedgridboundaryCheckBox.Value
3891             % Disable the items under the 'Import gauge list' option
3892             app.ImportgaugeListCheckBox.Value = false;
3893             app.LoadFileEditField.Enable = "off";
3894             app.Button_7.Enable = "off";
3895
3896             % Enable fields under the nested grid option
3897             for i = 1:length(fieldsToToggle)
3898                 fieldsToToggle{i}.Enable = "on";
3899             end
3900         else
3901             % Set font color to gray
3902             grayColor = '0.65,0.65,0.65';
3903             app.SpacingEditField.FontColor = grayColor;
3904             app.WestEditField_2.FontColor = grayColor;
3905             app.SouthEditField_2.FontColor = grayColor;
3906             app.EastEditField_2.FontColor = grayColor;
3907             app.NorthEditField_2.FontColor = grayColor;
3908
3909             %Disable fields under the nested grid option
3910             for i = 1:length(fieldsToToggle)
3911                 fieldsToToggle{i}.Enable = "off";
3912             end
3913         end
3914
3915     end
3916
3917     % Value changing function: DepthFileEditField
3918     function DepthFileEditFieldValueChanging(app, event)
3919
3920
3921     end
3922
3923     % Value changed function: ImportgaugeListCheckBox
3924     function ImportgaugeListCheckBoxValueChanged(app, event)

```



```

3922     app.WestEditField_2.Enable = "off";
3923     app.SouthEditField_2.Enable = "off";
3924     app.EastEditField_2.Enable = "off";
3925     app.NorthEditField_2.Enable = "off";
3926     app.BoundarylimitsButton.Enable = "off";
3927
3928     if app.ImportgaugelistCheckBox.Value
3929         app.CreatenestedgridboundaryCheckBox.Enable = "on";
3930         app.LoadFileEditField.Enable = "on";
3931         app.Button_7.Enable = "on";
3932         app.CreatenestedgridboundaryCheckBox.Value = false;
3933         app.SpacingEditFieldLabel.Enable = "off";
3934         app.SpacingEditField.Enable = "off";
3935         app.SpacingEditFieldLabel.Enable = 'off';
3936
3937     else
3938         app.LoadFileEditField.Enable = "off";
3939         app.Button_7.Enable = "off"; %To upload gauge data
3940         app.SpacingEditFieldLabel.Enable = "off";
3941         app.SpacingEditField.Enable = "off";
3942         app.SpacingEditFieldLabel.Enable = "off";
3943
3944
3945     end
3946 end
3947
3948 % Value changed function: LoadFileEditField
3949 function LoadFileEditFieldValueChanged(app, event)
3950
3951
3952 end
3953
3954 % Value changing function: LoadFileEditField
3955 function LoadFileEditFieldValueChanging(app, event)
3956
3957
3958 end
3959
3960 % Selection change function: TabGroup2
3961 function TabGroup2SelectionChanged(app, event)
3962     selectedTab = app.TabGroup2.SelectedTab;
3963
3964 end
3965
3966 % Value changed function: UseSmagorinskyCheckBox
3967 function UseSmagorinskyCheckBoxValueChanged2(app, event)
3968     if app.UseSmagorinskyCheckBox.Value
3969         app.TimeintervalsecEditFieldLabel.Visible = "on";
3970         app.TimeintervalsecEditField.Visible = "on";
3971         app.SteadyTimeEditFieldLabel.Visible = "on";
3972         app.SteadyTimeEditField.Visible = "on";
3973     else
3974         app.TimeintervalsecEditFieldLabel.Visible = "off";
3975         app.TimeintervalsecEditField.Visible = "off";
3976         app.SteadyTimeEditFieldLabel.Visible = "off";
3977         app.SteadyTimeEditField.Visible = "off";
3978     end
3979 end

```

```

3980
3981 % Value changed function: SavefilestoEditField
3982 function SavefilestoEditFieldValueChanged(app, event)
3983
3984
3985 end
3986
3987 % Button pushed function: Button_41
3988 function Button_41Pushed(app, event)
3989     workingfolder = uigetdir;
3990     workingfolder = fullfile(workingfolder, ' ');
3991     app.Mpif90EditField.Value = deblank(string(workingfolder));
3992     app.Mpif90EditField.FontColor = 'k';
3993 end
3994
3995 % Callback function
3996 function Button_42Pushed(app, event)
3997     workingfolder = uigetdir;
3998     workingfolder = fullfile(workingfolder, ' ');
3999     app.MakefileEditField.Value = deblank(string(workingfolder));
4000     app.MakefileEditField.FontColor = 'k';
4001     app.MakefileEditField.BackgroundColor = 'w';
4002 end
4003
4004 % Callback function
4005 function IncludefunwaveexeintheGenerationCheckBoxValueChanged(app, event)
4006
4007 end
4008
4009 % Callback function
4010 function MakefileEditFieldValueChanging(app, event)
4011     app.MakefileEditField.FontColor = 'k';
4012     app.MakefileEditField.BackgroundColor = 'w';
4013 end
4014
4015 % Value changed function: Mpif90EditField
4016 function Mpif90EditFieldValueChanged(app, event)
4017
4018
4019 end
4020
4021 % Value changing function: Mpif90EditField
4022 function Mpif90EditFieldValueChanging(app, event)
4023     app.Mpif90EditField.FontColor = 'k';
4024     app.Mpif90EditField.BackgroundColor = 'w';
4025 end
4026
4027 % Button pushed function: StartsimulationButton
4028 function StartsimulationButtonPushed(app, event)
4029     %Show the To disable the button whilst the rest of the code is
running
4030
4031     app.TabGroup3.SelectedTab = app.LogReportTab;
4032     % app.StartsimulationButton.Enable = 'off';
4033     drawnow;
4034
4035     cd(fullfile(app.directory));

```

```

4036
4037         if exist(app.ResultsfolderEditField.Value, 'dir')
4038             rmdir(app.ResultsfolderEditField.Value, 's'); % Delete the
pre-existing folder and its contents
4039         end
4040
4041         LogFolder = fullfile(app.directory, 'Log_Files');
4042
4043         % Check if the 'INPUTFOLDER' directory exists, create it if it
doesn't
4044         if ~exist(LogFolder, 'dir')
4045             mkdir(LogFolder); % Creates the 'INPUTFOLDER' directory
4046         end
4047
4048         logFilePath = fullfile(LogFolder, 'Log_Report_InputPrep.txt'); %
Define the log file path correctly
4049         currentLogContent = fileread(logFilePath); % Read the current
content of the log file
4050
4051         app.Preview_LogReport.Value = fileread(logFilePath);
4052         drawnow;
4053
4054         % Ensure the current directory is set to where funwave.exe resides
4055         cd(fullfile(string(app.directory)));
4056
4057
4058
4059         %In reclicking the button, delete existing Makefile and funwave.
exe
4060         sourceFileExe = fullfile(app.directory, "funwave");
4061         if exist(sourceFileExe, 'file')
4062             delete(sourceFileExe);
4063         end
4064
4065
4066         % Determine the MPIF90 path based on user input or system defaults
4067         if isempty(app.Mpif90EditField.Value) || strcmp(app.
Mpif90EditField.Value, '0') %|| ~exist(fullfile(app.Mpif90EditField.Value), "dir")
4068             % Set the default MPIF90 path based on the operating system
4069             if ismac
4070                 mpiPath = '/opt/homebrew/bin/'; % macOS default path
4071             elseif ispc
4072                 mpiPath = 'C:\Program Files (x86)\MPICH2\bin\'; % Windows
default path
4073             else
4074                 error('Unsupported OS'); % Handle other operating systems
or configurations
4075             end
4076             app.Mpif90EditField.Value = fullfile(mpiPath); % Update the
edit field with the default path
4077         else
4078             mpiPath = app.Mpif90EditField.Value; % Use the user-provided
MPIF90 path
4079         end
4080
4081
4082         if exist(fullfile(app.Mpif90EditField.Value), "dir") == 7 %Check if
the mpif90 folder exists

```

```

4083 %Update the system PATH environment variable to include the
MPI compiler path
4084 setenv('PATH', [getenv('PATH') ':' fileparts(mpiPath)]);
4085 app.Mpif90EditField.FontColor = 'k';
4086
4087 if exist(fullfile(app.MakefileEditField.Value), "dir") == 7 %
Check if the Makefile Source folder exists
4088 cd(fullfile(app.MakefileEditField.Value));
4089
4090 % Check if the "Make Clean Completed" message is already
in the log file
4091 makeCleanMessage = '> Status: Make Clean...';
4092 if ~contains(fileread(logFilePath), makeCleanMessage)
4093 % Append the message only if it's not already there
4094 fileId = fopen(logFilePath, 'a');
4095 fprintf(fileId, '%s', makeCleanMessage);
4096 drawnow
4097 end
4098
4099 system('make clean'); % Clean up previous files created
4100
4101 completedMessage = '...Completed!';
4102 if ~contains(logFilePath, completedMessage)
4103 fileId = fopen(logFilePath, 'a');
4104 fprintf(fileId, '%s\n\n', completedMessage);
4105 fclose(fileId);
4106 app.Preview_LogReport.Value = fileread(logFilePath); %
Update the log report preview panel
4107 drawnow; % Update GUI
4108 end
4109
4110 else
4111 fprintf(fileId, '%-30s-s\n', 'MAKEFILE:', ' Invalid path.
Missing .o, .F, and .f90 files');
4112 fprintf(fileId, '%s\n', '');
4113 app.Preview_LogReport.Value = fileread(logFilePath);
4114 drawnow
4115 %
4116 % app.MakefileEditField.Value = "INVALID PATH";
4117 % app.MakefileEditField.FontColor = 'w';
4118 % app.MakefileEditField.BackgroundColor = 'r';
4119 end
4120
4121 %--- Execute Makefile ---
4122 generatingFunwaveMessage = '> Status: Generating the funwave.
exe... ';
4123 if ~contains(logFilePath, generatingFunwaveMessage)
4124 fileId = fopen(logFilePath, 'a');
4125 fprintf(fileId, '%s', generatingFunwaveMessage);
4126 fclose(fileId);
4127 app.Preview_LogReport.Value = fileread(logFilePath);
4128 drawnow;
4129 end
4130
4131 [status, cmdout] = system('make'); % Execute 'make' command
4132
4133 if status ~= 0 %if make execution fails
4134 if ~contains(logFilePath, 'Failed.')

```

```

4135         fileId = fopen(logFilePath, 'a');
4136         fprintf(fileId, '%s\n\n', 'Failed.');
```

4137 app.Preview_LogReport.Value = fileread(logFilePath);

4138 fclose(fileId);

4139 drawnow

4140 end

4141

4142 makeErrorMessage = cmdout;

4143 if ~contains(logFilePath, makeErrorMessage) %show the

error message from the terminal

4144 fileId = fopen(logFilePath, 'a');

4145 fprintf(fileId, '%s\n', makeErrorMessage);

4146 fclose(fileId);

4147 app.Preview_LogReport.Value = fileread(logFilePath);

4148 drawnow;

4149 end

4150

4151 else %if successful, copy the funwave executable file to the

specified output directory

4152 sourceFile = fullfile(app.MakefileEditField.Value,

"funwave");

4153 if exist(sourceFile, 'file')

4154 destinationFile = fullfile(app.directory, "funwave");

4155 copyfile(sourceFile, destinationFile);

4156 delete(sourceFile); % Remove the source file after

copying

4157

4158 %Append "Completed" message for "make" execution

4159 if ~contains(logFilePath, 'Completed!')

4160 fileId = fopen(logFilePath, 'a');

4161 fprintf(fileId, '%s\n\n', 'Completed!');

4162 app.Preview_LogReport.Value = fileread

(logFilePath);

4163 fclose(fileId);

4164 drawnow

4165 end

4166 drawnow

4167

4168 MakefileMessage5 = '> Status: Starting the

simulation...';

4169 if ~contains(fileread(logFilePath), MakefileMessage5)

%Successful status printed

4170 % Append the message only if it's not already

there

4171 fileId = fopen(logFilePath, 'a');

4172 fprintf(fileId, '%s\n\n', MakefileMessage5);

4173 app.Preview_LogReport.Value = fileread

(logFilePath);

4174

4175 drawnow

4176 end

4177 else

4178 if ~contains(logFilePath, 'Failed.')

4179 fileId = fopen(logFilePath, 'a');

4180 fprintf(fileId, '%s\n\n', 'Failed.');

4181 app.Preview_LogReport.Value = fileread

(logFilePath);

4182 fclose(fileId);

```

4183             drawnow
4184         end
4185     end
4186 end
4187 else
4188     % app.Mpif90EditField.Value = "INVALID PATH";
4189     % app.Mpif90EditField.FontColor = 'w';
4190     % app.Mpif90EditField.BackgroundColor = 'r';
4191     fileId = fopen(logFilePath, 'a');
4192     fprintf(fileId, [
4193         '%-30s%-s\n' ...
4194         '%-30s%\n\n' ...
4195         '%-30s%\n' ...
4196         '%-30s%\n' ...
4197         '%-40s%\n\n' ...
4198         '%-30s%\n' ...
4199         '%-40s%\n\n' ...
4200         '%-30s%\n\n'
4201     ], ...
4202     'MPIF90:', 'Invalid MPI path - 'mpif90' not found', ...
4203     '', 'Failed to execute make', ...
4204     '', 'In your computer's terminal, type:', ...
4205     '', ' *Mac OS: "which mpif90"', ...
4206     '', '     Default: /opt/homebrew/bin/', ...
4207     '', ' *Windows: "where mpif90"', ...
4208     '', '     Default: C:\Program Files (x86)\MPICH2\bin', ...
4209     '', 'Copy that path into the Mpif90 Folder textbox');
4210
4211     app.Preview_LogReport.Value = fileread(logFilePath);
4212     fclose(fileId);
4213     drawnow
4214 end
4215
4216
4217
4218     cd(fullfile(app.directory));
4219
4220
4221     % Append the start message only if it's not already in the log
4222     startMessage = sprintf('> Status: Starting the tsunami ↵
4223     simulations...\n\n See the detailed progress in INPUT_FILES/LOG.txt');
4224     app.Preview_LogReport.Value = fileread(logFilePath);
4225     drawnow
4226     %
4227     if ~contains(logFilePath, startMessage)
4228         fileId = fopen(logFilePath, 'a');
4229         fprintf(fileId, '%s\n\n', startMessage);
4230         fclose(fileId); % Close the file after appending the start ↵
4231     message
4232     end
4233
4234     commandStr = ['mpirun -np ', num2str(app.ProcessorX.Value * app. ↵
4235     ProcessorY.Value), ' ./funwave'];
4236     [status, cmdout] = system(commandStr);
4237
4238     % Process cmdout to remove repeated lines and ensure unique ↵
4239     logging
4240     cmdoutLines = splitlines(string(cmdout));

```

```

4237         [uniqueLines, ~] = unique(cmdoutLines, 'stable');
4238         uniqueCmdout = join(uniqueLines, newline);
4239
4240         fileId = fopen(logFilePath, 'a'); % Reopen log file to append the
execution status and cmdout
4241         if status == 0
4242             runCompletedMessage = 'FUNWAVE.EXE: Run Completed';
4243             if ~contains(logFilePath, runCompletedMessage)
4244                 fprintf(fileId, '%-30s%-s\n\n', runCompletedMessage);
4245             end
4246             for line = uniqueLines'
4247                 if line ~= ""
4248                     fprintf(fileId, '    %-28s\n', line);
4249                 end
4250             end
4251         else
4252             executionFailedMessage = 'FUNWAVE.EXE: Execution failed';
4253             if ~contains(currentLogContent, executionFailedMessage)
4254                 fprintf(fileId, '%-30s%-s\n', executionFailedMessage);
4255             end
4256             for line = uniqueLines'
4257                 if line ~= ""
4258                     fprintf(fileId, '    Error: %-25s\n', line);
4259                 end
4260             end
4261         end
4262
4263         fclose(fileId); % Close the file
4264
4265
4266         app.Preview_LogReport.Value = fileread(logFilePath); % Display the
updated log content
4267
4268
4269
4270     end
4271
4272     % Button pushed function: CreateMakefileButton
4273     function CreateMakefileButtonPushed(app, event)
4274         %Disable the simulation button
4275         app.StartsimulationButton.Enable = 'off';
4276         drawnow
4277
4278         %Update the input.txt file
4279         GenerateinputtxtButtonPushed(app, event);
4280
4281         %Switch tab of the Preview Panel to the Log Report Tab
4282         app.TabGroup3.SelectedTab = app.LogReportTab;
4283
4284         %Locate the Log Report file
4285         cd(fullfile(app.directory));
4286         LogFolder = fullfile(app.directory, 'Log_Files');
4287
4288         %If 'INPUT_FILES' directory does not exist, create the folder
4289         if ~exist(LogFolder, 'dir')
4290             mkdir(LogFolder);
4291         end
4292

```

```

4293         %If the log text file does not exist, create the text file
4294         logFilePath = fullfile(LogFolder, 'Log_Report_InputPrep.txt');
4295         if ~exist(logFilePath, 'file')
4296             fileId=fopen(logFilePath, 'w');
4297         else
4298             %Update the log report
4299             MakefileMessage00 = '***** MAKEFILE SECTION *****'; %␣
Check if the "Creating Makefile completed" message is already in the log file
4300             if ~contains(fileread(logFilePath), MakefileMessage00)
4301                 % Append the message only if it's not already there
4302                 fileId = fopen(logFilePath, 'a');
4303                 fprintf(fileId, '%s\n\n', MakefileMessage00);
4304                 fprintf('');
4305                 app.Preview_LogReport.Value = fileread(logFilePath);
4306                 drawnow
4307             end
4308         end
4309
4310
4311
4312         %Remove pre-existing progress status under the Makefile section
4313         %Read the content of the log report
4314         fileContent = fileread(logFilePath);
4315
4316         % Find the index of 'Makefile Section' header
4317         startIndex = strfind(fileContent, MakefileMessage00);
4318
4319         %If there are pre-existing data in this section, overwrite them
4320         if ~isempty(startIndex)
4321             %Find the end of the line of the header
4322             endOfLineIndex = strfind(fileContent(startIndex:end), ␣
newline);
4323             endIndex = startIndex + endOfLineIndex(1) - 1;
4324
4325             %Trim the fileContent up to endIndex
4326             trimmedContent = fileContent(1:endIndex);
4327
4328             %Write the trimmed content back to the file
4329             fileId = fopen(logFilePath, 'w');
4330             fprintf(fileId, '%s', trimmedContent);
4331             fclose(fileId);
4332
4333             %Update app.Preview_LogReport.Value with the updated content
4334             app.Preview_LogReport.Value = trimmedContent;
4335         else
4336             fprintf('MakefileMessage00 not found in %s\n', logFilePath);
4337         end
4338
4339         %Update the log reporrt
4340         app.Preview_LogReport.Value = fileread(logFilePath);
4341
4342
4343         %␣
-----
4344         %-----TEMPLATE OF THE MAKEFILE from FUNWAVE-TVD-version_3.␣
6/src-----
4345         %␣
-----

```



```

4346     makefile_template = {
4347         '#-----BEGIN MAKEFILE-----'
4348         'FUNWAVE_DIR = .'
4349         'WORK_DIR    = .'
4350         'COMPILER    = #COMPILER_OPTION#'
4351         'PARALLEL    = #PARALLEL_OPTION#'
4352         'EXEC        = funwave'
4353         'PRECISION   = #PRECISION_OPTION#'
4354         ''
4355         '#-----DEFINE FLAGS-----'
4356         '#          uncomment to choose the model'
4357         '#FLAG_1  = -DCOUPLING'
4358         '#FLAG_2  = -DZALPHA'
4359         '#FLAG_3  = -DVESSEL'
4360         '#FLAG_4  = -DMETEO'
4361         '#FLAG_5  = -DWIND'
4362         '#FLAG_6  = -DSEDIMENT'
4363         '#FLAG_7  = -DCHECK_MASS_CONSERVATION'
4364         '#FLAG_8  = -DTMP'
4365         '#FLAG_9  = -DTRACKING'
4366         '#FLAG_10 = -DDEEP_DRAFT_VESSEL'
4367         '#FLAG_11 = -DMIXING'
4368         '#-----uncommon options-----'
4369         'DEF_FC      = mpif90'
4370         'DEF_FC_FLAG = '
4371         'SPHERICAL   = #COORDINATE_OPTION#'
4372         'MPI          = openmpi'
4373         'DEBUG        = true'
4374         'INCS         = $(IOINCS) $(GOTMINCS) '
4375         'LIBS         = $(PV3LIB) $(CLIB)  $(PARLIB) $(IOLIBS) ↵
4376         '$(MPILIB) $(GOTMLIB) '
4377         'CLIB         ='
4378         'MDEPFLAGS    = --cpp --fext=f90 --file=-'
4379         'RANLIB        = ranlib'
4380         ''
4381         '#-----include the essential makefiles-----'
4382         '#Copied file from $(FUNWAVE_DIR)/.. ↵
4383         /GNUmake/Essential/Make_Essential'
4384         '#-----Use shell-----'
4385         'SHELL        = /bin/sh'
4386         ''
4387         '#-----Eliminate the leading and ending white space-----'
4388         'FUNWAVE_DIR := $(strip $(FUNWAVE_DIR)) '
4389         'WORK_DIR    := $(strip $(WORK_DIR)) '
4390         'COMPILER    := $(strip $(COMPILER)) '
4391         'PARALLEL    := $(strip $(PARALLEL)) '
4392         'PRECISION   := $(strip $(PRECISION)) '
4393         'EXEC        := $(strip $(EXEC)) '
4394         'DEF_FC      := $(strip $(DEF_FC)) '
4395         'DEF_FC_FLAG := $(strip $(DEF_FC_FLAG)) '
4396         'MPI          := $(strip $(MPI)) '
4397         'SPHERICAL   := $(strip $(SPHERICAL)) '
4398         'DEBUG        := $(strip $(DEBUG)) '
4399         ''
4400         '#-----C Pre-processor-----'
4401         '# note: CPP is not C++'
4402         'CPP          = /usr/bin/cpp'

```

```

4402 'CPPFLAGS = -P -traditional'
4403 ''
4404 '#-----Fortran Compiler-----'
4405 '# if $(DEF_FC) is not empty'
4406 'ifneq ($(DEF_FC),$(filter $(DEF_FC), '''))'
4407 '    FC = $(DEF_FC)'
4408 '# if $(DEF_FC) is empty'
4409 'else'
4410 '    # intel'
4411 '    ifeq ($(COMPILER),$(filter $(COMPILER), intel))'
4412 '        USE_MOD = -module $(MOD_DIR)'
4413 '        ifeq ($(PARALLEL),$(filter $(PARALLEL), true))'
4414 '            ifeq ($(MPI),$(filter $(MPI), intelmpi))'
4415 '                FC = mpiifort'
4416 '            else'
4417 '                FC = mpif90'
4418 '            endif'
4419 '        else ifeq ($(PARALLEL),$(filter $(PARALLEL), false))'
4420 '            FC = ifort'
4421 '        endif'
4422 '    # gnu'
4423 '    else ifeq ($(COMPILER),$(filter $(COMPILER), gnu))'
4424 '        USE_MOD = -J $(MOD_DIR)'
4425 '        ifeq ($(PARALLEL),$(filter $(PARALLEL), true))'
4426 '            FC = mpif90'
4427 '        else ifeq ($(PARALLEL),$(filter $(PARALLEL), false))'
4428 '            FC = gfortran'
4429 '        endif'
4430 '    # pgi'
4431 '    else ifeq ($(COMPILER),$(filter $(COMPILER), pgi))'
4432 '        USE_MOD = -module $(MOD_DIR)'
4433 '        ifeq ($(PARALLEL),$(filter $(PARALLEL), true))'
4434 '            FC = mpif90'
4435 '        else ifeq ($(PARALLEL),$(filter $(PARALLEL), false))'
4436 '            FC = pgfortran'
4437 '        endif'
4438 '    # onyx hpc'
4439 '    else ifeq ($(COMPILER),$(filter $(COMPILER), onyx))'
4440 '        USE_MOD = -module $(MOD_DIR)'
4441 '        FC = ftn'
4442 '        # Removing -lmpi flag if defined in CLIB'
4443 '        ifneq (, $(findstring -lmpi, $(CLIB)))'
4444 '            $(warning WARNING: lmpi flag detected in CLIB, Onyx
does not require explicit linking of MPI library. Removing -lmpi from CLIB. )'
4445 '            CLIB := $(filter-out -lmpi, $(CLIB))'
4446 '        endif'
4447 '        # Removing -Bdynamic flat if defined in CLIB'
4448 '        ifneq (, $(findstring -Bdynamic, $(CLIB)))'
4449 '            $(warning WARNING: Bdynamic flag detected in CLIB,
Onyx executables do not work correctly with dynamic linking of shared libraries.
Removing -Bdynamic from CLIB. )'
4450 '            CLIB := $(filter-out -Bdynamic, $(CLIB))'
4451 '        endif'
4452
4453
4454 '    # koehr and gaffney'
4455 '    else ifeq ($(COMPILER),$(filter $(COMPILER), koehr
gaffney))'

```

```

4456         '        USE_MOD = -module $(MOD_DIR) '
4457         '        FC          = ifort'
4458         '        # invalid option'
4459         '        else'
4460         '        $(error Fatal ERROR: COMPILER=$(COMPILER) and DEF_FC is
empty, Please correct the COMPILER or customize the DEF_FC.) '
4461         '        endif'
4462     'endif'
4463     ''
4464     '#-----Debug and optimization flags-----'
4465     '# DEBUG=true'
4466     'ifeq ($(DEBUG),$(filter $(DEBUG), true))'
4467     '    ifeq ($(COMPILER),$(filter $(COMPILER), intel onyx gaffney
koehr)) '
4468     '        DEBFLGS  = -g -check -warn -fPIC'
4469     '        OPT       = -O0'
4470     '    else ifeq ($(COMPILER),$(filter $(COMPILER), gnu)) '
4471     '        DEBFLGS  = -g -fcheck=all -Wall -fPIC'
4472     '        OPT       = -O0'
4473     '    else ifeq ($(COMPILER),$(filter $(COMPILER), pgi)) '
4474     '        DEBFLGS  = -g -fPIC'
4475     '        OPT       = -O0'
4476     '    endif'
4477     'endif'
4478     '# DEBUG=false'
4479     'ifeq ($(DEBUG),$(filter $(DEBUG), false))'
4480     '    ifeq ($(COMPILER),$(filter $(COMPILER), intel onyx gaffney
koehr)) '
4481     '        DEBFLGS  = -fPIC'
4482     '        OPT       = -O2'
4483     '    else ifeq ($(COMPILER),$(filter $(COMPILER), gnu)) '
4484     '        DEBFLGS  = -fPIC'
4485     '        OPT       = -O3'
4486     '    else ifeq ($(COMPILER),$(filter $(COMPILER), pgi)) '
4487     '        DEBFLGS  = -w -fPIC'
4488     '        OPT       = -O2'
4489     '    endif'
4490     'endif'
4491     ''
4492     '#-----Pre-processor flags-----'
4493     'ifeq ($(PRECISION),$(filter $(PRECISION), double))'
4494     '    FLAG_PRECISION = -DDOUBLE_PRECISION '
4495     'else ifeq ($(PRECISION),$(filter $(PRECISION), single))'
4496     '    FLAG_PRECISION = '
4497     'else'
4498     '    $(error Fatal ERROR: PRECISION=$(PRECISION), should be
single or double.) '
4499     'endif'
4500     ''
4501     'ifeq ($(SPHERICAL),$(filter $(SPHERICAL), true))'
4502     '    FLAG_CARTESIAN = '
4503     'else'
4504     '    FLAG_CARTESIAN = -DCARTESIAN'
4505     'endif'
4506     ''
4507     'ifeq ($(PARALLEL),$(filter $(PARALLEL), true))'
4508     '    FLAG_PARALLEL = -DPARALLEL'
4509     'endif'

```

```

4510      ''
4511      'ifeq ($(COMPILER),$(filter $(COMPILER), intel onyx koehr
gaffney))'
4512      '    FLAG_INTEL = -DINTEL'
4513      'endif'
4514      ''
4515      'ifeq ($(DEBUG),$(filter $(DEBUG), true))'
4516      '    FLAG_DEBUG = -DDEBUG'
4517      'endif'
4518      '#-----FUNWAVE flags-----'
4519      'FLAG_ARGS = $(FLAG_1) $(FLAG_2) \'
4520      '    $(FLAG_3) $(FLAG_4) $(FLAG_5) $(FLAG_6) \'
4521      '    $(FLAG_7) $(FLAG_8) $(FLAG_9) $(FLAG_10) \'
4522      '    $(FLAG_11) $(FLAG_12) $(FLAG_13) $(FLAG_14) \'
4523      '    $(FLAG_15) $(FLAG_16) $(FLAG_17) $(FLAG_18) \'
4524      '    $(FLAG_19) $(FLAG_20) $(FLAG_21) $(FLAG_22) \'
4525      '    $(FLAG_23) $(FLAG_24) '
4526      'CPPARGS = $(CPPFLAGS) $(DEF_FLAGS) $(FLAG_ARGS) \'
4527      '    $(FLAG_PARALLEL) $(FLAG_INTEL) $(FLAG_DEBUG) \'
4528      '    $(FLAG_PRECISION) $(FLAG_CARTESIAN) '
4529      ''
4530      '# remove extra whitespace between flags'
4531      'CPPARGS := $(shell echo $(CPPARGS) | sed 's~ \+~ ~g')'
4532      ''
4533      '#-----EXEC name-----'
4534      'EXEC := $(WORK_DIR)/$(EXEC) '
4535      ''
4536      '#-----Library name-----'
4537      'ifneq ($(DEF_FC),$(filter $(DEF_FC), ''))'
4538      '    FCFLAGS := $(DEF_FC_FLAG) $(INCS) '
4539      'else'
4540      '    FCFLAGS := $(DEBFLGS) $(OPT) $(INCS) '
4541      'endif'
4542      ''
4543      '# remove extra whitespace between flags'
4544      'FCFLAGS := $(shell echo $(FCFLAGS) | sed 's~ \+~ ~g')'
4545      ''
4546      '#-----FUNWAVE source code-----'
4547      'MODS  = mod_param.F mod_global.F mod_input.F mod_vessel.F
mod_bathy_correction.F \'
4548      '    mod_meteo.F mod_parallel_field_io.F mod_sediment.F
mod_tide.F\'
4549      '    mod_tracer.F'
4550      ''
4551      'MAIN  = main.F bc.F fluxes.F init.F io.F tridiagnal.F
\'
4552      '    breaker.F derivatives.F dispersion.F etauv_solver.F
\'
4553      '    sponge.F sources.F masks.F parallel.F statistics.F \'
4554      '    wavemaker.F mixing.F nesting.F misc.F samples.F\'
4555      ''
4556      '#-----Compile process-----'
4557      'SRC_DIR  = $(FUNWAVE_DIR) '
4558      'PRE_DIR  = $(WORK_DIR)/build/pre'
4559      'OBJ_DIR  = $(WORK_DIR)/build/obj'
4560      'MOD_DIR  = $(WORK_DIR)/build/mod'
4561      ''
4562      'SRCS := $(MODS)  $(MAIN) '

```

```

4563         'SRCS := $(patsubst %, $(SRC_DIR) /%, $(SRCS)) '
4564         'PRES := $(patsubst $(SRC_DIR) /%.F, $(PRE_DIR) /%.f90, $(SRCS)) '
4565         'OBJS := $(patsubst $(PRE_DIR) /%.f90, $(OBJ_DIR) /%.o, $(PRES)) '
4566         ''
4567         '# keep the *.f90 files in $(PRE_DIR) after make'
4568         '# .PRECIOUS: $(PRE_DIR) /%.f90'
4569         ''
4570         '# pre-processors'
4571         '$(PRE_DIR) /%.f90 : $(SRC_DIR) /%.F'
4572         '    $(CPP) $(CPPARGS) $< $@'
4573         ''
4574         '# compile objective files'
4575         '$(OBJ_DIR) /%.o : $(PRE_DIR) /%.f90'
4576         '    $(FC) $(FCFLAGS) -c $< -o $@ $(USE_MOD) '
4577         ''
4578         '# link and generate the exec file'
4579         '$(EXEC): $(OBJS) '
4580         '    $(FC) $(FCFLAGS) -o $(EXEC) $(OBJS) $(LIBS) '
4581         ''
4582         '$(PRES): | $(PRE_DIR) '
4583         ''
4584         '$(PRE_DIR):'
4585         '    mkdir -p $(WORK_DIR) '
4586         '    mkdir -p $(PRE_DIR) '
4587         '    mkdir -p $(OBJ_DIR) '
4588         '    mkdir -p $(MOD_DIR) '
4589         '#-----Cleaning targets-----'
4590         'clean:'
4591         '    /bin/rm -rf $(WORK_DIR)/build'
4592         ''
4593         'clobber: clean'
4594         '    /bin/rm -f $(EXEC) '
4595         ''
4596         'extra-clobber: clean'
4597         '    /bin/rm -rf $(WORK_DIR) '
4598         '};'
4599
4600
4601 mkfile = sprintf('%s\n', makefile_template{:});
4602 mkfile = string(mkfile);
4603
4604
4605 %-----
4606 %-----Uncommenting the Flags in the Makefile-----
4607 %-----
4608
4609 %-----DOUBLE PRECISION-----
4610 if app.DoublePrecisionCheckBox.Value
4611     mkfile = strrep(mkfile, "#PRECISION_OPTION#", "double");
4612 else
4613     mkfile = strrep(mkfile, "#PRECISION_OPTION#", "single");
4614 end
4615
4616 %-----PARALLEL MODE-----
4617 if app.ParallelModeCheckBox.Value
4618     mkfile = strrep(mkfile, "#PARALLEL_OPTION#", "true");
4619 else
4620     mkfile = strrep(mkfile, "#PARALLEL_OPTION#", "false");

```

```

4621
4622     end
4623
4624     %-----INTEL COMPILER-----
4625     if app.IntelCompilerCheckBox.Value
4626         mkfile = strrep(mkfile, "#COMPILER_OPTION#", "intel");
4627     else
4628         mkfile = strrep(mkfile, "#COMPILER_OPTION#", "gnu");
4629     end
4630
4631
4632     %-----GRID-----
4633     if app.SphericalButton.Value
4634         %         mkfile = strrep(mkfile, "#FLAG_2", "FLAG_2"); %Ask
Masashi if it is always true -DZALPHA
4635         mkfile = strrep(mkfile, "#COORDINATE_OPTION#", "true");
4636     elseif app.CartesianButton.Value
4637         mkfile = strrep(mkfile, "#COORDINATE_OPTION#", "false");
4638     end
4639
4640
4641     %-----TURBULENT MIXING-----
4642     if app.UseSmagorinskyCheckBox.Value
4643         mkfile = strrep(mkfile, "#FLAG_12", "FLAG_12");
4644     end
4645
4646
4647     %-----METEOTSUNAMI-----
4648     if app.MeteotsunamiButton.Value
4649         mkfile = strrep(mkfile, "#FLAG_5", "FLAG_5");
4650         % Checking if any of the switches are "On" to perform the
operation once
4651         if strcmp(app.WindforceSwitch.Value, "On") || strcmp(app.
HollandstormmodelSwitch.Value, "On") || strcmp(app.ConstantwindfieldSwitch.Value,
"On")
4652             mkfile = strrep(mkfile, "#FLAG_6", "FLAG_6");
4653         end
4654     end
4655
4656     %-----SEDIMENT-----
4657     if app.SedimentTransportCheckBox.Value
4658         mkfile = strrep(mkfile, "#FLAG_7", "FLAG_7");
4659     end
4660
4661     %-----VESSEL-----
4662     if app.VesselButton.Value
4663         mkfile = strrep(mkfile, "#FLAG_4", "FLAG_4");
4664
4665         if app.IncludeSedimentEffectintheModelCheckBox.Value
4666             mkfile = strrep(mkfile, "#FLAG_7", "FLAG_7");
4667         end
4668
4669         % Deep-draft Vessel
4670         if app.ActivateCheckBox.Value
4671             mkfile = strrep(mkfile, "#FLAG_11", "FLAG_11");
4672         end
4673     end
4674
4675

```

```

4675
4676 %-----
4677 %-----Create the Makefile using the Makefile template-----
4678 %-----
4679 outputFilePath3 = fullfile(app.directory, 'Makefile');
4680 outputFile3 = fopen(outputFilePath3, 'w');
4681
4682 for i = 1:length(mkfile)
4683     fprintf(outputFile3, '%s\n',mkfile{i});
4684 end
4685 fclose(outputFile3);
4686 app.Preview_Makefile.Value = string(mkfile);
4687
4688 %Update the log report - Check if the "Creating Makefile
completed" message is already in the log file
4689 MakefileMessage = '> Status: Creating the Makefile in the
INPUT_FILES folder... Completed!';
4690 if ~contains(fileread(logFilePath), MakefileMessage)
4691     % Append the message only if it's not already there
4692     fileId = fopen(logFilePath, 'a');
4693     fprintf(fileId, '\n %s\n\n', MakefileMessage);
4694     app.Preview_LogReport.Value = fileread(logFilePath);
4695     drawnow
4696 end
4697
4698 %Texts to include in the log report
4699 MakefileMessage2 = '> Status: Copying the Makefile into the
Makefile source folder... Completed!';
4700 MakefileMessage3 = '> Status: Copying the Makefile into the
Makefile source folder... Failed.';
4701 MakefileMessage4 = '> Invalid path for the Makefile source';
4702
4703 %-----
4704 %-----Copy the generated Makefile into the source folder-----
4705 %-----
4706 %If the Makefile source folder exists, copy the in-app generated
Makefile to the Makefile source folder
4707 if exist(fullfile(app.MakefileEditField.Value), "dir") == 7
4708     cd(fullfile(app.MakefileEditField.Value));
4709     sourceFile2 = fullfile(app.directory, "Makefile");
4710     destinationFile2 = fullfile(app.MakefileEditField.Value);
4711     copyfile(sourceFile2, destinationFile2);
4712
4713     %Enable the MPIf90 folder & the 'Start Simulation' button
4714     app.Mpif90EditField.Enable = 'on';
4715     app.Mpif90EditFieldLabel.Enable = 'on';
4716     app.Button_41.Enable = 'on';
4717     app.StartsimulationButton.Enable = "on";
4718     drawnow
4719
4720 %Update the log report to show that the copying of file is
completed
4721 if ~contains(fileread(logFilePath), MakefileMessage2)
4722     fileId = fopen(logFilePath, 'a');
4723     fprintf(fileId, '%s\n\n', MakefileMessage2);
4724     app.Preview_LogReport.Value = fileread(logFilePath);
4725     drawnow
4726 end

```

```

4727
4728     else %Print error status in the report
4729         if ~contains(fileread(logFilePath), MakefileMessage3)
4730             fileId = fopen(logFilePath, 'a');
4731             fprintf(fileId, '%s\n\n', MakefileMessage3);
4732             fprintf(fileId, '%s\n\n', MakefileMessage4);
4733             app.Preview_LogReport.Value = fileread(logFilePath);
4734             drawnow
4735         end
4736
4737         app.Preview_LogReport.Value = fileread(logFilePath);
4738         app.StartsimulationButton.Enable = "off";
4739
4740         app.MakefileEditField.Value = "INVALID PATH";
4741         app.MakefileEditField.FontColor = 'w';
4742         app.MakefileEditField.BackgroundColor = 'r';
4743         drawnow
4744     end
4745
4746     %Enable the Makefile and input.txt buttons
4747     app.CreateMakefileButton.Enable = 'on';
4748     app.GenerateinputtxtButton.Enable = 'on';
4749     drawnow
4750 end
4751
4752 % Value changed function: FrictionCoefficientEditField
4753 function FrictionCoefficientEditFieldValueChanged(app, event)
4754     app.FrictionCoefficientEditField.FontColor = 'k';
4755
4756 end
4757
4758 % Value changed function: ValueEditField
4759 function ValueEditFieldValueChanged(app, event)
4760     app.ValueEditField.FontColor = 'k';
4761
4762 end
4763
4764 % Value changed function: FrictionCheckBox_2
4765 function FrictionCheckBox_2ValueChanged(app, event)
4766     if app.FrictionCheckBox_2.Value
4767         app.FrictionCoefficientEditField.Enable = 'on';
4768         app.FrictionCoefficientEditFieldLabel.Enable = 'on';
4769     else
4770         app.FrictionCoefficientEditField.Enable = 'off';
4771         app.FrictionCoefficientEditFieldLabel.Enable = 'off';
4772     end
4773 end
4774
4775 % Value changed function: ViscosityCheckBox
4776 function ViscosityCheckBoxValueChanged(app, event)
4777     if app.ViscosityCheckBox.Value
4778         app.ValueEditField.Enable = 'on';
4779         app.ValueEditFieldLabel_2.Enable = 'on';
4780     else
4781         app.ValueEditField.Enable = 'off';
4782         app.ValueEditFieldLabel_2.Enable = 'off';
4783     end
4784

```



```

4785     end
4786
4787     % Value changed function: ActivateCheckBox
4788     function ActivateCheckBoxValueChanged(app, event)
4789         if app.ActivateCheckBox.Value
4790             if app.FrictionCheckBox_2.Value
4791                 app.FrictionCoefficientEditField.Enable = 'on';
4792                 app.FrictionCoefficientEditFieldLabel.Enable = 'on';
4793             else
4794                 app.FrictionCoefficientEditField.Enable = 'off';
4795                 app.FrictionCoefficientEditFieldLabel.Enable = 'off';
4796             end
4797
4798             if app.ViscosityCheckBox.Value
4799                 app.ValueEditField.Enable = 'on';
4800                 app.ValueEditFieldLabel_2.Enable = 'on';
4801             else
4802                 app.ValueEditField.Enable = 'off';
4803                 app.ValueEditFieldLabel_2.Enable = 'off';
4804             end
4805
4806             app.SelectMethodLabel.Enable = 'on';
4807             app.MinimumClearanceEditField.Enable = 'on';
4808             app.MinimumClearanceLabel.Enable = 'on';
4809             app.ViscosityCheckBox.Enable = 'on';
4810             app.FrictionCheckBox_2.Enable = 'on';
4811             app.ShockcapturingCheckBox.Enable = 'on';
4812         else
4813             app.SelectMethodLabel.Enable = 'off';
4814             app.MinimumClearanceEditField.Enable = 'off';
4815             app.MinimumClearanceLabel.Enable = 'off';
4816             app.ValueEditField.Enable = 'off';
4817             app.ValueEditFieldLabel_2.Enable = 'off';
4818             app.FrictionCoefficientEditField.Enable = 'off';
4819             app.FrictionCoefficientEditFieldLabel.Enable = 'off';
4820             app.ViscosityCheckBox.Enable = 'off';
4821             app.FrictionCheckBox_2.Enable = 'off';
4822             app.ShockcapturingCheckBox.Enable = 'off';
4823         end
4824     end
4825
4826     % Value changed function: MinimumClearanceEditField
4827     function MinimumClearanceEditFieldValueChanged(app, event)
4828         app.MinimumClearanceEditField.FontColor = 'k';
4829
4830     end
4831
4832     % Button pushed function: GenerateinputtxtButton_2
4833     function GenerateinputtxtButton_2Pushed(app, event)
4834         % app.GenerateinputtxtButtonPushed(app, []);
4835         GenerateinputtxtButtonPushed(app, event);
4836     end
4837
4838     % Button pushed function: Button_42
4839     function Button_42Pushed2(app, event)
4840         workingfolder = uigetdir;
4841         workingfolder = fullfile(workingfolder, ' ');
4842         app.MakefileEditField.Value = deblank(string(workingfolder));

```

```

4843         app.MakefileEditField.FontColor = 'k';
4844         app.MakefileEditField.BackgroundColor = 'w';
4845     end
4846
4847     % Value changed function: WaterlevelmEditField
4848     function WaterlevelmEditFieldValueChanged(app, event)
4849         app.WaterlevelmEditField.FontColor = 'k';
4850
4851     end
4852
4853     % Callback function
4854     function GenerateinputtxtButton_3Pushed(app, event)
4855         GenerateinputtxtButtonPushed(app, event);
4856     end
4857
4858     % Selection changed function: ButtonGroup_23
4859     function ButtonGroup_23SelectionChanged(app, event)
4860
4861         if app.ShockwavecapturingButton.Value
4862             app.C1EditField.Value = 0.65;
4863             app.C2EditField.Visible= 'off';
4864             app.C2EditFieldLabel.Visible = 'off';
4865             app.C1EditField.Visible = 'off';
4866             app.C1EditFieldLabel.Visible = 'off';
4867         elseif app.ViscositybreakingButton.Value
4868             app.C2EditField.Visible = "on";
4869             app.C2EditField.Enable = "on";
4870             app.C2EditFieldLabel.Visible = "on";
4871             app.C2EditFieldLabel.Enable = "on";
4872             app.C1EditField.Visible = "on";
4873             app.C1EditField.Enable = "on";
4874             app.C1EditFieldLabel.Visible = "on";
4875             app.C1EditFieldLabel.Enable = "on";
4876
4877         end
4878     end
4879
4880     % Value changed function: InvertvaluesCheckBox
4881     function InvertvaluesCheckBoxValueChanged(app, event)
4882
4883
4884     end
4885
4886     % Callback function
4887     function ImportInfoButtonPushed(app, event)
4888         [bathymetryfile,path2] = uigetfile('*.tiff;*.tif');
4889         bathymetryinputdata = string(path2) + string(bathymetryfile);
4890
4891
4892         if contains(bathymetryinputdata, '.tif')
4893             try
4894                 app.depthtype = 2;
4895                 app.LatitudeEditField.Enable = "on";
4896                 app.LongitudeEditField.Enable = "on";
4897                 %
4898
4899                 [A1,R2] = readgeoraster(bathymetryinputdata);
4900                 A = size(A1);

```

```

4901
4902         %Extract M and N global dimenstions
4903         app.XEditField_2.Value = A(1,2);
4904         app.YEditField_2.Value = A(1,1);
4905         app.XEditField_2.FontColor = 'k';
4906         app.YEditField_2.FontColor = 'k';
4907
4908         %Extract lat and long of the southwest corner
4909
4910         app.LongitudeEditField.Value = min(R2.LongitudeLimits);
4911         app.LatitudeEditField.Value = min(R2.LatitudeLimits);
4912         app.LongitudeEditField.FontColor = 'k';
4913         app.LatitudeEditField.FontColor = 'k';
4914
4915         %Extract grid size of the bathymetry
4916         app.XEditField.Value = R2.CellExtentInLatitude;
4917         app.YEditField.Value = R2.CellExtentInLongitude;
4918         app.XEditField.FontColor = 'k';
4919         app.YEditField.FontColor = 'k';
4920
4921         if R2.AngleUnits == "degrees"
4922             app.ButtonGroup_4.SelectedObject = app.↵
SphericalButton;
4923             app.SphericalButton.FontColor = 'k';
4924             app.CartesianButton.FontColor = 'k';
4925         end
4926         app.rasterdegrees = R2;
4927
4928         catch
4929             app.ButtonGroup_4.SelectedObject = app.Button_2;
4930         end
4931     end
4932 end
4933
4934 % Value changed function: DoublePrecisionCheckBox
4935 function DoublePrecisionCheckBoxValueChanged(app, event)
4936
4937
4938 end
4939
4940 % Button pushed function: BoundarylimitsButton
4941 function BoundarylimitsButtonPushed(app, event)
4942     [bathymetryfile,path2] = uigetfile(['*.tif;*.asc']);
4943
4944     bathymetryinputdata = string(path2) + string(bathymetryfile);
4945
4946
4947     if contains(bathymetryinputdata, '.tif')
4948         %This works with a .tif file generated using QGIS
4949         %             try
4950             app.depthtype = 2;
4951
4952             [A1,R2] = readgeoraster(bathymetryinputdata);
4953             A = size(A1);
4954
4955
4956             if strcmpi(R2.CoordinateSystemType, 'geographic')
4957                 southwestLongitude = min(R2.LongitudeLimits);

```

```

4958         southwestLatitude = min(R2.LatitudeLimits);
4959         gridSize = R2.CellExtentInLatitude;
4960         % gridSize = R2.CellExtentInLongitude;
4961
4962     elseif strcmpi(R2.CoordinateSystemType, 'planar')
4963         southwestLongitude = min(R2.XWorldLimits);
4964         southwestLatitude = min(R2.YWorldLimits);
4965         gridSize = R2.CellExtentInWorldX;
4966         % gridSize = R2.CellExtentInWorldY;
4967
4968     end
4969
4970     %Extract M and N global dimensions
4971     app.CouplingX = R2.RasterSize(1,2);
4972     app.CouplingY = R2.RasterSize(1,1);
4973
4974     %Extract grid size of the bathymetry
4975     cellsize = gridSize;
4976     app.SpacingEditField.Value = cellsize;
4977     app.SpacingEditField.FontColor = 'k';
4978
4979     % Extract and round the southwest corner coordinates
4980     rawSouth = southwestLatitude;
4981     westValue = southwestLongitude;
4982
4983     % Process based on input type
4984     if isnumeric(rawSouth) && isscalar(rawSouth)
4985         % Case 1: Scalar numeric input
4986         southValue = rawSouth;
4987
4988     elseif ischar(rawSouth) || isstring(rawSouth)
4989         % Case 2: String input
4990         southValue = str2double(rawSouth);
4991
4992     elseif isnumeric(rawSouth) && ~isscalar(rawSouth)
4993         % Case 3: Numeric array input
4994         southValue = double(rawSouth);
4995     end
4996
4997     % Assign to SouthEditField_2 if valid and within limits
4998     if exist('southValue', 'var') && isnumeric(southValue) && ~
isscalar(southValue)
4999         southLimits = app.SouthEditField_2.Limits;
5000         if southValue >= southLimits(1) && southValue <=
southLimits(2)
5001             app.SouthEditField_2.Value = southValue;
5002         end
5003     end
5004
5005     % Validate and assign west value
5006     if isnumeric(westValue) && isscalar(westValue)
5007         westLimits = app.WestEditField_2.Limits;
5008         if westValue >= westLimits(1) && westValue <= westLimits
(2)
5009             app.WestEditField_2.Value = westValue;
5010         end
5011     end
5012

```

```

5013         app.EastEditField_2.Value = westValue + (cellsize*app.
CouplingX);
5014         app.NorthEditField_2.Value = southValue + (cellsize*app.
CouplingY);
5015
5016     end
5017     % % Initialize Xmatrix and Ymatrix
5018     % Xmatrix = 1:app.CouplingX;
5019     % Ymatrix = 1:app.CouplingY;
5020     %
5021     % % Set the first element to the southwest corner
5022     % Xmatrix(Xmatrix == 1) = min(R2.LongitudeLimits);
5023     % Ymatrix(Ymatrix == 1) = min(R2.LatitudeLimits);
5024     %
5025     % % Calculate CoupMatrixX and CoupMatrixY
5026     % CoupMatrixX = Xmatrix * cellsize + min(R2.LongitudeLimits);
5027     % CoupMatrixY = Ymatrix * cellsize + min(R2.LatitudeLimits);
5028     %
5029     % % Recalculate Xmatrix and CoupMatrixX
5030     % Xmatrix = 1:app.CouplingX;
5031     % Xmatrix(Xmatrix == 1) = 0;
5032     % for i = 1:length(Xmatrix)
5033     %     if Xmatrix(i) ~= 0
5034     %         Xmatrix(i) = Xmatrix(i) * cellsize;
5035     %     end
5036     %     CoupMatrixX(i) = Xmatrix(i) + min(R2.LongitudeLimits);
5037     % end
5038     %
5039     % % Recalculate Ymatrix and CoupMatrixY
5040     % Ymatrix = 1:app.CouplingY;
5041     % Ymatrix(Ymatrix == 1) = 0;
5042     % for i = 1:length(Ymatrix)
5043     %     if Ymatrix(i) > 1
5044     %         Ymatrix(i) = Ymatrix(i) * cellsize;
5045     %     end
5046     %     CoupMatrixY(i) = Ymatrix(i) + min(R2.LatitudeLimits);
5047     % end
5048     %
5049     % % Store matrices
5050     % app.CoupX = CoupMatrixX;
5051     % app.CoupY = CoupMatrixY;
5052     % app.CoupMatrixON = 0; % Flag for nested grid boundary setup
5053     %
5054     % % Validate and assign east value
5055     % eastValue = max(CoupMatrixX(:));
5056     % if isnumeric(eastValue) && isscalar(eastValue)
5057     %     eastLimits = app.EastEditField_2.Limits;
5058     %     if eastValue >= eastLimits(1) && eastValue <= eastLimits
(2)
5059     %         app.EastEditField_2.Value = eastValue;
5060     %     end
5061     % end
5062     %
5063     % % Validate and assign north value
5064     % northValue = max(CoupMatrixY(:));
5065     % if isnumeric(northValue) && isscalar(northValue)
5066     %     northLimits = app.NorthEditField_2.Limits;
5067     %     if northValue >= northLimits(1) && northValue <=

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```

northLimits(2)
5068         %             app.NorthEditField_2.Value = northValue;
5069         %         end
5070         %     end
5071         %
5072         app.SouthEditField_2.FontColor = 'k';
5073         app.WestEditField_2.FontColor = 'k';
5074         app.NorthEditField_2.FontColor = 'k';
5075         app.EastEditField_2.FontColor = 'k';
5076         drawnow
5077     % end
5078     %
5079
5080
5081
5082
5083 end
5084
5085 % Value changed function: EastEditField_2
5086 function EastEditField_2ValueChanged(app, event)
5087     app.EastEditField_2.FontColor = 'k';
5088
5089 end
5090
5091 % Value changed function: NorthEditField_2
5092 function NorthEditField_2ValueChanged(app, event)
5093     app.NorthEditField_2.FontColor = 'k';
5094
5095 end
5096
5097 % Value changed function: SouthEditField_2
5098 function SouthEditField_2ValueChanged(app, event)
5099     app.SouthEditField_2.FontColor = 'k';
5100 end
5101
5102 % Value changed function: WestEditField_2
5103 function WestEditField_2ValueChanged(app, event)
5104     app.WestEditField_2.FontColor = 'k';
5105
5106 end
5107
5108 % Value changed function: SpacingEditField
5109 function SpacingEditFieldValueChanged(app, event)
5110     app.SpacingEditField.FontColor = 'k';
5111
5112 end
5113
5114 % Size changed function: Panel_32
5115 function Panel_32SizeChanged(app, event)
5116
5117
5118 end
5119
5120 % Button pushed function: GenerateinputtxtButton_4
5121 function GenerateinputtxtButton_4Pushed(app, event)
5122     GenerateinputtxtButtonPushed(app, event);
5123 end
5124

```

```

5125 % Button pushed function: Button_43
5126 function Button_43Pushed(app, event)
5127     [file,path2] = uigetfile('*.');
5128     app.PressurefileEditField.Value = string(file);
5129     app.PressurefileEditField.FontColor = 'k';
5130     app.PressurefileEditField.BackgroundColor = 'w';
5131     app.PressureFileLOC = string(path2) + string(file);
5132     app.PressureFileNAME = string(file);
5133 end
5134
5135 % Value changed function: SedimentTransportCheckBox
5136 function SedimentTransportCheckBoxValueChanged(app, event)
5137     %Set some 'Expected Output' parameters to T or F
5138     if app.SedimentTransportCheckBox.Value
5139         app.TabGroup2.SelectedTab = app.Tab_3;
5140         app.DepthCheckBox.Value = 1;
5141         app.UVelocityCheckBox.Value = 1;
5142         app.VVelocityCheckBox.Value = 1;
5143         app.WetdrymaskCheckBox.Value = 1;
5144         app.WaveHeightCheckBox.Value = 1;
5145     else
5146         app.DepthCheckBox.Value = 0;
5147         app.UVelocityCheckBox.Value = 0;
5148         app.VVelocityCheckBox.Value = 0;
5149         app.WetdrymaskCheckBox.Value = 0;
5150         app.WaveHeightCheckBox.Value = 0;
5151     end
5152
5153
5154 end
5155
5156 % Value changed function: MinimumEditField
5157 function MinimumEditFieldValueChanged(app, event)
5158     app.MinimumEditField.FontColor = 'k';
5159
5160 end
5161
5162 % Value changed function: MaximumEditField
5163 function MaximumEditFieldValueChanged(app, event)
5164     app.MaximumEditField.FontColor = 'k';
5165 end
5166
5167 % Value changed function: PeakEditField
5168 function PeakEditFieldValueChanged(app, event)
5169     app.PeakEditField.FontColor = 'k';
5170
5171 end
5172
5173 % Value changed function: WaveHeightmEditField
5174 function WaveHeightmEditFieldValueChanged(app, event)
5175     app.WaveHeightmEditField.FontColor = 'k';
5176
5177 end
5178
5179 % Value changed function: FroudeCapEditField
5180 function FroudeCapEditFieldValueChanged(app, event)
5181     app.FroudeCapEditField.FontColor = 'k';
5182

```

```

5183         end
5184
5185         % Value changed function: ThresholdEditField
5186         function ThresholdEditFieldValueChanged(app, event)
5187             app.ThresholdEditField.FontColor = 'k';
5188
5189         end
5190     end
5191
5192     % Component initialization
5193     methods (Access = private)
5194
5195         % Create UIFigure and components
5196         function createComponents(app)
5197
5198             % Create UIFigure and hide until all components are created
5199             app UIFigure = uifigure('Visible', 'off');
5200             app UIFigure.Position = [100 100 1146 676];
5201             app UIFigure.Name = 'MATLAB App';
5202             app UIFigure.Scrollable = 'on';
5203
5204             % Create TabGroup3
5205             app.TabGroup3 = uitabgroup(app UIFigure);
5206             app.TabGroup3.Position = [743 128 399 547];
5207
5208             % Create LogReportTab
5209             app.LogReportTab = uitab(app.TabGroup3);
5210             app.LogReportTab.Title = 'Log Report';
5211
5212             % Create Preview_LogReport
5213             app.Preview_LogReport = uitextarea(app.LogReportTab);
5214             app.Preview_LogReport.HandleVisibility = 'off';
5215             app.Preview_LogReport.Editable = 'off';
5216             app.Preview_LogReport.WordWrap = 'off';
5217             app.Preview_LogReport.Position = [1 0 397 511];
5218
5219             % Create inputtxtTab
5220             app.inputtxtTab = uitab(app.TabGroup3);
5221             app.inputtxtTab.Title = 'input.txt';
5222
5223             % Create Preview_Input
5224             app.Preview_Input = uitextarea(app.inputtxtTab);
5225             app.Preview_Input.Editable = 'off';
5226             app.Preview_Input.Position = [1 0 397 511];
5227
5228             % Create Makefile
5229             app.Makefile = uitab(app.TabGroup3);
5230             app.Makefile.Title = 'Makefile';
5231
5232             % Create Preview_Makefile
5233             app.Preview_Makefile = uitextarea(app.Makefile);
5234             app.Preview_Makefile.Editable = 'off';
5235             app.Preview_Makefile.Position = [1 0 397 511];
5236
5237             % Create TabGroup
5238             app.TabGroup = uitabgroup(app UIFigure);
5239             app.TabGroup.Position = [1 1 737 674];
5240

```



```

5241         % Create InputTab
5242         app.InputTab = uitab(app.TabGroup);
5243         app.InputTab.SizeChangedFcn = createCallbackFcn(app, ↵
@InputTabSizeChanged, true);
5244         app.InputTab.Title = 'Input';
5245         app.InputTab.BackgroundColor = [0.9412 0.9412 0.9412];
5246         app.InputTab.ForegroundColor = [0.149 0.149 0.149];
5247
5248         % Create Panel_27
5249         app.Panel_27 = uipanel(app.InputTab);
5250         app.Panel_27.Position = [371 396 356 42];
5251
5252         % Create WetDrySchemeEditFieldLabel
5253         app.WetDrySchemeEditFieldLabel = uilabel(app.Panel_27);
5254         app.WetDrySchemeEditFieldLabel.Tooltip = {'Minimum Depth (m) for↵
wet-dry scheme.'; ''; 'Suggested values:'; 'Lab scale: 0.001'; 'Field scale:↵
0.01'};
5255         app.WetDrySchemeEditFieldLabel.Position = [25 7 97 22];
5256         app.WetDrySchemeEditFieldLabel.Text = 'Wet-Dry Scheme';
5257
5258         % Create WetDrySchemeEditField
5259         app.WetDrySchemeEditField = uieditfield(app.Panel_27, 'numeric');
5260         app.WetDrySchemeEditField.Limits = [0 Inf];
5261         app.WetDrySchemeEditField.ValueDisplayFormat = '%.2f';
5262         app.WetDrySchemeEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@WetDrySchemeEditFieldValueChanged, true);
5263         app.WetDrySchemeEditField.FontColor = [0.651 0.651 0.651];
5264         app.WetDrySchemeEditField.Tooltip = {' '};
5265         app.WetDrySchemeEditField.Position = [124 10 42 16];
5266         app.WetDrySchemeEditField.Value = 10;
5267
5268         % Create BottomFrictionEditFieldLabel
5269         app.BottomFrictionEditFieldLabel = uilabel(app.Panel_27);
5270         app.BottomFrictionEditFieldLabel.HorizontalAlignment = 'right';
5271         app.BottomFrictionEditFieldLabel.FontColor = [0.149 0.149 0.149];
5272         app.BottomFrictionEditFieldLabel.Tooltip = {'Minimum Depth (m) to↵
limit bottom friction.'; ''; 'Suggested values:'; 'Lab scale: 0.01'; 'Field scale:↵
0.1'};
5273         app.BottomFrictionEditFieldLabel.Position = [188 7 89 22];
5274         app.BottomFrictionEditFieldLabel.Text = 'Bottom Friction';
5275
5276         % Create BottomFrictionEditField
5277         app.BottomFrictionEditField = uieditfield(app.Panel_27, ↵
'numeric');
5278         app.BottomFrictionEditField.Limits = [0 Inf];
5279         app.BottomFrictionEditField.ValueDisplayFormat = '%11.2f';
5280         app.BottomFrictionEditField.ValueChangedFcn = createCallbackFcn ↵
(app, @BottomFrictionEditFieldValueChanged, true);
5281         app.BottomFrictionEditField.FontColor = [0.651 0.651 0.651];
5282         app.BottomFrictionEditField.Tooltip = {' '};
5283         app.BottomFrictionEditField.Position = [284 10 41 16];
5284         app.BottomFrictionEditField.Value = 0.1;
5285
5286         % Create Panel_25
5287         app.Panel_25 = uipanel(app.InputTab);
5288         app.Panel_25.Position = [371 522 356 116];
5289
5290         % Create ButtonGroup 3

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```

5291         app.ButtonGroup_3 = uibuttongroup(app.Panel_25);
5292         app.ButtonGroup_3.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_3SelectionChanged, true);
5293         app.ButtonGroup_3.BorderType = 'none';
5294         app.ButtonGroup_3.Position = [7 10 344 95];
5295
5296         % Create LinearShallowWaterEquationButton
5297         app.LinearShallowWaterEquationButton = uiradiobutton(app. ↵
ButtonGroup_3);
5298         app.LinearShallowWaterEquationButton.Tooltip = {'Use linear ↵
shallow water equation.'; ''; 'When Boussinesq equation is not used, the ↵
''HIGH_ORDER'' value under ''Numerics'' in the input.txt is set from ''THIRD'' to ↵
''SECOND'' '};
5299         app.LinearShallowWaterEquationButton.Text = ' Linear Shallow Water ↵
Equation';
5300         app.LinearShallowWaterEquationButton.Position = [27 72 191 22];
5301
5302         % Create FullyNonlinearBoussinesqEquationButton
5303         app.FullyNonlinearBoussinesqEquationButton = uiradiobutton(app. ↵
ButtonGroup_3);
5304         app.FullyNonlinearBoussinesqEquationButton.Text = ' Fully ↵
Nonlinear Boussinesq Equation';
5305         app.FullyNonlinearBoussinesqEquationButton.Position = [27 24 224 ↵
22];
5306
5307         % Create Button
5308         app.Button = uiradiobutton(app.ButtonGroup_3);
5309         app.Button.Visible = 'off';
5310         app.Button.Text = '';
5311         app.Button.Position = [160 73 25 22];
5312         app.Button.Value = true;
5313
5314         % Create NonlinearShallowWaterEquationButton
5315         app.NonlinearShallowWaterEquationButton = uiradiobutton(app. ↵
ButtonGroup_3);
5316         app.NonlinearShallowWaterEquationButton.Tooltip = {'Use nonlinear ↵
shallow water equation.'; ''; ''; 'When Boussinesq equation is not used, the ↵
''HIGH_ORDER'' value under ''Numerics'' in the input.txt is set from ''THIRD'' to ↵
''SECOND'' '};
5317         app.NonlinearShallowWaterEquationButton.Text = ' Nonlinear Shallow ↵
Water Equation';
5318         app.NonlinearShallowWaterEquationButton.Position = [27 48 209 22];
5319
5320         % Create FroudeCapEditFieldLabel
5321         app.FroudeCapEditFieldLabel = uilabel(app.ButtonGroup_3);
5322         app.FroudeCapEditFieldLabel.HorizontalAlignment = 'right';
5323         app.FroudeCapEditFieldLabel.Tooltip = {'Froude number limit for ↵
velocity efficiency'};
5324         app.FroudeCapEditFieldLabel.Position = [41 0 69 22];
5325         app.FroudeCapEditFieldLabel.Text = 'Froude Cap';
5326
5327         % Create FroudeCapEditField
5328         app.FroudeCapEditField = uieditfield(app.ButtonGroup_3, ↵
'numeric');
5329         app.FroudeCapEditField.Limits = [0 Inf];
5330         app.FroudeCapEditField.ValueDisplayFormat = '%8.2f';
5331         app.FroudeCapEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@FroudeCapEditFieldValueChanged, true);

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```

5332     app.FroudeCapEditField.FontColor = [0.651 0.651 0.651];
5333     app.FroudeCapEditField.Position = [119 3 41 16];
5334     app.FroudeCapEditField.Value = 10;
5335
5336     % Create Panel_17
5337     app.Panel_17 = uipanel(app.InputTab);
5338     app.Panel_17.Position = [10 596 351 42];
5339
5340     % Create Panel_5
5341     app.Panel_5 = uipanel(app.Panel_17);
5342     app.Panel_5.BorderType = 'none';
5343     app.Panel_5.Position = [43 4 264 30];
5344
5345     % Create ProcessornumbersXLabel
5346     app.ProcessornumbersXLabel = uilabel(app.Panel_5);
5347     app.ProcessornumbersXLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5348     app.ProcessornumbersXLabel.HorizontalAlignment = 'right';
5349     app.ProcessornumbersXLabel.Tooltip = {'Must be consistent with mpirun -np n (px*py)'};
5350     app.ProcessornumbersXLabel.Position = [3 3 134 22];
5351     app.ProcessornumbersXLabel.Text = 'Processor numbers: X';
5352
5353     % Create ProcessorX
5354     app.ProcessorX = uieditfield(app.Panel_5, 'numeric');
5355     app.ProcessorX.Limits = [0 Inf];
5356     app.ProcessorX.ValueChangedFcn = createCallbackFcn(app, @ProcessorXValueChanged, true);
5357     app.ProcessorX.FontColor = [0.651 0.651 0.651];
5358     app.ProcessorX.Tooltip = {' '};
5359     app.ProcessorX.Position = [146 6 34 16];
5360     app.ProcessorX.Value = 2;
5361
5362     % Create YLabel
5363     app.YLabel = uilabel(app.Panel_5);
5364     app.YLabel.HorizontalAlignment = 'right';
5365     app.YLabel.Tooltip = {'Must be consistent with mpirun -np n (px*py)'};
5366     app.YLabel.Position = [180 3 35 22];
5367     app.YLabel.Text = 'Y';
5368
5369     % Create ProcessorY
5370     app.ProcessorY = uieditfield(app.Panel_5, 'numeric');
5371     app.ProcessorY.ValueChangedFcn = createCallbackFcn(app, @ProcessorYValueChanged, true);
5372     app.ProcessorY.FontColor = [0.651 0.651 0.651];
5373     app.ProcessorY.Tooltip = {' '};
5374     app.ProcessorY.Position = [223 6 32 16];
5375     app.ProcessorY.Value = 2;
5376
5377     % Create ParallelInfoLabel
5378     app.ParallelInfoLabel = uilabel(app.InputTab);
5379     app.ParallelInfoLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5380     app.ParallelInfoLabel.FontSize = 13;
5381     app.ParallelInfoLabel.FontWeight = 'bold';
5382     app.ParallelInfoLabel.FontColor = [0.0314 0.3686 0.6];
5383     app.ParallelInfoLabel.Position = [21 626 89 22];
5384     app.ParallelInfoLabel.Text = ' Parallel Info ';

```

```

5385
5386         % Create Friction_____Label_2
5387         app.Friction_____Label_2 = uilabel(app. InputTab);
5388         app.Friction_____Label_2. BackgroundColor = [0.9412 0.9412 0.9412];
5389         app.Friction_____Label_2.FontSize = 13;
5390         app.Friction_____Label_2.FontWeight = 'bold';
5391         app.Friction_____Label_2.FontColor = [0.0314 0.3686 0.6];
5392         app.Friction_____Label_2.Position = [383 427 111 22];
5393         app.Friction_____Label_2.Text = 'Minimum Depth';
5394
5395         % Create Panel_18
5396         app.Panel_18 = uipanel(app.InputTab);
5397         app.Panel_18.Position = [10 501 351 84];
5398
5399         % Create ButtonGroup_7
5400         app.ButtonGroup_7 = uibuttongroup(app.Panel_18);
5401         app.ButtonGroup_7.SelectionChangedFcn = createCallbackFcn(app, @ButtonGroup_7SelectionChanged, true);
5402         app.ButtonGroup_7.ForegroundColor = [0.149 0.149 0.149];
5403         app.ButtonGroup_7.BorderType = 'none';
5404         app.ButtonGroup_7.Position = [25 9 100 64];
5405
5406         % Create ImportDataButton
5407         app.ImportDataButton = uiradiobutton(app.ButtonGroup_7);
5408         app.ImportDataButton.Tooltip = {'Import the bathymetry.'; ''; 'Accepted formats: .txt, .tif'; ''; 'If a .tif file is uploaded, it will be automatically converted to a text file. In the conversion, the array is flipped up to down as FUNWAVE reads the depth file starting from the southwest corner.'};
5409         app.ImportDataButton.Text = ' Import Data';
5410         app.ImportDataButton.Position = [5 41 89 22];
5411
5412         % Create FlatButton
5413         app.FlatButton = uiradiobutton(app.ButtonGroup_7);
5414         app.FlatButton.Tooltip = {'Create and use a flat bottom bathymetry'};
5415         app.FlatButton.Text = ' Flat';
5416         app.FlatButton.Position = [5 19 45 22];
5417
5418         % Create SlopeButton
5419         app.SlopeButton = uiradiobutton(app.ButtonGroup_7);
5420         app.SlopeButton.Tooltip = {'Create and use a sloping bathymetry'};
5421         app.SlopeButton.Text = ' Slope';
5422         app.SlopeButton.Position = [5 -3 56 22];
5423
5424         % Create Button4
5425         app.Button4 = uiradiobutton(app.ButtonGroup_7);
5426         app.Button4.Visible = 'off';
5427         app.Button4.Text = 'Button4';
5428         app.Button4.Position = [57 -4 27 22];
5429         app.Button4.Value = true;
5430

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```

5431         % Create DepthFileEditField
5432         app.DepthFileEditField = uieditfield(app.Panel_18, 'text');
5433         app.DepthFileEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@DepthFileEditFieldValueChanged, true);
5434         app.DepthFileEditField.ValueChangingFcn = createCallbackFcn(app, ↵
@DepthFileEditFieldValueChanging, true);
5435         app.DepthFileEditField.Editable = 'off';
5436         app.DepthFileEditField.FontSize = 11;
5437         app.DepthFileEditField.FontColor = [0 0 1];
5438         app.DepthFileEditField.Visible = 'off';
5439         app.DepthFileEditField.Position = [133 52 144 17];
5440
5441         % Create Button_8
5442         app.Button_8 = uibutton(app.Panel_18, 'push');
5443         app.Button_8.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_8Pushed, true);
5444         app.Button_8.VerticalAlignment = 'top';
5445         app.Button_8.FontSize = 10;
5446         app.Button_8.FontAngle = 'italic';
5447         app.Button_8.FontColor = [0 0.451 0.7412];
5448         app.Button_8.Visible = 'off';
5449         app.Button_8.Tooltip = {'Import the bathymetry.'; ''; 'Accepted↵
formats: .txt, .tif'; ''; 'If a .tif file is uploaded, it will be automatically↵
converted to a text file. This is tested with .tif files generated using QGIS.';↵
''; 'The dimensions and grid details are automatically extracted from the file.';↵
''; 'The values in the .txt file should include decimal places.'};
5450         app.Button_8.Position = [285 51 19 18];
5451         app.Button_8.Text = '...';
5452
5453         % Create WaterlevelmEditField
5454         app.WaterlevelmEditField = uieditfield(app.Panel_18, 'numeric');
5455         app.WaterlevelmEditField.Limits = [0 Inf];
5456         app.WaterlevelmEditField.ValueDisplayFormat = '%5.2f';
5457         app.WaterlevelmEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@WaterlevelmEditFieldValueChanged, true);
5458         app.WaterlevelmEditField.FontColor = [0.651 0.651 0.651];
5459         app.WaterlevelmEditField.Visible = 'off';
5460         app.WaterlevelmEditField.Tooltip = {''};
5461         app.WaterlevelmEditField.Position = [233 9 44 17];
5462
5463         % Create WaterlevelmEditFieldLabel
5464         app.WaterlevelmEditFieldLabel = uilabel(app.Panel_18);
5465         app.WaterlevelmEditFieldLabel.Visible = 'off';
5466         app.WaterlevelmEditFieldLabel.Tooltip = {'Adjust the initial water↵
level of the input bathymetry.'; ''; 'For example, to add surge or tide effects.'};
5467         app.WaterlevelmEditFieldLabel.Position = [133 6 96 22];
5468         app.WaterlevelmEditFieldLabel.Text = 'Initial waver level';
5469
5470         % Create InvertvaluesCheckBox
5471         app.InvertvaluesCheckBox = uicheckbox(app.Panel_18);
5472         app.InvertvaluesCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@InvertvaluesCheckBoxValueChanged, true);
5473         app.InvertvaluesCheckBox.Visible = 'off';
5474         app.InvertvaluesCheckBox.Tooltip = {'Convert bathymetry values to↵
positive and land values to negative'};
5475         app.InvertvaluesCheckBox.Text = 'Invert values';
5476         app.InvertvaluesCheckBox.WordWrap = 'on';
5477         app.InvertvaluesCheckBox.Position = [133 28 105 22];

```

```

5478
5479 % Create DepthmEditField
5480 app.DepthmEditField = uieditfield(app.Panel_18, 'numeric');
5481 app.DepthmEditField.Limits = [0 Inf];
5482 app.DepthmEditField.ValueDisplayFormat = '%5.2f';
5483 app.DepthmEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@DepthmEditFieldValueChanged, true);
5484 app.DepthmEditField.FontColor = [0.651 0.651 0.651];
5485 app.DepthmEditField.Visible = 'off';
5486 app.DepthmEditField.Tooltip = {' '};
5487 app.DepthmEditField.Position = [190 31 48 17];
5488 app.DepthmEditField.Value = 100;
5489
5490 % Create DepthmEditFieldLabel
5491 app.DepthmEditFieldLabel = uilabel(app.Panel_18);
5492 app.DepthmEditFieldLabel.HorizontalAlignment = 'right';
5493 app.DepthmEditFieldLabel.Visible = 'off';
5494 app.DepthmEditFieldLabel.Tooltip = {'Unit: meters'};
5495 app.DepthmEditFieldLabel.Position = [126 29 41 22];
5496 app.DepthmEditFieldLabel.Text = ' Depth';
5497
5498 % Create StartatXEditFieldLabel
5499 app.StartatXEditFieldLabel = uilabel(app.Panel_18);
5500 app.StartatXEditFieldLabel.HorizontalAlignment = 'right';
5501 app.StartatXEditFieldLabel.Visible = 'off';
5502 app.StartatXEditFieldLabel.Tooltip = {'The location where the ↵
sloping beach begins, with a zero value indicating the deeper part of the water.'; ↵
' '; 'Unit: meters'};
5503 app.StartatXEditFieldLabel.Position = [127 6 56 22];
5504 app.StartatXEditFieldLabel.Text = 'Start at X';
5505
5506 % Create StartatXEditField
5507 app.StartatXEditField = uieditfield(app.Panel_18, 'numeric');
5508 app.StartatXEditField.ValueDisplayFormat = '%3.2f';
5509 app.StartatXEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@StartatXEditFieldValueChanged, true);
5510 app.StartatXEditField.FontColor = [0.651 0.651 0.651];
5511 app.StartatXEditField.Visible = 'off';
5512 app.StartatXEditField.Tooltip = {' '};
5513 app.StartatXEditField.Position = [190 9 48 17];
5514
5515 % Create AngleEditFieldLabel
5516 app.AngleEditFieldLabel = uilabel(app.Panel_18);
5517 app.AngleEditFieldLabel.HorizontalAlignment = 'right';
5518 app.AngleEditFieldLabel.Visible = 'off';
5519 app.AngleEditFieldLabel.Tooltip = {'Slope angle'};
5520 app.AngleEditFieldLabel.Position = [248 8 36 22];
5521 app.AngleEditFieldLabel.Text = 'Angle';
5522
5523 % Create AngleEditField
5524 app.AngleEditField = uieditfield(app.Panel_18, 'numeric');
5525 app.AngleEditField.ValueDisplayFormat = '%3.2f';
5526 app.AngleEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@AngleEditFieldValueChanged, true);
5527 app.AngleEditField.FontColor = [0.651 0.651 0.651];
5528 app.AngleEditField.Visible = 'off';
5529 app.AngleEditField.Tooltip = {' '};
5530 app.AngleEditField.Position = [289 11 42 16];

```

```

5531
5532 % Create Panel_19
5533 app.Panel_19 = uipanel(app.InputTab);
5534 app.Panel_19.Position = [10 454 351 36];
5535
5536 % Create YEditField_2Label
5537 app.YEditField_2Label = uilabel(app.Panel_19);
5538 app.YEditField_2Label.HorizontalAlignment = 'right';
5539 app.YEditField_2Label.Tooltip = {'Matrix size in Y direction'};
5540 app.YEditField_2Label.Position = [182 6 16 22];
5541 app.YEditField_2Label.Text = 'Y';
5542
5543 % Create YEditField_2
5544 app.YEditField_2 = uieditfield(app.Panel_19, 'numeric');
5545 app.YEditField_2.ValueDisplayFormat = '%.0f';
5546 app.YEditField_2.ValueChangedFcn = createCallbackFcn(app, @YEditField_2ValueChanged, true);
5547 app.YEditField_2.FontColor = [0.651 0.651 0.651];
5548 app.YEditField_2.Tooltip = {' '};
5549 app.YEditField_2.Position = [204 9 45 16];
5550
5551 % Create XEditField_2Label
5552 app.XEditField_2Label = uilabel(app.Panel_19);
5553 app.XEditField_2Label.HorizontalAlignment = 'right';
5554 app.XEditField_2Label.Tooltip = {'Matrix size in X direction'};
5555 app.XEditField_2Label.Position = [97 6 18 22];
5556 app.XEditField_2Label.Text = 'X ';
5557
5558 % Create XEditField_2
5559 app.XEditField_2 = uieditfield(app.Panel_19, 'numeric');
5560 app.XEditField_2.Limits = [0 Inf];
5561 app.XEditField_2.ValueDisplayFormat = '%.0f';
5562 app.XEditField_2.ValueChangedFcn = createCallbackFcn(app, @XEditField_2ValueChanged, true);
5563 app.XEditField_2.FontColor = [0.651 0.651 0.651];
5564 app.XEditField_2.Tooltip = {' '};
5565 app.XEditField_2.Position = [119 9 45 16];
5566
5567 % Create Panel_20
5568 app.Panel_20 = uipanel(app.InputTab);
5569 app.Panel_20.Position = [10 403 351 39];
5570
5571 % Create ResultsfolderEditFieldLabel
5572 app.ResultsfolderEditFieldLabel = uilabel(app.Panel_20);
5573 app.ResultsfolderEditFieldLabel.Tooltip = {'Folder where the tsunami simulations will be saved'};
5574 app.ResultsfolderEditFieldLabel.Position = [53 6 79 22];
5575 app.ResultsfolderEditFieldLabel.Text = 'Results folder';
5576
5577 % Create ResultsfolderEditField
5578 app.ResultsfolderEditField = uieditfield(app.Panel_20, 'text');
5579 app.ResultsfolderEditField.ValueChangedFcn = createCallbackFcn(app, @ResultsfolderEditFieldValueChanged, true);
5580 app.ResultsfolderEditField.ValueChangingFcn = createCallbackFcn(app, @ResultsfolderEditFieldValueChanging, true);
5581 app.ResultsfolderEditField.FontSize = 11;
5582 app.ResultsfolderEditField.FontColor = [0 0 1];
5583 app.ResultsfolderEditField.Tooltip = {' '};

```



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5584         app.ResultsfolderEditField.Placeholder = 'Default:↵
"Simulation_Results/";
5585         app.ResultsfolderEditField.Position = [139 9 178 16];
5586
5587         % Create PrintLabel
5588         app.PrintLabel = uilabel(app.InputTab);
5589         app.PrintLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5590         app.PrintLabel.FontSize = 13;
5591         app.PrintLabel.FontWeight = 'bold';
5592         app.PrintLabel.FontColor = [0.0314 0.3686 0.6];
5593         app.PrintLabel.Position = [21 432 42 21];
5594         app.PrintLabel.Text = ' Print';
5595
5596         % Create Panel_21
5597         app.Panel_21 = uipanel(app.InputTab);
5598         app.Panel_21.Position = [10 332 351 60];
5599
5600         % Create ScreenLoadingIntervalLabel
5601         app.ScreenLoadingIntervalLabel = uilabel(app.Panel_21);
5602         app.ScreenLoadingIntervalLabel.WordWrap = 'on';
5603         app.ScreenLoadingIntervalLabel.Tooltip = {'Initila time'};
5604         app.ScreenLoadingIntervalLabel.Position = [55 27 31 22];
5605         app.ScreenLoadingIntervalLabel.Text = 'Start';
5606
5607         % Create StartEditField
5608         app.StartEditField = uieditfield(app.Panel_21, 'numeric');
5609         app.StartEditField.Limits = [0 Inf];
5610         app.StartEditField.ValueDisplayFormat = '%.1f';
5611         app.StartEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@StartEditFieldValueChanged, true);
5612         app.StartEditField.FontColor = [0.651 0.651 0.651];
5613         app.StartEditField.Tooltip = {''};
5614         app.StartEditField.Position = [90 30 52 16];
5615
5616         % Create GaugeIntervalEditFieldLabel
5617         app.GaugeIntervalEditFieldLabel = uilabel(app.Panel_21);
5618         app.GaugeIntervalEditFieldLabel.HorizontalAlignment = 'right';
5619         app.GaugeIntervalEditFieldLabel.Tooltip = {'Time interval between↵
station recordings.'; ''; 'Use 0.0 when the stations are for the nesting data.';↵
''; 'Unit: seconds'};
5620         app.GaugeIntervalEditFieldLabel.Position = [154 3 110 28];
5621         app.GaugeIntervalEditFieldLabel.Text = 'Station Plot Interval';
5622
5623         % Create GaugeIntervalEditField
5624         app.GaugeIntervalEditField = uieditfield(app.Panel_21, 'numeric');
5625         app.GaugeIntervalEditField.Limits = [0 Inf];
5626         app.GaugeIntervalEditField.ValueDisplayFormat = '%.1f';
5627         app.GaugeIntervalEditField.ValueChangedFcn = createCallbackFcn ↵
(app, @GaugeIntervalEditFieldValueChanged, true);
5628         app.GaugeIntervalEditField.FontColor = [0.651 0.651 0.651];
5629         app.GaugeIntervalEditField.Tooltip = {''};
5630         app.GaugeIntervalEditField.Position = [270 9 47 16];
5631
5632         % Create SimulationIntervalEditFieldLabel
5633         app.SimulationIntervalEditFieldLabel = uilabel(app.Panel_21);
5634         app.SimulationIntervalEditFieldLabel.HorizontalAlignment = ↵
'right';
5635         app.SimulationIntervalEditFieldLabel.WordWrap = 'on';

```



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5636         app.SimulationIntervalEditFieldLabel.Tooltip = {'Interval for generating simulation output'; ''; 'Unit: seconds'};
5637         app.SimulationIntervalEditFieldLabel.Position = [154 29 110 18];
5638         app.SimulationIntervalEditFieldLabel.Text = 'Simulation Interval';
5639
5640         % Create SimulationIntervalEditField
5641         app.SimulationIntervalEditField = uieditfield(app.Panel_21, 'numeric');
5642         app.SimulationIntervalEditField.Limits = [0 Inf];
5643         app.SimulationIntervalEditField.ValueDisplayFormat = '%.1f';
5644         app.SimulationIntervalEditField.ValueChangedFcn = createCallbackFcn(app, @SimulationIntervalEditFieldValueChanged, true);
5645         app.SimulationIntervalEditField.FontColor = [0.651 0.651 0.651];
5646         app.SimulationIntervalEditField.Tooltip = {''; ''};
5647         app.SimulationIntervalEditField.Position = [270 30 47 16];
5648
5649         % Create TotalEditFieldLabel
5650         app.TotalEditFieldLabel = uilabel(app.Panel_21);
5651         app.TotalEditFieldLabel.WordWrap = 'on';
5652         app.TotalEditFieldLabel.Tooltip = {'Total simulation time'; ''; 'Unit: seconds'};
5653         app.TotalEditFieldLabel.Position = [55 6 34 22];
5654         app.TotalEditFieldLabel.Text = 'Total';
5655
5656         % Create TotalEditField
5657         app.TotalEditField = uieditfield(app.Panel_21, 'numeric');
5658         app.TotalEditField.Limits = [0 Inf];
5659         app.TotalEditField.ValueDisplayFormat = '%.1f';
5660         app.TotalEditField.ValueChangedFcn = createCallbackFcn(app, @TotalEditFieldValueChanged2, true);
5661         app.TotalEditField.FontColor = [0.651 0.651 0.651];
5662         app.TotalEditField.Tooltip = {''; ''};
5663         app.TotalEditField.Position = [90 9 52 16];
5664         app.TotalEditField.Value = 3600;
5665
5666         % Create TimeLabel_2
5667         app.TimeLabel_2 = uilabel(app.InputTab);
5668         app.TimeLabel_2.BackgroundColor = [0.9412 0.9412 0.9412];
5669         app.TimeLabel_2.FontSize = 13;
5670         app.TimeLabel_2.FontWeight = 'bold';
5671         app.TimeLabel_2.FontColor = [0.0314 0.3686 0.6];
5672         app.TimeLabel_2.Position = [22 381 47 21];
5673         app.TimeLabel_2.Text = 'Time';
5674
5675         % Create Panel_22
5676         app.Panel_22 = uipanel(app.InputTab);
5677         app.Panel_22.Position = [10 214 351 107];
5678
5679         % Create XEditFieldLabel
5680         app.XEditFieldLabel = uilabel(app.Panel_22);
5681         app.XEditFieldLabel.HorizontalAlignment = 'right';
5682         app.XEditFieldLabel.Tooltip = {''; ''};
5683         app.XEditFieldLabel.Position = [214 29 25 22];
5684         app.XEditFieldLabel.Text = 'X';
5685
5686         % Create XEditField
5687         app.XEditField = uieditfield(app.Panel_22, 'numeric');
5688         app.XEditField.Limits = [0 Inf];

```

```

5689         app.XEditField.ValueDisplayFormat = '%8.5f';
5690         app.XEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@XEditFieldValueChanged, true);
5691         app.XEditField.FontColor = [0.651 0.651 0.651];
5692         app.XEditField.Tooltip = {' '};
5693         app.XEditField.Position = [245 33 57 16];
5694
5695         % Create YgridsizeLabel
5696         app.YgridsizeLabel = uilabel(app.Panel_22);
5697         app.YgridsizeLabel.HorizontalAlignment = 'right';
5698         app.YgridsizeLabel.Position = [211 7 25 22];
5699         app.YgridsizeLabel.Text = 'Y';
5700
5701         % Create YEditField
5702         app.YEditField = uieditfield(app.Panel_22, 'numeric');
5703         app.YEditField.Limits = [0 Inf];
5704         app.YEditField.ValueDisplayFormat = '%8.5f';
5705         app.YEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@YEditFieldValueChanged, true);
5706         app.YEditField.FontColor = [0.651 0.651 0.651];
5707         app.YEditField.Tooltip = {' '};
5708         app.YEditField.Position = [245 11 57 16];
5709
5710         % Create LongitudeEditFieldLabel
5711         app.LongitudeEditFieldLabel = uilabel(app.Panel_22);
5712         app.LongitudeEditFieldLabel.WordWrap = 'on';
5713         app.LongitudeEditFieldLabel.Tooltip = {' '};
5714         app.LongitudeEditFieldLabel.Position = [29 31 60 22];
5715         app.LongitudeEditFieldLabel.Text = 'Longitude';
5716
5717         % Create LongitudeEditField
5718         app.LongitudeEditField = uieditfield(app.Panel_22, 'numeric');
5719         app.LongitudeEditField.ValueDisplayFormat = '%11.5f';
5720         app.LongitudeEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@LongitudeEditFieldValueChanged, true);
5721         app.LongitudeEditField.FontColor = [0.651 0.651 0.651];
5722         app.LongitudeEditField.Tooltip = {' '};
5723         app.LongitudeEditField.Position = [88 34 76 16];
5724
5725         % Create LatitudeEditFieldLabel
5726         app.LatitudeEditFieldLabel = uilabel(app.Panel_22);
5727         app.LatitudeEditFieldLabel.WordWrap = 'on';
5728         app.LatitudeEditFieldLabel.Tooltip = {' '};
5729         app.LatitudeEditFieldLabel.Position = [29 6 46 28];
5730         app.LatitudeEditFieldLabel.Text = 'Latitude';
5731
5732         % Create LatitudeEditField
5733         app.LatitudeEditField = uieditfield(app.Panel_22, 'numeric');
5734         app.LatitudeEditField.ValueDisplayFormat = '%11.5f';
5735         app.LatitudeEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@LatitudeEditFieldValueChanged, true);
5736         app.LatitudeEditField.FontColor = [0.651 0.651 0.651];
5737         app.LatitudeEditField.Tooltip = {' '};
5738         app.LatitudeEditField.Position = [88 11 76 17];
5739
5740         % Create ButtonGroup_4
5741         app.ButtonGroup_4 = uibuttongroup(app.Panel_22);
5742         app.ButtonGroup_4.SelectionChangedFcn = createCallbackFcn(app, ↵

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```

@ButtonGroup_4SelectionChanged, true);
5743         app.ButtonGroup_4.Tooltip = {'Grid coordinate'};
5744         app.ButtonGroup_4.BorderType = 'none';
5745         app.ButtonGroup_4.Position = [82 69 187 30];
5746
5747         % Create SphericalButton
5748         app.SphericalButton = uiradiobutton(app.ButtonGroup_4);
5749         app.SphericalButton.Tooltip = {'Unit: degrees'};
5750         app.SphericalButton.Text = 'Spherical';
5751         app.SphericalButton.Position = [11 7 76 22];
5752
5753         % Create CartesianButton
5754         app.CartesianButton = uiradiobutton(app.ButtonGroup_4);
5755         app.CartesianButton.Tooltip = {'Unit: meters'};
5756         app.CartesianButton.Text = 'Cartesian';
5757         app.CartesianButton.FontColor = [0.149 0.149 0.149];
5758         app.CartesianButton.Position = [99 7 76 22];
5759
5760         % Create Button_2
5761         app.Button_2 = uiradiobutton(app.ButtonGroup_4);
5762         app.Button_2.Visible = 'off';
5763         app.Button_2.Text = '';
5764         app.Button_2.Position = [181 7 25 22];
5765         app.Button_2.Value = true;
5766
5767         % Create SouthwestcornerLabel
5768         app.SouthwestcornerLabel = uilabel(app.Panel_22);
5769         app.SouthwestcornerLabel.Tooltip = {'The corner coordinates of the
numerical grid.'; ''; 'Unit: degrees / meters'};
5770         app.SouthwestcornerLabel.Position = [28 50 104 22];
5771         app.SouthwestcornerLabel.Text = 'Southwest corner: ';
5772
5773         % Create GridSizeLabel
5774         app.GridSizeLabel = uilabel(app.Panel_22);
5775         app.GridSizeLabel.Tooltip = {'Spatial resolution of the numerical
grid.'; ''; 'Unit: meters'};
5776         app.GridSizeLabel.Position = [227 50 58 22];
5777         app.GridSizeLabel.Text = 'Grid Size: ';
5778
5779         % Create GridLabel
5780         app.GridLabel = uilabel(app.InputTab);
5781         app.GridLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5782         app.GridLabel.FontSize = 13;
5783         app.GridLabel.FontWeight = 'bold';
5784         app.GridLabel.FontColor = [0.0314 0.3686 0.6];
5785         app.GridLabel.Position = [22 310 46 21];
5786         app.GridLabel.Text = 'Grid';
5787
5788         % Create Panel_23
5789         app.Panel_23 = uipanel(app.InputTab);
5790         app.Panel_23.Position = [10 59 351 144];
5791
5792         % Create LayerwidthLabel
5793         app.LayerwidthLabel = uilabel(app.Panel_23);
5794         app.LayerwidthLabel.Tooltip = {'Sponge layer thickness relative to
model boundary. The width should be at least half of the wavelength.'; ''; 'Unit:
meters'; ''; 'Example: '; 'Entering 10,000 in the 'North' textbox extends the
sponge layer 10,000 meters southward from the North boundary.'};

```

```

5795     app.LayerwidthLabel.Position = [140 46 71 23];
5796     app.LayerwidthLabel.Text = 'Layer width:';
5797
5798     % Create DirectCheckBox
5799     app.DirectCheckBox = ucheckbox(app.Panel_23);
5800     app.DirectCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@DirectCheckBoxValueChanged, true);
5801     app.DirectCheckBox.Tooltip = {'L-D type sponge'};
5802     app.DirectCheckBox.Text = ' Direct';
5803     app.DirectCheckBox.Position = [29 107 57 22];
5804     app.DirectCheckBox.Value = true;
5805
5806     % Create FrictionCheckBox
5807     app.FrictionCheckBox = ucheckbox(app.Panel_23);
5808     app.FrictionCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@FrictionCheckBoxValueChanged, true);
5809     app.FrictionCheckBox.Text = ' Friction';
5810     app.FrictionCheckBox.Position = [138 107 65 22];
5811
5812     % Create DiffusionCheckBox
5813     app.DiffusionCheckBox = ucheckbox(app.Panel_23);
5814     app.DiffusionCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@DiffusionCheckBoxValueChanged, true);
5815     app.DiffusionCheckBox.Text = ' Diffusion';
5816     app.DiffusionCheckBox.Position = [255 107 72 22];
5817
5818     % Create NorthEditField
5819     app.NorthEditField = ueditfield(app.Panel_23, 'numeric');
5820     app.NorthEditField.Limits = [0 Inf];
5821     app.NorthEditField.ValueDisplayFormat = '%5.1f';
5822     app.NorthEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@NorthEditFieldValueChanged, true);
5823     app.NorthEditField.FontColor = [0.651 0.651 0.651];
5824     app.NorthEditField.Tooltip = {' '};
5825     app.NorthEditField.Position = [93 29 61 16];
5826     app.NorthEditField.Value = 100000;
5827
5828     % Create EastEditField
5829     app.EastEditField = ueditfield(app.Panel_23, 'numeric');
5830     app.EastEditField.Limits = [0 Inf];
5831     app.EastEditField.ValueDisplayFormat = '%5.1f';
5832     app.EastEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@EastEditFieldValueChanged, true);
5833     app.EastEditField.FontColor = [0.651 0.651 0.651];
5834     app.EastEditField.Tooltip = {' '};
5835     app.EastEditField.Position = [218 29 62 16];
5836     app.EastEditField.Value = 100000;
5837
5838     % Create SouthEditField
5839     app.SouthEditField = ueditfield(app.Panel_23, 'numeric');
5840     app.SouthEditField.Limits = [0 Inf];
5841     app.SouthEditField.ValueDisplayFormat = '%5.1f';
5842     app.SouthEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@SouthEditFieldValueChanged, true);
5843     app.SouthEditField.FontColor = [0.651 0.651 0.651];
5844     app.SouthEditField.Tooltip = {' '};
5845     app.SouthEditField.Position = [93 8 61 16];
5846     app.SouthEditField.Value = 100000;

```

```

5847
5848 % Create WestEditField
5849 app.WestEditField = uieditfield(app.Panel_23, 'numeric');
5850 app.WestEditField.Limits = [0 Inf];
5851 app.WestEditField.ValueDisplayFormat = '%5.1f';
5852 app.WestEditField.ValueChangedFcn = createCallbackFcn(app, ⚡
@WestEditFieldValueChanged, true);
5853 app.WestEditField.FontColor = [0.651 0.651 0.651];
5854 app.WestEditField.Tooltip = {' '};
5855 app.WestEditField.Position = [218 8 62 16];
5856 app.WestEditField.Value = 100000;
5857
5858 % Create NorthLabel
5859 app.NorthLabel = uilabel(app.Panel_23);
5860 app.NorthLabel.Tooltip = {'Unit: meters'};
5861 app.NorthLabel.Position = [53 26 35 22];
5862 app.NorthLabel.Text = 'North';
5863
5864 % Create EastLabel
5865 app.EastLabel = uilabel(app.Panel_23);
5866 app.EastLabel.Tooltip = {'Unit: meters'};
5867 app.EastLabel.Position = [185 26 29 22];
5868 app.EastLabel.Text = 'East';
5869
5870 % Create WestLabel
5871 app.WestLabel = uilabel(app.Panel_23);
5872 app.WestLabel.Tooltip = {'Unit: meters'};
5873 app.WestLabel.Position = [185 5 32 22];
5874 app.WestLabel.Text = 'West';
5875
5876 % Create SouthLabel
5877 app.SouthLabel = uilabel(app.Panel_23);
5878 app.SouthLabel.Tooltip = {'Unit: meters'};
5879 app.SouthLabel.Position = [53 5 37 22];
5880 app.SouthLabel.Text = 'South';
5881
5882 % Create DecayTypeDropDownLabel
5883 app.DecayTypeDropDownLabel = uilabel(app.Panel_23);
5884 app.DecayTypeDropDownLabel.Position = [29 87 66 22];
5885 app.DecayTypeDropDownLabel.Text = 'Decay type';
5886
5887 % Create DecayTypeDropDown
5888 app.DecayTypeDropDown = uidropdown(app.Panel_23);
5889 app.DecayTypeDropDown.Items = {'Very mild', 'Mild', 'Sharp'};
5890 app.DecayTypeDropDown.ValueChangedFcn = createCallbackFcn(app, ⚡
@DecayTypeDropDownValueChanged, true);
5891 app.DecayTypeDropDown.Position = [29 73 72 16];
5892 app.DecayTypeDropDown.Value = 'Sharp';
5893
5894 % Create MaximumCdEditFieldLabel
5895 app.MaximumCdEditFieldLabel = uilabel(app.Panel_23);
5896 app.MaximumCdEditFieldLabel.Visible = 'off';
5897 app.MaximumCdEditFieldLabel.Tooltip = {'Cd = drag coefficient'};
5898 app.MaximumCdEditFieldLabel.Position = [138 87 78 22];
5899 app.MaximumCdEditFieldLabel.Text = 'Maximum Cd';
5900
5901 % Create MaximumCdEditField
5902 app.MaximumCdEditField = uieditfield(app.Panel_23, 'numeric');

```

```

5903         app.MaximumCdEditField.Limits = [0 Inf];
5904         app.MaximumCdEditField.ValueDisplayFormat = '%11.3f';
5905         app.MaximumCdEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@MaximumCdEditFieldValueChanged, true);
5906         app.MaximumCdEditField.FontColor = [0.651 0.651 0.651];
5907         app.MaximumCdEditField.Visible = 'off';
5908         app.MaximumCdEditField.Tooltip = {''};
5909         app.MaximumCdEditField.Position = [138 73 51 16];
5910         app.MaximumCdEditField.Value = 10;
5911
5912         % Create MaximumCspEditFieldLabel
5913         app.MaximumCspEditFieldLabel = uilabel(app.Panel_23);
5914         app.MaximumCspEditFieldLabel.Visible = 'off';
5915         app.MaximumCspEditFieldLabel.Tooltip = {'Csd = maximum diffusion ↵
coefficient'};
5916         app.MaximumCspEditFieldLabel.Position = [255 86 84 22];
5917         app.MaximumCspEditFieldLabel.Text = 'Maximum Csp';
5918
5919         % Create MaximumCspEditField
5920         app.MaximumCspEditField = uieditfield(app.Panel_23, 'numeric');
5921         app.MaximumCspEditField.Limits = [0 Inf];
5922         app.MaximumCspEditField.ValueDisplayFormat = '%11.3f';
5923         app.MaximumCspEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@MaximumCspEditFieldValueChanged, true);
5924         app.MaximumCspEditField.FontColor = [0.651 0.651 0.651];
5925         app.MaximumCspEditField.Visible = 'off';
5926         app.MaximumCspEditField.Tooltip = {''};
5927         app.MaximumCspEditField.Position = [255 73 42 16];
5928         app.MaximumCspEditField.Value = 1;
5929
5930         % Create SpongeBoundaryLabel
5931         app.SpongeBoundaryLabel = uilabel(app.InputTab);
5932         app.SpongeBoundaryLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5933         app.SpongeBoundaryLabel.FontSize = 13;
5934         app.SpongeBoundaryLabel.FontWeight = 'bold';
5935         app.SpongeBoundaryLabel.FontColor = [0.0314 0.3686 0.6];
5936         app.SpongeBoundaryLabel.Tooltip = {'Use sponge layer for wave ↵
damping'};
5937         app.SpongeBoundaryLabel.Position = [22 192 130 21];
5938         app.SpongeBoundaryLabel.Text = '  Sponge Boundary  ';
5939
5940         % Create Panel_24
5941         app.Panel_24 = uipanel(app.InputTab);
5942         app.Panel_24.Position = [10 10 351 39];
5943
5944         % Create ObstacleLoadFile
5945         app.ObstacleLoadFile = uibutton(app.Panel_24, 'push');
5946         app.ObstacleLoadFile.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_19Pushed, true);
5947         app.ObstacleLoadFile.VerticalAlignment = 'top';
5948         app.ObstacleLoadFile.Visible = 'off';
5949         app.ObstacleLoadFile.Tooltip = {''};
5950         app.ObstacleLoadFile.Position = [287 7 19 19];
5951         app.ObstacleLoadFile.Text = '...';
5952
5953         % Create ObstacleFilename
5954         app.ObstacleFilename = uieditfield(app.Panel_24, 'text');
5955         app.ObstacleFilename.Editable = 'off';

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5956         app.ObstacleFilename.FontSize = 11;
5957         app.ObstacleFilename.Visible = 'off';
5958         app.ObstacleFilename.Position = [161 8 123 16];
5959
5960         % Create AddobstaclesCheckBox
5961         app.AddobstaclesCheckBox = uicheckbox(app.Panel_24);
5962         app.AddobstaclesCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@AddobstaclesCheckBoxValueChanged, true);
5963         app.AddobstaclesCheckBox.Tooltip = {'Integrate obstacles with a ↵
file format matching the depth data: 1 represents water, and 0 represents dry ↵
points.'; ''; 'File size must be the same as the input bathymetry.'};
5964         app.AddobstaclesCheckBox.Text = ' Add obstacles';
5965         app.AddobstaclesCheckBox.Position = [52 5 102 22];
5966
5967         % Create ObstaclesLabel
5968         app.ObstaclesLabel = uilabel(app.InputTab);
5969         app.ObstaclesLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5970         app.ObstaclesLabel.FontSize = 13;
5971         app.ObstaclesLabel.FontWeight = 'bold';
5972         app.ObstaclesLabel.FontColor = [0.0314 0.3686 0.6];
5973         app.ObstaclesLabel.Position = [22 37 78 21];
5974         app.ObstaclesLabel.Text = ' Obstacles ';
5975
5976         % Create PhysicsLabel
5977         app.PhysicsLabel = uilabel(app.InputTab);
5978         app.PhysicsLabel.BackgroundColor = [0.9412 0.9412 0.9412];
5979         app.PhysicsLabel.HorizontalAlignment = 'center';
5980         app.PhysicsLabel.FontSize = 13;
5981         app.PhysicsLabel.FontWeight = 'bold';
5982         app.PhysicsLabel.FontColor = [0.0314 0.3686 0.6];
5983         app.PhysicsLabel.Position = [383 627 73 21];
5984         app.PhysicsLabel.Text = ' Physics ';
5985
5986         % Create Panel_26
5987         app.Panel_26 = uipanel(app.InputTab);
5988         app.Panel_26.Position = [371 450 356 60];
5989
5990         % Create ButtonGroup_8
5991         app.ButtonGroup_8 = uibuttongroup(app.Panel_26);
5992         app.ButtonGroup_8.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_8SelectionChanged, true);
5993         app.ButtonGroup_8.BorderType = 'none';
5994         app.ButtonGroup_8.Position = [18 9 159 41];
5995
5996         % Create ConstantCoefficientButton
5997         app.ConstantCoefficientButton = uiradiobutton(app.ButtonGroup_8);
5998         app.ConstantCoefficientButton.Tooltip = {'Set a constant drag ↵
friction value throughout the domain'};
5999         app.ConstantCoefficientButton.Text = ' Constant Coefficient';
6000         app.ConstantCoefficientButton.Position = [16 20 135 22];
6001         app.ConstantCoefficientButton.Value = true;
6002
6003         % Create NonconstantButton
6004         app.NonconstantButton = uiradiobutton(app.ButtonGroup_8);
6005         app.NonconstantButton.Tooltip = {'Assign a non-uniform drag ↵
friction value across the domain.'};
6006         app.NonconstantButton.Text = ' Nonconstant';
6007         app.NonconstantButton.Position = [16 0 94 22];

```



```

6008
6009     % Create FrictionFileEditFieldLabel
6010     app.FrictionFileEditFieldLabel = uilabel(app.Panel_26);
6011     app.FrictionFileEditFieldLabel.HorizontalAlignment = 'right';
6012     app.FrictionFileEditFieldLabel.Visible = 'off';
6013     app.FrictionFileEditFieldLabel.Tooltip = {'Import the friction
file.'; ''; 'Dimension should be the same with the input bathymetry.'};
6014     app.FrictionFileEditFieldLabel.Position = [189 9 25 22];
6015     app.FrictionFileEditFieldLabel.Text = 'File';
6016
6017     % Create FrictionFileEditField
6018     app.FrictionFileEditField = uieditfield(app.Panel_26, 'text');
6019     app.FrictionFileEditField.ValueChangingFcn = createCallbackFcn(
app, @FrictionFileEditFieldValueChanging, true);
6020     app.FrictionFileEditField.Editable = 'off';
6021     app.FrictionFileEditField.FontSize = 11;
6022     app.FrictionFileEditField.Visible = 'off';
6023     app.FrictionFileEditField.Tooltip = {'Dimension should be the same
with the input bathymetry'};
6024     app.FrictionFileEditField.Position = [222 12 84 16];
6025
6026     % Create Button_10
6027     app.Button_10 = uibutton(app.Panel_26, 'push');
6028     app.Button_10.ButtonPushedFcn = createCallbackFcn(app,
@Button_10Pushed, true);
6029     app.Button_10.VerticalAlignment = 'top';
6030     app.Button_10.Visible = 'off';
6031     app.Button_10.Tooltip = {' '};
6032     app.Button_10.Position = [311 11 19 19];
6033     app.Button_10.Text = '...';
6034
6035     % Create ValueEditFieldLabel
6036     app.ValueEditFieldLabel = uilabel(app.Panel_26);
6037     app.ValueEditFieldLabel.HorizontalAlignment = 'right';
6038     app.ValueEditFieldLabel.Tooltip = {'Set to 0 for no friction'};
6039     app.ValueEditFieldLabel.Position = [180 29 34 22];
6040     app.ValueEditFieldLabel.Text = 'Value';
6041
6042     % Create ConstantFrictionValue
6043     app.ConstantFrictionValue = uieditfield(app.Panel_26, 'numeric');
6044     app.ConstantFrictionValue.ValueDisplayFormat = '%8.3f';
6045     app.ConstantFrictionValue.ValueChangedFcn = createCallbackFcn(app,
@ConstantFrictionValueValueChanged, true);
6046     app.ConstantFrictionValue.FontColor = [0.651 0.651 0.651];
6047     app.ConstantFrictionValue.Tooltip = {' '};
6048     app.ConstantFrictionValue.Position = [222 32 41 16];
6049
6050     % Create BottomFrictionLabel
6051     app.BottomFrictionLabel = uilabel(app.InputTab);
6052     app.BottomFrictionLabel.BackgroundColor = [0.9412 0.9412 0.9412];
6053     app.BottomFrictionLabel.FontSize = 13;
6054     app.BottomFrictionLabel.FontWeight = 'bold';
6055     app.BottomFrictionLabel.FontColor = [0.0314 0.3686 0.6];
6056     app.BottomFrictionLabel.Tooltip = {' '};
6057     app.BottomFrictionLabel.Position = [383 500 114 21];
6058     app.BottomFrictionLabel.Text = ' Bottom Friction ';
6059
6060     % Create Panel 28

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6061         app.Panel_28 = uipanel(app.InputTab);
6062         app.Panel_28.Position = [371 108 356 148];
6063
6064         % Create CreatenestedgridboundaryCheckBox
6065         app.CreatenestedgridboundaryCheckBox = uicheckbox(app.Panel_28);
6066         app.CreatenestedgridboundaryCheckBox.ValueChangedFcn = ↵
createCallbackFcn(app, @CreatenestedgridboundaryCheckBoxValueChanged, true);
6067         app.CreatenestedgridboundaryCheckBox.Tooltip = {'Use gauges as ↵
points to define the boundary of the subgrid layer'};
6068         app.CreatenestedgridboundaryCheckBox.Text = 'Create nested grid ↵
boundary';
6069         app.CreatenestedgridboundaryCheckBox.WordWrap = 'on';
6070         app.CreatenestedgridboundaryCheckBox.Position = [33 84 182 23];
6071
6072         % Create Button_7
6073         app.Button_7 = uibutton(app.Panel_28, 'push');
6074         app.Button_7.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_7Pushed2, true);
6075         app.Button_7.Enable = 'off';
6076         app.Button_7.Tooltip = {'Import the file containing a list of ↵
coordinates representing the gauge locations.'; ''; 'Text file format: Tab-↵
delimited, with columns for latitude and longitude.'};
6077         app.Button_7.Position = [230 114 19 19];
6078         app.Button_7.Text = '...';
6079
6080         % Create LoadFileEditField
6081         app.LoadFileEditField = uieditfield(app.Panel_28, 'text');
6082         app.LoadFileEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@LoadFileEditFieldValueChanged, true);
6083         app.LoadFileEditField.ValueChangingFcn = createCallbackFcn(app, ↵
@LoadFileEditFieldValueChanging, true);
6084         app.LoadFileEditField.Editable = 'off';
6085         app.LoadFileEditField.FontSize = 11;
6086         app.LoadFileEditField.Enable = 'off';
6087         app.LoadFileEditField.Placeholder = '.shp or .txt';
6088         app.LoadFileEditField.Position = [152 115 74 16];
6089
6090         % Create ImportgaugeListCheckBox
6091         app.ImportgaugeListCheckBox = uicheckbox(app.Panel_28);
6092         app.ImportgaugeListCheckBox.ValueChangedFcn = createCallbackFcn ↵
(app, @ImportgaugeListCheckBoxValueChanged, true);
6093         app.ImportgaugeListCheckBox.Tooltip = {'Add stations at specific ↵
locations to record wave patterns throughout the simulation'};
6094         app.ImportgaugeListCheckBox.Text = 'Import gauge list';
6095         app.ImportgaugeListCheckBox.Position = [33 112 112 22];
6096
6097         % Create NorthEditField_2
6098         app.NorthEditField_2 = uieditfield(app.Panel_28, 'numeric');
6099         app.NorthEditField_2.ValueDisplayFormat = '%5.4f';
6100         app.NorthEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@NorthEditField_2ValueChanged, true);
6101         app.NorthEditField_2.FontColor = [0.651 0.651 0.651];
6102         app.NorthEditField_2.Enable = 'off';
6103         app.NorthEditField_2.Tooltip = {'North limit'};
6104         app.NorthEditField_2.Position = [147 37 64 16];
6105
6106         % Create EastEditField_2
6107         app.EastEditField_2 = uieditfield(app.Panel_28, 'numeric');

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6108         app.EastEditField_2.ValueDisplayFormat = '%5.4f';
6109         app.EastEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@EastEditField_2ValueChanged, true);
6110         app.EastEditField_2.FontColor = [0.651 0.651 0.651];
6111         app.EastEditField_2.Enable = 'off';
6112         app.EastEditField_2.Tooltip = {'East limit'};
6113         app.EastEditField_2.Position = [217 24 63 16];
6114
6115         % Create SouthEditField_2
6116         app.SouthEditField_2 = uieditfield(app.Panel_28, 'numeric');
6117         app.SouthEditField_2.ValueDisplayFormat = '%5.4f';
6118         app.SouthEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@SouthEditField_2ValueChanged, true);
6119         app.SouthEditField_2.FontColor = [0.651 0.651 0.651];
6120         app.SouthEditField_2.Enable = 'off';
6121         app.SouthEditField_2.Tooltip = {'South limit'};
6122         app.SouthEditField_2.Position = [147 13 64 16];
6123
6124         % Create WestEditField_2
6125         app.WestEditField_2 = uieditfield(app.Panel_28, 'numeric');
6126         app.WestEditField_2.ValueDisplayFormat = '%5.4f';
6127         app.WestEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@WestEditField_2ValueChanged, true);
6128         app.WestEditField_2.FontColor = [0.651 0.651 0.651];
6129         app.WestEditField_2.Enable = 'off';
6130         app.WestEditField_2.Tooltip = {'West limit'};
6131         app.WestEditField_2.Position = [79 24 63 16];
6132
6133         % Create BoundarylimitsButton
6134         app.BoundarylimitsButton = uibutton(app.Panel_28, 'push');
6135         app.BoundarylimitsButton.ButtonPushedFcn = createCallbackFcn(app, ↵
@BoundarylimitsButtonPushed, true);
6136         app.BoundarylimitsButton.Enable = 'off';
6137         app.BoundarylimitsButton.Tooltip = {'Extract coordinate and grid↵
spacing information from a tif file. '};
6138         app.BoundarylimitsButton.Position = [79 58 201 22];
6139         app.BoundarylimitsButton.Text = 'Boundary limits: ';
6140
6141         % Create SpacingEditFieldLabel
6142         app.SpacingEditFieldLabel = uilabel(app.Panel_28);
6143         app.SpacingEditFieldLabel.WordWrap = 'on';
6144         app.SpacingEditFieldLabel.Enable = 'off';
6145         app.SpacingEditFieldLabel.Tooltip = {'Set the gauge spacing'};
6146         app.SpacingEditFieldLabel.Position = [216 84 77 22];
6147         app.SpacingEditFieldLabel.Text = 'Grid spacing: ';
6148
6149         % Create SpacingEditField
6150         app.SpacingEditField = uieditfield(app.Panel_28, 'numeric');
6151         app.SpacingEditField.ValueDisplayFormat = '%5.4f';
6152         app.SpacingEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@SpacingEditFieldValueChanged, true);
6153         app.SpacingEditField.FontColor = [0.651 0.651 0.651];
6154         app.SpacingEditField.Enable = 'off';
6155         app.SpacingEditField.Tooltip = {' '};
6156         app.SpacingEditField.Position = [294 87 48 16];
6157
6158         % Create Friction_____Label_5
6159         app.Friction_____Label_5 = uilabel(app.↵

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InputTab);
6160         app.Friction_____Label_5.↵
BackgroundColor = [0.9412 0.9412 0.9412];
6161         app.Friction_____Label_5.FontSize = 13;
6162         app.Friction_____Label_5.FontWeight = ↵
'bold';
6163         app.Friction_____Label_5.FontColor = ↵
[0.0314 0.3686 0.6];
6164         app.Friction_____Label_5.Position = ↵
[383 246 65 21];
6165         app.Friction_____Label_5.Text = '↵
Gauges  ';
6166
6167         % Create Panel_29
6168         app.Panel_29 = uipanel(app.InputTab);
6169         app.Panel_29.Position = [371 60 356 37];
6170
6171         % Create Button_14
6172         app.Button_14 = uibutton(app.Panel_29, 'push');
6173         app.Button_14.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_14Pushed, true);
6174         app.Button_14.VerticalAlignment = 'top';
6175         app.Button_14.FontSize = 10;
6176         app.Button_14.FontAngle = 'italic';
6177         app.Button_14.Tooltip = {''};
6178         app.Button_14.Position = [288 7 19 18];
6179         app.Button_14.Text = '...';
6180
6181         % Create SavefilestoLabel
6182         app.SavefilestoLabel = uilabel(app.Panel_29);
6183         app.SavefilestoLabel.Tooltip = {'Specify the directory where ↵
'INPUT_FILES', containing all generated files, will be saved.'};
6184         app.SavefilestoLabel.Position = [37 5 74 22];
6185         app.SavefilestoLabel.Text = 'Save files to:';
6186
6187         % Create SavefilestoEditField
6188         app.SavefilestoEditField = uieditfield(app.Panel_29, 'text');
6189         app.SavefilestoEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@SavefilestoEditFieldValueChanged, true);
6190         app.SavefilestoEditField.ValueChangingFcn = createCallbackFcn(app, ↵
@SavefilestoEditFieldValueChanging, true);
6191         app.SavefilestoEditField.Editable = 'off';
6192         app.SavefilestoEditField.FontSize = 11;
6193         app.SavefilestoEditField.FontColor = [0 0 1];
6194         app.SavefilestoEditField.Tooltip = {'Specify the directory where ↵
'OUTPUT_FILES' that contians all the files generated in this app will be saved'};
6195         app.SavefilestoEditField.Placeholder = 'Default: Desktop';
6196         app.SavefilestoEditField.Position = [113 8 170 16];
6197
6198         % Create Friction_____Label_6
6199         app.Friction_____Label_6 = uilabel(app.↵
InputTab);
6200         app.Friction_____Label_6.↵
BackgroundColor = [0.9412 0.9412 0.9412];
6201         app.Friction_____Label_6.FontSize = 13;
6202         app.Friction_____Label_6.FontWeight = ↵
'bold';
6203         app.Friction_____Label_6.FontColor = ↵

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[0.0314 0.3686 0.6];
6204         app.Friction_____Label_6.Position = 
[383 87 73 21];
6205         app.Friction_____Label_6.Text = '
Directory  ';
6206
6207         % Create DepthLabel
6208         app.DepthLabel = uilabel(app.InputTab);
6209         app.DepthLabel.BackgroundColor = [0.9412 0.9412 0.9412];
6210         app.DepthLabel.FontSize = 13;
6211         app.DepthLabel.FontWeight = 'bold';
6212         app.DepthLabel.FontColor = [0.0314 0.3686 0.6];
6213         app.DepthLabel.Position = [21 574 52 21];
6214         app.DepthLabel.Text = '  Depth  ';
6215
6216         % Create DimensionLabel
6217         app.DimensionLabel = uilabel(app.InputTab);
6218         app.DimensionLabel.BackgroundColor = [0.9412 0.9412 0.9412];
6219         app.DimensionLabel.FontSize = 13;
6220         app.DimensionLabel.FontWeight = 'bold';
6221         app.DimensionLabel.FontColor = [0.0314 0.3686 0.6];
6222         app.DimensionLabel.Position = [21 479 86 21];
6223         app.DimensionLabel.Text = '  Dimension  ';
6224
6225         % Create Panel_30
6226         app.Panel_30 = uipanel(app.InputTab);
6227         app.Panel_30.Position = [371 329 356 55];
6228
6229         % Create ButtonGroup_23
6230         app.ButtonGroup_23 = uibuttongroup(app.Panel_30);
6231         app.ButtonGroup_23.SelectionChangedFcn = createCallbackFcn(app, 
@ButtonGroup_23SelectionChanged, true);
6232         app.ButtonGroup_23.BorderType = 'none';
6233         app.ButtonGroup_23.Position = [26 2 155 42];
6234
6235         % Create ShockwavecapturingButton
6236         app.ShockwavecapturingButton = uiradiobutton(app.ButtonGroup_23);
6237         app.ShockwavecapturingButton.Enable = 'off';
6238         app.ShockwavecapturingButton.Text = ' Shock wave capturing';
6239         app.ShockwavecapturingButton.Position = [7 22 145 22];
6240         app.ShockwavecapturingButton.Value = true;
6241
6242         % Create ViscositybreakingButton
6243         app.ViscositybreakingButton = uiradiobutton(app.ButtonGroup_23);
6244         app.ViscositybreakingButton.Enable = 'off';
6245         app.ViscositybreakingButton.Text = ' Viscosity breaking';
6246         app.ViscositybreakingButton.Position = [7 4 122 22];
6247
6248         % Create C1EditField
6249         app.C1EditField = uieditfield(app.Panel_30, 'numeric');
6250         app.C1EditField.Limits = [0 Inf];
6251         app.C1EditField.ValueDisplayFormat = '%.2f';
6252         app.C1EditField.ValueChangedFcn = createCallbackFcn(app, 
@C1EditFieldValueChanged, true);
6253         app.C1EditField.FontColor = [0.651 0.651 0.651];
6254         app.C1EditField.Visible = 'off';
6255         app.C1EditField.Tooltip = { ''};
6256         app.C1EditField.Position = [222 27 41 16];

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6257         app.C1EditField.Value = 0.65;
6258
6259         % Create C1EditFieldLabel
6260         app.C1EditFieldLabel = uilabel(app.Panel_30);
6261         app.C1EditFieldLabel.HorizontalAlignment = 'right';
6262         app.C1EditFieldLabel.Visible = 'off';
6263         app.C1EditFieldLabel.Tooltip = {'Parameter value from Kennedy et al. (2000)'};
6264         app.C1EditFieldLabel.Position = [192 23 25 22];
6265         app.C1EditFieldLabel.Text = 'C1';
6266
6267         % Create C2EditFieldLabel
6268         app.C2EditFieldLabel = uilabel(app.Panel_30);
6269         app.C2EditFieldLabel.HorizontalAlignment = 'right';
6270         app.C2EditFieldLabel.Visible = 'off';
6271         app.C2EditFieldLabel.Tooltip = {'Parameter value from Kennedy et al. (2000)'};
6272         app.C2EditFieldLabel.Position = [192 4 25 22];
6273         app.C2EditFieldLabel.Text = 'C2';
6274
6275         % Create C2EditField
6276         app.C2EditField = uieditfield(app.Panel_30, 'numeric');
6277         app.C2EditField.Limits = [0 Inf];
6278         app.C2EditField.ValueDisplayFormat = '%.2f';
6279         app.C2EditField.ValueChangedFcn = createCallbackFcn(app, @C2EditFieldValueChanged, true);
6280         app.C2EditField.FontColor = [0.651 0.651 0.651];
6281         app.C2EditField.Visible = 'off';
6282         app.C2EditField.Tooltip = {' '};
6283         app.C2EditField.Position = [222 8 41 16];
6284         app.C2EditField.Value = 0.35;
6285
6286         % Create Friction_____Label_7
6287         app.Friction_____Label_7 = uilabel(app.InputTab);
6288         app.Friction_____Label_7.BackgroundColor = [0.9412 0.9412 0.9412];
6289         app.Friction_____Label_7.FontSize = 13;
6290         app.Friction_____Label_7.FontWeight = 'bold';
6291         app.Friction_____Label_7.FontColor = [0.0314 0.3686 0.6];
6292         app.Friction_____Label_7.Position = [383 373 109 22];
6293         app.Friction_____Label_7.Text = ' Wave Breaking ' ;
6294
6295         % Create GenerateinputtxtButton
6296         app.GenerateinputtxtButton = uibutton(app.InputTab, 'push');
6297         app.GenerateinputtxtButton.ButtonPushedFcn = createCallbackFcn(app, @GenerateinputtxtButtonPushed, true);
6298         app.GenerateinputtxtButton.BackgroundColor = [0.9216 0.9216 0.9216];
6299         app.GenerateinputtxtButton.FontSize = 15;
6300         app.GenerateinputtxtButton.FontWeight = 'bold';
6301         app.GenerateinputtxtButton.Position = [370 10 357 39];
6302         app.GenerateinputtxtButton.Text = 'Generate input.txt';
6303

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```

6304         % Create Panel_33
6305         app.Panel_33 = uipanel(app.InputTab);
6306         app.Panel_33.Position = [371 268 356 50];
6307
6308         % Create UseSmagorinskyCheckBox
6309         app.UseSmagorinskyCheckBox = uicheckbox(app.Panel_33);
6310         app.UseSmagorinskyCheckBox.ValueChangedFcn = createCallbackFcn(
6311 app, @UseSmagorinskyCheckBoxValueChanged2, true);
6312         app.UseSmagorinskyCheckBox.Tooltip = {'Apply a Smagorinsky-like
method for subgrid turbulent mixing'};
6313         app.UseSmagorinskyCheckBox.Text = ' Smagorinsky mixing';
6314         app.UseSmagorinskyCheckBox.Position = [33 12 134 22];
6315
6316         % Create TimeintervalsecEditFieldLabel
6317         app.TimeintervalsecEditFieldLabel = uilabel(app.Panel_33);
6318         app.TimeintervalsecEditFieldLabel.Visible = 'off';
6319         app.TimeintervalsecEditFieldLabel.Tooltip = {'Set time interval
for calculating values'; ''; 'Unit: seconds'};
6320         app.TimeintervalsecEditFieldLabel.Position = [196 23 74 22];
6321         app.TimeintervalsecEditFieldLabel.Text = 'Time interval';
6322
6323         % Create TimeintervalsecEditField
6324         app.TimeintervalsecEditField = uieditfield(app.Panel_33,
'numeric');
6325         app.TimeintervalsecEditField.Limits = [0 Inf];
6326         app.TimeintervalsecEditField.FontColor = [0.651 0.651 0.651];
6327         app.TimeintervalsecEditField.Visible = 'off';
6328         app.TimeintervalsecEditField.Position = [273 26 41 16];
6329         app.TimeintervalsecEditField.Value = 20;
6330
6331         % Create SteadyTimeEditFieldLabel
6332         app.SteadyTimeEditFieldLabel = uilabel(app.Panel_33);
6333         app.SteadyTimeEditFieldLabel.Visible = 'off';
6334         app.SteadyTimeEditFieldLabel.Tooltip = {'Set the initial time for
calculating mean values'; ''; 'Unit: seconds'};
6335         app.SteadyTimeEditFieldLabel.Position = [196 3 70 22];
6336         app.SteadyTimeEditFieldLabel.Text = 'Steady time';
6337
6338         % Create SteadyTimeEditField
6339         app.SteadyTimeEditField = uieditfield(app.Panel_33, 'numeric');
6340         app.SteadyTimeEditField.Limits = [0 Inf];
6341         app.SteadyTimeEditField.FontColor = [0.651 0.651 0.651];
6342         app.SteadyTimeEditField.Visible = 'off';
6343         app.SteadyTimeEditField.Position = [273 6 41 16];
6344
6345         % Create Friction_____Label_10
6346         app.Friction_____Label_10 = uilabel(
6347 app.InputTab);
6348         app.Friction_____Label_10.
BackgroundColor = [0.9412 0.9412 0.9412];
6349         app.Friction_____Label_10.FontSize =
13;
6350         app.Friction_____Label_10.FontWeight =
'bold';
6351         app.Friction_____Label_10.FontColor =
[0.0314 0.3686 0.6];
6352         app.Friction_____Label_10.Position =
[383 306 116 22];

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```

6351         app.Friction_____Label_10.Text = '↵
Turbulent Mixing';
6352
6353         % Create InitialConditionTab
6354         app.InitialConditionTab = uitab(app.TabGroup);
6355         app.InitialConditionTab.Title = 'Initial Condition';
6356         app.InitialConditionTab.BackgroundColor = [0.9412 0.9412 0.9412];
6357         app.InitialConditionTab.Interruptible = 'off';
6358
6359         % Create TabGroup2
6360         app.TabGroup2 = uitabgroup(app.InitialConditionTab);
6361         app.TabGroup2.TabLocation = 'left';
6362         app.TabGroup2.SelectionChangedFcn = createCallbackFcn(app, ↵
@TabGroup2SelectionChanged, true);
6363         app.TabGroup2.Position = [-69 84 806 515];
6364
6365         % Create Tab_5
6366         app.Tab_5 = uitab(app.TabGroup2);
6367
6368         % Create Tab
6369         app.Tab = uitab(app.TabGroup2);
6370
6371         % Create Label_9
6372         app.Label_9 = uilabel(app.Tab);
6373         app.Label_9.BackgroundColor = [0.651 0.651 0.651];
6374         app.Label_9.Position = [1 495 738 22];
6375         app.Label_9.Text = '';
6376
6377         % Create ImportUVZfilesLabel
6378         app.ImportUVZfilesLabel = uilabel(app.Tab);
6379         app.ImportUVZfilesLabel.FontSize = 14;
6380         app.ImportUVZfilesLabel.FontWeight = 'bold';
6381         app.ImportUVZfilesLabel.FontColor = [1 1 1];
6382         app.ImportUVZfilesLabel.Position = [24 495 131 22];
6383         app.ImportUVZfilesLabel.Text = 'Import U, V, Z files';
6384
6385         % Create Button_32
6386         app.Button_32 = uibutton(app.Tab, 'push');
6387         app.Button_32.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_4Pushed, true);
6388         app.Button_32.VerticalAlignment = 'top';
6389         app.Button_32.Tooltip = {'Import the Z file. '; ''; 'Accepted↵
format: .txt'; ''; 'If you only have .tif file, convert this file to .txt in↵
''Input'' tab > Depth > Import Data'; ''; 'The values in the .txt file should↵
include decimal places.'};
6390         app.Button_32.Position = [523 371 22 22];
6391         app.Button_32.Text = '...';
6392
6393         % Create Button_34
6394         app.Button_34 = uibutton(app.Tab, 'push');
6395         app.Button_34.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_5Pushed, true);
6396         app.Button_34.Tooltip = {'Import the U vector file. '; '';↵
'Accepted format: .txt'; ''; 'If you only have .tif file, convert this file to .txt↵
in ''Input'' tab > Depth > Import Data'; ''; 'The values in the .txt file should↵
include decimal places.'};
6397         app.Button_34.Position = [523 339 22 22];
6398         app.Button 34.Text = '...';

```



```

6399
6400         % Create SurfaceHeightEditFieldLabel
6401         app.SurfaceHeightEditFieldLabel = uilabel(app.Tab);
6402         app.SurfaceHeightEditFieldLabel.HorizontalAlignment = 'right';
6403         app.SurfaceHeightEditFieldLabel.Tooltip = {'Import file containing
the sea surface displacement matrix.'; ''; 'The values in the .txt file should
include decimal places.'};
6404         app.SurfaceHeightEditFieldLabel.Position = [196 371 85 22];
6405         app.SurfaceHeightEditFieldLabel.Text = 'Surface Height';
6406
6407         % Create SurfaceHeightEditField
6408         app.SurfaceHeightEditField = uieditfield(app.Tab, 'text');
6409         app.SurfaceHeightEditField.ValueChangingFcn = createCallbackFcn
(app, @SurfaceHeightEditFieldValueChanging, true);
6410         app.SurfaceHeightEditField.Editable = 'off';
6411         app.SurfaceHeightEditField.FontSize = 11;
6412         app.SurfaceHeightEditField.Tooltip = {'';};
6413         app.SurfaceHeightEditField.Position = [307 371 207 22];
6414
6415         % Create UVelocityEditFieldLabel
6416         app.UVelocityEditFieldLabel = uilabel(app.Tab);
6417         app.UVelocityEditFieldLabel.HorizontalAlignment = 'right';
6418         app.UVelocityEditFieldLabel.Tooltip = {'Import file with x-
direction velocity matrix.'; ''; 'The values in the .txt file should include
decimal places.'};
6419         app.UVelocityEditFieldLabel.Position = [196 339 59 22];
6420         app.UVelocityEditFieldLabel.Text = 'U Velocity';
6421
6422         % Create UVelocityEditField
6423         app.UVelocityEditField = uieditfield(app.Tab, 'text');
6424         app.UVelocityEditField.ValueChangingFcn = createCallbackFcn(app,
@UVelocityEditFieldValueChanging, true);
6425         app.UVelocityEditField.Editable = 'off';
6426         app.UVelocityEditField.FontSize = 11;
6427         app.UVelocityEditField.Tooltip = {'';};
6428         app.UVelocityEditField.Position = [307 339 207 22];
6429
6430         % Create VVelocityEditFieldLabel
6431         app.VVelocityEditFieldLabel = uilabel(app.Tab);
6432         app.VVelocityEditFieldLabel.HorizontalAlignment = 'right';
6433         app.VVelocityEditFieldLabel.Tooltip = {'Import file with y-
direction velocity matrix.'; ''; 'The values in the .txt file should include
decimal places.'};
6434         app.VVelocityEditFieldLabel.Position = [197 307 58 22];
6435         app.VVelocityEditFieldLabel.Text = 'V Velocity';
6436
6437         % Create VVelocityEditField
6438         app.VVelocityEditField = uieditfield(app.Tab, 'text');
6439         app.VVelocityEditField.ValueChangingFcn = createCallbackFcn(app,
@VVelocityEditFieldValueChanging, true);
6440         app.VVelocityEditField.Editable = 'off';
6441         app.VVelocityEditField.FontSize = 11;
6442         app.VVelocityEditField.Tooltip = {'';};
6443         app.VVelocityEditField.Position = [307 306 207 22];
6444
6445         % Create Button_33
6446         app.Button_33 = uibutton(app.Tab, 'push');
6447         app.Button_33.ButtonPushedFcn = createCallbackFcn(app,

```



```

@Button_6Pushed, true);
6448         app.Button_33.Tooltip = {'Import the V vector file. '; ''};
'Accepted format: .txt'; ''; 'If you only have .tif file, convert this file to .txt'
in 'Input' tab > Depth > Import Data'; ''; 'The values in the .txt file should
include decimal places.'};
6449         app.Button_33.Position = [523 306 22 22];
6450         app.Button_33.Text = '...';
6451
6452         % Create Tab_2
6453         app.Tab_2 = uitab(app.TabGroup2);
6454
6455         % Create Label_10
6456         app.Label_10 = uilabel(app.Tab_2);
6457         app.Label_10.BackgroundColor = [0.651 0.651 0.651];
6458         app.Label_10.Position = [1 495 751 22];
6459         app.Label_10.Text = '';
6460
6461         % Create WavemakerParametersLabel_2
6462         app.WavemakerParametersLabel_2 = uilabel(app.Tab_2);
6463         app.WavemakerParametersLabel_2.FontSize = 14;
6464         app.WavemakerParametersLabel_2.FontWeight = 'bold';
6465         app.WavemakerParametersLabel_2.FontColor = [1 1 1];
6466         app.WavemakerParametersLabel_2.Position = [24 495 165 22];
6467         app.WavemakerParametersLabel_2.Text = 'Wavemaker Parameters';
6468
6469         % Create Button_20
6470         app.Button_20 = uibutton(app.Tab_2, 'push');
6471         app.Button_20.ButtonPushedFcn = createCallbackFcn(app,
@Button_20Pushed, true);
6472         app.Button_20.Enable = 'off';
6473         app.Button_20.Tooltip = {''};
6474         app.Button_20.Position = [404 77 22 18];
6475         app.Button_20.Text = '...';
6476
6477         % Create WavemakertypeDropDownLabel
6478         app.WavemakertypeDropDownLabel = uilabel(app.Tab_2);
6479         app.WavemakertypeDropDownLabel.HorizontalAlignment = 'right';
6480         app.WavemakertypeDropDownLabel.Position = [161 459 95 22];
6481         app.WavemakertypeDropDownLabel.Text = 'Wavemaker type';
6482
6483         % Create WavemakertypeDropDown
6484         app.WavemakertypeDropDown = uidropdown(app.Tab_2);
6485         app.WavemakertypeDropDown.Items = {'See options', 'INI_REC',
'INI_GAUSSIAN', 'LEF_SOL', 'INI_SOL', 'JON_1D', 'JON_2D', 'WK_IRR', 'WK_REG',
'WK_TIME_SERIES', 'WK_DATA2D', 'TMA_1D'};
6486         app.WavemakertypeDropDown.DropDownOpeningFcn = createCallbackFcn
(app, @WavemakertypeDropDownOpening, true);
6487         app.WavemakertypeDropDown.ValueChangedFcn = createCallbackFcn(app,
@WavemakertypeDropDownValueChanged, true);
6488         app.WavemakertypeDropDown.Position = [310 460 134 22];
6489         app.WavemakertypeDropDown.Value = 'See options';
6490
6491         % Create AmplitudemEditFieldLabel
6492         app.AmplitudemEditFieldLabel = uilabel(app.Tab_2);
6493         app.AmplitudemEditFieldLabel.HorizontalAlignment = 'right';
6494         app.AmplitudemEditFieldLabel.FontColor = [0.149 0.149 0.149];
6495         app.AmplitudemEditFieldLabel.Tooltip = {'Height of the initial sea
surface displacement'};

```

```

6496     app.AmplitudemEditFieldLabel.Position = [161 427 80 22];
6497     app.AmplitudemEditFieldLabel.Text = 'Amplitude (m)';
6498
6499     % Create AmplitudemEditField
6500     app.AmplitudemEditField = uieditfield(app.Tab_2, 'numeric');
6501     app.AmplitudemEditField.Limits = [0 Inf];
6502     app.AmplitudemEditField.ValueDisplayFormat = '%11.2f';
6503     app.AmplitudemEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@AmplitudemEditFieldValueChanged, true);
6504     app.AmplitudemEditField.FontColor = [0.8 0.8 0.8];
6505     app.AmplitudemEditField.Enable = 'off';
6506     app.AmplitudemEditField.Tooltip = {' '};
6507     app.AmplitudemEditField.Position = [310 429 70 18];
6508
6509     % Create WaterdepthmEditFieldLabel
6510     app.WaterdepthmEditFieldLabel = uilabel(app.Tab_2);
6511     app.WaterdepthmEditFieldLabel.HorizontalAlignment = 'right';
6512     app.WaterdepthmEditFieldLabel.FontColor = [0.149 0.149 0.149];
6513     app.WaterdepthmEditFieldLabel.Tooltip = {'Water depth at the ↵
location of the wavemaker'};
6514     app.WaterdepthmEditFieldLabel.Position = [161 395 91 22];
6515     app.WaterdepthmEditFieldLabel.Text = 'Water depth (m)';
6516
6517     % Create WaterdepthmEditField
6518     app.WaterdepthmEditField = uieditfield(app.Tab_2, 'numeric');
6519     app.WaterdepthmEditField.Limits = [0 Inf];
6520     app.WaterdepthmEditField.ValueDisplayFormat = '%11.2f';
6521     app.WaterdepthmEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@WaterdepthmEditFieldValueChanged, true);
6522     app.WaterdepthmEditField.FontColor = [0.8 0.8 0.8];
6523     app.WaterdepthmEditField.Enable = 'off';
6524     app.WaterdepthmEditField.Tooltip = {' '};
6525     app.WaterdepthmEditField.Position = [310 397 70 18];
6526
6527     % Create LagtimesecEditFieldLabel
6528     app.LagtimesecEditFieldLabel = uilabel(app.Tab_2);
6529     app.LagtimesecEditFieldLabel.HorizontalAlignment = 'right';
6530     app.LagtimesecEditFieldLabel.FontColor = [0.149 0.149 0.149];
6531     app.LagtimesecEditFieldLabel.Tooltip = {'Lag time associated with ↵
the solitary wave generated at the left boundary'};
6532     app.LagtimesecEditFieldLabel.Position = [161 139 80 22];
6533     app.LagtimesecEditFieldLabel.Text = 'Lag time (sec)';
6534
6535     % Create LagtimesecEditField
6536     app.LagtimesecEditField = uieditfield(app.Tab_2, 'numeric');
6537     app.LagtimesecEditField.Limits = [0 Inf];
6538     app.LagtimesecEditField.ValueDisplayFormat = '%.2f';
6539     app.LagtimesecEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@LagtimesecEditFieldValueChanged, true);
6540     app.LagtimesecEditField.FontColor = [0.8 0.8 0.8];
6541     app.LagtimesecEditField.Enable = 'off';
6542     app.LagtimesecEditField.Tooltip = {' '};
6543     app.LagtimesecEditField.Position = [310 141 70 18];
6544
6545     % Create XcoordinatemEditFieldLabel
6546     app.XcoordinatemEditFieldLabel = uilabel(app.Tab_2);
6547     app.XcoordinatemEditFieldLabel.HorizontalAlignment = 'right';
6548     app.XcoordinatemEditFieldLabel.FontColor = [0.149 0.149 0.149];

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6549         app.XcoordinatemEditFieldLabel.Tooltip = {'Coordinate in X↵
direction.'; ''; 'For INI_REC, it refers to the location of the center of the↵
rectangular hump'};
6550         app.XcoordinatemEditFieldLabel.Position = [161 363 93 22];
6551         app.XcoordinatemEditFieldLabel.Text = 'X coordinate (m)';
6552
6553         % Create XcoordinatemEditField
6554         app.XcoordinatemEditField = uieditfield(app.Tab_2, 'numeric');
6555         app.XcoordinatemEditField.ValueDisplayFormat = '%.4f';
6556         app.XcoordinatemEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@XcoordinatemEditFieldValueChanged, true);
6557         app.XcoordinatemEditField.FontColor = [0.8 0.8 0.8];
6558         app.XcoordinatemEditField.Enable = 'off';
6559         app.XcoordinatemEditField.Tooltip = {'';
6560         app.XcoordinatemEditField.Position = [310 365 70 18];
6561
6562         % Create YcoordinatemEditFieldLabel
6563         app.YcoordinatemEditFieldLabel = uilabel(app.Tab_2);
6564         app.YcoordinatemEditFieldLabel.HorizontalAlignment = 'right';
6565         app.YcoordinatemEditFieldLabel.FontColor = [0.149 0.149 0.149];
6566         app.YcoordinatemEditFieldLabel.Tooltip = {'Coordinate in Y↵
direction'; ''; 'For INI_REC, it refers to the location of the center of the↵
rectangular hump'};
6567         app.YcoordinatemEditFieldLabel.Position = [161 331 94 22];
6568         app.YcoordinatemEditFieldLabel.Text = 'Y coordinate (m)';
6569
6570         % Create YcoordinatemEditField
6571         app.YcoordinatemEditField = uieditfield(app.Tab_2, 'numeric');
6572         app.YcoordinatemEditField.ValueDisplayFormat = '%.4f';
6573         app.YcoordinatemEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@YcoordinatemEditFieldValueChanged, true);
6574         app.YcoordinatemEditField.FontColor = [0.8 0.8 0.8];
6575         app.YcoordinatemEditField.Enable = 'off';
6576         app.YcoordinatemEditField.Tooltip = {'';
6577         app.YcoordinatemEditField.Position = [310 333 70 18];
6578
6579         % Create WidthmEditFieldLabel
6580         app.WidthmEditFieldLabel = uilabel(app.Tab_2);
6581         app.WidthmEditFieldLabel.HorizontalAlignment = 'right';
6582         app.WidthmEditFieldLabel.FontColor = [0.149 0.149 0.149];
6583         app.WidthmEditFieldLabel.Tooltip = {'For INI_REC or INI_GAUSSIAN,↵
it refers to the width of the rectangular hump '; ''; 'For WK_REG, it is the width↵
in Y direction'};
6584         app.WidthmEditFieldLabel.Position = [161 299 57 22];
6585         app.WidthmEditFieldLabel.Text = 'Width (m)';
6586
6587         % Create WidthmEditField
6588         app.WidthmEditField = uieditfield(app.Tab_2, 'numeric');
6589         app.WidthmEditField.ValueDisplayFormat = '%.2f';
6590         app.WidthmEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@WidthmEditFieldValueChanged, true);
6591         app.WidthmEditField.FontColor = [0.8 0.8 0.8];
6592         app.WidthmEditField.Enable = 'off';
6593         app.WidthmEditField.Tooltip = {'';
6594         app.WidthmEditField.Position = [310 301 70 18];
6595
6596         % Create TimerampsecEditFieldLabel
6597         app.TimerampsecEditFieldLabel = uilabel(app.Tab 2);

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6598         app.TimerampsecEditFieldLabel.HorizontalAlignment = 'right';
6599         app.TimerampsecEditFieldLabel.FontColor = [0.149 0.149 0.149];
6600         app.TimerampsecEditFieldLabel.Tooltip = {'The period during which wave height gradually increases from zero to its maximum'};
6601         app.TimerampsecEditFieldLabel.Position = [161 203 91 22];
6602         app.TimerampsecEditFieldLabel.Text = 'Time ramp (sec)';
6603
6604         % Create TimerampsecEditField
6605         app.TimerampsecEditField = uicontrol(app.Tab_2, 'numeric');
6606         app.TimerampsecEditField.Limits = [0 Inf];
6607         app.TimerampsecEditField.ValueDisplayFormat = '%.2f';
6608         app.TimerampsecEditField.ValueChangedFcn = createCallbackFcn(app, @TimerampsecEditFieldValueChanged, true);
6609         app.TimerampsecEditField.FontColor = [0.8 0.8 0.8];
6610         app.TimerampsecEditField.Enable = 'off';
6611         app.TimerampsecEditField.Tooltip = {''};
6612         app.TimerampsecEditField.Position = [310 205 70 18];
6613
6614         % Create WidthDeltaEditFieldLabel
6615         app.WidthDeltaEditFieldLabel = uicontrol(app.Tab_2);
6616         app.WidthDeltaEditFieldLabel.HorizontalAlignment = 'right';
6617         app.WidthDeltaEditFieldLabel.FontColor = [0.149 0.149 0.149];
6618         app.WidthDeltaEditFieldLabel.Tooltip = {'Delta width parameter'; ''};
6619         app.WidthDeltaEditFieldLabel.Position = [161 267 68 22];
6620         app.WidthDeltaEditFieldLabel.Text = 'Width Delta';
6621
6622         % Create WidthDeltaEditField
6623         app.WidthDeltaEditField = uicontrol(app.Tab_2, 'numeric');
6624         app.WidthDeltaEditField.ValueDisplayFormat = '%.2f';
6625         app.WidthDeltaEditField.ValueChangedFcn = createCallbackFcn(app, @WidthDeltaEditFieldValueChanged, true);
6626         app.WidthDeltaEditField.FontColor = [0.8 0.8 0.8];
6627         app.WidthDeltaEditField.Enable = 'off';
6628         app.WidthDeltaEditField.Tooltip = {''};
6629         app.WidthDeltaEditField.Position = [310 269 70 18];
6630
6631         % Create PeriodsecEditFieldLabel
6632         app.PeriodsecEditFieldLabel = uicontrol(app.Tab_2);
6633         app.PeriodsecEditFieldLabel.HorizontalAlignment = 'right';
6634         app.PeriodsecEditFieldLabel.FontColor = [0.149 0.149 0.149];
6635         app.PeriodsecEditFieldLabel.Tooltip = {'Time period of the regular wave'};
6636         app.PeriodsecEditFieldLabel.Position = [161 171 69 22];
6637         app.PeriodsecEditFieldLabel.Text = 'Period (sec)';
6638
6639         % Create PeriodsecEditField
6640         app.PeriodsecEditField = uicontrol(app.Tab_2, 'numeric');
6641         app.PeriodsecEditField.Limits = [0 Inf];
6642         app.PeriodsecEditField.ValueDisplayFormat = '%.2f';
6643         app.PeriodsecEditField.ValueChangedFcn = createCallbackFcn(app, @PeriodsecEditFieldValueChanged, true);
6644         app.PeriodsecEditField.FontColor = [0.8 0.8 0.8];
6645         app.PeriodsecEditField.Enable = 'off';
6646         app.PeriodsecEditField.Tooltip = {''};
6647         app.PeriodsecEditField.Position = [310 173 70 18];
6648
6649         % Create ThetadeegreesEditFieldLabel

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6650     app.ThetadegreesEditFieldLabel = uilabel(app.Tab_2);
6651     app.ThetadegreesEditFieldLabel.HorizontalAlignment = 'right';
6652     app.ThetadegreesEditFieldLabel.FontColor = [0.149 0.149 0.149];
6653     app.ThetadegreesEditFieldLabel.Tooltip = {'Direction of regular
wave'};
6654     app.ThetadegreesEditFieldLabel.Position = [161 235 88 22];
6655     app.ThetadegreesEditFieldLabel.Text = 'Theta (degrees)';
6656
6657     % Create ThetadegreesEditField
6658     app.ThetadegreesEditField = uieditfield(app.Tab_2, 'numeric');
6659     app.ThetadegreesEditField.ValueDisplayFormat = '%.2f';
6660     app.ThetadegreesEditField.ValueChangedFcn = createCallbackFcn(app,
@ThetadegreesEditFieldValueChanged, true);
6661     app.ThetadegreesEditField.FontColor = [0.8 0.8 0.8];
6662     app.ThetadegreesEditField.Enable = 'off';
6663     app.ThetadegreesEditField.Tooltip = {' '};
6664     app.ThetadegreesEditField.Position = [310 237 70 18];
6665
6666     % Create WavecomponentfileEditFieldLabel
6667     app.WavecomponentfileEditFieldLabel = uilabel(app.Tab_2);
6668     app.WavecomponentfileEditFieldLabel.HorizontalAlignment = 'right';
6669     app.WavecomponentfileEditFieldLabel.FontColor = [0.149 0.149
0.149];
6670     app.WavecomponentfileEditFieldLabel.Position = [161 75 118 22];
6671     app.WavecomponentfileEditFieldLabel.Text = 'Wave component file';
6672
6673     % Create WavecomponentfileEditField
6674     app.WavecomponentfileEditField = uieditfield(app.Tab_2, 'text');
6675     app.WavecomponentfileEditField.ValueChangedFcn = createCallbackFcn
(app, @WavecomponentfileEditFieldValueChanged, true);
6676     app.WavecomponentfileEditField.Editable = 'off';
6677     app.WavecomponentfileEditField.FontColor = [0.8 0.8 0.8];
6678     app.WavecomponentfileEditField.Enable = 'off';
6679     app.WavecomponentfileEditField.Tooltip = {'Import the file that
contains the wave component'};
6680     app.WavecomponentfileEditField.Position = [311 77 89 18];
6681
6682     % Create WavecomponentcountEditFieldLabel
6683     app.WavecomponentcountEditFieldLabel = uilabel(app.Tab_2);
6684     app.WavecomponentcountEditFieldLabel.HorizontalAlignment =
'right';
6685     app.WavecomponentcountEditFieldLabel.FontColor = [0.149 0.149
0.149];
6686     app.WavecomponentcountEditFieldLabel.Tooltip = {'Number of the
wave components'};
6687     app.WavecomponentcountEditFieldLabel.Position = [161 107 133 22];
6688     app.WavecomponentcountEditFieldLabel.Text = 'Wave component
count';
6689
6690     % Create WaveCompCountEditField
6691     app.WaveCompCountEditField = uieditfield(app.Tab_2, 'numeric');
6692     app.WaveCompCountEditField.ValueDisplayFormat = '%.2f';
6693     app.WaveCompCountEditField.FontColor = [0.8 0.8 0.8];
6694     app.WaveCompCountEditField.Enable = 'off';
6695     app.WaveCompCountEditField.Tooltip = {' '};
6696     app.WaveCompCountEditField.Position = [311 109 70 18];
6697
6698     % Create FrequencyLabel

```

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6699     app.FrequencyLabel = uilabel(app.Tab_2);
6700     app.FrequencyLabel.Position = [166 13 62 22];
6701     app.FrequencyLabel.Text = 'Frequency';
6702
6703     % Create FrequencyPeakLabel
6704     app.FrequencyPeakLabel = uilabel(app.Tab_2);
6705     app.FrequencyPeakLabel.HorizontalAlignment = 'right';
6706     app.FrequencyPeakLabel.Position = [550 13 32 22];
6707     app.FrequencyPeakLabel.Text = 'Peak';
6708
6709     % Create PeakEditField
6710     app.PeakEditField = uieditfield(app.Tab_2, 'numeric');
6711     app.PeakEditField.ValueDisplayFormat = '%11.2f';
6712     app.PeakEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@PeakEditFieldValueChanged, true);
6713     app.PeakEditField.FontColor = [0.8 0.8 0.8];
6714     app.PeakEditField.Enable = 'off';
6715     app.PeakEditField.Position = [588 15 48 18];
6716
6717     % Create MinimumEditFieldLabel
6718     app.MinimumEditFieldLabel = uilabel(app.Tab_2);
6719     app.MinimumEditFieldLabel.HorizontalAlignment = 'right';
6720     app.MinimumEditFieldLabel.Position = [307 13 55 22];
6721     app.MinimumEditFieldLabel.Text = 'Minimum';
6722
6723     % Create MinimumEditField
6724     app.MinimumEditField = uieditfield(app.Tab_2, 'numeric');
6725     app.MinimumEditField.ValueDisplayFormat = '%11.2f';
6726     app.MinimumEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@MinimumEditFieldValueChanged, true);
6727     app.MinimumEditField.FontColor = [0.8 0.8 0.8];
6728     app.MinimumEditField.Enable = 'off';
6729     app.MinimumEditField.Position = [369 15 48 18];
6730
6731     % Create MaximumEditFieldLabel
6732     app.MaximumEditFieldLabel = uilabel(app.Tab_2);
6733     app.MaximumEditFieldLabel.HorizontalAlignment = 'right';
6734     app.MaximumEditFieldLabel.Position = [427 13 58 22];
6735     app.MaximumEditFieldLabel.Text = 'Maximum';
6736
6737     % Create MaximumEditField
6738     app.MaximumEditField = uieditfield(app.Tab_2, 'numeric');
6739     app.MaximumEditField.ValueDisplayFormat = '%11.2f';
6740     app.MaximumEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@MaximumEditFieldValueChanged, true);
6741     app.MaximumEditField.FontColor = [0.8 0.8 0.8];
6742     app.MaximumEditField.Enable = 'off';
6743     app.MaximumEditField.Position = [491 14 48 18];
6744
6745     % Create WaveHeightmEditFieldLabel
6746     app.WaveHeightmEditFieldLabel = uilabel(app.Tab_2);
6747     app.WaveHeightmEditFieldLabel.HorizontalAlignment = 'right';
6748     app.WaveHeightmEditFieldLabel.Tooltip = {'Significant wave height↵
in Wei and Kirby (1999)'};
6749     app.WaveHeightmEditFieldLabel.Position = [161 44 93 22];
6750     app.WaveHeightmEditFieldLabel.Text = 'Wave Height (m)';
6751
6752     % Create WaveHeightmEditField

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6753     app.WaveHeightmEditField = uieditfield(app.Tab_2, 'numeric');
6754     app.WaveHeightmEditField.ValueDisplayFormat = '%11.2f';
6755     app.WaveHeightmEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@WaveHeightmEditFieldValueChanged, true);
6756     app.WaveHeightmEditField.FontColor = [0.8 0.8 0.8];
6757     app.WaveHeightmEditField.Enable = 'off';
6758     app.WaveHeightmEditField.Position = [311 46 48 18];
6759
6760     % Create Tab_4
6761     app.Tab_4 = uitab(app.TabGroup2);
6762
6763     % Create Label_11
6764     app.Label_11 = uilabel(app.Tab_4);
6765     app.Label_11.BackgroundColor = [0.651 0.651 0.651];
6766     app.Label_11.Position = [1 495 738 22];
6767     app.Label_11.Text = '';
6768
6769     % Create MeteotsunamiLabel
6770     app.MeteotsunamiLabel = uilabel(app.Tab_4);
6771     app.MeteotsunamiLabel.FontSize = 14;
6772     app.MeteotsunamiLabel.FontWeight = 'bold';
6773     app.MeteotsunamiLabel.FontColor = [1 1 1];
6774     app.MeteotsunamiLabel.Position = [24 495 101 22];
6775     app.MeteotsunamiLabel.Text = 'Meteotsunami';
6776
6777     % Create Button_29
6778     app.Button_29 = uibutton(app.Tab_4, 'push');
6779     app.Button_29.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_29Pushed, true);
6780     app.Button_29.Enable = 'off';
6781     app.Button_29.Tooltip = {' '};
6782     app.Button_29.Position = [575 345 22 20];
6783     app.Button_29.Text = '...';
6784
6785     % Create Button_30
6786     app.Button_30 = uibutton(app.Tab_4, 'push');
6787     app.Button_30.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_30Pushed, true);
6788     app.Button_30.Enable = 'off';
6789     app.Button_30.Tooltip = {' '};
6790     app.Button_30.Position = [575 226 22 20];
6791     app.Button_30.Text = '...';
6792
6793     % Create WindforceSwitchLabel
6794     app.WindforceSwitchLabel = uilabel(app.Tab_4);
6795     app.WindforceSwitchLabel.HorizontalAlignment = 'center';
6796     app.WindforceSwitchLabel.Tooltip = {'Use wind effect'};
6797     app.WindforceSwitchLabel.Position = [81 303 64 22];
6798     app.WindforceSwitchLabel.Text = 'Wind force';
6799
6800     % Create WindforceSwitch
6801     app.WindforceSwitch = uiswitch(app.Tab_4, 'slider');
6802     app.WindforceSwitch.ValueChangedFcn = createCallbackFcn(app, ↵
@WindforceSwitchValueChanged, true);
6803     app.WindforceSwitch.Tooltip = {' '};
6804     app.WindforceSwitch.Position = [243 307 32 14];
6805
6806     % Create AirpressureSwitchLabel

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6807     app.AirpressureSwitchLabel = uilabel(app.Tab_4);
6808     app.AirpressureSwitchLabel.HorizontalAlignment = 'center';
6809     app.AirpressureSwitchLabel.Position = [81 429 70 22];
6810     app.AirpressureSwitchLabel.Text = 'Air pressure';
6811
6812     % Create AirpressureSwitch
6813     app.AirpressureSwitch = uiswitch(app.Tab_4, 'slider');
6814     app.AirpressureSwitch.ValueChangedFcn = createCallbackFcn(app, @AirpressureSwitchValueChanged, true);
6815     app.AirpressureSwitch.Tooltip = {'Use air pressure effect'};
6816     app.AirpressureSwitch.Position = [243 433 32 14];
6817     app.AirpressureSwitch.Value = 'On';
6818
6819     % Create WindstresscoefficientEditField
6820     app.WindstresscoefficientEditField = uieditfield(app.Tab_4, 'numeric');
6821     app.WindstresscoefficientEditField.Limits = [0 Inf];
6822     app.WindstresscoefficientEditField.ValueDisplayFormat = '%11.3g';
6823     app.WindstresscoefficientEditField.ValueChangedFcn = createCallbackFcn(app, @WindstresscoefficientEditFieldValueChanged, true);
6824     app.WindstresscoefficientEditField.FontColor = [0.651 0.651 0.651];
6825     app.WindstresscoefficientEditField.Enable = 'off';
6826     app.WindstresscoefficientEditField.Tooltip = {''};
6827     app.WindstresscoefficientEditField.Position = [518 305 50 18];
6828     app.WindstresscoefficientEditField.Value = 0.002;
6829
6830     % Create WindstresscoefficientEditFieldLabel
6831     app.WindstresscoefficientEditFieldLabel = uilabel(app.Tab_4);
6832     app.WindstresscoefficientEditFieldLabel.HorizontalAlignment = 'right';
6833     app.WindstresscoefficientEditFieldLabel.Enable = 'off';
6834     app.WindstresscoefficientEditFieldLabel.Tooltip = {'Value used in the quadratic formula'};
6835     app.WindstresscoefficientEditFieldLabel.Position = [341 303 127 22];
6836     app.WindstresscoefficientEditFieldLabel.Text = 'Wind stress coefficient';
6837
6838     % Create CrestRatioEditFieldLabel
6839     app.CrestRatioEditFieldLabel = uilabel(app.Tab_4);
6840     app.CrestRatioEditFieldLabel.HorizontalAlignment = 'right';
6841     app.CrestRatioEditFieldLabel.Enable = 'off';
6842     app.CrestRatioEditFieldLabel.Tooltip = {'The ratio of the forced wave crest height to the maximum surface elevation.'; ''; 'For storm surges, default value is 1.'};
6843     app.CrestRatioEditFieldLabel.Position = [341 264 140 22];
6844     app.CrestRatioEditFieldLabel.Text = 'Wind crest percent (ratio)';
6845
6846     % Create CrestRatioEditField
6847     app.CrestRatioEditField = uieditfield(app.Tab_4, 'numeric');
6848     app.CrestRatioEditField.Limits = [0 Inf];
6849     app.CrestRatioEditField.ValueDisplayFormat = '%11.2f';
6850     app.CrestRatioEditField.ValueChangedFcn = createCallbackFcn(app, @CrestRatioEditFieldValueChanged, true);
6851     app.CrestRatioEditField.FontColor = [0.651 0.651 0.651];
6852     app.CrestRatioEditField.Enable = 'off';
6853     app.CrestRatioEditField.Tooltip = {'Ratio of the wave crest height'}

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to the maximum surface height. '; '; 'Default for storm surge: 1.0'};
6854         app.CrestRatioEditField.Position = [519 266 49 18];
6855         app.CrestRatioEditField.Value = 1;
6856
6857         % Create ConstantwindfieldSwitchLabel
6858         app.ConstantwindfieldSwitchLabel = uilabel(app.Tab_4);
6859         app.ConstantwindfieldSwitchLabel.HorizontalAlignment = 'center';
6860         app.ConstantwindfieldSwitchLabel.Tooltip = {'Use a constant wind
field'};
6861         app.ConstantwindfieldSwitchLabel.Position = [81 343 109 22];
6862         app.ConstantwindfieldSwitchLabel.Text = 'Constant wind field';
6863
6864         % Create ConstantwindfieldSwitch
6865         app.ConstantwindfieldSwitch = uiswitch(app.Tab_4, 'slider');
6866         app.ConstantwindfieldSwitch.ValueChangedFcn = createCallbackFcn
(app, @ConstantwindfieldSwitchValueChanged, true);
6867         app.ConstantwindfieldSwitch.Tooltip = {'';
6868         app.ConstantwindfieldSwitch.Position = [243 347 32 15];
6869
6870         % Create ConstantwindfileEditFieldLabel
6871         app.ConstantwindfileEditFieldLabel = uilabel(app.Tab_4);
6872         app.ConstantwindfileEditFieldLabel.HorizontalAlignment = 'right';
6873         app.ConstantwindfileEditFieldLabel.Enable = 'off';
6874         app.ConstantwindfileEditFieldLabel.Position = [341 344 102 22];
6875         app.ConstantwindfileEditFieldLabel.Text = 'Constant wind file';
6876
6877         % Create ConstantwindfileEditField
6878         app.ConstantwindfileEditField = uieditfield(app.Tab_4, 'text');
6879         app.ConstantwindfileEditField.Eitable = 'off';
6880         app.ConstantwindfileEditField.Enable = 'off';
6881         app.ConstantwindfileEditField.Tooltip = {'Import the wind file'};
6882         app.ConstantwindfileEditField.Position = [453 345 115 20];
6883
6884         % Create HollandstormmodelSwitchLabel
6885         app.HollandstormmodelSwitchLabel = uilabel(app.Tab_4);
6886         app.HollandstormmodelSwitchLabel.HorizontalAlignment = 'center';
6887         app.HollandstormmodelSwitchLabel.Tooltip = {'Use the Holland
model'};
6888         app.HollandstormmodelSwitchLabel.Position = [81 225 118 22];
6889         app.HollandstormmodelSwitchLabel.Text = 'Holland storm model';
6890
6891         % Create HollandstormmodelSwitch
6892         app.HollandstormmodelSwitch = uiswitch(app.Tab_4, 'slider');
6893         app.HollandstormmodelSwitch.ValueChangedFcn = createCallbackFcn
(app, @HollandstormmodelSwitchValueChanged, true);
6894         app.HollandstormmodelSwitch.Tooltip = {'';
6895         app.HollandstormmodelSwitch.Position = [243 229 32 14];
6896
6897         % Create WindwaveinteractionSwitchLabel
6898         app.WindwaveinteractionSwitchLabel = uilabel(app.Tab_4);
6899         app.WindwaveinteractionSwitchLabel.HorizontalAlignment = 'center';
6900         app.WindwaveinteractionSwitchLabel.Enable = 'off';
6901         app.WindwaveinteractionSwitchLabel.Tooltip = {'Activate wind-wave
interaction by Chen et al. (2003)'};
6902         app.WindwaveinteractionSwitchLabel.Position = [81 264 125 22];
6903         app.WindwaveinteractionSwitchLabel.Text = 'Wind-wave interaction';
6904
6905         % Create WindwaveinteractionSwitch

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6906         app.WindwaveinteractionSwitch = uiswitch(app.Tab_4, 'slider');
6907         app.WindwaveinteractionSwitch.ValueChangedFcn = createCallbackFcn
(app, @WindwaveinteractionSwitchValueChanged, true);
6908         app.WindwaveinteractionSwitch.Enable = 'off';
6909         app.WindwaveinteractionSwitch.Tooltip = {''};
6910         app.WindwaveinteractionSwitch.Position = [243 268 32 14];
6911         app.WindwaveinteractionSwitch.Value = 'On';
6912
6913         % Create StormfileEditField
6914         app.StormfileEditField = uieditfield(app.Tab_4, 'text');
6915         app.StormfileEditField.Eitable = 'off';
6916         app.StormfileEditField.Enable = 'off';
6917         app.StormfileEditField.Tooltip = {'Import the storm file'};
6918         app.StormfileEditField.Position = [427 226 141 20];
6919
6920         % Create StormfileEditFieldLabel
6921         app.StormfileEditFieldLabel = uilabel(app.Tab_4);
6922         app.StormfileEditFieldLabel.HorizontalAlignment = 'right';
6923         app.StormfileEditFieldLabel.Enable = 'off';
6924         app.StormfileEditFieldLabel.Tooltip = {'It must contain the
variables used in the Holland model'};
6925         app.StormfileEditFieldLabel.Position = [341 225 57 22];
6926         app.StormfileEditFieldLabel.Text = 'Storm file';
6927
6928         % Create PressurefileEditFieldLabel
6929         app.PressurefileEditFieldLabel = uilabel(app.Tab_4);
6930         app.PressurefileEditFieldLabel.HorizontalAlignment = 'right';
6931         app.PressurefileEditFieldLabel.Position = [344 429 71 22];
6932         app.PressurefileEditFieldLabel.Text = 'Pressure file';
6933
6934         % Create PressurefileEditField
6935         app.PressurefileEditField = uieditfield(app.Tab_4, 'text');
6936         app.PressurefileEditField.Position = [427 429 141 20];
6937
6938         % Create Button_43
6939         app.Button_43 = uibutton(app.Tab_4, 'push');
6940         app.Button_43.ButtonPushedFcn = createCallbackFcn(app,
@Button_43Pushed, true);
6941         app.Button_43.Position = [575 429 22 20];
6942         app.Button_43.Text = '...';
6943
6944         % Create PressureLabel
6945         app.PressureLabel = uilabel(app.Tab_4);
6946         app.PressureLabel.FontWeight = 'bold';
6947         app.PressureLabel.Position = [50 454 56 22];
6948         app.PressureLabel.Text = 'Pressure';
6949
6950         % Create WindLabel
6951         app.WindLabel = uilabel(app.Tab_4);
6952         app.WindLabel.FontWeight = 'bold';
6953         app.WindLabel.Position = [50 370 34 22];
6954         app.WindLabel.Text = 'Wind';
6955
6956         % Create Tab_3
6957         app.Tab_3 = uitab(app.TabGroup2);
6958
6959         % Create Label_6
6960         app.Label 6 = uilabel(app.Tab 3);

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6961     app.Label_6.BackgroundColor = [0.651 0.651 0.651];
6962     app.Label_6.Position = [1 62 738 22];
6963     app.Label_6.Text = '';
6964
6965     % Create Label_5
6966     app.Label_5 = uilabel(app.Tab_3);
6967     app.Label_5.BackgroundColor = [0.651 0.651 0.651];
6968     app.Label_5.Position = [1 155 739 22];
6969     app.Label_5.Text = '';
6970
6971     % Create Label_12
6972     app.Label_12 = uilabel(app.Tab_3);
6973     app.Label_12.BackgroundColor = [0.651 0.651 0.651];
6974     app.Label_12.Position = [2 495 738 22];
6975     app.Label_12.Text = '';
6976
6977     % Create SedimentLabel
6978     app.SedimentLabel = uilabel(app.Tab_3);
6979     app.SedimentLabel.FontSize = 14;
6980     app.SedimentLabel.FontWeight = 'bold';
6981     app.SedimentLabel.FontColor = [1 1 1];
6982     app.SedimentLabel.Position = [35 495 69 22];
6983     app.SedimentLabel.Text = 'Sediment';
6984
6985     % Create ButtonGroup_19
6986     app.ButtonGroup_19 = uibuttongroup(app.Tab_3);
6987     app.ButtonGroup_19.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_19SelectionChanged, true);
6988     app.ButtonGroup_19.BorderType = 'none';
6989     app.ButtonGroup_19.Position = [306 459 168 30];
6990
6991     % Create UpwindingButton
6992     app.UpwindingButton = uiradiobutton(app.ButtonGroup_19);
6993     app.UpwindingButton.Text = 'Upwinding';
6994     app.UpwindingButton.Position = [10 6 80 22];
6995     app.UpwindingButton.Value = true;
6996
6997     % Create TVDButton
6998     app.TVDButton = uiradiobutton(app.ButtonGroup_19);
6999     app.TVDButton.Tooltip = {'Total Variation Diminishing '};
7000     app.TVDButton.Text = 'TVD';
7001     app.TVDButton.Position = [102 6 45 22];
7002
7003     % Create NumericalschemeLabel
7004     app.NumericalschemeLabel = uilabel(app.Tab_3);
7005     app.NumericalschemeLabel.Tooltip = {'Scheme to use for solving the ↵
advection-diffusion equation'};
7006     app.NumericalschemeLabel.Position = [187 463 105 22];
7007     app.NumericalschemeLabel.Text = 'Numerical scheme';
7008
7009     % Create MediangrainDMEditFieldLabel
7010     app.MediangrainDMEditFieldLabel = uilabel(app.Tab_3);
7011     app.MediangrainDMEditFieldLabel.Tooltip = {'D50 of the grain ↵
diameter'};
7012     app.MediangrainDMEditFieldLabel.Position = [187 428 156 22];
7013     app.MediangrainDMEditFieldLabel.Text = 'Median grain diameter ↵
(mm)';
7014

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7015         % Create MediangrainDMEditField
7016         app.MediagrainsDMEditField = uicontrol(app.Tab_3, 'numeric');
7017         app.MediagrainsDMEditField.Limits = [0 Inf];
7018         app.MediagrainsDMEditField.ValueDisplayFormat = '%11.3f';
7019         app.MediagrainsDMEditField.ValueChangedFcn = createCallbackFcn(
7020             (app, @MediagrainsDMEditFieldValueChanged, true);
7021         app.MediagrainsDMEditField.FontColor = [0.651 0.651 0.651];
7022         app.MediagrainsDMEditField.Tooltip = {' '};
7023         app.MediagrainsDMEditField.Position = [490 429 63 19];
7024         app.MediagrainsDMEditField.Value = 0.5;
7025
7026         % Create SedimentdensityEditFieldLabel
7027         app.SedimentdensityEditFieldLabel = uicontrol(app.Tab_3);
7028         app.SedimentdensityEditFieldLabel.Position = [187 398 99 22];
7029         app.SedimentdensityEditFieldLabel.Text = 'Sediment density';
7030
7031         % Create SedimentdensityEditField
7032         app.SedimentdensityEditField = uicontrol(app.Tab_3, 'numeric');
7033         app.SedimentdensityEditField.Limits = [0 Inf];
7034         app.SedimentdensityEditField.ValueDisplayFormat = '%11.3f';
7035         app.SedimentdensityEditField.ValueChangedFcn = createCallbackFcn(
7036             (app, @SedimentdensityEditFieldValueChanged, true);
7037         app.SedimentdensityEditField.FontColor = [0.651 0.651 0.651];
7038         app.SedimentdensityEditField.Position = [490 400 63 19];
7039         app.SedimentdensityEditField.Value = 2.68;
7040
7041         % Create SedimentporosityEditFieldLabel
7042         app.SedimentporosityEditFieldLabel = uicontrol(app.Tab_3);
7043         app.SedimentporosityEditFieldLabel.Position = [187 369 103 22];
7044         app.SedimentporosityEditFieldLabel.Text = 'Sediment porosity';
7045
7046         % Create SedimentporosityEditField
7047         app.SedimentporosityEditField = uicontrol(app.Tab_3, 'numeric');
7048         app.SedimentporosityEditField.Limits = [0 Inf];
7049         app.SedimentporosityEditField.ValueDisplayFormat = '%11.3f';
7050         app.SedimentporosityEditField.ValueChangedFcn = createCallbackFcn(
7051             (app, @SedimentporosityEditFieldValueChanged, true);
7052         app.SedimentporosityEditField.FontColor = [0.651 0.651 0.651];
7053         app.SedimentporosityEditField.Position = [490 370 63 19];
7054         app.SedimentporosityEditField.Value = 0.47;
7055
7056         % Create SettlingvelocitymsEditFieldLabel
7057         app.SettlingvelocitymsEditFieldLabel = uicontrol(app.Tab_3);
7058         app.SettlingvelocitymsEditFieldLabel.Tooltip = {'Rate at which
7059             sediment particles settles'};
7060         app.SettlingvelocitymsEditFieldLabel.Position = [187 340 120 22];
7061         app.SettlingvelocitymsEditFieldLabel.Text = 'Settling velocity
7062             (m/s)';
7063
7064         % Create SettlingvelocitymsEditField
7065         app.SettlingvelocitymsEditField = uicontrol(app.Tab_3,
7066             'numeric');
7067         app.SettlingvelocitymsEditField.Limits = [0 Inf];
7068         app.SettlingvelocitymsEditField.ValueDisplayFormat = '%11.3f';
7069         app.SettlingvelocitymsEditField.ValueChangedFcn =
7070             createCallbackFcn(app, @SettlingvelocitymsEditFieldValueChanged, true);
7071         app.SettlingvelocitymsEditField.FontColor = [0.651 0.651 0.651];
7072         app.SettlingvelocitymsEditField.Tooltip = {' '};

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7066         app.SettlingvelocitymsEditField.Position = [490 342 63 19];
7067         app.SettlingvelocitymsEditField.Value = 0.125;
7068
7069         % Create CriticalShieldsEditFieldLabel
7070         app.CriticalShieldsEditFieldLabel = uilabel(app.Tab_3);
7071         app.CriticalShieldsEditFieldLabel.Tooltip = {'Value defining when sediment particles begin to move under the influence of fluid flow'};
7072         app.CriticalShieldsEditFieldLabel.Position = [187 311 241 22];
7073         app.CriticalShieldsEditFieldLabel.Text = 'Critical Shields parameter (suspended load)';
7074
7075         % Create CriticalShieldsEditField
7076         app.CriticalShieldsEditField = uieditfield(app.Tab_3, 'numeric');
7077         app.CriticalShieldsEditField.Limits = [0 Inf];
7078         app.CriticalShieldsEditField.ValueDisplayFormat = '%11.3f';
7079         app.CriticalShieldsEditField.ValueChangedFcn = createCallbackFcn(app, @CriticalShieldsEditFieldValueChanged, true);
7080         app.CriticalShieldsEditField.FontColor = [0.651 0.651 0.651];
7081         app.CriticalShieldsEditField.Tooltip = {' '};
7082         app.CriticalShieldsEditField.Position = [490 313 63 19];
7083         app.CriticalShieldsEditField.Value = 0.05;
7084
7085         % Create ShieldsparameterbedloadEditFieldLabel
7086         app.ShieldsparameterbedloadEditFieldLabel = uilabel(app.Tab_3);
7087         app.ShieldsparameterbedloadEditFieldLabel.Tooltip = {'Default value is based on Meyer-Peter and Muller (1984)'};
7088         app.ShieldsparameterbedloadEditFieldLabel.Position = [187 282 156 22];
7089         app.ShieldsparameterbedloadEditFieldLabel.Text = 'Shields parameter (bedload)';
7090
7091         % Create ShieldsparamBedloadEditField
7092         app.ShieldsparamBedloadEditField = uieditfield(app.Tab_3, 'numeric');
7093         app.ShieldsparamBedloadEditField.Limits = [0 Inf];
7094         app.ShieldsparamBedloadEditField.ValueDisplayFormat = '%11.3f';
7095         app.ShieldsparamBedloadEditField.ValueChangedFcn = createCallbackFcn(app, @ShieldsparamBedloadEditFieldValueChanged, true);
7096         app.ShieldsparamBedloadEditField.FontColor = [0.651 0.651 0.651];
7097         app.ShieldsparamBedloadEditField.Tooltip = {' '};
7098         app.ShieldsparamBedloadEditField.Position = [490 284 63 19];
7099         app.ShieldsparamBedloadEditField.Value = 0.047;
7100
7101         % Create RungeKuttaparameter1EditFieldLabel
7102         app.RungeKuttaparameter1EditFieldLabel = uilabel(app.Tab_3);
7103         app.RungeKuttaparameter1EditFieldLabel.Position = [187 253 142 22];
7104         app.RungeKuttaparameter1EditFieldLabel.Text = 'Runge-Kutta parameter 1';
7105
7106         % Create RungeKuttaparameter1EditField
7107         app.RungeKuttaparameter1EditField = uieditfield(app.Tab_3, 'numeric');
7108         app.RungeKuttaparameter1EditField.Limits = [0 Inf];
7109         app.RungeKuttaparameter1EditField.ValueDisplayFormat = '%11.4f';
7110         app.RungeKuttaparameter1EditField.ValueChangedFcn = createCallbackFcn(app, @RungeKuttaparameter1EditFieldValueChanged, true);
7111         app.RungeKuttaparameter1EditField.FontColor = [0.651 0.651 0.651];

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7112     app.RungeKuttaparameter1EditField.Position = [490 254 63 19];
7113     app.RungeKuttaparameter1EditField.Value = 0.3333;
7114
7115     % Create RungeKuttaparameter2EditFieldLabel
7116     app.RungeKuttaparameter2EditFieldLabel = uilabel(app.Tab_3);
7117     app.RungeKuttaparameter2EditFieldLabel.Position = [187 224 142 22];
7118     app.RungeKuttaparameter2EditFieldLabel.Text = 'Runge-Kutta
parameter 2';
7119
7120     % Create RungeKuttaparameter2EditField
7121     app.RungeKuttaparameter2EditField = uieditfield(app.Tab_3,
'numeric');
7122     app.RungeKuttaparameter2EditField.Limits = [0 Inf];
7123     app.RungeKuttaparameter2EditField.ValueDisplayFormat = '%11.3f';
7124     app.RungeKuttaparameter2EditField.ValueChangedFcn =
createCallbackFcn(app, @RungeKuttaparameter2EditFieldValueChanged, true);
7125     app.RungeKuttaparameter2EditField.FontColor = [0.651 0.651 0.651];
7126     app.RungeKuttaparameter2EditField.Position = [490 226 63 19];
7127     app.RungeKuttaparameter2EditField.Value = 1;
7128
7129     % Create MinDepthSedPickUpEditFieldLabel
7130     app.MinDepthSedPickUpEditFieldLabel = uilabel(app.Tab_3);
7131     app.MinDepthSedPickUpEditFieldLabel.Position = [187 195 259 22];
7132     app.MinDepthSedPickUpEditFieldLabel.Text = 'Minimum depth for
sediment pickup action (m)';
7133
7134     % Create MinDepthSedPickUpEditField
7135     app.MinDepthSedPickUpEditField = uieditfield(app.Tab_3,
'numeric');
7136     app.MinDepthSedPickUpEditField.Limits = [0 Inf];
7137     app.MinDepthSedPickUpEditField.ValueDisplayFormat = '%11.3f';
7138     app.MinDepthSedPickUpEditField.ValueChangedFcn = createCallbackFcn
(app, @MinDepthSedPickUpEditFieldValueChanged, true);
7139     app.MinDepthSedPickUpEditField.FontColor = [0.651 0.651 0.651];
7140     app.MinDepthSedPickUpEditField.Position = [490 197 63 19];
7141     app.MinDepthSedPickUpEditField.Value = 0.1;
7142
7143     % Create ButtonGroup_20
7144     app.ButtonGroup_20 = uibuttongroup(app.Tab_3);
7145     app.ButtonGroup_20.SelectionChangedFcn = createCallbackFcn(app,
@ButtonGroup_20SelectionChanged, true);
7146     app.ButtonGroup_20.BorderType = 'none';
7147     app.ButtonGroup_20.Position = [266 120 241 30];
7148
7149     % Create UpdateddepthButton
7150     app.UpdateddepthButton = uiradiobutton(app.ButtonGroup_20);
7151     app.UpdateddepthButton.Tooltip = {'Bathymetry is updated in the
simulation'};
7152     app.UpdateddepthButton.Text = ' Update depth';
7153     app.UpdateddepthButton.Position = [6 6 99 22];
7154
7155     % Create NobedchangeButton
7156     app.NobedchangeButton = uiradiobutton(app.ButtonGroup_20);
7157     app.NobedchangeButton.Tooltip = {'Bathymetry remains constant'};
7158     app.NobedchangeButton.Text = ' No bed change';
7159     app.NobedchangeButton.Position = [121 6 108 22];
7160

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7161         % Create Button_36
7162         app.Button_36 = uiradiobutton(app.ButtonGroup_20);
7163         app.Button_36.Visible = 'off';
7164         app.Button_36.Text = '';
7165         app.Button_36.Position = [232 6 25 22];
7166         app.Button_36.Value = true;
7167
7168         % Create BedChangeLabel
7169         app.BedChangeLabel = uilabel(app.Tab_3);
7170         app.BedChangeLabel.Position = [187 125 76 22];
7171         app.BedChangeLabel.Text = 'Bed Change: ';
7172
7173         % Create EditFieldLabel
7174         app.EditFieldLabel = uilabel(app.Tab_3);
7175         app.EditFieldLabel.HorizontalAlignment = 'right';
7176         app.EditFieldLabel.Enable = 'off';
7177         app.EditFieldLabel.Visible = 'off';
7178         app.EditFieldLabel.Position = [282 -4 341 22];
7179         app.EditFieldLabel.Text = 'Time step ratio: Boussinesq model to
morphological updating';
7180
7181         % Create TimeStepBouss
7182         app.TimeStepBouss = uieditfield(app.Tab_3, 'numeric');
7183         app.TimeStepBouss.Limits = [0 Inf];
7184         app.TimeStepBouss.ValueDisplayFormat = '%11.2f';
7185         app.TimeStepBouss.ValueChangedFcn = createCallbackFcn(app,
@TimeStepBoussValueChanged, true);
7186         app.TimeStepBouss.Editable = 'off';
7187         app.TimeStepBouss.FontColor = [0.651 0.651 0.651];
7188         app.TimeStepBouss.Enable = 'off';
7189         app.TimeStepBouss.Visible = 'off';
7190         app.TimeStepBouss.Position = [636 -2 63 19];
7191         app.TimeStepBouss.Value = 25;
7192
7193         % Create AvalancheLabel
7194         app.AvalancheLabel = uilabel(app.Tab_3);
7195         app.AvalancheLabel.FontSize = 14;
7196         app.AvalancheLabel.FontWeight = 'bold';
7197         app.AvalancheLabel.FontColor = [1 1 1];
7198         app.AvalancheLabel.Position = [35 62 74 22];
7199         app.AvalancheLabel.Text = 'Avalanche';
7200
7201         % Create MorphologicalChangeLabel_2
7202         app.MorphologicalChangeLabel_2 = uilabel(app.Tab_3);
7203         app.MorphologicalChangeLabel_2.FontSize = 14;
7204         app.MorphologicalChangeLabel_2.FontWeight = 'bold';
7205         app.MorphologicalChangeLabel_2.FontColor = [1 1 1];
7206         app.MorphologicalChangeLabel_2.Position = [35 155 157 22];
7207         app.MorphologicalChangeLabel_2.Text = 'Morphological Change';
7208
7209         % Create ConsideravalancheLabel
7210         app.ConsideravalancheLabel = uilabel(app.Tab_3);
7211         app.ConsideravalancheLabel.Position = [57 30 118 22];
7212         app.ConsideravalancheLabel.Text = 'Consider avalanche?';
7213
7214         % Create ButtonGroup_22
7215         app.ButtonGroup_22 = uibuttongroup(app.Tab_3);
7216         app.ButtonGroup_22.SelectionChangedFcn = createCallbackFcn(app,

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@ButtonGroup_22SelectionChanged, true);
7217         app.ButtonGroup_22.BorderType = 'none';
7218         app.ButtonGroup_22.Position = [181 27 150 30];
7219
7220         % Create YesButton_8
7221         app.YesButton_8 = uiradiobutton(app.ButtonGroup_22);
7222         app.YesButton_8.Text = 'Yes';
7223         app.YesButton_8.Position = [9 5 41 22];
7224
7225         % Create NoButton_7
7226         app.NoButton_7 = uiradiobutton(app.ButtonGroup_22);
7227         app.NoButton_7.Text = 'No';
7228         app.NoButton_7.Position = [61 5 38 22];
7229         app.NoButton_7.Value = true;
7230
7231         % Create Button_38
7232         app.Button_38 = uiradiobutton(app.ButtonGroup_22);
7233         app.Button_38.Visible = 'off';
7234         app.Button_38.Text = '';
7235         app.Button_38.Position = [103 2 25 22];
7236
7237         % Create RungeKuttaparam2EditFieldLabel_2
7238         app.RungeKuttaparam2EditFieldLabel_2 = uilabel(app.Tab_3);
7239         app.RungeKuttaparam2EditFieldLabel_2.Enable = 'off';
7240         app.RungeKuttaparam2EditFieldLabel_2.Tooltip = {'Steepest angle↵
which the sediments remain stable'; ''; 'Default value 0.7 = 35 deg'};
7241         app.RungeKuttaparam2EditFieldLabel_2.Position = [392 30 154 22];
7242         app.RungeKuttaparam2EditFieldLabel_2.Text = 'Tangent of the repose↵
angle';
7243
7244         % Create RungeKuttaparam2EditField_2
7245         app.RungeKuttaparam2EditField_2 = uieditfield(app.Tab_3, ↵
'numeric');
7246         app.RungeKuttaparam2EditField_2.ValueDisplayFormat = '%11.2f';
7247         app.RungeKuttaparam2EditField_2.ValueChangedFcn = ↵
createCallbackFcn(app, @RungeKuttaparam2EditField_2ValueChanged, true);
7248         app.RungeKuttaparam2EditField_2.FontColor = [0.651 0.651 0.651];
7249         app.RungeKuttaparam2EditField_2.Enable = 'off';
7250         app.RungeKuttaparam2EditField_2.Tooltip = {'';
7251         app.RungeKuttaparam2EditField_2.Position = [558 32 46 19];
7252         app.RungeKuttaparam2EditField_2.Value = 0.7;
7253
7254         % Create FileEditFieldLabel
7255         app.FileEditFieldLabel = uilabel(app.Tab_3);
7256         app.FileEditFieldLabel.HorizontalAlignment = 'right';
7257         app.FileEditFieldLabel.Visible = 'off';
7258         app.FileEditFieldLabel.Position = [558 102 27 22];
7259         app.FileEditFieldLabel.Text = ' File';
7260
7261         % Create FileEditField
7262         app.FileEditField = uieditfield(app.Tab_3, 'text');
7263         app.FileEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@FileEditFieldValueChanged, true);
7264         app.FileEditField.Editable = 'off';
7265         app.FileEditField.Visible = 'off';
7266         app.FileEditField.Tooltip = {'Import the hard bottom file'; '';↵
'File format and grid dimension must be the same as the input bathymetry'};
7267         app.FileEditField.Position = [592 103 96 20];

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7268
7269         % Create Button_35
7270         app.Button_35 = uibutton(app.Tab_3, 'push');
7271         app.Button_35.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_35Pushed, true);
7272         app.Button_35.Visible = 'off';
7273         app.Button_35.Tooltip = {''};
7274         app.Button_35.Position = [694 103 22 20];
7275         app.Button_35.Text = '...';
7276
7277         % Create BottomTypeLabel
7278         app.BottomTypeLabel = uilabel(app.Tab_3);
7279         app.BottomTypeLabel.Position = [187 100 77 22];
7280         app.BottomTypeLabel.Text = 'Bottom Type';
7281
7282         % Create ButtonGroup_21
7283         app.ButtonGroup_21 = uibuttongroup(app.Tab_3);
7284         app.ButtonGroup_21.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_21SelectionChanged, true);
7285         app.ButtonGroup_21.ForegroundColor = [0.651 0.651 0.651];
7286         app.ButtonGroup_21.BorderType = 'none';
7287         app.ButtonGroup_21.Position = [266 96 274 30];
7288
7289         % Create HardbednoerosionButton
7290         app.HardbednoerosionButton = uiradiobutton(app.ButtonGroup_21);
7291         app.HardbednoerosionButton.Text = ' Hard bed, no erosion';
7292         app.HardbednoerosionButton.Position = [121 5 138 22];
7293
7294         % Create SedimentbedButton
7295         app.SedimentbedButton = uiradiobutton(app.ButtonGroup_21);
7296         app.SedimentbedButton.Tooltip = {'Allow bed erosion'};
7297         app.SedimentbedButton.Text = ' Sediment bed';
7298         app.SedimentbedButton.Position = [6 5 100 22];
7299
7300         % Create Button_37
7301         app.Button_37 = uiradiobutton(app.ButtonGroup_21);
7302         app.Button_37.Visible = 'off';
7303         app.Button_37.Text = '';
7304         app.Button_37.Position = [261 4 25 22];
7305         app.Button_37.Value = true;
7306
7307         % Create TimeStepSedToMorpEditFieldLabel
7308         app.TimeStepSedToMorpEditFieldLabel = uilabel(app.Tab_3);
7309         app.TimeStepSedToMorpEditFieldLabel.HorizontalAlignment = 'right';
7310         app.TimeStepSedToMorpEditFieldLabel.Enable = 'off';
7311         app.TimeStepSedToMorpEditFieldLabel.Visible = 'off';
7312         app.TimeStepSedToMorpEditFieldLabel.Position = [331 194 237 22];
7313         app.TimeStepSedToMorpEditFieldLabel.Text = 'Time step ratio:↵
sediment to morphological';
7314
7315         % Create TimeStepSedToMorphlEditField
7316         app.TimeStepSedToMorphlEditField = uieditfield(app.Tab_3, ↵
'numeric');
7317         app.TimeStepSedToMorphlEditField.Limits = [0 Inf];
7318         app.TimeStepSedToMorphlEditField.ValueChangedFcn = ↵
createCallbackFcn(app, @TimeStepSedToMorphlEditFieldValueChanged, true);
7319         app.TimeStepSedToMorphlEditField.Editable = 'off';
7320         app.TimeStepSedToMorphlEditField.FontColor = [0.651 0.651 0.651];

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7321 app.TimeStepSedToMorphlEditField.Enable = 'off';
7322 app.TimeStepSedToMorphlEditField.Visible = 'off';
7323 app.TimeStepSedToMorphlEditField.Position = [491 187 63 19];
7324 app.TimeStepSedToMorphlEditField.Value = 10;
7325
7326 % Create AddrollereffectsCheckBox
7327 app.AddrollereffectsCheckBox = uicheckbox(app.Tab_3);
7328 app.AddrollereffectsCheckBox.Text = 'Add roller effects';
7329 app.AddrollereffectsCheckBox.Position = [490 463 113 22];
7330
7331 % Create Tab_6
7332 app.Tab_6 = uitab(app.TabGroup2);
7333
7334 % Create Label_13
7335 app.Label_13 = uilabel(app.Tab_6);
7336 app.Label_13.BackgroundColor = [0.651 0.651 0.651];
7337 app.Label_13.Position = [1 495 738 22];
7338 app.Label_13.Text = '';
7339
7340 % Create ImportVesselFilesLabel
7341 app.ImportVesselFilesLabel = uilabel(app.Tab_6);
7342 app.ImportVesselFilesLabel.Tooltip = {' '};
7343 app.ImportVesselFilesLabel.Position = [77 452 106 22];
7344 app.ImportVesselFilesLabel.Text = 'Import vessel files:';
7345
7346 % Create ImportvesselfilesTextArea
7347 app.ImportvesselfilesTextArea = uitextarea(app.Tab_6);
7348 app.ImportvesselfilesTextArea.Editable = 'off';
7349 app.ImportvesselfilesTextArea.Position = [77 324 154 120];
7350
7351 % Create Button_40
7352 app.Button_40 = uibutton(app.Tab_6, 'push');
7353 app.Button_40.ButtonPushedFcn = createCallbackFcn(app, @
7354 @Button_40Pushed, true);
7355 app.Button_40.Tooltip = {'Find vessel files'};
7356 app.Button_40.Position = [240 374 26 20];
7357 app.Button_40.Text = '...';
7358
7359 % Create VesselLabel
7360 app.VesselLabel = uilabel(app.Tab_6);
7361 app.VesselLabel.FontSize = 14;
7362 app.VesselLabel.FontWeight = 'bold';
7363 app.VesselLabel.FontColor = [1 1 1];
7364 app.VesselLabel.Position = [24 495 48 22];
7365 app.VesselLabel.Text = 'Vessel';
7366
7367 % Create Label_8
7368 app.Label_8 = uilabel(app.Tab_6);
7369 app.Label_8.BackgroundColor = [0.651 0.651 0.651];
7370 app.Label_8.Position = [1 197 738 22];
7371 app.Label_8.Text = '';
7372
7373 % Create DeepDraftVesselsLabel
7374 app.DeepDraftVesselsLabel = uilabel(app.Tab_6);
7375 app.DeepDraftVesselsLabel.FontWeight = 'bold';
7376 app.DeepDraftVesselsLabel.FontColor = [1 1 1];
7377 app.DeepDraftVesselsLabel.Position = [24 197 114 22];
7378 app.DeepDraftVesselsLabel.Text = 'Deep Draft Vessels';

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7378
7379     % Create MinimumClearanceLabel
7380     app.MinimumClearanceLabel = uilabel(app.Tab_6);
7381     app.MinimumClearanceLabel.Enable = 'off';
7382     app.MinimumClearanceLabel.Position = [80 132 116 22];
7383     app.MinimumClearanceLabel.Text = 'Minimum Clearance: ';
7384
7385     % Create MinimumClearanceEditField
7386     app.MinimumClearanceEditField = uieditfield(app.Tab_6, 'numeric');
7387     app.MinimumClearanceEditField.Limits = [0 Inf];
7388     app.MinimumClearanceEditField.ValueDisplayFormat = '%5.1f';
7389     app.MinimumClearanceEditField.ValueChangedFcn = createCallbackFcn(
7390 (app, @MinimumClearanceEditFieldValueChanged, true);
7391     app.MinimumClearanceEditField.FontColor = [0.651 0.651 0.651];
7392     app.MinimumClearanceEditField.Enable = 'off';
7393     app.MinimumClearanceEditField.Position = [198 134 35 19];
7394     app.MinimumClearanceEditField.Value = 1;
7395
7396     % Create IncludeSedimentEffectintheModelCheckBox
7397     app.IncludeSedimentEffectintheModelCheckBox = uicheckbox(app.
7398 Tab_6);
7399     app.IncludeSedimentEffectintheModelCheckBox.Text = 'Include
7400 Sediment Effect in the Model';
7401     app.IncludeSedimentEffectintheModelCheckBox.Position = [75 296 221
7402 22];
7403
7404     % Create ActivateCheckBox
7405     app.ActivateCheckBox = uicheckbox(app.Tab_6);
7406     app.ActivateCheckBox.ValueChangedFcn = createCallbackFcn(app,
7407 @ActivateCheckBoxValueChanged, true);
7408     app.ActivateCheckBox.Tooltip = {'Check the box to address the
7409 instability issue that may arise when modelling a large vessel with a draft close
7410 to the channel depth'};
7411     app.ActivateCheckBox.Text = ' Activate to reduce high-frequency
7412 spikes occurring close to the ship';
7413     app.ActivateCheckBox.Position = [81 163 401 22];
7414
7415     % Create ShockcapturingCheckBox
7416     app.ShockcapturingCheckBox = uicheckbox(app.Tab_6);
7417     app.ShockcapturingCheckBox.Enable = 'off';
7418     app.ShockcapturingCheckBox.Tooltip = {'Utilizes shock-capturing
7419 method'};
7420     app.ShockcapturingCheckBox.Text = 'Shock capturing';
7421     app.ShockcapturingCheckBox.Position = [451 72 110 22];
7422     app.ShockcapturingCheckBox.Value = true;
7423
7424     % Create FrictionCheckBox_2
7425     app.FrictionCheckBox_2 = uicheckbox(app.Tab_6);
7426     app.FrictionCheckBox_2.ValueChangedFcn = createCallbackFcn(app,
7427 @FrictionCheckBox_2ValueChanged, true);
7428     app.FrictionCheckBox_2.Enable = 'off';
7429     app.FrictionCheckBox_2.Tooltip = {'Utilizes dissipation method for
7430 damping waves similar to a sponge layer'};
7431     app.FrictionCheckBox_2.Text = 'Friction';
7432     app.FrictionCheckBox_2.Position = [80 72 62 22];
7433     app.FrictionCheckBox_2.Value = true;
7434
7435     % Create ViscosityCheckBox

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7425         app.ViscosityCheckBox = uicheckbox(app.Tab_6);
7426         app.ViscosityCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@ViscosityCheckBoxValueChanged, true);
7427         app.ViscosityCheckBox.Enable = 'off';
7428         app.ViscosityCheckBox.Tooltip = {'Similar to friction method but ↵
its wave damping rate is usually lower'};
7429         app.ViscosityCheckBox.Text = 'Viscosity';
7430         app.ViscosityCheckBox.Position = [279 72 70 22];
7431
7432         % Create FrictionCoefficientEditFieldLabel
7433         app.FrictionCoefficientEditFieldLabel = uilabel(app.Tab_6);
7434         app.FrictionCoefficientEditFieldLabel.HorizontalAlignment = ↵
'right';
7435         app.FrictionCoefficientEditFieldLabel.Enable = 'off';
7436         app.FrictionCoefficientEditFieldLabel.Tooltip = {'Suggested ↵
values: 0.1 - 1'};
7437         app.FrictionCoefficientEditFieldLabel.Position = [136 74 63 19];
7438         app.FrictionCoefficientEditFieldLabel.Text = 'Coefficient';
7439
7440         % Create FrictionCoefficientEditField
7441         app.FrictionCoefficientEditField = uieditfield(app.Tab_6, ↵
'numeric');
7442         app.FrictionCoefficientEditField.Limits = [0 Inf];
7443         app.FrictionCoefficientEditField.ValueDisplayFormat = '%5.1f';
7444         app.FrictionCoefficientEditField.ValueChangedFcn = ↵
createCallbackFcn(app, @FrictionCoefficientEditFieldValueChanged, true);
7445         app.FrictionCoefficientEditField.FontColor = [0.651 0.651 0.651];
7446         app.FrictionCoefficientEditField.Enable = 'off';
7447         app.FrictionCoefficientEditField.Tooltip = {''};
7448         app.FrictionCoefficientEditField.Position = [209 74 35 19];
7449         app.FrictionCoefficientEditField.Value = 0.1;
7450
7451         % Create ValueEditFieldLabel_2
7452         app.ValueEditFieldLabel_2 = uilabel(app.Tab_6);
7453         app.ValueEditFieldLabel_2.HorizontalAlignment = 'right';
7454         app.ValueEditFieldLabel_2.Enable = 'off';
7455         app.ValueEditFieldLabel_2.Tooltip = {'Suggested values: 0.1 - 5'};
7456         app.ValueEditFieldLabel_2.Position = [343 74 34 19];
7457         app.ValueEditFieldLabel_2.Text = 'Value';
7458
7459         % Create ValueEditField
7460         app.ValueEditField = uieditfield(app.Tab_6, 'numeric');
7461         app.ValueEditField.Limits = [0 Inf];
7462         app.ValueEditField.ValueDisplayFormat = '%5.1f';
7463         app.ValueEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@ValueEditFieldValueChanged, true);
7464         app.ValueEditField.Editable = 'off';
7465         app.ValueEditField.FontColor = [0.651 0.651 0.651];
7466         app.ValueEditField.Enable = 'off';
7467         app.ValueEditField.Tooltip = {''};
7468         app.ValueEditField.Position = [384 74 35 19];
7469         app.ValueEditField.Value = 1;
7470
7471         % Create SelectMethodLabel
7472         app.SelectMethodLabel = uilabel(app.Tab_6);
7473         app.SelectMethodLabel.FontWeight = 'bold';
7474         app.SelectMethodLabel.Enable = 'off';
7475         app.SelectMethodLabel.Position = [81 97 92 22];

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7476     app.SelectMethodLabel.Text = 'Select Method:';
7477
7478     % Create SelectInitialConditonLabel
7479     app.SelectInitialConditonLabel = uilabel(app.InitialConditionTab);
7480     app.SelectInitialConditonLabel.FontWeight = 'bold';
7481     app.SelectInitialConditonLabel.Position = [21 614 135 22];
7482     app.SelectInitialConditonLabel.Text = 'Select Initial Conditon:';
7483
7484     % Create ButtonGroup_18
7485     app.ButtonGroup_18 = uibuttongroup(app.InitialConditionTab);
7486     app.ButtonGroup_18.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_6SelectionChanged, true);
7487     app.ButtonGroup_18.BorderType = 'none';
7488     app.ButtonGroup_18.FontSize = 14;
7489     app.ButtonGroup_18.Position = [167 610 412 30];
7490
7491     % Create ImportUVZButton
7492     app.ImportUVZButton = uiradiobutton(app.ButtonGroup_18);
7493     app.ImportUVZButton.Text = 'Import UVZ';
7494     app.ImportUVZButton.Position = [3 5 84 22];
7495
7496     % Create WavemakerButton
7497     app.WavemakerButton = uiradiobutton(app.ButtonGroup_18);
7498     app.WavemakerButton.Text = 'Wavemaker';
7499     app.WavemakerButton.Position = [109 5 85 22];
7500
7501     % Create Button_31
7502     app.Button_31 = uiradiobutton(app.ButtonGroup_18);
7503     app.Button_31.Enable = 'off';
7504     app.Button_31.Visible = 'off';
7505     app.Button_31.Text = '';
7506     app.Button_31.Position = [395 5 25 22];
7507     app.Button_31.Value = true;
7508
7509     % Create MeteotsunamiButton
7510     app.MeteotsunamiButton = uiradiobutton(app.ButtonGroup_18);
7511     app.MeteotsunamiButton.Tooltip = {'Initiate the tsunami using ↵
atmospheric disturbances'};
7512     app.MeteotsunamiButton.Text = 'Meteotsunami';
7513     app.MeteotsunamiButton.Position = [215 5 98 22];
7514
7515     % Create VesselButton
7516     app.VesselButton = uiradiobutton(app.ButtonGroup_18);
7517     app.VesselButton.Tooltip = {'Use the ship-wake model in the ↵
simulation'};
7518     app.VesselButton.Text = 'Vessel';
7519     app.VesselButton.Position = [334 5 56 22];
7520
7521     % Create GenerateinputtxtButton_4
7522     app.GenerateinputtxtButton_4 = uibutton(app.InitialConditionTab, ↵
'push');
7523     app.GenerateinputtxtButton_4.ButtonPushedFcn = createCallbackFcn ↵
(app, @GenerateinputtxtButton_4Pushed, true);
7524     app.GenerateinputtxtButton_4.HandleVisibility = 'off';
7525     app.GenerateinputtxtButton_4.BackgroundColor = [0.9216 0.9216 ↵
0.9216];
7526     app.GenerateinputtxtButton_4.FontSize = 15;
7527     app.GenerateinputtxtButton_4.FontWeight = 'bold';

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7528         app.GenerateinputtxtButton_4.Position = [186 23 355 43];
7529         app.GenerateinputtxtButton_4.Text = 'Generate input.txt';
7530
7531         % Create SedimentTransportCheckBox
7532         app.SedimentTransportCheckBox = uicheckbox(app. InitialConditionTab);
7533         app.SedimentTransportCheckBox.ValueChangedFcn = createCallbackFcn(
7534         (app, @SedimentTransportCheckBoxValueChanged, true);
7535         app.SedimentTransportCheckBox.Text = 'Sediment Transport';
7536         app.SedimentTransportCheckBox.Position = [584 614 127 22];
7537
7538         % Create ExpectedOutputTab
7539         app.ExpectedOutputTab = uitab(app.TabGroup);
7540         app.ExpectedOutputTab.Tooltip = {'Flux F in X direction'};
7541         app.ExpectedOutputTab.Title = 'Expected Output';
7542         app.ExpectedOutputTab.BackgroundColor = [0.9412 0.9412 0.9412];
7543
7544         % Create GridLayout
7545         app.GridLayout = uigridlayout(app.ExpectedOutputTab);
7546         app.GridLayout.ColumnWidth = {'1x', 31, 45, 36, 39, 38, 54, 67,
66, 76, 57, '2.84x'};
7547         app.GridLayout.RowHeight = {22, 22, 22, 22, 22, 22, '1x', 22, 22,
22, 22, 22, 22, 22, 22, '1.75x', 43};
7548
7549         % Create DepthCheckBox
7550         app.DepthCheckBox = uicheckbox(app.GridLayout);
7551         app.DepthCheckBox.ValueChangedFcn = createCallbackFcn(app,
@DepthCheckBoxValueChanged, true);
7552         app.DepthCheckBox.Tooltip = {'DEPTH_OUT'};
7553         app.DepthCheckBox.Text = 'Depth';
7554         app.DepthCheckBox.Layout.Row = 6;
7555         app.DepthCheckBox.Layout.Column = 7;
7556
7557         % Create MaxVorticityCheckBox
7558         app.MaxVorticityCheckBox = uicheckbox(app.GridLayout);
7559         app.MaxVorticityCheckBox.ValueChangedFcn = createCallbackFcn(app,
@MaxVorticityCheckBoxValueChanged, true);
7560         app.MaxVorticityCheckBox.Tooltip = {'VORMAX'};
7561         app.MaxVorticityCheckBox.Text = 'Maximum Vorticity';
7562         app.MaxVorticityCheckBox.Layout.Row = 9;
7563         app.MaxVorticityCheckBox.Layout.Column = [7 8];
7564
7565         % Create UVelocityCheckBox
7566         app.UVelocityCheckBox = uicheckbox(app.GridLayout);
7567         app.UVelocityCheckBox.ValueChangedFcn = createCallbackFcn(app,
@UVelocityCheckBoxValueChanged, true);
7568         app.UVelocityCheckBox.Tooltip = {'U: Velocity in X direction'};
7569         app.UVelocityCheckBox.Text = 'U Velocity';
7570         app.UVelocityCheckBox.Layout.Row = 2;
7571         app.UVelocityCheckBox.Layout.Column = 10;
7572
7573         % Create MaxMomentumFluxCheckBox
7574         app.MaxMomentumFluxCheckBox = uicheckbox(app.GridLayout);
7575         app.MaxMomentumFluxCheckBox.ValueChangedFcn = createCallbackFcn(
7576         (app, @MaxMomentumFluxCheckBoxValueChanged, true);
7577         app.MaxMomentumFluxCheckBox.Tooltip = {'MFMAX'};
7578         app.MaxMomentumFluxCheckBox.Text = 'Maximum Momentum Flux';
7579         app.MaxMomentumFluxCheckBox.Layout.Row = 11;

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7578         app.MaxMomentumFluxCheckBox.Layout.Column = [2 5];
7579
7580         % Create VVelocityCheckBox
7581         app.VVelocityCheckBox = uicheckbox(app.GridLayout);
7582         app.VVelocityCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@VVelocityCheckBoxValueChanged, true);
7583         app.VVelocityCheckBox.Tooltip = {' in y direction'};
7584         app.VVelocityCheckBox.Text = 'V Velocity';
7585         app.VVelocityCheckBox.Layout.Row = 3;
7586         app.VVelocityCheckBox.Layout.Column = 10;
7587
7588         % Create ArrivalCheckBox
7589         app.ArrivalCheckBox = uicheckbox(app.GridLayout);
7590         app.ArrivalCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@ArrivalCheckBoxValueChanged, true);
7591         app.ArrivalCheckBox.Tooltip = {'OUT_Time'};
7592         app.ArrivalCheckBox.Text = 'Arrival';
7593         app.ArrivalCheckBox.Layout.Row = 2;
7594         app.ArrivalCheckBox.Layout.Column = [7 8];
7595
7596         % Create SurfaceElevationattimetCheckBox
7597         app.SurfaceElevationattimetCheckBox = uicheckbox(app.GridLayout);
7598         app.SurfaceElevationattimetCheckBox.ValueChangedFcn = ↵
createCallbackFcn(app, @SurfaceElevationattimetCheckBoxValueChanged, true);
7599         app.SurfaceElevationattimetCheckBox.Tooltip = {'ETA: Surface ↵
elevation at time t'};
7600         app.SurfaceElevationattimetCheckBox.Text = 'Surface Elevation at ↵
time t';
7601         app.SurfaceElevationattimetCheckBox.Layout.Row = 4;
7602         app.SurfaceElevationattimetCheckBox.Layout.Column = [2 5];
7603         app.SurfaceElevationattimetCheckBox.Value = true;
7604
7605         % Create WetdrymaskCheckBox
7606         app.WetdrymaskCheckBox = uicheckbox(app.GridLayout);
7607         app.WetdrymaskCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@WetdrymaskCheckBoxValueChanged, true);
7608         app.WetdrymaskCheckBox.Tooltip = {'MASK'};
7609         app.WetdrymaskCheckBox.Text = 'Wet-dry mask';
7610         app.WetdrymaskCheckBox.Layout.Row = 9;
7611         app.WetdrymaskCheckBox.Layout.Column = [10 11];
7612
7613         % Create WaveHeightCheckBox
7614         app.WaveHeightCheckBox = uicheckbox(app.GridLayout);
7615         app.WaveHeightCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@WaveHeightCheckBoxValueChanged, true);
7616         app.WaveHeightCheckBox.Tooltip = {'Wave height: Hsig,Hrms, Havg'};
7617         app.WaveHeightCheckBox.Text = 'Wave Height';
7618         app.WaveHeightCheckBox.Layout.Row = 5;
7619         app.WaveHeightCheckBox.Layout.Column = [2 3];
7620
7621         % Create WetdrymaskforBoussinesqNSWECheckBox
7622         app.WetdrymaskforBoussinesqNSWECheckBox = uicheckbox(app. ↵
GridLayout);
7623         app.WetdrymaskforBoussinesqNSWECheckBox.ValueChangedFcn = ↵
createCallbackFcn(app, @WetdrymaskforBoussinesqNSWECheckBoxValueChanged, true);
7624         app.WetdrymaskforBoussinesqNSWECheckBox.Tooltip = {'MASK9: Crate a ↵
wetting-drying mask for Boussinesq/NSWE modelling'};
7625         app.WetdrymaskforBoussinesqNSWECheckBox.Text = 'Wet-dry mask for ↵

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Boussinesq/NSWE';
7626         app.WetdrymaskforBoussinesqNSWECheckBox.Layout.Row = 10;
7627         app.WetdrymaskforBoussinesqNSWECheckBox.Layout.Column = [10 12];
7628
7629         % Create RollerinducedFluxCheckBox
7630         app.RollerinducedFluxCheckBox = uicontrol(app.GridLayout);
7631         app.RollerinducedFluxCheckBox.ValueChangedFcn = createCallbackFcn(
7632 (app, @RollerinducedFluxCheckBoxValueChanged, true);
7633         app.RollerinducedFluxCheckBox.Tooltip = {'ROLLER: Roller-induced
mass flux'};
7634         app.RollerinducedFluxCheckBox.Text = 'Roller-induced Flux';
7635         app.RollerinducedFluxCheckBox.Layout.Row = 16;
7636         app.RollerinducedFluxCheckBox.Layout.Column = [2 4];
7637
7638         % Create XSourceCheckBox
7639         app.XSourceCheckBox = uicontrol(app.GridLayout);
7640         app.XSourceCheckBox.ValueChangedFcn = createCallbackFcn(app,
7641 @XSourceCheckBoxValueChanged, true);
7642         app.XSourceCheckBox.Tooltip = {'SoureceX: Source terms in X
direction'};
7643         app.XSourceCheckBox.Text = 'X Source';
7644         app.XSourceCheckBox.Layout.Row = 4;
7645         app.XSourceCheckBox.Layout.Column = [7 8];
7646
7647         % Create PressureFieldCheckBox
7648         app.PressureFieldCheckBox = uicontrol(app.GridLayout);
7649         app.PressureFieldCheckBox.ValueChangedFcn = createCallbackFcn(app,
7650 @PressureFieldCheckBoxValueChanged, true);
7651         app.PressureFieldCheckBox.Tooltip = {'OUT_METEO'};
7652         app.PressureFieldCheckBox.Text = 'Pressure Field';
7653         app.PressureFieldCheckBox.Layout.Row = 13;
7654         app.PressureFieldCheckBox.Layout.Column = [7 8];
7655
7656         % Create YSourceCheckBox
7657         app.YSourceCheckBox = uicontrol(app.GridLayout);
7658         app.YSourceCheckBox.ValueChangedFcn = createCallbackFcn(app,
7659 @YSourceCheckBoxValueChanged, true);
7660         app.YSourceCheckBox.Tooltip = {'SoureceY: Source terms in Y
direction'};
7661         app.YSourceCheckBox.Text = 'Y Source';
7662         app.YSourceCheckBox.Layout.Row = 5;
7663         app.YSourceCheckBox.Layout.Column = [7 8];
7664
7665         % Create BreakingLocationCheckBox
7666         app.BreakingLocationCheckBox = uicontrol(app.GridLayout);
7667         app.BreakingLocationCheckBox.ValueChangedFcn = createCallbackFcn(
7668 (app, @BreakingLocationCheckBoxValueChanged, true);
7669         app.BreakingLocationCheckBox.Tooltip = {'OUT_NU'};
7670         app.BreakingLocationCheckBox.Text = 'Breaking Location';
7671         app.BreakingLocationCheckBox.Layout.Row = 11;
7672         app.BreakingLocationCheckBox.Layout.Column = [7 8];
7673
7674         % Create PMomentumFluxCheckBox
7675         app.PMomentumFluxCheckBox = uicontrol(app.GridLayout);
7676         app.PMomentumFluxCheckBox.ValueChangedFcn = createCallbackFcn(app,
7677 @PMomentumFluxCheckBoxValueChanged, true);
7678         app.PMomentumFluxCheckBox.Tooltip = {'P: Momentum flux in X
direction'};

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7673         app.PMomentumFluxCheckBox.Text = 'P Momentum Flux';
7674         app.PMomentumFluxCheckBox.Layout.Row = 9;
7675         app.PMomentumFluxCheckBox.Layout.Column = [2 4];
7676
7677         % Create QmomentumFluxCheckBox
7678         app.QmomentumFluxCheckBox = ucheckbox(app.GridLayout);
7679         app.QmomentumFluxCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@QmomentumFluxCheckBoxValueChanged, true);
7680         app.QmomentumFluxCheckBox.Tooltip = {'Q: Momentum flux in Y ↵
direction'};
7681         app.QmomentumFluxCheckBox.Text = 'Q momentum Flux';
7682         app.QmomentumFluxCheckBox.Layout.Row = 10;
7683         app.QmomentumFluxCheckBox.Layout.Column = [2 4];
7684
7685         % Create FxFluxCheckBox
7686         app.FxFluxCheckBox = ucheckbox(app.GridLayout);
7687         app.FxFluxCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@FxFluxCheckBoxValueChanged, true);
7688         app.FxFluxCheckBox.Tooltip = {'Flux F in X direction'};
7689         app.FxFluxCheckBox.Text = 'Fx Flux';
7690         app.FxFluxCheckBox.Layout.Row = 12;
7691         app.FxFluxCheckBox.Layout.Column = [2 3];
7692
7693         % Create FyFluxCheckBox
7694         app.FyFluxCheckBox = ucheckbox(app.GridLayout);
7695         app.FyFluxCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@FyFluxCheckBoxValueChanged, true);
7696         app.FyFluxCheckBox.Tooltip = {'Flux F in Y direction'};
7697         app.FyFluxCheckBox.Text = 'Fy Flux';
7698         app.FyFluxCheckBox.Layout.Row = 13;
7699         app.FyFluxCheckBox.Layout.Column = [2 3];
7700
7701         % Create GxFluxCheckBox
7702         app.GxFluxCheckBox = ucheckbox(app.GridLayout);
7703         app.GxFluxCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@GxFluxCheckBoxValueChanged, true);
7704         app.GxFluxCheckBox.Tooltip = {'Gx: Flux G in X direction'};
7705         app.GxFluxCheckBox.Text = 'Gx Flux';
7706         app.GxFluxCheckBox.Layout.Row = 14;
7707         app.GxFluxCheckBox.Layout.Column = [2 3];
7708
7709         % Create GyFluxCheckBox
7710         app.GyFluxCheckBox = ucheckbox(app.GridLayout);
7711         app.GyFluxCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@GyFluxCheckBoxValueChanged, true);
7712         app.GyFluxCheckBox.Tooltip = {'Gy: Flux G in Y direction'};
7713         app.GyFluxCheckBox.Text = 'Gy Flux';
7714         app.GyFluxCheckBox.Layout.Row = 15;
7715         app.GyFluxCheckBox.Layout.Column = [2 3];
7716
7717         % Create BreakingAgeCheckBox
7718         app.BreakingAgeCheckBox = ucheckbox(app.GridLayout);
7719         app.BreakingAgeCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@BreakingAgeCheckBoxValueChanged, true);
7720         app.BreakingAgeCheckBox.Tooltip = {'AGE: Breaking Age'};
7721         app.BreakingAgeCheckBox.Text = 'Breaking Age';
7722         app.BreakingAgeCheckBox.Layout.Row = 10;
7723         app.BreakingAgeCheckBox.Layout.Column = [7 8];

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7724
7725     % Create MaximumVelocityCheckBox
7726     app.MaximumVelocityCheckBox = uicheckbox(app.GridLayout);
7727     app.MaximumVelocityCheckBox.ValueChangedFcn = createCallbackFcn(
7728     (app, @MaximumVelocityCheckBoxValueChanged, true);
7729     app.MaximumVelocityCheckBox.Tooltip = {'Umax: Maximum velocity'};
7730     app.MaximumVelocityCheckBox.Text = 'Maximum Velocity ';
7731     app.MaximumVelocityCheckBox.Layout.Row = 6;
7732     app.MaximumVelocityCheckBox.Layout.Column = [10 11];
7733
7734     % Create UndertowCheckBox
7735     app.UndertowCheckBox = uicheckbox(app.GridLayout);
7736     app.UndertowCheckBox.ValueChangedFcn = createCallbackFcn(app,
7737     @UndertowCheckBoxValueChanged, true);
7738     app.UndertowCheckBox.Tooltip = {'UNDERTOW: Roller-induced extra
7739     undertow flux'};
7740     app.UndertowCheckBox.Text = 'Undertow';
7741     app.UndertowCheckBox.Layout.Row = 12;
7742     app.UndertowCheckBox.Layout.Column = [7 8];
7743
7744     % Create VelocityLabel
7745     app.VelocityLabel = uilabel(app.GridLayout);
7746     app.VelocityLabel.FontSize = 13;
7747     app.VelocityLabel.FontWeight = 'bold';
7748     app.VelocityLabel.FontColor = [0.0314 0.3686 0.6];
7749     app.VelocityLabel.Layout.Row = 1;
7750     app.VelocityLabel.Layout.Column = 10;
7751     app.VelocityLabel.Text = 'Velocity';
7752
7753     % Create WaveHeightLabel
7754     app.WaveHeightLabel = uilabel(app.GridLayout);
7755     app.WaveHeightLabel.FontSize = 13;
7756     app.WaveHeightLabel.FontWeight = 'bold';
7757     app.WaveHeightLabel.FontColor = [0.0314 0.3686 0.6];
7758     app.WaveHeightLabel.Layout.Row = 1;
7759     app.WaveHeightLabel.Layout.Column = [2 3];
7760     app.WaveHeightLabel.Text = 'Wave Height';
7761
7762     % Create SourceLabel
7763     app.SourceLabel = uilabel(app.GridLayout);
7764     app.SourceLabel.FontSize = 13;
7765     app.SourceLabel.FontWeight = 'bold';
7766     app.SourceLabel.FontColor = [0.0314 0.3686 0.6];
7767     app.SourceLabel.Layout.Row = 3;
7768     app.SourceLabel.Layout.Column = 7;
7769     app.SourceLabel.Text = 'Source';
7770
7771     % Create MaskLabel
7772     app.MaskLabel = uilabel(app.GridLayout);
7773     app.MaskLabel.FontSize = 13;
7774     app.MaskLabel.FontWeight = 'bold';
7775     app.MaskLabel.FontColor = [0.0314 0.3686 0.6];
7776     app.MaskLabel.Layout.Row = 8;
7777     app.MaskLabel.Layout.Column = 10;
7778     app.MaskLabel.Text = 'Mask';
7779
7780     % Create FluxLabel
7781     app.FluxLabel = uilabel(app.GridLayout);

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7779     app.FluxLabel.FontSize = 13;
7780     app.FluxLabel.FontWeight = 'bold';
7781     app.FluxLabel.FontColor = [0.0314 0.3686 0.6];
7782     app.FluxLabel.Layout.Row = 8;
7783     app.FluxLabel.Layout.Column = 2;
7784     app.FluxLabel.Text = 'Flux';
7785
7786     % Create TimeLabel
7787     app.TimeLabel = uilabel(app.GridLayout);
7788     app.TimeLabel.FontSize = 13;
7789     app.TimeLabel.FontWeight = 'bold';
7790     app.TimeLabel.FontColor = [0.0314 0.3686 0.6];
7791     app.TimeLabel.Layout.Row = 1;
7792     app.TimeLabel.Layout.Column = 7;
7793     app.TimeLabel.Text = 'Time';
7794
7795     % Create OthersLabel
7796     app.OthersLabel = uilabel(app.GridLayout);
7797     app.OthersLabel.FontSize = 13;
7798     app.OthersLabel.FontWeight = 'bold';
7799     app.OthersLabel.FontColor = [0.0314 0.3686 0.6];
7800     app.OthersLabel.Layout.Row = 8;
7801     app.OthersLabel.Layout.Column = 7;
7802     app.OthersLabel.Text = 'Others';
7803
7804     % Create GenerateinputtxtButton_2
7805     app.GenerateinputtxtButton_2 = uibutton(app.GridLayout, 'push');
7806     app.GenerateinputtxtButton_2.ButtonPushedFcn = createCallbackFcn(
7807 (app, @GenerateinputtxtButton_2Pushed, true);
7807     app.GenerateinputtxtButton_2.HandleVisibility = 'off';
7808     app.GenerateinputtxtButton_2.BackgroundColor = [0.9216 0.9216
0.9216];
7809     app.GenerateinputtxtButton_2.FontSize = 15;
7810     app.GenerateinputtxtButton_2.FontWeight = 'bold';
7811     app.GenerateinputtxtButton_2.Layout.Row = 18;
7812     app.GenerateinputtxtButton_2.Layout.Column = [4 10];
7813     app.GenerateinputtxtButton_2.Text = 'Generate input.txt';
7814
7815     % Create UVelocitymeanCheckBox
7816     app.UVelocitymeanCheckBox = uicheckbox(app.GridLayout);
7817     app.UVelocitymeanCheckBox.Tooltip = {'Umean: Mean velocity in X
direction'};
7818     app.UVelocitymeanCheckBox.Text = 'U Velocity (mean)';
7819     app.UVelocitymeanCheckBox.Layout.Row = 4;
7820     app.UVelocitymeanCheckBox.Layout.Column = [10 11];
7821
7822     % Create VVelocitymeanCheckBox
7823     app.VVelocitymeanCheckBox = uicheckbox(app.GridLayout);
7824     app.VVelocitymeanCheckBox.Tooltip = {'Vmean: Mean velocity in Y
direction'};
7825     app.VVelocitymeanCheckBox.Text = 'V Velocity (mean)';
7826     app.VVelocitymeanCheckBox.Layout.Row = 5;
7827     app.VVelocitymeanCheckBox.Layout.Column = [10 11];
7828
7829     % Create MinimumWaveHeightCheckBox
7830     app.MinimumWaveHeightCheckBox = uicheckbox(app.GridLayout);
7831     app.MinimumWaveHeightCheckBox.ValueChangedFcn = createCallbackFcn(
7832 (app, @MinimumWaveHeightCheckBoxValueChanged, true);

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7832         app.MinimumWaveHeightCheckBox.Tooltip = {'Hmin: Minimum height
within the total simulation time'};
7833         app.MinimumWaveHeightCheckBox.Text = 'Minimum Wave Height';
7834         app.MinimumWaveHeightCheckBox.Layout.Row = 3;
7835         app.MinimumWaveHeightCheckBox.Layout.Column = [2 5];
7836
7837         % Create MaximumWaveHeightCheckBox
7838         app.MaximumWaveHeightCheckBox = uicheckbox(app.GridLayout);
7839         app.MaximumWaveHeightCheckBox.ValueChangedFcn = createCallbackFcn
(app, @MaximumWaveHeightCheckBoxValueChanged, true);
7840         app.MaximumWaveHeightCheckBox.Tooltip = {'Hmax: Maximum height
within the total simulation time'};
7841         app.MaximumWaveHeightCheckBox.Text = 'Maximum Wave Height';
7842         app.MaximumWaveHeightCheckBox.Layout.Row = 2;
7843         app.MaximumWaveHeightCheckBox.Layout.Column = [2 5];
7844         app.MaximumWaveHeightCheckBox.Value = true;
7845
7846         % Create ThresholdEditFieldLabel
7847         app.ThresholdEditFieldLabel = uilabel(app.GridLayout);
7848         app.ThresholdEditFieldLabel.VerticalAlignment = 'bottom';
7849         app.ThresholdEditFieldLabel.Visible = 'off';
7850         app.ThresholdEditFieldLabel.Layout.Row = 1;
7851         app.ThresholdEditFieldLabel.Layout.Column = 8;
7852         app.ThresholdEditFieldLabel.Text = 'Threshold';
7853
7854         % Create ThresholdEditField
7855         app.ThresholdEditField = uieditfield(app.GridLayout, 'numeric');
7856         app.ThresholdEditField.Limits = [0.0001 Inf];
7857         app.ThresholdEditField.ValueChangedFcn = createCallbackFcn(app,
@ThresholdEditFieldValueChanged, true);
7858         app.ThresholdEditField.FontColor = [0.651 0.651 0.651];
7859         app.ThresholdEditField.Visible = 'off';
7860         app.ThresholdEditField.Tooltip = {'Threshold to record the arrival
time of the tsunami wave'; ''; 'Unit: meters'};
7861         app.ThresholdEditField.Layout.Row = 2;
7862         app.ThresholdEditField.Layout.Column = 8;
7863         app.ThresholdEditField.Value = 0.0001;
7864
7865         % Create Panel_32
7866         app.Panel_32 = uipanel(app.UIFigure);
7867         app.Panel_32.SizeChangedFcn = createCallbackFcn(app,
@Panel_32SizeChanged, true);
7868         app.Panel_32.Position = [744 11 398 100];
7869
7870         % Create CreateMakefileButton
7871         app.CreateMakefileButton = uibutton(app.Panel_32, 'push');
7872         app.CreateMakefileButton.ButtonPushedFcn = createCallbackFcn(app,
@CreateMakefileButtonPushed, true);
7873         app.CreateMakefileButton.VerticalAlignment = 'top';
7874         app.CreateMakefileButton.BackgroundColor = [0.9216 0.9216 0.9216];
7875         app.CreateMakefileButton.FontWeight = 'bold';
7876         app.CreateMakefileButton.Enable = 'off';
7877         app.CreateMakefileButton.Tooltip = {''};
7878         app.CreateMakefileButton.Position = [277 40 106 22];
7879         app.CreateMakefileButton.Text = 'Create Makefile';
7880
7881         % Create StartsimulationButton
7882         app.StartsimulationButton = uibutton(app.Panel_32, 'push');

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7883         app.StartsimulationButton.ButtonPushedFcn = createCallbackFcn(app, ↵
@StartsimulationButtonPushed, true);
7884         app.StartsimulationButton.BackgroundColor = [0.9216 0.9216 ↵
0.9216];
7885         app.StartsimulationButton.FontWeight = 'bold';
7886         app.StartsimulationButton.Enable = 'off';
7887         app.StartsimulationButton.Tooltip = {'Compile the Makefile to ↵
create 'funwave.exe' and start the tsunami simulation.'};
7888         app.StartsimulationButton.Position = [277 14 107 22];
7889         app.StartsimulationButton.Text = 'Start simulation';
7890
7891         % Create Button_42
7892         app.Button_42 = uibutton(app.Panel_32, 'push');
7893         app.Button_42.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_42Pushed2, true);
7894         app.Button_42.VerticalAlignment = 'top';
7895         app.Button_42.FontSize = 10;
7896         app.Button_42.FontAngle = 'italic';
7897         app.Button_42.Enable = 'off';
7898         app.Button_42.Tooltip = {''};
7899         app.Button_42.Position = [250 41 19 18];
7900         app.Button_42.Text = '...';
7901
7902         % Create MakefileEditFieldLabel
7903         app.MakefileEditFieldLabel = uilabel(app.Panel_32);
7904         app.MakefileEditFieldLabel.Enable = 'off';
7905         app.MakefileEditFieldLabel.Tooltip = {'Specify the folder ↵
directory containing all the .o, .F, and .f90 files'};
7906         app.MakefileEditFieldLabel.Position = [15 39 106 22];
7907         app.MakefileEditFieldLabel.Text = 'Source folder';
7908
7909         % Create MakefileEditField
7910         app.MakefileEditField = uieditfield(app.Panel_32, 'text');
7911         app.MakefileEditField.FontSize = 10;
7912         app.MakefileEditField.FontColor = [0.502 0.502 0.502];
7913         app.MakefileEditField.Enable = 'off';
7914         app.MakefileEditField.Tooltip = {''};
7915         app.MakefileEditField.Placeholder = 'FUNWAVE-TVD-Version_3.6/src';
7916         app.MakefileEditField.Position = [94 42 154 16];
7917
7918         % Create ParallelModeCheckBox
7919         app.ParallelModeCheckBox = uicheckbox(app.Panel_32);
7920         app.ParallelModeCheckBox.Enable = 'off';
7921         app.ParallelModeCheckBox.Tooltip = {'Use parallel mode'; ''}; ↵
'Unchecked box - serial mode'};
7922         app.ParallelModeCheckBox.Text = 'Parallel Mode';
7923         app.ParallelModeCheckBox.Position = [158 66 95 22];
7924
7925         % Create IntelCompilerCheckBox
7926         app.IntelCompilerCheckBox = uicheckbox(app.Panel_32);
7927         app.IntelCompilerCheckBox.Enable = 'off';
7928         app.IntelCompilerCheckBox.Tooltip = {'Use intel compiler'};
7929         app.IntelCompilerCheckBox.Text = 'Intel Compiler';
7930         app.IntelCompilerCheckBox.Position = [271 66 97 22];
7931
7932         % Create DoublePrecisionCheckBox
7933         app.DoublePrecisionCheckBox = uicheckbox(app.Panel_32);
7934         app.DoublePrecisionCheckBox.ValueChangedFcn = createCallbackFcn ↵

```

```

(app, @DoublePrecisionCheckBoxValueChanged, true);
7935         app.DoublePrecisionCheckBox.Enable = 'off';
7936         app.DoublePrecisionCheckBox.Tooltip = {'Use double precision'; ''};
'Unchecked box - single precision';
7937         app.DoublePrecisionCheckBox.Text = 'Double Precision';
7938         app.DoublePrecisionCheckBox.Position = [26 66 113 22];
7939
7940         % Create Mpif90EditFieldLabel
7941         app.Mpif90EditFieldLabel = uilabel(app.Panel_32);
7942         app.Mpif90EditFieldLabel.Enable = 'off';
7943         app.Mpif90EditFieldLabel.Tooltip = {'Specify the directory
containing the MPI F90 Compiler.'; ''; 'Mac OS default: '/opt/homebrew/bin/'; '';
'Windows OS: 'C:\Program Files (x86)\MPICH2\bin'};
7944         app.Mpif90EditFieldLabel.Position = [15 13 77 22];
7945         app.Mpif90EditFieldLabel.Text = 'Mpif90 folder';
7946
7947         % Create Mpif90EditField
7948         app.Mpif90EditField = uieditfield(app.Panel_32, 'text');
7949         app.Mpif90EditField.ValueChangedFcn = createCallbackFcn(app,
@Mpif90EditFieldValueChanged, true);
7950         app.Mpif90EditField.ValueChangingFcn = createCallbackFcn(app,
@Mpif90EditFieldValueChanging, true);
7951         app.Mpif90EditField.FontSize = 11;
7952         app.Mpif90EditField.FontColor = [0.502 0.502 0.502];
7953         app.Mpif90EditField.Enable = 'off';
7954         app.Mpif90EditField.Tooltip = {' '};
7955         app.Mpif90EditField.Placeholder = 'C:\Program Files (x86)
\MPICH2\bin\';
7956         app.Mpif90EditField.Position = [94 16 154 16];
7957
7958         % Create Button_41
7959         app.Button_41 = uibutton(app.Panel_32, 'push');
7960         app.Button_41.ButtonPushedFcn = createCallbackFcn(app,
@Button_41Pushed, true);
7961         app.Button_41.VerticalAlignment = 'top';
7962         app.Button_41.FontAngle = 'italic';
7963         app.Button_41.Enable = 'off';
7964         app.Button_41.Tooltip = {' '};
7965         app.Button_41.Position = [250 15 19 19];
7966         app.Button_41.Text = '...';
7967
7968         % Create Friction_____Label_9
7969         app.Friction_____Label_9 = uilabel(app.
UIFigure);
7970         app.Friction_____Label_9.
BackgroundColor = [0.9412 0.9412 0.9412];
7971         app.Friction_____Label_9.FontWeight =
'bold';
7972         app.Friction_____Label_9.Tooltip =
{'This section becomes enabled once Makefile-related sections on the left panel are
completed.'};
7973         app.Friction_____Label_9.Position =
[758 100 67 22];
7974         app.Friction_____Label_9.Text = '
Makefile '
7975
7976         % Show the figure after all components are created
7977         app.UIFigure.Visible = 'on';

```

```

7978         end
7979     end
7980
7981     % App creation and deletion
7982     methods (Access = public)
7983
7984         % Construct app
7985         function app = FUNMAP_Input
7986             runningApp = getRunningApp(app);
7987
7988             % Check for running singleton app
7989             if isempty(runningApp)
7990
7991                 % Create UIFigure and components
7992                 createComponents(app)
7993
7994                 % Register the app with App Designer
7995                 registerApp(app, app.UIFigure)
7996
7997                 % Execute the startup function
7998                 runStartupFcn(app, @startupFcn)
7999             else
8000
8001                 % Focus the running singleton app
8002                 figure(runningApp.UIFigure)
8003
8004                 app = runningApp;
8005             end
8006
8007             if nargin == 0
8008                 clear app
8009             end
8010         end
8011
8012         % Code that executes before app deletion
8013         function delete(app)
8014
8015             % Delete UIFigure when app is deleted
8016             delete(app.UIFigure)
8017         end
8018     end
8019 end

```